US Patent & Trademark Office
Patent Public Search | Text View

United States Patent	12414967
Kind Code	B2
Date of Patent	September 16, 2025
Inventor(s)	Takebe; Takanori et al.

Compositions and methods of treating liver disease

Abstract

Disclosed are methods of treating or reducing the occurrence of a steatohepatitis disorder. The disorder may include, for example, NASH, parenteral nutrition associated liver disease (PNALD), or genetic forms of liver disease. The method may comprise the step of administering a composition comprising obeticholic acid to an individual in need thereof.

Inventors: **Takebe; Takanori (Cincinnati, OH), Ouchi; Rie (Cincinnati, OH)**

Applicant: **Children's Hospital Medical Center** (Cincinnati, OH)

Family ID: **1000008821197**

Assignee: **Children's Hospital Medical Center (Cincinnati, OH)**

Appl. No.: **16/857338**

Filed: **April 24, 2020**

Prior Publication Data

Document Identifier	Publication Date
US 20210008123 A1	Jan. 14, 2021

Related U.S. Application Data

continuation parent-doc US 16343157 US 10668108 WO PCT/US2017/059865 20171103 child-doc US 16857338
us-provisional-application US 62517414 20170609
us-provisional-application US 62417371 20161104

Publication Classification

Int. Cl.: **A61K35/407** (20150101); **A61K9/00** (20060101); **A61K31/575** (20060101); **A61K45/06** (20060101); **A61P1/16** (20060101); **C12N5/071** (20100101); **C12N5/074** (20100101); **C12N15/01** (20060101); **G01N33/50** (20060101)

U.S. Cl.:

CPC **A61K35/407** (20130101); **A61K9/0029** (20130101); **A61K31/575** (20130101); **A61P1/16** (20180101); **C12N5/0671** (20130101); **C12N5/0672** (20130101); **C12N5/0696** (20130101); **C12N5/0697** (20130101); **C12N15/01** (20130101); **G01N33/5008** (20130101); A61K45/06 (20130101); C12N2500/36 (20130101); C12N2501/119 (20130101); C12N2506/45 (20130101)

Field of Classification Search

CPC: A61K (35/407); A61K (9/0029); A61K (31/575); A61K (45/06); A61P (1/16); C12N (5/0671); C12N (5/0672); C12N (5/0696); C12N (5/0697); C12N (15/01); C12N (2500/36); C12N (2501/119); C12N (2506/45); C12N (2500/38); C12N (2501/727); C12N (2533/54); G01N (33/5008); G01N (33/5082); G01N (2800/52)

USPC: 514/169

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
5912227	12/1998	Croom, Jr. et al.	N/A	N/A
5942435	12/1998	Wheeler	N/A	N/A
6607501	12/2002	Gorsuch	N/A	N/A
7160719	12/2006	Nyberg	N/A	N/A
7291626	12/2006	Beachy et al.	N/A	N/A
7326572	12/2007	Fisk et al.	N/A	N/A
7510876	12/2008	D'Amour et al.	N/A	N/A

7514185	12/2008	Fukushima et al.	N/A	N/A
7541185	12/2008	D'Amour et al.	N/A	N/A
7625753	12/2008	Kelly et al.	N/A	N/A
7695958	12/2009	Funatsu et al.	N/A	N/A
7704738	12/2009	D'Amour et al.	N/A	N/A
7727998	12/2009	Moriya et al.	N/A	N/A
7776592	12/2009	Wandinger-Ness et al.	N/A	N/A
7927869	12/2010	Rosero	N/A	N/A
7985585	12/2010	D'Amour et al.	N/A	N/A
7993916	12/2010	Agulnick et al.	N/A	N/A
8187878	12/2011	Dalton et al.	N/A	N/A
8216826	12/2011	Lee et al.	N/A	N/A
8216836	12/2011	D'Amour et al.	N/A	N/A
8298822	12/2011	Kruse et al.	N/A	N/A
8318492	12/2011	Choo et al.	N/A	N/A
8501476	12/2012	Morgan et al.	N/A	N/A
8586357	12/2012	D'Amour et al.	N/A	N/A
8603809	12/2012	Kruse	N/A	N/A
8609406	12/2012	Subramanian et al.	N/A	N/A
8609413	12/2012	Suter et al.	N/A	N/A
8623645	12/2013	D'Amour et al.	N/A	N/A
8632645	12/2013	Daitou et al.	N/A	N/A
8633024	12/2013	D'Amour et al.	N/A	N/A
8642339	12/2013	Sato et al.	N/A	N/A
8647873	12/2013	D'Amour et al.	N/A	N/A
8658151	12/2013	Kelly et al.	N/A	N/A
8685386	12/2013	West et al.	N/A	N/A
8685730	12/2013	Odorico et al.	N/A	N/A
8758323	12/2013	Michaud et al.	N/A	N/A
9127254	12/2014	Cohen et al.	N/A	N/A
9133439	12/2014	Davis et al.	N/A	N/A
9181301	12/2014	Carlson et al.	N/A	N/A
9200258	12/2014	Mezghanni et al.	N/A	N/A
9206393	12/2014	Kruse	N/A	N/A
9234170	12/2015	Snoeck et al.	N/A	N/A
9334479	12/2015	Herrera Sanchez et al.	N/A	N/A
9375514	12/2015	Kruse et al.	N/A	N/A
9381181	12/2015	Roberts et al.	N/A	N/A
9394522	12/2015	Brolen et al.	N/A	N/A
9446076	12/2015	Gaussin et al.	N/A	N/A
9447380	12/2015	Subramanian et al.	N/A	N/A
9476030	12/2015	Gadue et al.	N/A	N/A
9499795	12/2015	D'Amour et al.	N/A	N/A
9605243	12/2016	D'Amour et al.	N/A	N/A
9616039	12/2016	Roberts et al.	N/A	N/A
9618500	12/2016	Giselbrecht et al.	N/A	N/A
9650609	12/2016	Nyberg	N/A	N/A
9675646	12/2016	Bitar	N/A	N/A
9677085	12/2016	Guye et al.	N/A	N/A
9719067	12/2016	Snoeck et al.	N/A	N/A
9719068	12/2016	Wells et al.	N/A	N/A
9732116	12/2016	Steiner et al.	N/A	N/A
9752124	12/2016	Sato et al.	N/A	N/A
9763964	12/2016	Pellicciari et al.	N/A	N/A
9765301	12/2016	Huch Ortega et al.	N/A	N/A
9771562	12/2016	Shen et al.	N/A	N/A
9790470	12/2016	Vallier et al.	N/A	N/A
9828583	12/2016	Rajagopal et al.	N/A	N/A
9849104	12/2016	Bisgaier et al.	N/A	N/A
9850461	12/2016	Rizzi et al.	N/A	N/A
9856458	12/2017	Rosowski et al.	N/A	N/A
9878005	12/2017	Johns et al.	N/A	N/A
9914920	12/2017	Goodwin et al.	N/A	N/A
9926532	12/2017	Esteban et al.	N/A	N/A
9938499	12/2017	Slukvin et al.	N/A	N/A
10000740	12/2017	Vallier et al.	N/A	N/A
10023922	12/2017	Stelzer et al.	N/A	N/A
10045977	12/2017	Wu et al.	N/A	N/A
10047341	12/2017	Yu et al.	N/A	N/A
10052337	12/2017	Lancaster et al.	N/A	N/A
10087416	12/2017	Chan et al.	N/A	N/A
10087417	12/2017	Freed et al.	N/A	N/A
10100279	12/2017	Nicholas et al.	N/A	N/A
10130748	12/2017	Nyberg et al.	N/A	N/A
10172889	12/2018	Sokal et al.	N/A	N/A

10174289	12/2018	Wells et al.	N/A	N/A
10179176	12/2018	Kay et al.	N/A	N/A
10220386	12/2018	Williamson et al.	N/A	N/A
10222370	12/2018	Keshavarzian et al.	N/A	N/A
10260039	12/2018	Bhatia et al.	N/A	N/A
10265153	12/2018	La Francesca et al.	N/A	N/A
10265453	12/2018	Flieg et al.	N/A	N/A
10301303	12/2018	Liu	N/A	N/A
10350147	12/2018	Kyrkanides et al.	N/A	N/A
10369254	12/2018	Yanagawa et al.	N/A	N/A
10407664	12/2018	Knoblich et al.	N/A	N/A
10426757	12/2018	Sabatini et al.	N/A	N/A
10449221	12/2018	Kotton et al.	N/A	N/A
10472612	12/2018	Ingber et al.	N/A	N/A
10479977	12/2018	Wang et al.	N/A	N/A
10487314	12/2018	Accili et al.	N/A	N/A
10532111	12/2019	Kay et al.	N/A	N/A
10538741	12/2019	Sokal et al.	N/A	N/A
10545133	12/2019	Ewald et al.	N/A	N/A
10555929	12/2019	Mantzoros	N/A	N/A
10668108	12/2019	Takebe et al.	N/A	N/A
10781425	12/2019	Wells et al.	N/A	N/A
11053477	12/2020	Wells et al.	N/A	N/A
11066650	12/2020	Wells et al.	N/A	N/A
2003/0129751	12/2002	Grikscheit et al.	N/A	N/A
2003/0228685	12/2002	Nyberg	N/A	N/A
2005/0266554	12/2004	D'Amour et al.	N/A	N/A
2006/0110369	12/2005	Funatsu et al.	N/A	N/A
2006/0236415	12/2005	Silversides et al.	N/A	N/A
2007/0238169	12/2006	Abilez et al.	N/A	N/A
2007/0239083	12/2006	Voss	N/A	N/A
2008/0193421	12/2007	Kruse et al.	N/A	N/A
2008/0195224	12/2007	Teitelbaum et al.	N/A	N/A
2008/0286366	12/2007	Fischer et al.	N/A	N/A
2009/0011502	12/2008	D'Amour et al.	N/A	N/A
2009/0042287	12/2008	D'Amour et al.	N/A	N/A
2009/0220959	12/2008	D'Amour et al.	N/A	N/A
2009/0253202	12/2008	D'Amour et al.	N/A	N/A
2009/0263357	12/2008	Sayre et al.	N/A	N/A
2009/0311765	12/2008	Maguire et al.	N/A	N/A
2010/0016410	12/2009	Wagner et al.	N/A	N/A
2010/0041150	12/2009	Kelly et al.	N/A	N/A
2010/0048871	12/2009	Cho et al.	N/A	N/A
2010/0075295	12/2009	Dryden et al.	N/A	N/A
2010/0151568	12/2009	D'Amour et al.	N/A	N/A
2011/0125286	12/2010	Selden et al.	N/A	N/A
2011/0151564	12/2010	Menu et al.	N/A	N/A
2011/0218512	12/2010	Tullis et al.	N/A	N/A
2011/0231942	12/2010	He et al.	N/A	N/A
2011/0294735	12/2010	Marsh et al.	N/A	N/A
2011/0300543	12/2010	Wang	N/A	N/A
2012/0009086	12/2011	Nyberg et al.	N/A	N/A
2012/0009618	12/2011	Yu et al.	N/A	N/A
2012/0070419	12/2011	Christiansen-Weber	N/A	N/A
2012/0071451	12/2011	Spenard et al.	N/A	N/A
2012/0135519	12/2011	Ameri et al.	N/A	N/A
2012/0149630	12/2011	Zugates et al.	N/A	N/A
2012/0196275	12/2011	Mezghanni et al.	N/A	N/A
2012/0196312	12/2011	Sato et al.	N/A	N/A
2012/0201890	12/2011	Williams et al.	N/A	N/A
2012/0264209	12/2011	Odorico et al.	N/A	N/A
2012/0270295	12/2011	Choo et al.	N/A	N/A
2012/0291096	12/2011	Boldyrev et al.	N/A	N/A
2013/0031645	12/2012	Touboul et al.	N/A	N/A
2013/0095567	12/2012	Brolen et al.	N/A	N/A
2013/0115673	12/2012	West et al.	N/A	N/A
2013/0137130	12/2012	Wells et al.	N/A	N/A
2013/0189327	12/2012	Ortega et al.	N/A	N/A
2013/0217005	12/2012	Snoeck et al.	N/A	N/A
2013/0281374	12/2012	Levy et al.	N/A	N/A
2013/0316442	12/2012	Meurville et al.	N/A	N/A
2013/0330823	12/2012	Rezania	N/A	N/A
2014/0038279	12/2013	Ingber et al.	N/A	N/A
2014/0044713	12/2013	De Lau et al.	N/A	N/A
2014/0141509	12/2013	Gadue et al.	N/A	N/A

2014/0193905	12/2013	Kelly et al.	N/A	N/A
2014/0212910	12/2013	Bhatia et al.	N/A	N/A
2014/0234953	12/2013	Vacanti et al.	N/A	N/A
2014/0242693	12/2013	Fryer et al.	N/A	N/A
2014/0243227	12/2013	Clevers et al.	N/A	N/A
2014/0273210	12/2013	Baker et al.	N/A	N/A
2014/0302491	12/2013	Nadauld et al.	N/A	N/A
2014/0308695	12/2013	Bruce et al.	N/A	N/A
2014/0328808	12/2013	Watanabe et al.	N/A	N/A
2014/0336282	12/2013	Ewald et al.	N/A	N/A
2014/0369973	12/2013	Bernstein et al.	N/A	N/A
2015/0017140	12/2014	Bhatia et al.	N/A	N/A
2015/0151297	12/2014	Williamson et al.	N/A	N/A
2015/0153326	12/2014	Kogel et al.	N/A	N/A
2015/0185714	12/2014	Geveci	N/A	N/A
2015/0197802	12/2014	Zink et al.	N/A	N/A
2015/0201588	12/2014	Kamb et al.	N/A	N/A
2015/0238656	12/2014	Orlando et al.	N/A	N/A
2015/0247124	12/2014	Snoeck et al.	N/A	N/A
2015/0273071	12/2014	Green et al.	N/A	N/A
2015/0273127	12/2014	Flieg et al.	N/A	N/A
2015/0290154	12/2014	Roberts et al.	N/A	N/A
2015/0330970	12/2014	Knoblich et al.	N/A	N/A
2015/0343018	12/2014	Sansonetti et al.	N/A	N/A
2015/0359849	12/2014	Greenberg et al.	N/A	N/A
2015/0361393	12/2014	Nicholas et al.	N/A	N/A
2016/0002602	12/2015	Almeida-Porada et al.	N/A	N/A
2016/0022873	12/2015	Besner et al.	N/A	N/A
2016/0046905	12/2015	Inoue et al.	N/A	N/A
2016/0060707	12/2015	Goldenberg et al.	N/A	N/A
2016/0068805	12/2015	Martin et al.	N/A	N/A
2016/0101133	12/2015	Basu et al.	N/A	N/A
2016/0102289	12/2015	Yu et al.	N/A	N/A
2016/0121023	12/2015	Edelman et al.	N/A	N/A
2016/0122722	12/2015	Ejiri et al.	N/A	N/A
2016/0143949	12/2015	Ingber et al.	N/A	N/A
2016/0177270	12/2015	Takebe et al.	N/A	N/A
2016/0184387	12/2015	Charmot et al.	N/A	N/A
2016/0186140	12/2015	Dalton et al.	N/A	N/A
2016/0206664	12/2015	Sokal et al.	N/A	N/A
2016/0215014	12/2015	Steiner	N/A	A61K 9/2054
2016/0237400	12/2015	Xian	N/A	N/A
2016/0237401	12/2015	Vallier et al.	N/A	N/A
2016/0237409	12/2015	Little et al.	N/A	N/A
2016/0244724	12/2015	Ferro	N/A	N/A
2016/0245653	12/2015	Park et al.	N/A	N/A
2016/0256672	12/2015	Arumugaswami et al.	N/A	N/A
2016/0257937	12/2015	Wauthier et al.	N/A	N/A
2016/0263098	12/2015	Mantzoros	N/A	N/A
2016/0289635	12/2015	Sasai et al.	N/A	N/A
2016/0296599	12/2015	Dinh et al.	N/A	N/A
2016/0298087	12/2015	Qu et al.	N/A	N/A
2016/0312181	12/2015	Freed et al.	N/A	N/A
2016/0312190	12/2015	Ghaedi et al.	N/A	N/A
2016/0312191	12/2015	Spence et al.	N/A	N/A
2016/0319240	12/2015	Chan et al.	N/A	N/A
2016/0340645	12/2015	D'Amour et al.	N/A	N/A
2016/0340749	12/2015	Stelzer et al.	N/A	N/A
2016/0354408	12/2015	Hariri et al.	N/A	N/A
2016/0361466	12/2015	Yanagawa et al.	N/A	N/A
2016/0376557	12/2015	Dubart Kupperschmitt et al.	N/A	N/A
2017/0002330	12/2016	Vunjak-Novakovic et al.	N/A	N/A
2017/0027994	12/2016	Kotton et al.	N/A	N/A
2017/0035661	12/2016	Kyrkanides et al.	N/A	N/A
2017/0035784	12/2016	Lancaster et al.	N/A	N/A
2017/0037043	12/2016	Liu	N/A	N/A
2017/0067014	12/2016	Takebe et al.	N/A	N/A
2017/0101628	12/2016	Ingber et al.	N/A	N/A
2017/0107469	12/2016	Costa et al.	N/A	N/A
2017/0107483	12/2016	Pendergraft et al.	N/A	N/A
2017/0107498	12/2016	Sareen et al.	N/A	N/A
2017/0128625	12/2016	Bhatia et al.	N/A	N/A
2017/0151049	12/2016	La Francesca et al.	N/A	N/A
2017/0152486	12/2016	Shen et al.	N/A	N/A
2017/0152528	12/2016	Zhang	N/A	N/A

2017/0184569	12/2016	Keshavarzian et al.	N/A	N/A
2017/0191030	12/2016	Huch Ortega et al.	N/A	N/A
2017/0198261	12/2016	Sabaawy et al.	N/A	N/A
2017/0202885	12/2016	Agulnick	N/A	N/A
2017/0204375	12/2016	Accili et al.	N/A	N/A
2017/0205396	12/2016	Izpisua Belmonte et al.	N/A	N/A
2017/0205398	12/2016	Bruce et al.	N/A	N/A
2017/0239262	12/2016	Lefebvre	N/A	N/A
2017/0240863	12/2016	Sokal et al.	N/A	N/A
2017/0240866	12/2016	Wells et al.	N/A	N/A
2017/0240964	12/2016	Leung et al.	N/A	N/A
2017/0242035	12/2016	Kiehntopf et al.	N/A	N/A
2017/0258772	12/2016	Sabatini et al.	N/A	N/A
2017/0260501	12/2016	Semechkin et al.	N/A	N/A
2017/0260509	12/2016	Hung et al.	N/A	N/A
2017/0266145	12/2016	Nahmias et al.	N/A	N/A
2017/0267970	12/2016	Gupta et al.	N/A	N/A
2017/0267977	12/2016	Huang et al.	N/A	N/A
2017/0275592	12/2016	Sachs et al.	N/A	N/A
2017/0285002	12/2016	Taniguchi et al.	N/A	N/A
2017/0292116	12/2016	Wells et al.	N/A	N/A
2017/0296621	12/2016	Sansonetti et al.	N/A	N/A
2017/0304294	12/2016	Wu et al.	N/A	N/A
2017/0304369	12/2016	Ang et al.	N/A	N/A
2017/0319548	12/2016	Lefebvre	N/A	N/A
2017/0321188	12/2016	Viczian et al.	N/A	N/A
2017/0321191	12/2016	Kojima	N/A	N/A
2017/0335283	12/2016	Wang et al.	N/A	N/A
2017/0342385	12/2016	Sachs et al.	N/A	N/A
2017/0348433	12/2016	Kay et al.	N/A	N/A
2017/0349659	12/2016	Garcia et al.	N/A	N/A
2017/0349884	12/2016	Karp et al.	N/A	N/A
2017/0360962	12/2016	Kay et al.	N/A	N/A
2017/0362573	12/2016	Wells et al.	N/A	N/A
2017/0362574	12/2016	Sareen et al.	N/A	N/A
2018/0021341	12/2017	Harriman et al.	N/A	N/A
2018/0030409	12/2017	Lewis et al.	N/A	N/A
2018/0042970	12/2017	Rossen et al.	N/A	N/A
2018/0043357	12/2017	Bocchi et al.	N/A	N/A
2018/0059119	12/2017	Tak et al.	N/A	N/A
2018/0112187	12/2017	Smith et al.	N/A	N/A
2018/0142206	12/2017	Kime et al.	N/A	N/A
2018/0171302	12/2017	Accili	N/A	N/A
2018/0179496	12/2017	Rajesh et al.	N/A	N/A
2018/0193421	12/2017	Soula	N/A	N/A
2018/0250410	12/2017	Borros Gomez et al.	N/A	N/A
2018/0258400	12/2017	Ng et al.	N/A	N/A
2018/0344901	12/2017	Spence et al.	N/A	N/A
2019/0031992	12/2018	Kerns et al.	N/A	N/A
2019/0078055	12/2018	Wells et al.	N/A	N/A
2019/0093076	12/2018	Schulz	N/A	N/A
2019/0153395	12/2018	Barrett et al.	N/A	N/A
2019/0153397	12/2018	Wells et al.	N/A	N/A
2019/0298775	12/2018	Takebe et al.	N/A	N/A
2019/0300849	12/2018	Carpino et al.	N/A	N/A
2019/0314387	12/2018	Takebe et al.	N/A	N/A
2019/0367882	12/2018	Wells et al.	N/A	N/A
2020/0040309	12/2019	Takebe et al.	N/A	N/A
2020/0056157	12/2019	Takebe et al.	N/A	N/A
2020/0102543	12/2019	Okazaki et al.	N/A	N/A
2020/0149004	12/2019	Spence et al.	N/A	N/A
2020/0190478	12/2019	Wells et al.	N/A	N/A
2020/0199537	12/2019	Takebe et al.	N/A	N/A
2020/0199538	12/2019	Ng et al.	N/A	N/A
2021/0030811	12/2020	Kim et al.	N/A	N/A
2021/0096126	12/2020	Takebe et al.	N/A	N/A
2021/0115366	12/2020	Mahe et al.	N/A	N/A
2021/0180026	12/2020	Takebe et al.	N/A	N/A
2021/0189349	12/2020	Wells et al.	N/A	N/A
2021/0292714	12/2020	Takebe et al.	N/A	N/A
2021/0324334	12/2020	Takebe et al.	N/A	N/A
2021/0363490	12/2020	Yoshihara et al.	N/A	N/A
2021/0371815	12/2020	Holloway et al.	N/A	N/A
2021/0395695	12/2020	Kim et al.	N/A	N/A
2022/0041684	12/2021	Patterson	N/A	N/A

2022/0056420	12/2021	Wells et al.	N/A	N/A
2022/0090011	12/2021	Ngan et al.	N/A	N/A
2022/0275345	12/2021	Mayhew et al.	N/A	N/A

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
2968065	12/2015	CA	N/A
1299408	12/2000	CN	N/A
101600461	12/2008	CN	N/A
101855554	12/2009	CN	N/A
102307990	12/2011	CN	N/A
102439135	12/2011	CN	N/A
102459574	12/2011	CN	N/A
102740888	12/2011	CN	N/A
103068970	12/2012	CN	N/A
103154237	12/2012	CN	N/A
103561751	12/2013	CN	N/A
104387451	12/2014	CN	N/A
104995294	12/2014	CN	N/A
105209605	12/2014	CN	N/A
105985395	12/2015	CN	N/A
109415685	12/2018	CN	N/A
110371967	12/2018	CN	N/A
110381967	12/2018	CN	N/A
110582564	12/2018	CN	N/A
1063289	12/1999	EP	N/A
2393917	12/2010	EP	N/A
2412800	12/2011	EP	N/A
2393917	12/2015	EP	N/A
3228306	12/2016	EP	N/A
2003521673	12/2002	JP	N/A
2004166717	12/2003	JP	N/A
2008503203	12/2007	JP	N/A
2008505638	12/2007	JP	N/A
2008539697	12/2007	JP	N/A
2012516685	12/2011	JP	N/A
2012254081	12/2011	JP	N/A
2013066414	12/2012	JP	N/A
2013511969	12/2012	JP	N/A
2013521810	12/2012	JP	N/A
2013528397	12/2012	JP	N/A
2013535201	12/2012	JP	N/A
2014514918	12/2013	JP	N/A
2014516562	12/2013	JP	N/A
2014233281	12/2013	JP	N/A
2016514968	12/2015	JP	N/A
2019000014	12/2018	JP	N/A
2020516247	12/2019	JP	N/A
2020523000	12/2019	JP	N/A
7068305	12/2021	JP	N/A
7148552	12/2021	JP	N/A
20060114355	12/2005	KR	N/A
WO-9207615	12/1991	WO	N/A
WO-9821312	12/1997	WO	N/A
9945100	12/1998	WO	N/A
WO-9949807	12/1998	WO	N/A
03046141	12/2002	WO	N/A
WO-03082201	12/2002	WO	N/A
WO-2004020614	12/2003	WO	N/A
WO-2005001072	12/2004	WO	N/A
2005063971	12/2004	WO	N/A
WO-2005081970	12/2004	WO	N/A
WO-2005097974	12/2004	WO	N/A
WO-2005113747	12/2004	WO	N/A
WO-2006126236	12/2005	WO	N/A
2008073352	12/2007	WO	N/A
WO-2008075339	12/2007	WO	N/A
WO-2009022907	12/2008	WO	N/A
WO-2009086596	12/2008	WO	N/A
WO-2009146911	12/2008	WO	N/A
WO-2010008905	12/2009	WO	N/A
WO-2010090513	12/2009	WO	N/A
WO-2010094694	12/2009	WO	N/A
WO-2010127399	12/2009	WO	N/A
2010136583	12/2009	WO	N/A

WO-2010143747	12/2009	WO	N/A
WO-2011050672	12/2010	WO	N/A
2011064309	12/2010	WO	N/A
WO-2011116930	12/2010	WO	N/A
WO-2011139628	12/2010	WO	N/A
WO-2011140441	12/2010	WO	N/A
WO-2012014076	12/2011	WO	N/A
WO-2012027474	12/2011	WO	N/A
WO-2012089669	12/2011	WO	N/A
2012126013	12/2011	WO	N/A
WO-2012118799	12/2011	WO	N/A
WO-2012154834	12/2011	WO	N/A
WO-2012155110	12/2011	WO	N/A
WO-2012166903	12/2011	WO	N/A
WO-2012168930	12/2011	WO	N/A
WO-2012178215	12/2011	WO	N/A
WO-2013040087	12/2012	WO	N/A
WO-2013067498	12/2012	WO	N/A
WO-2013086486	12/2012	WO	N/A
WO-2013086502	12/2012	WO	N/A
WO-2013093812	12/2012	WO	N/A
WO-2013096741	12/2012	WO	N/A
WO-2013127921	12/2012	WO	N/A
WO-2013155060	12/2012	WO	N/A
WO-2013174794	12/2012	WO	N/A
WO-2013176772	12/2012	WO	N/A
WO-2013192290	12/2012	WO	N/A
WO-2014013334	12/2013	WO	N/A
WO-2014018691	12/2013	WO	N/A
WO-2014048637	12/2013	WO	N/A
WO-2014053596	12/2013	WO	N/A
WO-2014062138	12/2013	WO	N/A
WO-2014082096	12/2013	WO	N/A
WO-2014083132	12/2013	WO	N/A
WO-2014090993	12/2013	WO	N/A
WO-2014093595	12/2013	WO	N/A
WO-2014093622	12/2013	WO	N/A
WO-2014093655	12/2013	WO	N/A
WO-2014093661	12/2013	WO	N/A
WO-2014093712	12/2013	WO	N/A
WO-2014127170	12/2013	WO	N/A
2014148646	12/2013	WO	N/A
WO-2014151921	12/2013	WO	N/A
WO-2014153230	12/2013	WO	N/A
WO-2014153294	12/2013	WO	N/A
WO-2014159356	12/2013	WO	N/A
WO-2014173907	12/2013	WO	N/A
WO-2014182885	12/2013	WO	N/A
WO-2014197934	12/2013	WO	N/A
WO-2014199622	12/2013	WO	N/A
WO-2014204728	12/2013	WO	N/A
WO-2014204729	12/2013	WO	N/A
WO-2015021358	12/2014	WO	N/A
WO-2015060790	12/2014	WO	N/A
WO-2015071474	12/2014	WO	N/A
WO-2015075175	12/2014	WO	N/A
WO-2015076388	12/2014	WO	N/A
WO-2015108893	12/2014	WO	N/A
WO-2015123183	12/2014	WO	N/A
WO-2015129822	12/2014	WO	N/A
WO-2015130919	12/2014	WO	N/A
WO-2015135893	12/2014	WO	N/A
WO-2015138032	12/2014	WO	N/A
WO-2015152954	12/2014	WO	N/A
WO-2015156929	12/2014	WO	N/A
WO-2015157163	12/2014	WO	N/A
WO-2015168022	12/2014	WO	N/A
WO-2015173425	12/2014	WO	N/A
2015189320	12/2014	WO	N/A
WO-2015183920	12/2014	WO	N/A
WO-2015184273	12/2014	WO	N/A
WO-2015184375	12/2014	WO	N/A
WO-2015185714	12/2014	WO	N/A
WO-2015196012	12/2014	WO	N/A
WO-2015200901	12/2014	WO	N/A

WO-2016011377	12/2015	WO	N/A
WO-2016015158	12/2015	WO	N/A
WO-2016030525	12/2015	WO	N/A
WO-2016033163	12/2015	WO	N/A
WO-2016056999	12/2015	WO	N/A
WO-2016057571	12/2015	WO	N/A
WO-2016061464	12/2015	WO	N/A
WO-2016073989	12/2015	WO	N/A
WO-2016083612	12/2015	WO	N/A
WO-2016083613	12/2015	WO	N/A
WO-2016085765	12/2015	WO	N/A
WO-2016094948	12/2015	WO	N/A
WO-2016103002	12/2015	WO	N/A
WO-2016103269	12/2015	WO	N/A
WO-2016115326	12/2015	WO	N/A
WO-2016121512	12/2015	WO	N/A
2016141131	12/2015	WO	N/A
2016148253	12/2015	WO	N/A
WO-2016140716	12/2015	WO	N/A
WO-2016141084	12/2015	WO	N/A
WO-2016141137	12/2015	WO	N/A
WO-2016141224	12/2015	WO	N/A
WO-2016144769	12/2015	WO	N/A
WO-2016164413	12/2015	WO	N/A
WO-2016168950	12/2015	WO	N/A
WO-2016174604	12/2015	WO	N/A
WO-2016176208	12/2015	WO	N/A
WO-2016183143	12/2015	WO	N/A
2016204809	12/2015	WO	N/A
WO-2016193441	12/2015	WO	N/A
WO-2016207621	12/2015	WO	N/A
WO-2016210313	12/2015	WO	N/A
WO-2016210416	12/2015	WO	N/A
WO-2017009263	12/2016	WO	N/A
WO-2017023803	12/2016	WO	N/A
WO-2017036533	12/2016	WO	N/A
WO-2017037295	12/2016	WO	N/A
WO-2017041041	12/2016	WO	N/A
WO-2017048193	12/2016	WO	N/A
WO-2017048322	12/2016	WO	N/A
WO-2017049243	12/2016	WO	N/A
WO-2017059171	12/2016	WO	N/A
WO-2017060884	12/2016	WO	N/A
WO-2017066507	12/2016	WO	N/A
WO-2017066659	12/2016	WO	N/A
WO-2017070007	12/2016	WO	N/A
WO-2017070224	12/2016	WO	N/A
WO-2017070337	12/2016	WO	N/A
WO-2017070471	12/2016	WO	N/A
WO-2017070506	12/2016	WO	N/A
WO-2017070633	12/2016	WO	N/A
2017083696	12/2016	WO	N/A
WO-2017075389	12/2016	WO	N/A
WO-2017077535	12/2016	WO	N/A
WO-2017079632	12/2016	WO	N/A
WO-2017083705	12/2016	WO	N/A
WO-2017083838	12/2016	WO	N/A
WO-2017096192	12/2016	WO	N/A
WO-2017096282	12/2016	WO	N/A
WO-2017112901	12/2016	WO	N/A
WO-2017115982	12/2016	WO	N/A
WO-2017117333	12/2016	WO	N/A
WO-2017117547	12/2016	WO	N/A
WO-2017117571	12/2016	WO	N/A
WO-2017120543	12/2016	WO	N/A
WO-2017121754	12/2016	WO	N/A
WO-2017123791	12/2016	WO	N/A
WO-2017136462	12/2016	WO	N/A
WO-2017136479	12/2016	WO	N/A
WO-2017139455	12/2016	WO	N/A
WO-2017139638	12/2016	WO	N/A
WO-2017142069	12/2016	WO	N/A
WO-2017143100	12/2016	WO	N/A
WO-2017149025	12/2016	WO	N/A
WO-2017153992	12/2016	WO	N/A

WO-2017160234	12/2016	WO	N/A
WO-2017160671	12/2016	WO	N/A
WO-2017172638	12/2016	WO	N/A
WO-2017174609	12/2016	WO	N/A
WO-2017175876	12/2016	WO	N/A
WO-2017176810	12/2016	WO	N/A
WO-2017184586	12/2016	WO	N/A
WO-2017192997	12/2016	WO	N/A
WO-2017205511	12/2016	WO	N/A
WO-2017218287	12/2016	WO	N/A
WO-2017220586	12/2016	WO	N/A
WO-2018011558	12/2017	WO	N/A
WO-2018019704	12/2017	WO	N/A
WO-2018026947	12/2017	WO	N/A
WO-2018027023	12/2017	WO	N/A
WO-2018027112	12/2017	WO	N/A
WO-2018035574	12/2017	WO	N/A
WO-2018038042	12/2017	WO	N/A
WO-2018044685	12/2017	WO	N/A
WO-2018044885	12/2017	WO	N/A
WO-2018044937	12/2017	WO	N/A
WO-2018044940	12/2017	WO	N/A
WO-2018085615	12/2017	WO	N/A
WO-2018085622	12/2017	WO	N/A
WO-2018085623	12/2017	WO	N/A
WO-2018091677	12/2017	WO	N/A
WO-2018094522	12/2017	WO	N/A
WO-2018106628	12/2017	WO	N/A
WO-2018115852	12/2017	WO	N/A
2018170280	12/2017	WO	N/A
WO-2018191673	12/2017	WO	N/A
WO-2018197544	12/2017	WO	N/A
WO-2018200481	12/2017	WO	N/A
2018229251	12/2017	WO	N/A
WO-2018226267	12/2017	WO	N/A
WO-2019060336	12/2018	WO	N/A
WO-2019074793	12/2018	WO	N/A
WO-2019126626	12/2018	WO	N/A
2019140151	12/2018	WO	N/A
WO-2020023245	12/2019	WO	N/A
WO-2020056158	12/2019	WO	N/A
WO-2020069285	12/2019	WO	N/A
WO-2020100481	12/2019	WO	N/A
WO-2020154374	12/2019	WO	N/A
2020160371	12/2019	WO	N/A
WO-2020227711	12/2019	WO	N/A
2020247528	12/2019	WO	N/A
WO-2020243633	12/2019	WO	N/A
WO-2021030373	12/2020	WO	N/A
WO-2021041443	12/2020	WO	N/A
2021087508	12/2020	WO	N/A
WO-2021262676	12/2020	WO	N/A
2022261471	12/2021	WO	N/A
2023023180	12/2022	WO	N/A
2023030158	12/2022	WO	N/A

OTHER PUBLICATIONS

Tetri et al., Lancet 2015, 385, 956-965 (Year: 2015). cited by examiner
Sanchez-Valle et al Current Medicinal Chemistry, 2012, 19, 4850-4860 (Year: 2012). cited by examiner
Burrin et al Hepatology, 2015, Paper No. 194, 62(1)Suppl:307A (Year: 2015). cited by examiner
Bain C.C., et al., "Constant Replenishment from Circulating Monocytes Maintains the Macrophage Pool in Adult Intestine," Nat Immunol, Oct. 2014, vol. 15
applicant
Bain C.C., et al., "Resident and Pro-Inflammatory Macrophages in the Colon Represent Alternative Context-Dependent Fates of the Same Ly6Chi Monocyte
Immunology, May 2013, vol. 6 (3), pp. 498-510. cited by applicant
Bayha E., et al., "Retinoic Acid Signaling Organizes Endodermal Organ Specification Along the Entire Antero-Posterior Axis," PLoS one, Jun. 10, 2009, vol.
applicant
Bort R., et al., "Hex Homeobox Gene-Dependent Tissue Positioning is Required for Organogenesis of the Ventral Pancreas," Development, Jan. 2004, vol. 13
applicant
Bujko A., et al., "Transcriptional and Functional Profiling Defines Human Small Intestinal Macrophage Subsets," Journal of Experimental Medicine, 2018, vo
applicant
Bulmer J.N., et al., "Macrophage Populations in the Human Placenta and Amniochorion," Clinical Experimental Immunology, 1984, vol. 57 (2), pp. 393-403.
Camp J.G., et al., "Multilineage Communication Regulates Human Liver Bud Development from Pluripotency," Nature, 2017, vol. 546 (7659), pp. 533-538. c
Campbell E.L., et al., "Transmigrating Neutrophils Shape the Mucosal Microenvironment Through Localized Oxygen Depletion to Influence Resolution of In
vol. 40 (1), pp. 66-77. cited by applicant
Choi K.D., et al., "Identification of the Hemogenic Endothelial Progenitor and Its Direct Precursor in Human Pluripotent Stem Cell Differentiation Cultures,"

2(3), pp. 553-567. cited by applicant
Cumano A., et al., "Lymphoid Potential, Probed before Circulation in Mouse, Is Restricted to Caudal Intraembryonic Splanchnopleura," Cell, Sep. 20, 1996, v
applicant
Davies L.C., et al., "Tissue-Resident Macrophages," Nat Immunol, Oct. 2013, vol. 14 (10), pp. 986-995. cited by applicant
Dekkers R., et al., "A Bioassay Using Intestinal Organoids to Measure CFTR Modulators in Human Plasma," Journal of Cystic Fibrosis, 2015, vol. 14 (2), pp.
DeSchepper S., et al., "Self-Maintaining Gut Macrophages Are Essential for Intestinal Homeostasis," Cell, Oct. 4, 2018, vol. 175 (2), pp. 400-415. cited by ap
Fukuda A., et al., "Ectopic Pancreas Formation in Hes1-Knockout Mice Reveals Plasticity of Endodermal Progenitors of the Gut, Bile Duct, and Pancreas," T
Investigation, Jun. 2006, vol. 116 (6), pp. 1484-1493. cited by applicant
Glocker E.O., et al., "Inflammatory Bowel Disease and Mutations Affecting the Interleukin-10 Receptor," N Engl J Med, Nov. 19, 2009, vol. 361 (21), pp. 20
Hentsch B., et al., "Hlx Homeo Box Gene is Essential for an Inductive Tissue Interaction that Drives Expansion of Embryonic Liver and Gut," Genes & Deve
70-79. cited by applicant
Higashiyama H., et al., "Embryonic Cholecystitis and Defective Gallbladder Contraction in the Sox17-Haploinsufficient Model of Biliary Atresia," Developm
1906-1917. cited by applicant
Hoeffel G., et al., "C-Myb+ Erythro-Myeloid Progenitor-Derived Fetal Monocytes Give Rise to Adult Tissue-Resident Macrophages," Immunity, Apr. 21, 201
by applicant
Iacovino M., et al., "HoxA3 is an Apical Regulator of Hemogenic Endothelium," Nat Cell Biol, Jan. 2011, vol. 13 (1), pp. 72-78. cited by applicant
Jørgensen M.C., et al., "Neurog3-Dependent Pancreas Dysgenesis Causes Ectopic Pancreas in Hes1 Mutant Mice," Development, 2018, vol. 145 (17), 11 page
Kennedy M., et al., "T Lymphocyte Potential Marks the Emergence of Definitive Hematopoietic Progenitors in Human Pluripotent Stem Cell Differentiation C
2012, vol. 2 (6), pp. 1722-1735. cited by applicant
Kuci Z., et al., "Mesenchymal Stromal Cells from Pooled Mononuclear Cells of Multiple Bone Marrow Donors as Rescue Therapy in Pediatric Severe Steroid
Disease: A Multicenter Survey," Haematologica, 2016, vol. 101 (8), pp. 985-994. cited by applicant
Lanctot P.M., et al., "The Glycans of Stem Cells," Curr Opin Chem Biol, Aug. 2007, vol. 11(4), pp. 373-380. cited by applicant
Maeno M., et al., "The Role of BMP-4 and GATA-2 in the Induction and Differentiation of Hematopoietic Mesoderm in Xenopus Laevis," Blood, Sep. 15, 19
cited by applicant
Maheshwari A., et al., "TGF-β2 Suppresses Macrophage Cytokine Production and Mucosal Inflammatory Responses in the Developing Intestine," Gastroente
242-253. cited by applicant
Man A.L., et al., "CX3CR1+ Cell-Mediated *Salmonella* Exclusion Protects the Intestinal Mucosa during the Initial Stage of Infection," The Journal Immunolo
343. cited by applicant
Martin M.J., et al., "Human Embryonic Stem Cells Express an Immunogenic Nonhuman Sialic Acid," Nature Medicine, Feb. 2005, vol. 11(2), pp. 228-232. ci
Miller A.J., et al., "Generation of Lung Organoids from Human Pluripotent Stem Cells in Vitro," Nature Protocols, Feb. 28, 2019, vol. 14, No. 2, pp. 518-540.
Montalbano G., et al., "Synthesis of Bioinspired Collagen/Alginate/Fibrin Based Hydrogels for Soft Tissue Engineering," Material Science & Engineering, C
applicant
Nissim S., et al., "Iterative Use of Nuclear Receptor Nr5a2 Regulates Multiple Stages of Liver and Pancreas Development," Development Biology, Jul. 26, 20
cited by applicant
Palaria A., et al., "Patterning of the Hepato-Pancreatobiliary Boundary by BMP Reveals Heterogeneity Within the Murine Liver Bud," Hepatology, Jul. 2018,
applicant
Perdiguero E.G., et al., "Development and Maintenance of Resident Macrophages," Nature Immunology, Jan. 2016, vol. 17 (1), pp. 2-8. cited by applicant
Perdiguero E.G., et al., "Tissue-Resident Macrophages Originate from Yolk-Sac-Derived Erythro-Myeloid Progenitors," Nature, Feb. 26, 2015, vol. 518 (754
Rankin S.A., et al., "A Retinoic Acid-Hedgehog Cascade Coordinates Mesoderm-Inducing Signals and Endoderm Competence During Lung Specification," C
16 (1), pp. 66-78. cited by applicant
San Roman A.K., et al., "Boundaries, Junctions and Transitions in the Gastrointestinal Tract," Exp Cell Res, Nov. 15, 2011, vol. 317 (19), pp. 2711-2718. cite
Shaw T.N., et al., "Tissue-Resident Macrophages in the Intestine are Long Lived and Defined by Tim-4 and CD4 Expression," Journal of Experimental Medic
1518. cited by applicant
Sheng J., et al., "Most Tissue-Resident Macrophages Except Microglia Are Derived from Fetal Hematopoietic Stem Cells," Immunity, Aug. 18, 2015, vol. 43
applicant
Shibata Y., et al., "Prediction of Hepatic Clearance and Availability by Cryopreserved Human Hepatocytes: An Application of Serum Incubation Method," Dru
2002, vol. 30(8), pp. 892-896. cited by applicant
Shih H.P., et al., "A Gene Regulatory Network Cooperatively Controlled by Pdx1 and Sox9 Governs Lineage Allocation of Foregut Progenitor Cells," Cell Re
326-336. cited by applicant
Smith D.M., et al., "Roles of BMP Signaling and Nkx2.5 in Patterning at the Chick Midgut-Foregut Boundary," Development, 2000, vol. 127 (17), pp. 3671-3
Smith P.D., et al., "Intestinal Macrophages Lack CD14 and CD89 and Consequently are Down-Regulated for LPS- and IgA-Mediated Activities," The Journal
(5), pp. 2651-2656. cited by applicant
Spence J.R., et al., "Sox17 Regulates Organ Lineage Segregation of Ventral Foregut Progenitor Cells," Dev Cell, Jul. 2009, vol. 17 (1), pp. 62-74. cited by app
Stresser D.M., et al., "Validation of Pooled Cryopreserved Human Hepatocytes as a Model for Metabolism Studies," BD Biosciences, Jan. 1, 2004, Retrieved
https://www.researchgate.net/profile/David-
Stresser/publication/268359224_Validation_of_Pooled_Cryopreserved_Human_Hepatocytes_as_a_Model_for_Metabolism_Studies/links/54ed49710cf2465f
Cryopreserved-Human-Hepatocytes-as-a-Model-for-Metabolism-Studies.pdf on Jan. 15, 2021, 2 pages. cited by applicant
Sturgeon C.M., et al., "Wnt Signaling Controls the Specification of Definitive and Primitive Hematopoiesis from Human Pluripotent Stem Cells," Natural Bio
pp. 554-561. cited by applicant
Sumazaki R., et al., "Conversion of Biliary System to Pancreatic Tissue in Hes1-Deficient Mice," Nature Genetics, Jan. 2004, vol. 36 (1), pp. 83-87. cited by a
Takata K., et al., "Induced-Pluripotent-Stem-Cell-Derived Primitive Macrophages Provide a Platform for Modeling Tissue-Resident Macrophage Differentiati
18, 2017, vol. 47 (1), pp. 183-198. cited by applicant
Tepass U., et al., "Epithelium Formation in the Drosophila Midgut Depends on the Interaction of Endoderm and Mesoderm," Development, 1994, vol. 120 (3
Thamm K., et al., "Notch Signaling During Larval and Juvenile Development in the Polychaete Annelid *Capitella* sp. I," Developmental Biology, 2008, vol. 3
applicant
Tugizov S.M., et al., "Differential Transmission of HIV Traversing Fetal Oral/Intestinal Epithelia and Adult Oral Epithelia," Journal of Virology, 2012, vol. 86
applicant
Udager A., et al., "Dividing the Tubular Gut: Generation of Organ Boundaries at the Pylorus," Progress in Molecular Biology and Translational Science, 2010
applicant
Uhlén M., et al., "A Human Protein Atlas for Normal and Cancer Tissues Based on Antibody Proteomics," Molecular & and Cellular Proteomics, Aug. 27, 200
cited by applicant
Zhang Y., et al., "Development and Stem Cells of the Esophagus," Seminars in Cell & Developmental Biology, Dec. 19, 2016, vol. 66, pp. 25-35. cited by app
Zhang Z., et al., "Syndecan4 Coordinates Wnt/JNK and BMP Signaling to Regulate Foregut Progenitor Development," Developmental Biology, 2016, vol. 41
applicant
Adorini L., et al., "Farnesoid X Receptor Targeting to Treat Nonalcoholic Steatohepatitis," Drug Discovery Today, Sep. 2012, vol. 17 (17/18), pp. 988-997. cit

Ajmera V., et al., "Novel Plasma Biomarkers Associated with Liver Disease Severity in Adults with Nonalcoholic Fatty Liver Disease," Hepatology, Jan. 2017
applicant
Aleo M.D., et al., "Human Drug-Induced Liver Injury Severity is Highly Associated with Dual Inhibition of Liver Mitochondrial Function and Bile Salt Expo
60 (3), pp. 1015-1022. cited by applicant
Andrews P.W., et al., "Embryonic Stem (ES) cells and Embryonal Carcinoma (EC) Cells: Opposite Sides of the same Coin", Biochemical Society Transaction
1530. cited by applicant
Ang S.L., et al., "The Formation and Maintenance of the Definitive Endoderm Lineage in the Mouse: Involvement of HNF3/Forkhead Proteins," Developmen
1993, vol. 119, pp. 1301-1315. cited by applicant
Asai A., et al. "Paracrine Signals Regulate Human Liver Organoid Maturation from Induced Pluripotent Stem Cells," Human Development, 2017, vol. 144, pp
Halpern K.B., et al. "Single-cell Spatial Reconstruction Reveals Global Division of Labor in the Mammalian Liver," Nature, 2017, vol. 542, pp. 352-356. cited
Barth C.A., et al., "Transcellular Transport of Fluorescein in Hepatocyte Monolayers: Evidence for Functional Polarity of Cells in Culture," Proceedings of the
USA, 1982, vol. 79, pp. 4985-4987. cited by applicant
Begriche K., et al., "Drug-induced Toxicity on Mitochondria and Lipid Metabolism: Mechanistic Diversity and Deleterious Consequences for the Liver", Jour
pp. 773-794. cited by applicant
Bell L.N., et al., "Epidemiology of Idiosyncratic Drug-Induced Liver Injury," Seminars in Liver Disease, 2009, vol. 29, Issue 4, pp. 337-347. cited by applicar
Bernadi P., "The permeability transition pore. Control Points of a Cyclosporin A-Sensitive Mitochondrial Channel Involved in Cell Death," Biochimica et Bio
pp. 5-9. cited by applicant
Bohan T.P., et al., "Effect of L-Carnitine Treatment for Valproate-Induced Hepatotoxicity", Neurology, 2001, vol. 56, pp. 1405-1409. cited by applicant
Boullata J.I., et al., "A.S.P.E.N. Clinical Guidelines: Parental Nutrition Ordering, Order Review, Compounding, Labeling and Dispensing", The Journal of Par
2014, vol. 38, Issue 3, pp. 334-377. cited by applicant
Bravo P., et al., "Efficient In Vitro Vectorial Transport of a Fluorescent Conjugated Bile Acid Analogue by Polarized Hepatic Hybrid WIF-B and WIF-B9 Cell
576-583. cited by applicant
Browning J.D., et al., "Molecular Mediators of Hepatic Steatosis and Liver Injury", Journal of Clinical Investigation, 2004, vol. 114, Issue 2, pp. 147-152. cite
Cabezas J., et al., "Nonalcoholic Fatty Liver Disease: A Pathological View," Chapter 8, in Liver Biopsy-Indications, Procedures Results, N. Tagaya (Ed.), InTe
188. cited by applicant
Chang J.H., et al., "Evaluating the In Vitro Inhibition of UGT1A1, OATP1B1, OATP1B3, MRP2, and BSEP in Predicting Drug-Induced Hyperbilirubinemia",
vol. 10, pp. 3067-3075. cited by applicant
Chatterjee S., et al., "Hepatocyte-Based in Vitro Model for Assessment of Drug-Induced Cholestasis", Toxicology and Applied Pharmacology, 2014, vol. 274,
Chen Y., et al., "Retinoic Acid Signaling is Essential for Pancreas Development and Promotes Endocrine at the Expense of Exocrine Cell Differentiation in Xe
2004, vol. 271, pp. 144-160. cited by applicant
Chughlay M.F., et al., "N-Acetylcysteine for Non-Paracetamol Drug-Induced Liver Injury: a Systematic Review", British Journal of Clinical Pharmacology, 2
by applicant
Cincinnati Children's Hospital Medical Center, "Scientists Grow Human Esophagus in Lab: Tiny Organoids Enable Personalized Disease Diagnosis, Regenera
Press Release, Sep. 20, 2018, 2 pages. cited by applicant
Cutrin J.C., et al., "Reperfusion Damage to the Bile Canaliculi in Transplanted Human Liver", Hematology, 1996, vol. 24, pp. 1053-1057. cited by applicant
D'Amour K.A. et al., "Efficient Differentiation of Human Embryonic Stem Cells to Definitive Endoderm", Nature Biotechnology, Dec. 2005, vol. 23, No. 12,
Dash A., et al., "Pharmacotoxicology of Clinically-Relevant Concentrations of Obeticholic Acid in an Organotypic Human Hepatocyte System", Toxicol in Vi
cited by applicant
Davidson M.D., et al., "Long-Term Exposure to Abnormal Glucose Levels Alters Drug Metabolism Pathways and Insulin Sensitivity in Primary Human Hepa
2016, vol. 6, 28178, 11 pages. cited by applicant
De Santa Barbara P., et al., "Development and Differentiation of the Intestinal Epithelium", Cellular and Molecular Life Sciences, 2003, vol. 60, issue 7, pp. 1
Dessimoz J., et al., "FGF Signaling is Necessary for Establishing Gut Tube Domains along the Anterior-Posterior Axis in Vivo", Mech Dev, 2006, vol. 123, pp
Dumortier G., et al., "Tolérance hépatique des antipsychotiques atypiques, [Hepatic Tolerance of Atypical Antipsychotic Drugs]", L'Encephale, 2002, vol. 28,
Dvir-Ginzberg M., et al., "Liver Tissue Engineering within Alginate Scaffolds: Effects of Cell-Seeding Density on Hepatocyte Viability, Morphology, and Fun
vol. 9, pp. 757-766. cited by applicant
Edling Y., et al., "Increased Sensitivity for Troglitazone-Induced Cytotoxicity using a Human in Vitro Co-Culture Model," Toxicol in Vitro, 2009, vol. 23, pp.
El Kasmi K.C., et al., "Phytosterols Promote Liver Injury and Kupffer Cell Activation in Parenteral Nutrition-Associated Disease", Science Translational Med
pages. cited by applicant
El Taghdouini A., et al., "In Vitro Reversion of Activated Primary Human Hepatic Stellate Cells", Fibrogenesis & Tissue Repair, 2015, vol. 8(14), 15 pages. ci
Evans M.J., et al., "Establishment in Culture of Pluripotent Cells from Mouse Embryos", Nature, 1981, vol. 292, pp. 154-156. cited by applicant
Fahrmayr C., et al., "Phase I and II Metabolism and MRP2-Mediated Export of Bosentan in a MDCKII-OATP1B1-CYP3A4-UGT1A1-MRP2 quadruple-Tran
Journal of Pharmacology, 2013, vol. 169, pp. 21-33. cited by applicant
Falasca L., et al., "The Effect of Retinoic Acid on the re-establishment of Differentiated Hepatocyte Phenotype in Primary Culture", Cell Tissue Res, Aug. 199
applicant
Fisher A., et al., "Entacapone-Induced Hepatotoxicity and Hepatic Dysfunction", Movement Disorders, 2002, vol. 17, pp. 1362-1365. cited by applicant
Fromenty B., "Drug-induced liver injury in obesity", Journal of Hepatology, 2013, vol. 58, pp. 824-826. cited by applicant
Geerts A., et al., "Formation of Normal Desmin Intermediate Filaments in Mouse Hepatic Stellate Cells Requires Vimentin", Hepatology, Oct. 2001, vol. 33, p
Gori M., et al., "Investigating Nonalcoholic Fatty Liver Disease in a Liver-on-a-Chip Microfluidic Device", PLOS One, Jul. 2016, vol. 11, Issue 7, e0159729,
Hardy T., et al., "Nonalcoholic Fatty Liver Disease: New Treatments", Current Opinion in Gastroenterology, May 2015, vol. 31, Issue 3, pp. 175-183. cited by
Hassan W., et al., "Reduced Oxidative Stress Contributes to the Lipid Lowering Effects of Isoquercitrin in Free Fatty Acids Induced Hepatocytes", Oxidative
2014, vol. 2014, 313602, 18 pages. cited by applicant
Heidari R., et al., "Factors Affecting Drug-Induced Liver Injury: Antithyroid Drugs as Instances", Clinical and Molecular Hepatology, 2014, vol. 20, Issue 3, p
Hynds R.E., et al., "The Relevance of Human Stem Cell-Derived Organoid Models for Epithelial Translational Medicine", Stem Cells, 2013, Issue 31, vol. 3,
Ijpenberg., et al., "Wt1 and Retinoic Acid Signaling are Essential for Stellate cell development and Liver Morphogenesis", Developmental Biology, Dec. 2007
cited by applicant
Inoue H., et al., "IPS Cells: A game Changer for Future Medicine", EMBO Journal, 2014, issue 33, vol. 5, pp. 409-417. cited by applicant
Ito K., et al., "Temporal Transition of Mechanical Characteristics of HUVEC/MSC Spheroids Using a Microfluidic Chip with Force Sensor Probes", Microma
pages. cited by applicant
Jalan-Sakrikar N., et al., "Hedgehog Signaling Overcomes an EZH2-Dependent Epigenetic Barrier to Promote Cholangiocyte Expansion", PLoS One, 2016, v
cited by applicant
Kaji K., et al., "Virus Free Induction of Pluripotency and Subsequent Excision of Reprogramming Factors", Nature, Apr. 2009, vol. 458, Issue 7239, pp. 771-7
Kanuri G., et al., "In Vitro and in Vivo Models of Non-Alcoholic Fatty Liver Disease (NAFLD)", International Journal of Molecular Sciences, 2013, vol. 14, p
applicant
Kelly G.M., et al., "Retinoic Acid and the Development of the Endoderm", Journal Developmental Biology, 2015, vol. 3, pp. 25-56. cited by applicant
Klimamskaya I., et al., "Human Embryonic Stem Cells Derived without Feeder Cells", Lancet, 2005, vol. 365, Issue 9471, pp. 1636-1641. cited by applicant

Kock K., et al., “A Perspective on Efflux Transport Proteins in the Liver”, Clinical Pharmacology & Therapeutics, 2012, vol. 92, Issue 5, pp. 599-612. cited by
Koehler E.M., et al., “Presence of Diabetes Mellitus and Steatosis Is Associated with Liver Stiffness in a General Population: The Rotterdam Study”, Hepatolo
cited by applicant
Kordes C., et al., “Hepatic Stellate Cells Contribute to Progenitor Cells and Liver Regeneration”, Journal of Clinical Investigation, 2014, vol. 124, Issue 12, p
Krahenbuhl S., et al., “Toxicity of Bile Acids on the Electron Transport Chain of Isolated Rat Liver Mitochondria”, Hepatology, 1994, vol. 19, pp. 471-479. ci
Kubo A., et al., “Development of Definitive Endoderm from Embryonic Stem Cells in Culture,” Development, 2004, vol. 131, Issue 7, pp. 1651-1662. cited b
Kullak-Ublick G.A., et al., “Drug-Induced Liver Injury: Recent Advantages in Diagnosis and Risk Assessment,” Gut, 2017, vol. 66, pp. 1154-1164. cited by a
Kumar J.A., et al., “Controversies in the Mechanism of Total Parenteral Nutrition Induced Pathology,” Children, 2015, vol. 2, Issue 3, pp. 358-370. cited by a
Lancaster M.A., et al., “Organogenesis in a Dish: Modeling Development and Disease Using Organoid Technologies,” Science, 2014, vol. 345, Issue 6194, 12
applicant
Le S., et al., “FactoMineR: An R Package for Multivariate Analysis,” Journal of Statistical Software, 2008, vol. 25, Issue 1, pp. 1-18. cited by applicant
Le Vee M., et al., “Polarized Expression of Drug Transporters in Differentiated Human Hepatoma HepaRG Cells,” Toxicol in Vitro, 2013, vol. 27, pp. 1979-19
Lechner C., et al., “Development of a Fluorescence-Based Assay for Drug Interactions with Human Multidrug Resistance Related Protein (MRP2; ABCC2) in
Vesicles,” European Journal of Pharmaceutics and Biopharmaceutics, 2010, vol. 75, pp. 284-290. cited by applicant
Lee W.M., et al., “Intravenous N-Acetylcysteine Improves Transplant-Free Survival in Early Stage Non-Acetaminophen Acute Liver Failure,” Gastroenterolog
856-864. cited by applicant
Leslie E.M., et al., “Differential Inhibition of Rat and Human Na +-Dependent Taurocholate Cotransporting Polypeptide (NTCP/SLC10A1) by Bosentan: A M
in Hepatotoxicity,” Journal of Pharmacology and Experimental Therapeutics, 2007, vol. 321, Issue 3, pp. 1170-1178. cited by applicant
Li N., et al., “A Systematic Assessment of Mitochondrial Function Identified Novel Signatures for Drug-Induced Mitochondrial Disruption in Cells,” Toxicolo
Issue 1, pp. 261-273. cited by applicant
Makin A.J., et al., “A 7-Year Experience of Severe Acetaminophen-Induced Hepatotoxicity (1987-1993),” Gastroenterology, Dec. 1995, vol. 109, pp. 1907-19
Malinen M.M., et al., “Differentiation of Liver Progenitor Cell Line to Functional Organotypic Cultures in 3D Nanofibrillar Cellulose and Hyaluronan-gelatin
2014, vol. 35, pp. 5110-5121. cited by applicant
Marcum Z.A., et al., “Medication Adherence to Multi-Drug Regimens,” Clinics in Geriatric Medicine, May 2012, vol. 28, Issue 2, pp. 287-300. cited by appli
Markova S.M., et al., “Association of CYP2C9*2 with Bosentan-Induced Liver Injury,” Clinical Pharmacology & Therapeutics, Dec. 2013, vol. 94, Issue 6, p
Martin G.R., “Teratocarcinomas and Mammalian Embryogenesis,” Science, Aug. 1980, vol. 209, Issue 4458, pp. 768-776. cited by applicant
McCracken K.W., et al., “Wnt/beta-Catenin Promotes Gastric Fundus Specifications in Mice and Humans,” Nature, Jan. 2017, vol. 541, Issue 7636, pp. 182-1
McLin V.A., et al., “Repression of Wnt/beta-Catenin Signaling in the Anterior Endoderm is Essential for Liver and Pancreas Development,” Development, Ju
cited by applicant
Mercaldi C.J., et al., “Methods to Identify and Compare Parenteral Nutrition Administered from Hospital-Compound and Premixed Multichamber Bags in a R
Database,” Journal of Parenteral and Enteral Nutrition, May 2012, vol. 36, Issue 3, pp. 330-336. cited by applicant
Michaut A., et al., “A Cellular Model to Study Drug-Induced Liver Injury in Nonalcoholic Fatty Liver Disease: Application of Acetaminophen,” Toxicology a
2016, vol. 292, pp. 40-55. cited by applicant
Miki T., et al., “Hepatic Differentiation of Human Embryonic Stem Cells Is Promoted by Three-Dimensional Dynamic Perfusion Culture Conditions,” Tissue
May 2011, vol. 17, Issue 5, pp. 557-568. cited by applicant
Mork L.M., et al., “Comparison Culture Media for Bile Acid Transport Studies in Primary Human Hepatocytes,” Journal of Clinical and Experimental Hepato
cited by applicant
Mudaliar S., et al., “Efficacy and Safety of the Farnesoid X Receptor Agonist Obeticholic Acid in Patients with Type 2 Diabetes and Nonalcoholic Fatty Liver
2013, vol. 145, pp. 574-582. cited by applicant
Mullin E., “Tiny Human Esophagus Grown in the Lab-Here's Why: Miniature Version of the Organ that Guides Food to the Stomach could Help Scientists Tr
Ailments,” National Geographic, Sep. 20, 2018, downloaded from https://www.nationalgeographic.com/science/2018/news-human-esophagus-grow-n-lab-ste
pages. cited by applicant
Nandivada P., et al., “Treatment of Parenteral Nutrition Associated Liver Disease: The Role of Lipid Emulsions”, Advances in Nutrition, vol. 4, No. 6, Nov. 2
applicant
Navarro V J., et al., “Drug-Related Hepatotoxicity”, New England Journal of Medicine, 2006, vol. 354, pp. 731-739. cited by applicant
Negishi T., et al., “Retinoic Acid Signaling Positively Regulates Liver Specification by Inducing wnt2bb Gene Expression in Medaka”, Hepatology, 2010, vol
applicant
Neuschwander-Tetri B A., et al., “Farnesoid X Nuclear Receptor Ligand Obeticholic Acid for Non-Cirrhotic, Non-Alcoholic Steatohepatitis (Flint): A Multice
Controlled Trial”, Lancet, 2015, vol. 385, Issue 9972, pp. 956-965. cited by applicant
Ni X., et al. “Functional Human Induced Hepatocytes (hiHeps) with Bile Acid Synthesis and Transport Capacities: A Novel in Vitro Cholestatic Model”, Scie
38694, 16 pages. cited by applicant
Nishida T., et al., “Rat Liver Canalicular Membrane Vesicles Contain an ATP-Dependent Bile Acid Transport System”, Proceedings of the National Academy
88, Issue 15, pp. 6590-6594. cited by applicant
Okita K., et al., “Generation of Mouse Induced Pluripotent Stem Cells without Viral Vectors”, Science, Nov. 7, 2008, vol. 322, Issue 5903, pp. 949-953. cited
Oorts M., et al., “Drug-induced cholestasis risk assessment in sandwich-cultured human hepatocytes”, Toxicol in Vitro, 2016, vol. 34, pp. 179-186. cited by a
Orso G., et al., “Pediatric Parenteral Nutrition-Associated Liver Disease and Cholestasis: Novel Advances in Pathomechanisms-based Prevention and Treatm
Issue 3, pp. 215-222. cited by applicant
Park H.R., et al., “Lipotoxicity of Palmitic Acid on Neural Progenitor Cells and Hippocampal Neurogenesis”, Toxicological Research, Jun. 2011, vol. 27, Issu
applicant
Pessayre D., et al., “Central Role of Mitochondria in Drug-Induced Liver Injury”, Drug Metabolism Reviews, 2012, vol. 44, Issue 1, pp. 34-87. cited by appli
Pessayre D., et al., “Mitochondrial involvement in Drug-Induced Liver Injury”, in Adverse Drug Reaction, J. Uetrecht (ed.). Handbook of Experimental Pharm
cited by applicant
Polson J., et al., “AASLD Position Paper: The Management of Acute Liver Failure”, Hepatology, 2005, vol. 41, Issue 5, pp. 1179-1197. cited by applicant
Purton L.E., et al., “All-Trans Retinoic Acid Enhances the Long-Term Repopulating Activity of Cultured Hematopoietic Stem Cells”, Blood, 2000, vol. 95, Iss
applicant
Rachek Li., et al., “Troglitazone, but not Rosiglitazone, Damages Mitochondrial DNA and induces Mitochondrial Dysfunction and Cell Death in Human Hep
Pharmacology, 2009, vol. 240, Issue 3, pp. 348-354. cited by applicant
Ramachandran, et al., “In Vitro Generation of Functional Liver Organoid-Like Structures Using Adult Human Cells”, PLoS One, Oct. 21, 2015, vol. 10, No. 1
Rector, R.S., et al., “Mitochondrial Dysfunction Precedes Insulin Resistance and Hepatic Steatosis and Contributes to the Natural History of Non-Alcoholic F
Rodent Model,” Journal of Hepatology, 2010, vol. 52, Issue 5, pp. 727-736. cited by applicant
Reuben A., et al., “ Drug-Induced Acute Liver Failure: Results of a U.S. Multicenter, Prospective Study”, Hepatology, 2010, vol. 52, pp. 2065-2076. cited by a
Ronn R.E., et al., “Retinoic Acid Regulates Hematopoietic Development from Human Pluripotent Stem Cells”, Stem Cell Reports. 2015, vol. 4, pp. 269-281.
Russo M. W., et al., “Liver Transplantation for Acute Liver Failure From Drug Induced Liver Injury in the United States”, Liver Transplantation, 2004, vol. 1
applicant
Saini A., “Cystic Fibrosis Patients Benefit from Mini Guts,” Cell Stem Cell, 2016, vol. 19, pp. 425-427. cited by applicant

Sandoiu A., “Scientists Create Human Esophagus in Stem Cell First”, Medical News Today, Downloaded from https://www.medicalnewstoday.com/articles/32
cited by applicant
Schuppan D., et al., “Non-alcoholic Steatohepatitis: Pathogenesis and Novel Therapeutic Approaches”, Journal of Gastroenterology and Hepatology, Aug. 201
cited by applicant
Serviddio G., et al., “Ursodeoxycholic Acid Protects Against Secondary Bilary Cirrhosis in Rats by Preventing Mitochondrial Oxidative Stress”, Hepatology, 2
by applicant
Singh S., et al., “Comparative Effectiveness of Pharmacological Interventions for Nonalcoholic Steatohepatitis: A Systematic Review and Network Meta-analy
vol. 62, Issue 5, pp. 1417-1432. cited by applicant
Si-Tayeb K., et al., “Highly Efficient Generation of Human Hepatocyte-Like Cell from Induced Pluripotent Stem Cells”, Hepatology, Jan. 2010, vol. 51, Issue
applicant
Sloan C.A., et al., “ENCODE Data at the ENCODE portal”, Nucleic Acids Research, Jan. 4, 2016, vol. 44, Issue D1, pp. D726-D732. cited by applicant
Sneddon I.N., “The Relation between Load and Penetration in the Axisymmetric Boussinesq Problem for a Punch of Arbitrary Profile”, International Journal o
1965, vol. 3, Issue 1, pp. 47-57. cited by applicant
Song W., et al., “Engraftment of Human Induced Pluripotent Stem cell-Derived Hepatocytes in Immunocompetent Mice via 3D Co-aggregation and Encapsula
vol. 5, Issue 16884, 13 pages. cited by applicant
Song Z., et al., “Efficient Generation of Hepatocyte-like cells from Human Induced Pluripotent Stem Cells,” Cell Res, Nov. 2009, vol. 19, Issue 11, pp. 1233-1
Spence J.R., et al., “Directed Differentiation of Human Pluripotent Stem Cells into Intestinal Tissue in Vitro”, Nature (London), Feb. 3, 2011, vol. 470, Issue 7
applicant
Stadtfeld M., et al., “Induced Pluripotent Stem Cells Generated without Viral Integration”, Science, Nov. 7, 2008, vol. 322, issue 5903, pp. 945-949. cited by a
Stafford D., et al., “A Conserved Role for Retinoid Signaling in Vertebrate Pancreas Development”, Development Genes and Evolution, Sep. 2004, vol. 214, I
applicant
Stender S., et al., “Adiposity Amplifies the Genetic Risk of Fatty Liver Disease Conferred by Multiple Loci”, Nat Genet, Jun. 2017, vol. 49, Issue 6, pp. 842-8
Stevens J L., et al., “The Future of Drug Safety Testing: Expanding the View and Narrowing the Focus”, Drug Discovery Today, Feb. 2009, vol. 14, Issue 3-4,
Takahashi K., et al., “Induction of Pluripotent Stem Cells from Adult Human Fibroblasts by Defined Factors,” Cell, Nov. 30, 2007, vol. 131(5), pp. 861-872. c
Takebe T., et al., “Generation of a Vascularized and Functional Human Liver from an iPSC-derived Organ Bud Transplant,” Nature Protocols, Feb. 2014, vol.
applicant
Takebe T., et al., “Human iPSC-Derived Miniature Organs: A Tool for Drug Studies,” Clinical Pharmacology & Therapeutics, Sep. 2014, vol. 96(3), pp. 310-3
Takebe T., et al., “Vascularized and Complex Organ Buds from Diverse Tissues via Mesenchymal Cell-Driven Condensation,” Cell Stem Cell, May 7, 2015, v
applicant
Takebe T., et al., “Vascularized and Functional Human Liver from an iPSC-derived Organ bud Transplant,” Nature, Jul. 25, 2013, vol. 499(7459), pp. 481-484
Tesdensodnom O., et al. “ROS: Redux and Paradox in Fatty Liver Disease,” Hepatology, 2013, vol. 58(4), pp. 1210-1212. cited by applicant
The Encode Project Consortium, “An Integrated Encyclopedia of DNA Elements in the Human Genome,” Nature, Sep. 5, 2012, vol. 489, pp. 57-74. cited by a
The United States Pharmacopeia: The National Formulary (USP 24 NF 19), United States Pharmacopeial Convention, Inc., Rockville, MD, 1999, 4 pgs. cited
Thomson J.A., et al., “Embryonic Stem Cell Lines Derived from Human Blastocysts,” Science, Nov. 6, 1998, vol. 282(5391), pp. 1145-1147. cited by applicar
Tian X., et al., “Modulation of Multidrug Resistance-Associated Protein 2 (Mrp2) and Mrp3 Expression and Function with Small Interfering RNA in Sandwic
Molecular Pharmacology, Oct. 2004, vol. 66(4), pp. 1004-1010. cited by applicant
Troy D.B (ed.), “Remington: The Science and Practice of Pharmacy,” 21.sup.st Ed., 2006, Lippincott, Williams & Wilkens, Baltimore, MD, 6 pages. cited by a
Tsukada N., et al., “The Structure and Organization of the Bile Canalicular Cytoskeleton with Special Reference to Actin and Actin-Binding Proteins,” Hepato
1106-1113. cited by applicant
Van De Garde M.D., et al., “Liver Monocytes and Kupffer Cells Remain Transcriptionally Distinct during Chronic Viral Infection,” PLoS One, Nov. 3, 2016, v
applicant
Verma S., et al., “Diagnosis, Management and Prevention of Drug-Induced Liver Injury,” Gut, 2009, vol. 58, pp. 1555-1564. cited by applicant
Vosough M., et al. “Generation of Functional Hepatocyte-Like Cells from Human Pluripotent Stem Cells in a Scalable Suspension Culture,” Stem Cells and D
20, pp. 2693-2705. cited by applicant
Wang Y., et al., “Hepatic Stellate Cells, Liver Innate Immunity, and Hepatitis C Virus,” Journal of Gastroenterology and Hepatology, 2013, vol. 28(1), pp. 112
Warren C.R., et al., “Induced Pluripotent Stem Cell Differentiation Enables Functional Validation of GWAS Variant in Metabolic Disease,” Cell Stem Cell, Ap
cited by applicant
Warren C.R., et al., “The NextGen Genetic Association Studies Consortium: A Foray into In Vitro Population Genetics,” Cell Stem Cell, 2017, vol. 20, pp. 431
Wells J.M., et al., “Early Mouse Endoderm is Patterned by Soluble Factors from Adjacent Germ Layers,” Development, Mar. 21, 2000, vol. 127, pp. 1563-157
Woltjen K., et al., “piggyBac transposition reprograms fibroblasts to induced pluripotent stem cells,” Nature, Apr. 9, 2009, vol. 458, pp. 766-770. cited by appl
Workman M.J., et al., “Engineered Human Pluripotent-Stem-Cell-derived Intestinal tissues with a functional Enteric Nervous System,” Nature Medicine, Jan.
by applicant
Xu R., et al., “Association Between Patatin-Like Phospholipase Domain Containing 3 Gene (PNPLA3) Polymorphisms and Nonalcoholic Fatty Liver Disease
Analysis,” Scientific Reports, Mar. 20, 2015, vol. 5(9284), 11 pages. cited by applicant
Yanagimachi M.D., et al., “Robust and Highly-Efficient Differentiation of Functional Monocytic Cells from Human Pluripotent Stem Cells Under Serum- and
PLoS One, 2013, vol. 8(4), e59243, 9 pages. cited by applicant
Yang K., et al., “Systems Pharmacology Modeling Predicts Delayed Presentation and Species Differences in Bile Acid-Mediated Troglitazone Hepatotoxicity”
Therapeutics, 2014, vol. 96(5), pp. 589-598. cited by applicant
Yoneda M., et al., “Noninvasive Assessment of Liver Fibrosis by Measurement of Stiffness in Patients with Nonalcoholic Fatty Liver Disease (NAFLD)”, Dig
371-378. cited by applicant
Zain S.M., et al., “A Common Variant in the Glucokinase Regulatory Gene rs780094 and Risk of Nonalcoholic Fatty Liver Disease: A Meta-Analysis,” Journa
Hepatology, 2015, vol. 30, pp. 21-27. cited by applicant
Zambrano E., et al., “Total parenteral Nutrition Induced Liver Pathology: An Autopsy Series of 24 Newborn Cases,” Pediatric and Developmental Pathology, 2
by applicant
Zborowski J., et al., “Induction of Swelling of Liver Mitochondria by Fatty Acids of Various Chain Length,” Biochimica et Biophysica Acta, Oct. 22, 1963, vo
applicant
Zhou H., et al., “Generation of Induced Pluripotent Stem Cells Using Recombinant Proteins,” Cell Stem Cell, May 8, 2009, vol. 4, pp. 381-384. cited by appli
Zorn A.M., et al., “Vertebrate Endoderm Development and Organ Formation,” Annual Review of Cell and Developmental Biology, 2009, vol. 25, pp. 221-251
Ader M., et al., “Modeling Human Development in 3D Culture,” Current Opinion in Cell Biology, Dec. 2014, vol. 31, pp. 23-28. cited by applicant
Agopian V.G., et al., “Intestinal Stem Cell Organoid Transplantation Generates Neomucosa in Dogs,” Journal of Gastrointestinal Surgery, Jan. 23, 2009, vol. 1
applicant
Ahnfelt-Ronne J., et al., “An Improved Method for Three-Dimensional Reconstruction of Protein Expression Patterns in Intact Mouse and Chicken Embryos a
Histochemistry and Cytochemistry, 2007, vol. 55 (9), pp. 925-930. cited by applicant
Alessi D.R., et al., “LKB1-Dependent Signaling Pathways,” Annual Review of Biochemistry, 2006, vol. 75, pp. 137-163. cited by applicant
Alkhatatbeh M.J., et al., “Low Simvastatin Concentrations Reduce Oleic Acid-Induced Steatosis in HepG2 Cells: An In Vitro Model of Non-Alcoholic Fatty L

Therapeutic Medicine, 2016, vol. 11 (4), pp. 1487-1492. cited by applicant
Allard J., et al., "Immunohistochemical Toolkit for Tracking and Quantifying Xenotransplanted Human Stem Cells," Regenerative Medicine, 2014, vol. 9(4),
Altman G. H., et al., "Cell Differentiation by Mechanical Stress," The FASEB Journal, 2001, vol. 16 (2), pp. 270-272. cited by applicant
Ameri J., et al., "FGF2 Specifies HESC-Derived Definitive Endoderm into Foregut/Midgut Cell Lineages in a Concentration-Dependent Manner," Stem Cells
56. cited by applicant
Amieva M.R., et al., "Helicobacter Pylori Enter and Survive within Multivesicular Vacuoles of Epithelial cells," Cellular Microbiology, Oct. 4, 2002, vol. 4 (1
applicant
An W.F., et al., "Discovery of Potent and Highly Selective Inhibitors of GSK3b," Molecular Libraries, Probe Report, May 2014, 115 Pages. cited by applicant
Anderson G., et al., "Loss of Enteric Dopaminergic Neurons and Associated Changes in Colon Motility in an MPTP Mouse Model of Parkinson's Disease," E
2007, vol. 207 (1), 16 pages. cited by applicant
Anlauf M., et al., "Chemical Coding of the Human Gastrointestinal Nervous System: Cholinergic, VIPergic, and Catecholaminergic Phenotypes," The Journal
2003, vol. 459, pp. 90-111. cited by applicant
Aronson B.E., et al., "GATA4 Represses an ileal Program of Gene Expression in the Proximal Small Intestine by Inhibiting the Acetylation of Histone H3, Ly
Biophysica Acta, Nov. 2014, vol. 1839 (11), pp. 1273-1282. cited by applicant
Arora N., et al., "A Process Engineering Approach to Increase Organoid Yield," Technical and Resources, 2017, vol. 144, pp. 1128-1136. cited by applicant
Arroyo J.D., et al., "Argonaute2 Complexes Carry a Population of Circulating MicroRNAs Independent of Vesicles in Human Plasma," PNAS, 2011, vol. 108
applicant
Aurora M., et al., "hPSC-Derived Lung and Intestinal Organoids as Models of Human Fetal Tissue," Developmental Biology, 2016, vol. 420, pp. 230-238. cit
Avansino J.R., et al., "Orthotopic Transplantation of Intestinal Mucosal Organoids in Rodents," Surgery, Sep. 2006, vol. 140 (3), pp. 423-434. cited by applica
Baetge G., et al., "Transient Catecholaminergic (TC) Cells in the Vagus Nerves and Bowel of Fetal Mice: Relationship to the Development of Enteric Neuron
vol. 132, pp. 189-211. cited by applicant
Bain G., "Embryonic Stem Cells Express Neuronal Properties in Vitro," Developmental Biology, 1995, vol. 168, pp. 842-357. cited by applicant
Bajpai R., et al., "CHD7 Cooperates with PBAF to Control Multipotent Neural Crest Formation," Nature, Feb. 18, 2010, vol. 463, pp. 958-962. cited by applic
Bansal D., et al., "An Ex-Vivo Human Intestinal Model to Study Entamoeba Histolytica Pathogenesis," PLoS Neglected Tropical Diseases, Nov. 17, 2009, vo
applicant
Baptista P.M., et al., "The Use of Whole Organ Decellularization for the Generation of a Vascularized Liver Organoid," Hepatology, 2011, vol. 53 (2), pp. 604
Bar-Ephraim Y.E., et al., "Modelling Cancer Immunomodulation using Epithelial Organoid Cultures," bioRxiv, Aug. 7, 2018, pp. 1-13. cited by applicant
Barker N., et al., "Lgr5(+ve) Stem Cells Drive Self-Renewal in the Stomach and Build Long-Lived Gastric Units In Vitro," Cell Stem Cell, Jan. 8, 2010, vol.
Barker N., et al., "Tissue-Resident Adult Stem Cell Populations of Rapidly Self-Renewing Organs," Cell Stem Cell, Dec. 3, 2010, vol. 7, pp. 656-670. cited by
Barlow A.J., et al., "Critical Numbers of Neural Crest Cells are Required in the Pathways from the Neural Tube to the Foregut to Ensure Complete Enteric Ne
Development, 2008, vol. 135, pp. 1681-1691. cited by applicant
Bartfeld S., et al., "In Vitro Expansion of Human Gastric Epithelial Stem Cells and Their Responses to Bacterial Infection," Gastroenterology, Jan. 2015, vol.
applicant
Bartfeld S., et al., "Stem Cell-Derived Organoids and their Application for Medical Research and Patient Treatment," Journal of Molecular Medicine, 2017, v
applicant
Bastide P., et al., "Sox9 Regulates Cell Proliferation and is required for Paneth Cell Differentiation in the Intestinal Epithelium," JCB, 2007, vol. 178, Issue 4,
Battle M A., et al., "GATA4 is Essential for Jejunal Function in Mice," Gastroenterology, 2008, vol. 135, pp. 1676-1686. cited by applicant
Baumann K., "Colonic Organoids for Drug Testing and Colorectal Disease Modelling," Nature Reviews Molecular Cell Biology, Jul. 2017, vol. 18, No. 8, p.
Beck F., et al., "Expression of Cdx-2 in the Mouse Embryo and Placenta: Possible Role in Patterning of the Extra-Embryonic Membranes," Dev Dyn, 1995, v
applicant
Bergeles C., et al., "From Passive Tool Holders to Microsurgeons: Safer, Smaller, Smarter Surgical Robots," IEEE Transactions on Biomedical Engineering, 2
1576. cited by applicant
Bergner A.J., et al., "Birthdating of Myenteric Neuron Subtypes in the Small Intestine of the Mouse," The Journal of Comparative Neurology, 2014, vol. 522,
Bernstein B.E., et al., "The NIH Roadmap Epigenomics Mapping Consortium," Nature Biotechnology, 2010, vol. 28, Issue 10, pp. 1045-1048. cited by applic
Beuling E., et al., "Co-Localization of Gata4 and Hnfl alpha in the Gastrointestinal Tract is Restricted to the Distal Stomach and Proximal Small Intestine," G
Abstract T1933, 2007, vol. 132, page A586. cited by applicant
Beuling E., et al., "Conditional Gata4 Deletion in Mice Induces Bile Acid absorption in the Proximal Small Intestine," Gut, 2010, vol. 59, Issue 7, pp. 888-89
Beuling E., et al., "Fog Cofactors Partially Mediate Gata4 Function in the Adult Mouse Small Intestine," Gastroenterology, AGA Abstracts, Abstract W1467,
cited by applicant
Beuling E., et al., "GATA4 Mediates Gene Repression in the Mature Mouse Small Intestine through Interactions with Friend of GATA (FOG) Cofactors," Dev
322, Issue 1, pp. 179-189. cited by applicant
Beuling E., et al., "The Absence of GATA4 in the Distal Small Intestine Defines the Ileal Phenotype," Gastroenterology, ABA Abstract, Abstract 602, 2008, v
applicant
Bharadwaj S., et al., "Current Status of Intestinal and Multivisceral Transplantation," Gastroentrerol Rep (Oxf), 2017, vol. 5, Issue 1, pp. 20-28. cited by appli
Bhutani N., et al., "Reprogramming towards Pluripotency Requires AID-Dependent DNA Demethylation," Nature, 2010, vol. 463, Issue 7284, pp. 1042-1047
Bitar K.N., et al., "Intestinal Tissue Engineering: Current Concepts and Future Vision of Regenerative Medicine in the Gut," Neurogastroenterology & Motilit
7-19. cited by applicant
Blaugrund E., et al., "Distinct Subpopulations of Enteric Neuronal Progenitors Defined by Time of Development, Sympathoadrenal Lineage Markers and Mas
vol. 122, 1996, pp. 309-320. cited by applicant
Bohorquez D.V., et al., "An Enteroendocrine Cell—Enteric Glia Connection Revealed by 3D Electron Microscopy," PLOS One, Feb. 2014, vol. 9, Issue 2, e8
Bonilla-Claudio M., et al., "Bmp Signaling Regulates a Dose-Dependent Transcriptional Program to Control Facial Skeletal Development," Development, 20
applicant
Boroviak T., et al., "Single Cell Transcriptome Analysis of Human, Marmoset and Mouse Embryos Reveals Common and Divergent Features of Preimplantat
2018, vol. 145, No. 21, pp. 1-18. cited by applicant
Bort R., et al., "Diclofenac Toxicity to Hepatocytes: A Role for Drug Metabolism in Cell Toxicity," Journal of Pharmacology and Experimental Therapeutics,
72. cited by applicant
Bosse T., et al., "Gata4 and Hnfl Alpha are partially required for the Expression of Specific Intestinal Genes during Development," American Journal of Physi
Physiology, May 2007, vol. 292, pp. G1302-G1314. cited by applicant
Bouchi R., et al., "FOXO1 Inhibition Yields Functional Insulin-Producing Cells in Human Gut Organoid Cultures," Nature Communications, 2014, vol. 5, Iss
applicant
Bragdon B., et al., "Bone Morphogenetic Proteins: A Critical Review," Cellular Signalling, 2011, vol. 23, pp. 609-620. cited by applicant
Brevini T.A.L. et al., "No shortcuts to Pig Embryonic Stem Cells," Theriogenology, 2010, vol. 74, pp. 544-550. cited by applicant
Broda T.R., et al., "Generation of Human Antral and Fundic Gastric Organoids from Pluripotent Stem Cells," Nature Protocols, Nov. 2018. Vol. 14(1), pp. 28-
Bruens L., et al., "Expanding the Tissue Toolbox: Deriving Colon Tissue from Human Pluripotent Stem Cells," Cell Stem Cell, Jul. 2017, vol. 21, Issue 1, pp.
Brugmann S.A., et al., "Building Additional Complexity to in Vitro-Derived Intestinal Tissues," Stem Cell Research & Therapy, 2013, vol. 4, Issue Suppl 1, p

Burke P., et al., “Towards a Single-Chip, Implantable RFID System: is a Single-Cell Radio Possible?,” Biomed Microdevices, 2010, vol. 12, pp. 589-596. cite
Burn S.F., et al., “Left-right Asymmetry in Gut Development: what happens next?,” BioEssays, 2009, vol. 31, pp. 1026-1037. cited by applicant
Burnicka-Turek O., et al., “INSL5-Deficient Mice Display an Alteration in Glucose Homeostasis and an Impaired Fertility,” Endocrinology, Oct. 2012, vol. 15 by applicant
Burns A J., et al., “In Ovo Transplantation of Enteric Nervous System Precursors From Vagal to Sacral Neural Crest Results in Extensive Hindgut Colonisatio 129, pp. 2785-2796. cited by applicant
Burns A J., et al., “Neural Stem Cell Therapies for Enteric Nervous System Disorders,” Nature Reviews/Gastroenterology & Hepatology, May 2014, Issue11,
Burns A.J., et al., “Enteric Nervous System Development: Analysis of the Selective Developmental Potentialities of Vagal and Sacral Neural Crest Cells using Anatomical Record, 2001, vol. 262, pp. 16-28. cited by applicant
Burrin D., et al., “Enteral Obeticholic Acid Prevents Hepatic Cholestasis in Total Parenteral Nutrition-Fed Neonatal Pigs,” Hepatology, vol. 62, Oct. 2015, p. 3
Buta C., et al., “Reconsidering Pluripotency Tests: Do We Still Need Teratoma Assays?,” Stem Cell Research, 2013, vol. 11, pp. 552-562. cited by applicant
Campbell F.C., et al., “Transplantation of Cultured Small Bowel Enterocytes,” Gut, Sep. 1993, vol. 34, Issue 9, pp. 1153-1155. cited by applicant
Caneparo L., et al., “Intercellular Bridges in Vertebrate Gastrulation,” PloS One, 2011, vol. 6, Issue 5, e20230, 6 pages. cited by applicant
Cao L., et al., “Development of Intestinal Organoids as Tissue Surrogates: Cell Composition and the Epigenetic Control of Differentiation,” Molecular Carcin 202. cited by applicant
Capeling M.M., et al., “Nonadhesive Alginate Hydrogels Support Growth of Pluripotent Stem Cell-Derived Intestinal Organoids,” Stem Cell Reports, Feb. 20 cited by applicant
Chai P.R., et al., “Ingestible Biosensors for Real-Time Medical Adherence Monitoring: MyTMed,” Processing Hawaii International Conference on System Sc cited by applicant
Chai P.R., et al., “Utilizing an Ingestible Biosensor to Assess Real-Time Medication Adherence,” Journal of Medical Toxicology, 2015, vol. 11, pp. 439-444. c
Chang H.M., et al., “BMP15 Suppresses Progesterone Production by Down-Regulating STAR via ALK3 in Human Granulosa Cells,” Molecular Endocrinolog cited by applicant
Chauhan R.K., et al., “Genetic and Functional Studies of Hirschsprung Disease,” Doctoral Thesis: Department of Clinical Genetics, Erasmus University Rotte pages. cited by applicant
Chen B., et al., “Dynamic Imaging of Genomic Loci in Living Human Cells by an Optimized CRISPR/Cas System,” Cell, 2013, vol. 155, Issue 7, pp. 1479-14
Chen C., et al., “Pdx1 Inactivation Restricted to the Intestinal Epithelium in Mice alters Duodenal Gene Expression in Enterocytes and Enteroendocrine Cells, Gastrointestinal and Liver Physiology, 2009, vol. 297, pp. G1126-G1137. cited by applicant
Chen L.Y., et al., “Mass Fabrication and Delivery of 3D Multilayer μTags into Living cells,” Scientific Reports, 2013, vol. 3, Issue 2295, 6 pages. cited by app
Chen T.W., et al., “Ultrasensitive Fluorescent Proteins for Imaging Neuronal Activity,” Nature, Jul. 18, 2013, vol. 499, pp. 295-300. cited by applicant
Cheng X., et al., “Self-Renewing Endodermal Progenitor Lines Generated from Human Pluripotent Stem Cells,” Cell Stem Cell, Apr. 6, 2012, vol. 10, pp. 371
Choi E., et al., “Cell Lineage Distribution Atlas of the Human Stomach Reveals Heterogeneous Gland Populations in the Gastric Antrum,” Gut, 2014, vol. 63, applicant
Choi E., et al., “Expression of Activated Ras in Gastric Chief Cells of Mice Leads to the Full Spectrum of Metaplastic Lineage Transitions,” Gastroenterology 918-930. cited by applicant
Christoffersson J., et al., “Developing Organ-on-a-chip Concepts Using Bio-Mechatronic Design Methodology,” Biofabrication, May 26, 2017, vol. 9, Issue 2 applicant
Churin Y., et al., “Helicobacter Pylori CagA Protein Targets the c-Met Receptor and Enhances the Motogenic Response,” Journal of Cell Biology, 2003, vol. 1 applicant
Cieslar-Pobuda A., et al., “The Expression Pattern of PFKFB3 Enzyme Distinguishes Between Induced-Pluripotent Stem Cells and Cancer Stem Cells,” Onco Issue 30, pp. 29753-29770. cited by applicant
Clarke L.L., “A Guide to Using Chamber Studies of Mouse Intestine,” American Journal of Physiology: Gastrointestinal and Liver Physiology, Jun. 2009, vol cited by applicant
Clevers H., “Modeling Development and Disease with Organoids,” Cell, Jun. 16, 2016, vol. 165, Issue 7, pp. 1586-1597. cited by applicant
Coghlan M., et al., “Selective Small Molecule Inhibitors of Glycogen Synthase Kinase-3 Modulate Glycogen Metabolism and Gene Transcription,” Chemistry Issue 10, pp. 793-803. cited by applicant
Collier A.J., et al., “Comprehensive Cell Surface Protein Profiling Identifies Specific Markers of Human Naive and Primed Pluripotent States,” Cell Stem Cel 890. cited by applicant
Correia C., et al., “Combining Hypoxia and Bioreactor Hydrodynamics Boosts Induced Pluripotent Stem Cell Differentiation towards Cardiomyocytes,” Stem 2014, vol. 10, pp. 786-801. cited by applicant
Cortez A R., et al., “Transplantation of Human Intestinal Organoids into the Mouse Mesentery: A More Physiological and Anatomic Engraftment Site,” Surger cited by applicant
Costa M., et al., “A Method for Genetic Modification of Human Embryonic Stem Cells using Electroporation,” Nature Protocols, Apr. 5, 2007, vol. 2, No. 4, p
Couzin J., “Small RNAs Make Big Splash,” Science, Dec. 20, 2002, vol. 298, pp. 2296-2297. cited by applicant
Covacci A., et al., “Molecular Characterization of the 128-kDa Immunodominant Antigen of Helicobacter Pylori Associated with Cytotoxicity and Duodenal U National Academy of Sciences USA, Jun. 15, 1993, vol. 90, pp. 5791-5795. cited by applicant
Crespo M., et al., “Colonic Organoids Derived from Human Induced Pluripotent Stem Cells for Modeling Colorectal Cancer and Drug Testing,” Nature Medi pp. 878-884. cited by applicant
Crocenzi F.A., et al., “Ca(2+)-Dependent Protein Kinase C Isoforms are Critical to Estradiol 17beta-D-Glucuronide-Induced Cholestasis in the Rat,” Hepatolo 1895. cited by applicant
Cunningham T.J., et al., “Mechanisms of Retinoic Acid Signalling and its Roles in Organ and Limb Development,” Nature Reviews Molecular Cell Biology, v 110-123. cited by applicant
Curchoe C.L., et al., “Early Acquisition of Neural Crest Competence During hESCs Neuralization,” PloS One, Nov. 2010, vol. 5, pp. 1-17. cited by applicant
Dahl A., et al., “Translational Regenerative Medicine-Hepatic Systems,” Chapter 34, Clinical Aspects of Regenerative Medicine, eds. A. Atala, M.D. and J. Al 2015, pp. 469-484. cited by applicant
D'Amour K A., et al., “Production of Pancreatic Hormone-Expressing Endocrine Cells from Human Embryonic Stem Cells,” Nature Biotechnology, 2006, vol by applicant
Das R., “RFID Forecasts, Players and Opportunities 2017-2027,” IDTechEx, 2017, downloaded from https://www.idtechex.com/en/research-report/rfid-foreca 2017-2027/546, 8 pages. cited by applicant
Date S., et al., “Mini-Gut Organoids: Reconstitution of the Stem Cell Niche,” Annual Review of Cell and Developmental Biology, Nov. 2015, vol. 31, pp. 269
Davenport C., et al., “Anterior-Posterior Patterning of Definitive Endoderm Generated from Human Embryonic Stem Cells Depends on the Differential Signa BMP-Signaling,” Stem Cells, 2016, vol. 34, pp. 2635-2647. cited by applicant
De Santa Barbara P., et al., “Bone Morphogenetic Protein Signaling Pathway Plays Multiple Roles During Gastrointestinal Tract Development,” Developmen 312-322. cited by applicant
Dedhia P.H., et al., “Organoid Models of Human Gastrointestinal Development and Disease,” Gastroenterology, Jan. 14, 2016, vol. 150, pp. 1098-1112. cited b
Dekaney C.M., et al., “Expansion of Intestinal Stem Cells Associated with Long-Term Adaptation Following lleocecal Resection in Mice,” American Journal o

and Liver Physiology, Sep. 13, 2007, vol. 293, pp. G1013-G1022. cited by applicant
Dekkers J F., et al., “A Functional CFTR Assay Using Primary Cystic Fibrosis Intestinal Organoids,” Nature Medicine, Jul. 2013, vol. 19, No. 7, pp. 939-945.
Demehri F.R., et al., “Development of an Endoluminal Intestinal Attachment for Clinically Applicable Distraction Enterogenesis Device,” Journal of Pediatric
101-106. cited by applicant
Demehri F.R., et al., “Development of an Endoluminal Intestinal Lengthening Device using a Geometric Intestinal Attachment Approach,” Surgery, 2015, vol.
applicant
Deng H., et al., “Effects of All-Trans Retinoic Acid on the Differentiation of Neural Stem Cells and the Expression of c-myc Gene,” Chinese Journal of Tissue
2007, vol. 11, pp. 2039-2042. cited by applicant
Deng H., “Mechanisms of Retinoic Acid on the Induction of Differentiation of Neural Stem Cells for Newborn Rat Striatum,” Chinese Doctoral and Master D
(Doctoral) Basic Science, Issue 4, May 20, 2005, pp. 1-91. (Translation). cited by applicant
Denham M., et al., “Multipotent Caudal Neural Progenitors derived from Human Pluripotent Stem Cells that give Rise to Lineages of the Central and Periphe
Mar. 5, 2015, vol. 33, pp. 1759-1770. cited by applicant
DeWard A.D., et al., “Cellular Heterogeneity in the Mouse Esophagus Implicates the Presence of a Nonquiescent Epithelial Stem Cell Population,” Cell Repo
pp. 701-711. cited by applicant
Discher D.E., et al., “Growth Factors, Matrices, and Forces Combine and Control Stem Cells,” Science, Jun. 2009, vol. 324, pp. 1673-1677. cited by applicant
Dobreva G., et al., “SATB2 Is a Multifunctional Determinant of Craniofacial Patterning and Osteoblast Differentiation,” Cell, 2006, vol. 125, Issue 5, pp. 971-
Driver I., et al., “Specification of regional intestinal stem cell identity during *Drosophila* metamorphosis,” Development, 2014, vol. 141, pp. 1848-1856. cited
Duluc I., et al., “Fetal Endoderm Primarily Holds the Temporal and Positional Information Required for Mammalian Intestinal Development,” The Journal of
211-221. cited by applicant
Dunn, “Cationic Nanoparticles for the Targeting and Delivery of Nucleic Acids to the Pulmonary Endothelium,” University of Cincinnati, Sep. 19, 2018, Doct
https://etd.ohiolink.edu/apexprod/rws_olink/r/1501/10?clear=10&p10_accession_num=ucin1544098242321181; 160 p. cited by applicant
Eberhard J., et al., “A Cohort Study of the Prognostic and Treatment Predictive Value of SATB2 Expression in Colorectal Cancer,” British Journal of Cancer, 2
by applicant
Eicher A.K., et al., “Translating Developmental Principles to Generate Human Gastric Organoids,” Cellular and Molecular Gastroenterology and Hepatology,
by applicant
Ekser B., et al., “Comparable Outcomes in Intestinal Retransplantation: Single-Center Cohort Study,” The Journal of Clinical and Translational Research, 201
cited by applicant
Elbashir S.M., et al., “Functional Anatomy of siRNAs for Mediating Efficient RNAi in *Drosophila melanogaster* Embryo Lysate,” The EMBO Journal, 2001,
cited by applicant
Engmann J., et al., “Fluid Mechanics of Eating, Swallowing and Digestion-Overview and Perspectives,” Food & Function, 2013, vol. 4, pp. 443-447. cited by
Ezashi T., et al., “Low O2 Tensions and the Prevention of Differentiation of hES Cells,” PNAS, Mar. 2005, vol. 102, Issue 13, pp. 4783-4788. cited by applica
Fagerberg L., et al., “Analysis of the Human Tissue-specific Expression by Genome-wide Integration of Transcriptomics and Antibody-based proteomics,” Mo
2014, vol. 13, pp. 397-406. cited by applicant
Fatehullah A., et al., “Organoids as an in Vitro Model of Human Development and Disease,” Nature Cell Biology, Mar. 2016, vol. 18, Issue 3, pp. 246-254. cit
Feldstein A.E., et al., “Free Fatty Acids Promote Hepatic Lipotoxicity by Stimulating TNF-α Expression via a Lysosomal Pathway,” Hepatology, Jul. 2004, vo
applicant
Finkbeiner S.R., et al., “A Gutsy Task: Generating Intestinal Tissue from Human Pluripotent Stem Cells,” Digestive Diseases and Sciences, 2013, vol. 58, pp.
Finkbeiner S.R., et al., “Stem Cell-Derived Human Intestinal Organoids as an Infection Model for Rotaviruses,” mBio, Jul./Aug. 2012, vol. 3, issue 4, e00159
Finkbeiner S.R., et al., “Transcriptome-wide Analysis Reveals Hallmarks of Human Intestine Development and Maturation in Vitro and In Vivo,” Stem Cell R
1140-1155. cited by applicant
Finkenzeller K., “RFID Handbook: Fundamentals and Applications in Contactless Smart Cards, Radio Frequency Identification and Near-Field Communicatio
Sons, Ltd., 2010, 8 pages. cited by applicant
Fitzpatrick D R., et al., “Identification of SATB2 as the Cleft Palate Gene on 2q32-q33,” Human Molecular Genetics, 2003, vol. 12, Issue 19, pp. 2491-2501. c
Fon Tacer K., et al., “Research Resource: Comprehensive Expression Atlas of the Fibroblast Growth Factor System in Adult Mouse,” Molecular Endocrinolog
2050-2064. cited by applicant
Fordham R.P., et al., “Transplantation of Expanded Fetal Intestinal Progenitors Contributes to Colon Regeneration after Injury,” Cell Stem Cell, Dec. 5, 2013,
applicant
Fu M., et al., “Embryonic Development of the Ganglion Plexuses and the Concentric Layer Structure of Human Gut: a Topographical Study,” Anatomy and E
208, pp. 33-41. cited by applicant
Fu M., et al., “HOXB5 expression is spatially and temporarily Regulated in Human Embryonic Gut during Neural Crest Cell Colonization and Differentiation
Developmental Dynamics, 2003, vol. 228, pp. 1-10. cited by applicant
Furness J.B., “The Enteric Nervous System and Neurogastroenterology,” Nature Reviews/Gastroenterology & Hepatology, May 2012, vol. 9, pp. 286-294. cite
Gafni O., et al., “Derivation of Novel Human Ground State Naive Pluripotent Stem Cells,” Nature, Oct. 29, 2013, vol. 504, pp. 282-286; Supplementary Infor
applicant
Genthe J.R., et al., “Ventromorphins: A New Class of Small Molecule Activators of the Canonical BMP Signaling Pathways,” ACS Chemical Biology, 2017, v
cited by applicant
Georgas K M., et al., “An illustrated Anatomical Ontology of the Developing Mouse Lower Urogenital Tract,” Development, May 15, 2015, vol. 142, pp. 189
Gerdes H.H., et al., “Tunneling Nanotubes, an Emerging Intercellular Communication Route in Development,” 2013, vol. 130, pp. 381-387. cited by applicant
Gessner R.C., et al., “Functional Ultrasound Imaging for Assessment of Extracellular Matrix Scaffolds used for Liver Organoid Formation,” Biomaterials, 201
by applicant
Giles D.A., et al., “Thermoneutral Housing Exacerbates Nonalcoholic Fatty Liver Disease in Mice and Allows for Sex-Independent Disease Modeling,” Natur
7, pp. 829-838. cited by applicant
Ginestet C., Book Review in the Journal of the Royal Statistical Society. Series A (Statistics in Society) (2011), of ggplot2: Elegant Graphics for Data Analysis
174, Issue 1, p. 245 (2 pages). cited by applicant
Glorioso J.M., et al., “Pivotal Preclinical Trial of the Spheroid Reservoir Bioartificial Liver,” Journal of Hepatology, 2015, vol. 63, Issue 2, pp. 388-398. cited
Goldenring J.R., et al., “Differentiation of the Gastric Mucosa: III. Animal Models of Oxyntic Atrophy and Metaplasia,” American Journal of Physiology-Gas
Physiology, 2006, vol. 291, pp. G999-G1004. cited by applicant
Goldenring J.R., et al., “Overexpression of Transforming Growth Factor-alpha Alters Differentiation of Gastric Cell Lineages,” Digestive Diseases and Scienc
773-784. cited by applicant
Goldstein A.M., et al., “BMP Signaling is Necessary for Neural Crest Cell Migration and Ganglion Formation in the Enteric Nervous System,” Mechanisms o
pp. 821-833. cited by applicant
Gomez M.C., et al., “Derivation of Cat Embryonic Stem-Like Cells from in Vitro-Produced Blastocysts on Homologous and Heterologous Feeder Cells,” The
74, Issue 4, pp. 498-515. cited by applicant
Gomez-Pinilla P.J., et al., “Ano1 is a Selective Marker of Interstitial Cells of Cajal in the Human and Mouse Gastrointestinal Tract,” American Journal of Phys
Liver Physiology, 2009, vol. 296, Issue 6, pp. G1370-G1381. cited by applicant

Gouon-Evans V., et al., “BMP-4 is Required for Hepatic Specification of Mouse Embryonic Stem Cell-Derived Definitive Endoderm,” Nature Biotechnology, 1402-1411. cited by applicant
Gracz A.D., et al., “Brief report: CD24 and CD44 Mark Human Intestinal Epithelial Cell Populations with Characteristics of Active and Facultative Stem Cell 31, pp. 2024-2030. cited by applicant
Gracz A.D., et al., “Sox9 Expression Marks a Subset of CD24-Expressing Small Intestine Epithelial Stem Cells that Form Organoids in Vitro,” American Jour Gastrointestinal and Liver Physiology, 2010, vol. 298, Issue 5, pp. G590-G600. cited by applicant
Gradwohl G., et al., “Neurogenin3 is Required for the Development of the Four Endocrine Cell Lineages of the Pancreas,” Proceedings of the National Acade 2000, vol. 97, No. 4, pp. 1607-1611. cited by applicant
Grapin-Botton A., “Three-Dimensional Pancreas Organogenesis Models,” Diabetes Obesity and Metabolism, Sep. 2016, vol. 18 (Suppl 1), pp. 33-40. cited by
Green M.D., et al., “Generation of Anterior Foregut Endoderm from Human Embryonic and Induced pluripotent Stem Cells,” Nature Biotechnology, Mar. 201 cited by applicant
Gregersen H., et al., “The Zero-Stress State of the Gastrointestinal Tract: Biomechanical and Functional Implications,” Digestive Diseases and Sciences, Dec. 2281. cited by applicant
Gregorieff A., et al., “Wnt Signaling in the Intestinal Epithelium: from Endoderm to Cancer,” Genes & Development, 2005, vol. 19, pp. 877-890. cited by app
Groneberg D.A., et al., “Intestinal Peptide Transport: Ex Vivo Uptake Studies and Localization of Peptide Carrier PEPT1,” American Journal of Physiology: G Physiology, Sep. 2001, vol. 281, pp. G697-G704. cited by applicant
Grosse A.S., et al., “Cell Dynamics in Fetal Intestinal Epithelium: Implications for Intestinal Growth and Morphogenesis,” Development, 2011, vol. 138, pp. 4
Guan Y., et al., “Human Hepatic Organoids for the Analysis of Human Genetic Diseases,” JCI Insight, Sep. 7, 2017, vol. 2, Issue 17, e94954; 17 pages. cited b
Guilak F., et al., “Control of Stem Cell Fate by Physical Interactions with the Extracellular Matrix,” Cell Stem Cell, Jul. 2009, vol. 5, pp. 17-26. cited by appli
Guo G., et al., “Epigenetic Resetting of Human Pluripotency,” Development, 2017, vol. 144, pp. 2748-2763. cited by applicant
Guo Z., et al., “Injury-Induced BMP Signaling Negatively Regulates Drosophila Midgut Homeostasis,” Journal of Cell Biology, 2013, vol. 201, Issue 6, pp. 94
Gurdon J.B., “Adult Frogs Derived from the Nuclei of Single Somatic Cells,” Developmental Biology, 1962, vol. 4, pp. 256-273. cited by applicant
Gurken A. “Advances in Small Bowel Transplantation,” Turkish Journal of Surgery, 2017, vol. 33, Issue 3, pp. 135-141. cited by applicant
Gyorgy A.B., et al., “SATB2 Interacts with Chromatin-Remodeling Molecules in Differentiating Cortical Neurons,” European Journal of Neuroscience, 2008, applicant
Haimovich G., et al., “Intercellular mRNA Trafficking via Membrane Nanotube-Like Extensions in Mammalian Cells,” PNAS, 2017, vol. 114, Issue 46, pp. E
Han B., et al., “Microbiological Safety of a Novel Bio-Artificial Liver support System Based on Porcine Hepatocytes: an Experimental Study,” European Jour vol. 17, Issue 1, Journal 13, 8 pages. cited by applicant
Han M E., et al., “Gastric Stem Cells and Gastric Cancer Stem Cells,” Anatomy & Cell Biology, 2013, vol. 46, Issue 1, pp. 8-18. cited by applicant
Hannon G.J., “RNA Interference,” Nature, 2002, vol. 418, Issue 6894, pp. 244-251. cited by applicant
Hannon N.R.F., et al., “Generation of Multipotent Foregut Stem Cells from Human Pluripotent Stem Cells,” Stem Cell Reports, Oct. 2013, vol. 1, Issue 4, pp.
Hao M.M., et al., “Development of Enteric Neuron Diversity,” Journal of Cellular and Molecular Medicine, 2009, vol. 13, Issue 7, pp. 1193-1210. cited by app
Haramis A.P.G., et al., “De Novo Crypt Formation and Juvenile Polyposis on BMP Inhibition in Mouse Intestine,” Science, 2004, vol. 303, Issue 5664, pp. 16
Hardwick J.C.H., et al., “Bone Morphogenetic Protein 2 Is Expressed by, and Acts Upon, Mature Epithelial Cells in the Colon,” Gastroenterology, 2004, vol. 1 applicant
Haveri H., et al., “Transcription Factors GATA-4 and GATA-6 in Normal and Neoplastic Human Gastrointestinal Mucosa,” BMC Gastroenterology, 2008, vol applicant
He X.C., et al., “BMP Signaling Inhibits Intestinal Stem Cell Self-Renewal through Suppression of Wnt-beta-Catenin Signaling,” Nature Genetics, 2004, vol. by applicant
Hernandez F., et al., “Refining Indications for Intestinal Retransplantation,” International Small Bowel Symposium 2013; Abstract 12.241, retrieved from https://www.tts.org/component/tts/view=presentation&id=13241, Accessed, Jun. 12, 2017, 3 pages. cited by applicant
Higuchi Y., et al., “Gastrointestinal Fibroblasts Have Specialized, Diverse Transcriptional Phenotypes: A Comprehensive Gene Expression Analysis of Human 2015, vol. 10, Issue 6, 19 pages. cited by applicant
Hockemeyer D., et al., “Genetic Engineering of Human ES and iPS Cells using TALE Nucleases,” Nature Biotechnology, 2012, vol. 29, Issue 8, pp. 731-734.
Hoffmann W., “Current Status on Stem Cells and Cancers of the Gastric Epithelium,” International Journal of Molecular Sciences, 2015, vol. 16, Issue 8, pp. 1
Holland P.W.H., et al., “Classification and Nomenclature of all Human Homeobox Genes,” BMC Biology, 2007, vol. 5, Issue 47, pp. 1-28. cited by applicant
Hooton D., et al., “The Secretion and Action of Brush Border Enzymes in the Mammalian Small Intestine,” Reviews of Physiology, Biochemistry and Pharma 118. cited by applicant
Hou P., et al., “Pluripotent Stem Cells Induced from Mouse Somatic Cells by Small-Molecule Compounds,” Science, 2013, vol. 341, Issue 6146, pp. 651-654.
Howell J.C., et al., “Generating Intestinal tissue from Stem Cells: Potential for Research and Therapy,” Regenerative Medicine, Nov. 2011, vol. 6, Issue 6, pp.
Hsu F., et al., “The UCSC Known Genes,” Bioinformatics, 2006, vol. 22, Issue 9, pp. 1036-1046. cited by applicant
Hu H., et al., “Long-Term Expansion of Functional Mouse and Human Hepatocytes as 3D Organoids,” Cell, 2018, vol. 175, Issue 6, pp. 1591-1606. cited by a
Hu X., et al., “Micrometer-Scale Magnetic-Resonance-Coupled Radio-Frequency Identification and Transceivers for Wireless Sensors in Cells,” Physical Rev 1, 13 pages. cited by applicant
Huang H., “Differentiation of Human Embryonic Stem Cells into Smooth Muscle Cells in Adherent Monolayer Culture,” Biochemical and Biophysical Resea 351 pp. 321-327. cited by applicant
Huch M., et al., “Lgr5+ Liver Stem Cells, Hepatic Organoids and Regenerative medicine,” Regenerative Medicine, 2013, vol. 8, Issue 4, pp. 385-387. cited by
Huch M., et al., “Long-Term Culture of Genome-Stable Bipotent Stem Cells from Adult Human Liver,” Cell, 2015, Issue 160, pp. 299-312. cited by applicant
Huch M., et al., “Modeling mouse and Human development using Organoid cultures,” Development, 2015, Issue 142, pp. 3113-3125. cited by applicant
Huebsch N., et al., “Automated Video-Based Analysis of Contractility and Calcium Flux in Human-Induced Pluripotent Stem Cell-Derived Cardiomyocytes C Scales,” Tissue Engineering: Part C, 2015, vol. 21, No. 5, pp. 467-479. cited by applicant
Huh W.J., et al., “ Menetrier's Disease: Its Mimickers and Pathogenesis,” Journal of Pathology and Translational Medicine, 2016, vol. 50, Issue 1, pp. 10-16. c
Hutvagner G., et al., “A microRNA in a Multiple-Turnover RNAi Enzyme Complex,” Science, Sep. 20, 2002, vol. 297, No. 5589, pp. 2056-2060. cited by app
Jean C., et al., “Pluripotent Genes in Avian Stem Cells,” Develop. Growth Differ., 2013, vol. 55, pp. 41-51. cited by applicant
Jeejeebhoy K.N., “Short Bowel Syndrome: A Nutritional and Medical Approach,” CMAJ, 2002, vol. 166, Issue 10, pp. 1297-1302. cited by applicant
Jenny M., et al., “Neurogenin3 is differentially Required for Endocrine Cell Fate Specification in the Intestinal and Gastric Epithelium,” EMBO J, 2002, vol. 2 by applicant
Johannesson M., et al., “FGF4 and Retinoic Acid Direct Differentiation of hESCs into PDX1-Expressing Foregut Endoderm in a Time- and Concentration-De Mar. 2009, vol. 4, Issue 3, pp. 1-13. cited by applicant
Johansson K.A., et al., “Temporal Control of Neurogenin3 Activity in Pancreas Progenitors Reveals Competence Windows for the Generation of Different En 2007, vol. 12, pp. 457-465. cited by applicant
Johnson L.R., et al., “Stimulation of Rat Oxyntic Gland Mucosal Growth by Epidermal Growth Factor,” American Journal of Physiology, 1980, vol. 238, Issu
Johnston T.B., et al., “Extroversion of the Bladder, Complicated by the Presence of Intestinal Openings on the Surface of the Extroverted Area,” Journal of An 89-106. cited by applicant
Jones P., et al., “Stromal Expression of Jagged 1 Promotes Colony Formation by Fetal Hematopoietic Progenitor Cells,” Blood, Sep. 1, 1998, vol. 92, No. 5, p

Jung P., et al., "Isolation and in Vitro Expansion of Human Colonic Stem Cells," Nature Medicine, Oct. 2011, vol. 17, pp. 1225-1227. cited by applicant
Juno R J., et al., "A serum factor(s) after Small Bowel Resection Induces Intestinal Epithelial Cell Proliferation: Effects of Timing, Site, and Extent of Resecti
Jun. 2003, vol. 38, pp. 868-874. cited by applicant
Juno R.J., et al., "A Serum Factor after Intestinal Resection Stimulates Epidermal Growth Factor Receptor Signaling and Proliferation in Intestinal Epithelial C
132, pp. 377-383. cited by applicant
Kabouridis P S., et al., "Microbiota Controls the Homeostasis of Glial Cells in the Gut Lamina Propria," Neuron, Jan. 21, 2015, vol. 85, pp. 289-295. cited by
Karlikow M., et al., "*Drosophila* Cells Use Nanotube-like Structures to Transfer dsRNA and RNAi Machinery Between Cells," Scientific Reports, 2016, vol. 6
applicant
Katoh M., "WNT Signaling in Stem Cell Biology and Regenerative Medicine," Current Drug Targets, 2008, vol. 9, Issue 7, pp. 565-570. cited by applicant
Kawaguchi J., et al., "Isolation and propagation of enteric neural crest progenitor cells from mouse embryonic stem cells and embryos," Development, 2010, v
applicant
Kawaguchi Y., et al., "The Role of the Transcriptional Regulator Ptf1a in Converting Intestinal to Pancreatic Progenitors," Nature Genetics, 2002, vol. 32, pp.
Keeley T.M., et al., "Cytodifferentiation of the postnatal mouse stomach in normal and Huntingtin-interacting protein 1-related-deficient mice," American Jou
Gastrointestinal and Liver Physiology, 2010, vol. 299, pp. G1241-G1251. cited by applicant
Keitel V., et al., "De Novo Bile Salt Transporter Antibodies as a Possible Cause of Recurrent Graft Failure after Liver Transplantation: A Novel Mechanism of
vol. 50, pp. 510-517. cited by applicant
Keung A.J., et al., "Presentation Counts: Microenvironmental Regulation of Stem Cells by Biophysical and Material Cues," Annual Review of Cell and Devel
pp. 533-556. cited by applicant
Khan F.A., et al., "Overview of Intestinal and Multivisceral Transplantation," UpToDate, Sep. 2018 retrieved from https://www.uptodate.com/contents/overvie
l-transplantation/print], 32 pages. cited by applicant
Kilens S., et al., "Parallel Derivation of Isogenic Human Primed and Naive Induced Pluripotent Stem Cells," Nature Communications, 2018, vol. 9, Issue 360,
Kilpinen H., et al., "Common Genetic Variation Drives Molecular Heterogeneity in Human iPSCs," Nature, 2017, vol. 546, Issue 7658, pp. 370-375. cited by
Kim B.M., et al., "Regulation of Mouse Stomach Development and Barx1 Expression by specific microRNAs," Development, 2011, vol. 138, pp. 1081-1086.
Kim B.M., et al., "The Stomach Mesenchymal Transcription Factor Barx1 Specifies Gastric Epithelial Identity through Inhibition of Transient Wnt Signaling,"
8, pp. 611-622. cited by applicant
Kim D., et al., "HISAT: a Fast Spliced Aligner with Low Memory Requirements," Nature Methods, 2015, vol. 12, Issue 4, pp. 357-360. cited by applicant
Kim T.H., et al., "Stomach Development, Stem Cells and Disease," Development, 2016, vol. 143, pp. 554-565. cited by applicant
Kohlnhofer B.M., et al., "GATA4 Regulates Epithelial Cell Proliferation to Control Intestinal Growth and Development in Mice," Cellular and Molecular Gas
2016, vol. 2(2), pp. 189-209. cited by applicant
Koike M., et al., "Effects of mechanical strain on proliferation and differentiation of bone marrow stromal cell line ST2," Journal of Bone and Mineral Metabo
cited by applicant
Kolahchi A.R., et al., "Microfluidic-Based Multi-Organ Platforms for Drug Discovery," Micromachines, 2016, vol. 7, Issue 162, pp. 1-33. cited by applicant
Kolodny G.M., "Evidence for Transfer of Macromolecular RNA Between Mammalian Cells in Culture," Experimental Cell Research, 1971, vol. 65, pp. 313-3
Koo B-K., et al., "Controlled Gene Expression in Primary Lgr5 Organoid Cultures," Nature Methods, Jan. 1, 2012, vol. 9, No. 1, Jan. 1, 2012, pp. 81-83. cited
Kosinski C., et al., "Indian Hedgehog Regulates Intestinal Stem Cell Fate through Epithelial-Mesenchymal Interactions during Development," Gastroenterolo
903. cited by applicant
Kostrzewski T., et al., "Three-dimensional Perfused Human in Vitro Model of Non-Alcoholic Fatty Liver Disease," World Journal of Gastroenterology, 2017,
cited by applicant
Kovalenko P.L., et al., "The Correlation between the Expression of Differentiation Markers in rat Small Intestinal Mucosa and the Transcript Levels of Schlafe
2013, vol. 148, pp. 1013-1019. cited by applicant
Kraus M.R.C., et al., "Patterning and Shaping the Endoderm in Vivo and in Culture," Current Opinion Genetics & Development, 2012, vol. 22, pp. 347-353. c
Krausova M., et al., "Wnt Signaling in Adult Intestinal Stem Cells and Cancer," Cellular Signalling, 2014, vol. 26, pp. 570-579. cited by applicant
Kretzschmar K., et al., "Organoids: Modeling Development and the Stem Cell Niche in a Dish," Developmental Cell, Sep. 2016, vol. 38, pp. 590-600. cited by
Kroon E., et al., "Pancreatic Endoderm Derived From Human Embryonic Stem Cells Generates Glucose-Responsive Insulin Secreting Cells In Vivo," Nature
Issue 4, pp. 443-452. cited by applicant
Kruitwagen H.S., et al., "SCH-O-5 Long-Term Adult Feline Liver Organoid Cultures for Disease Modelling of Hepatic Lipidosis," Research Communications
Congress, Sep. 2016, ECVIM Abstracts pp. 203-204. cited by applicant
Kruitwagen H.S., et al., "Long-Term Adult Feline Liver Organoid Cultures for Disease Modeling of Hepatic Steatosis," Stem Cell Reports, Apr. 2017, vol. 8(4
Kubal C.A., et al., "Challenges with Intestine and Multivisceral Re-Transplantation: Importance of Timing of Re-Transplantation and Optimal Immunosuppre
23, pp. 98-104. cited by applicant
Kudoh T., et al., "Distinct Roles for Fgf, Wnt and Retinoic Acid in Posteriorizing the Neural Ectoderm," Development, 2002, vol. 129, pp. 4335-4346. cited b
Kumar M., et al., "Signals from Lateral Plate Mesoderm Instruct Endoderm toward a Pancreatic Fate," Dev Biol, 2003, vol. 259, Issue 1, pp. 109-122. cited b
Kuratnik A., et al., "Intestinal Organoids as Tissue Surrogates for Toxicological and Pharmacological Studies," Biochemical Pharmacology, 2013, vol. 85, Issu
applicant
Kurpios N.A., et al., "The Direction of Gut Looping is Established by Changes in the Extracellular Matrix and in Cell: Cell Adhesion," PNAS, 2008, vol. 105,
by applicant
Lachmann N., et al., "Large-Scale Hematopoietic Differentiation of Human Induced Pluripotent Stem Cells Provides Granulocytes or Macrophages for Cell R
Cell Report, Feb. 10, 2015, vol. 4, pp. 282-296. cited by applicant
Lahar N., et al., "Intestinal Subepithelial Myofibroblasts Support in Vitro and in Vivo Growth of Human Small Intestinal Epithelium," PLoS One, Nov. 2011,
by applicant
Lai F.P-L., et al., "Correction of Hirschsprung-Associated Mutations in Human Induced Pluripotent Stem Cells Via Clustered Regularly Interspaced Short Pal
Neural Crest Cell Function," Gastroenterology, 2017, vol. 153, No. 1, pp. 139-153. cited by applicant
Lambert P.F., et al., "Using an Immortalized Cell Line to Study the HPV Life Cycle in Organotypic "Raft" Cultures," Methods Molecular Medicine, 2005, vol
applicant
Lambrecht N.W.G., et al., "Identification of the K Efflux Channel Coupled to the Gastric H-K-Atpase During Acid Secretion," Physiological Genomics, 2005,
by applicant
Lameris A.L., et al., "Expression Profiling of Claudins in the Human Gastrointestinal Tract in Health and During Inflammatory Bowel Disease," Scandinavian
2013, vol. 48, Issue 1, pp. 58-69. cited by applicant
Langmead G., et al., "Ultrafast and Memory-Efficient Alignment of Short DNA Sequences to the Human Genome," Genome Biology, 2009, vol. 10, 10 pages
Lavial F., et al., "Chicken Embryonic Stem Cells as a Non-Mammalian Embryonic Stem Cell Model," Development, Growth & Differentiation, 2010, vol. 52,
Le Douarin N.M., et al., "Neural Crest Cell Plasticity and its Limits," Development 131, 2004, pp. 4637-4650. cited by applicant
Lee C.S., et al., "Neurogenin 3 Is Essential for the Proper Specification of Gastric Enteroendocrine Cells and the Maintenance of Gastric Epithelial Cell Identi
1488-1497. cited by applicant
Lee G., et al., "Isolation and Directed Differentiation of Neural Crest Stem Cells Derived from Human Embryonic Stem Cells," Nature Biotechnology, Dec. 2
by applicant

Lennerz J.K.M., et al., “The Transcription Factor MIST1 Is a Novel Human Gastric Chief Cell Marker Whose Expression Is Lost in Metaplasia, Dysplasia, an
Journal of Pathology, 2010, vol. 177, Issue 3, pp. 1514-1533. cited by applicant
Leung A.A., et al., “Tolerance Testing of Passive Radio Frequency Identification Tags for Solvent, Temperature, and Pressure Conditions Encountered in an A
Biorepository Setting,” Journal of Pathology Informatics, 2010, vol. 1, 6 pages. cited by applicant
Levin D.E., et al., “Human Tissue-Engineered Small Intestine Forms from Postnatal Progenitor Cells,” Journal of Pediatric Surgery, 2013, vol. 48, pp. 129-13
Li H., et al., “Treefam: A Curated Database of Phylogenetic Trees of Animal Gene Families,” Nucleic Acids Research, 2006, vol. 34, pp. D572-D580. cited by
Li L., “BMP Signaling Inhibits Intestinal Stem Cell Self-Renewal through Antagonizing Wnt Signaling,” Gastroenterology, AASLD Abstracts, Abstract S122
by applicant
Li Y., et al., “In Vitro Organogenesis from Pluripotent Stem Cells,” Organogenesis, Jun. 2014, vol. 10, Issue 2, pp. 159-163. cited by applicant
Li Z., et al., “SATB2 is a Sensitive Marker for Lower Gastrointestinal Well-Differentiated Neuroendocrine Tumors,” International Journal of Clinical and Exp
6, Jan. 2015, pp. 7072-7082. cited by applicant
Lim D.A., et al., “Noggin Antagonizes BMP Signaling to Create a Niche for Adult Neurogenesis,” Neuron, Dec. 2000, vol. 28, pp. 713-726. cited by applicant
Lin C., et al., “The Application of Engineered Liver Tissues for Novel Drug Discovery,” Expert Opinion on Drug Discovery, 2015, vol. 10, Issue 5, pp. 519-5
Lin Y., et al., “Differentiation, Evaluation, and Application of Human Induced Pluripotent Stem Cell-Derived Endothelial Cells,” Arteriosclerosis, Thrombosis
37, pp. 2014-2025. cited by applicant
Lindley R.M., et al., “Human and Mouse Enteric Nervous System Neurosphere Transplants Regulate the Function of Aganglionic Embryonic Distal Colon,” G
135, No. 1, pp. 205-216. cited by applicant
Liu J., et al., “A Small-Molecule Agonist of the Wnt Signaling Pathway,” Angewandte Chemie International Edition Engl., 2005, vol. 44, issue 13, pp. 1987-1
Liu J.A-J., et al., “Identification of GLI Mutations in Patients with Hirschsprung Disease that Disrupt Enteric Nervous System Development in Mice,” Gastro
pp. 1837-1848. cited by applicant
Liu L., et al., “A Review of Locomotion Systems for Capsule Endoscopy,” IEEE Reviews in Biomedical Engineering, 2015, vol. 8, pp. 138-151. cited by appl
Logan C.Y., et al., “The Wnt Signaling Pathway in Development and Disease,” Annual Review of Cell and Developmental Biology, 2004, vol. 20, pp. 781-81
Loike J.D., et al., “Opinion: Develop Organoids, Not Chimeras, for Transplantation,” The Scientist Magazine, Aug. 2019, (online: http://www.the-scientist.co
develop-Organoids--not--chimeras--for-transplantation-66339), 3 pages. cited by applicant
Longmire T.A., et al., “Efficient Derivation of Purified Lung and Thyroid Progenitors from Embryonic Stem Cells,” Stem Cell, 2012, vol. 10, pp. 398-411. cit
Lopez-Diaz L., et al., “Intestinal Neurogenin 3 Directs Differentiation of a Bipotential Secretory Progenitor to Endocrine Cell Rather than Goblet Cell Fate,” I
vol. 309, pp. 298-305. cited by applicant
Love M.I., et al., “Moderated Estimation of Fold Change and Dispersion for RNA-seq Data with DESeq2,” Genome Biology, 2014, vol. 15, No. 12, pp. 1-21.
Low L.A., et al., “Organs-on-Chips: Progress, Challenges, and Future Directions,” Experimental Biology and Medicine, 2017, vol. 242, pp. 1573-1578. cited
Lu Y., et al., “A Novel 3D Liver Organoid System for Elucidation of Hepatic Glucose Metabolism,” Biotechnology and Bioengineering, Feb. 2012, vol. 109, I
applicant
Ludwig T.E., et al., “Derivation of Human Embryonic Stem Cells in Defined Conditions,” Nature Biotechnology, Feb. 2006, vol. 24, pp. 185-187. cited by ap
Ludwig T.E., et al., “Feeder-Independent Culture of Human Embryonic Stem Cells,” Nat Methods, Aug. 2006, vol. 3, pp. 637-646. cited by applicant
Lui V.C.H., et al., “Perturbation of Hoxb5 Signaling in Vagal Neural Crests Down-regulates Ret Leading to Intestinal Hypoganglionosis in Mice,” Gastroente
1104-1115. cited by applicant
Luntz J., et al., “Mechanical Extension Implants for Short-Bowel Syndrome,” Smart Structures and Materials 2006: Smart Structures and Integrated Systems,
6173, pp. 617309-1-617309-11. cited by applicant
Luo X., et al., “Generation of Endoderm Lineages from Pluripotent Stem Cells,” Regenerative Medicine, 2017, vol. 12, Issue 1, pp. 77-89. cited by applicant
MacParland S.A., et al., “Single Cell RNA Sequencing of Human Liver Reveals Distinct Intrahepatic Macrophage Populations,” Nature Communications, Oc
pages. cited by applicant
Mahe M.M., et al., “Establishment of Gastrointestinal Epithelial Organoids,” Current Protocols in Mouse Biology, 2013, vol. 3, No. 4, 31 pages. cited by appl
Mahe M.M., et al., “In Vivo Model of Small Intestine,” Methods in Molecular Biology, 2017, vol. 1597, pp. 229-245. cited by applicant
Majumdar A.P., “Postnatal Undernutrition: Effect of Epidermal Growth Factor on Growth and Function of the Gastrointestinal Tract in Rats,” Journal of Pedia
Nutrition, 1984, vol. 3, pp. 618-625. cited by applicant
Mammoto A., et al., “Mechanosensitive Mechanisms in Transcriptional Regulation,” Journal of Cell Science, 2012, vol. 125, pp. 3061-3073. cited by applica
Marini F., et al., “pcaExplorer: an R/Bioconductor Package for Interacting with RNA-seq Principal Components,” BMC Bioinformatics, Jun. 2019, vol. 20, Is
Marini F., “pcaExplorer: Interactive Visualization of RNA-seq Data Using a Principal Components Approach,” Bioconductor.org, R package version 2.3.0, 20
Marsh M.N., et al., “A Study of the Small Intestinal Mucosa Using the Scanning Electron Microscope,” Gut, 1969, vol. 10, pp. 940-949. cited by applicant
Martin M., et al., “Dorsal Pancreas Agenesis in Retinoic Acid-Deficient Raldh2 Mutant Mice,” Developmental Biology, Aug. 2005, vol. 284, pp. 399-411. cite
McCauley H.A., et al., “Pluripotent Stem Cell-derived Organoids: Using Principles of Developmental Biology to Grow Human Tissues in a Dish,” Developme
962. cited by applicant
McCracken K.W., et al., “Erratum: Wnt/β-catenin promotes gastric fundus specification in mice and humans,” Nature, 2017, vol. 543, Issue 136, 1 page. cited
McCracken K.W., et al., “Generating Human Intestinal Tissue from Pluripotent Stem Cells in Vitro,” Nature Protocols, Nov. 2011, vol. 6, Issue 12, pp. 1920-1
McCracken K.W., et al., “Mechanism of Embryonic Stomach Development,” Seminars in Cell & Development Biology, 2017, vol. 66, pp. 36-42. cited by app
McCracken K.W., et al., “Modelling Human Development and Disease in Pluripotent Stem-Cell-Derived Gastric Organoids,” Nature, Oct. 2014, vol. 516, Issu
applicant
McCracken K.W., “Mechanisms of Endoderm Patterning and Directed Differentiation of Human Stem Cells into Foregut Tissues,” Dissertation, Graduate Sch
Cincinnati, Jun. 19, 2014, 185 pages. cited by applicant
McGovern D.P.B., et al., “Genome-Wide Association Identifies Multiple Ulcerative Colitis Susceptibility Loci,” Nature Genetics, Apr. 2010, vol. 42, Issue 4,
McKenzie T.J., et al., “Artificial and Bioartificial Liver Support,” Seminars in Liver Disease, May 2008, vol. 28, Issue 2, pp. 210-217. cited by applicant
McKeown S.J., et al., “Hirschsprung Disease: A Developmental Disorder of the Enteric Nervous System,” Wiley Interdisciplinary Reviews Developmental Bi
113-129. cited by applicant
McLin V.A., et al., “The Role of the Visceral Mesoderm in the Development of the Gastrointestinal Tract,” Gastroenterology, Jun. 2009, vol. 136, pp. 2074-20
McMahon J.A., et al., “Noggin-Mediated Antagonism of BMP Signaling is required for Growth and Patterning of the Neural Tube and Somite,” Genes & Dev
1438-1452. cited by applicant
McManus M.T., et al., “Gene Silencing in Mammals by Small Interfering RNAs,” Nature Reviews Genetics, Oct. 2002, vol. 3, pp. 737-747. cited by applican
Meerbrey K.L., et al., “The pINDUCER Lentiviral Toolkit for Inducible RNA Interference in Vitro and in Vivo,” Proceedings of the National Academy of Sci
pp. 3665-3670. cited by applicant
Merker S.R., et al., “Gastrointestinal Organoids: How they Gut it out,” Developmental Biology, Dec. 2016, vol. 420, pp. 239-250. cited by applicant
Mica Y., et al., “Modeling Neural Crest Induction, Melanocyte Specification and Disease-Related Pigmentation Defects in hESCs and Patient-Specific iPSCs,
3, pp. 1140-1152. cited by applicant
Micallef S.J., et al., “Endocrine Cells Develop within Pancreatic Bud-like Structures Derived from Mouse ES Cells Differentiated in Response to BMP4 and
Research, Oct. 2007, vol. 1, pp. 25-36. cited by applicant
Mills J C., et al., “Gastric Epithelial Stem Cells,” Gastroenterology, Feb. 2011, vol. 140, pp. 412-424. cited by applicant
Miyabayashi T., et al., “Wnt/beta-Catenin/CBP Signaling Maintains Long-Term Murine Embryonic Stem Cell Pluripotency,” Proceedings of the National Aca

2007, vol. 104, issue 13, pp. 5668-5673. cited by applicant
Múnera J.O., et al., “Differentiation of Human Pluripotent Stem Cells into Colonic Organoids via Transient Activation of BMP Signaling,” Cell Stem Cell, Jul
64. cited by applicant
Molodecky N.A., et al., “Increasing Incidence and Prevalence of the Inflammatory Bowel Diseases With Time, Based on Systematic Review,” Gastroenterolog
cited by applicant
Molotkov A., et al., “Retinoic Acid Generated by Raldh2 in Mesoderm is required for Mouse Dorsal Endodermal Pancreas Development,” Developmental Dy
950-957. cited by applicant
Mori R., et al., “Micropatterned Organoid Culture of Rat Hepatocytes and HepG2 Cells,” Journal of Bioscience and Bioengineering, Sep. 2008, vol. 106(3), p
Moser A.R., et al., “A Dominant Mutation that Predisposes to Multiple Intestinal Neoplasia in the Mouse,” Science, Jan. 1990, vol. 247, Issue 4940, pp. 322-3
Mosher J T., et al., “Intrinsic Differences among Spatially Distinct Neural Crest Stem Cells in Terms of Migratory Properties, Fate-Determination, and Ability
System,” Developmental Biology, Mar. 2007, vol. 303, issue 1, pp. 1-15. cited by applicant
Mou H., et al., “Generation of Multipotent Lung and Airway Progenitors from Mouse ESCs and Patient-Specific Cystic Fibrosis iPSCs,” Stem Cell, Apr. 2012
applicant
Munera J.O., et al., “Generation of Gastrointestinal Organoids from Human Pluripotent Stem Cells, Organ Regeneration,” In: Tsuji T., (eds), Organ Regenerat
Biology, Humana Press, New York, NY, 2017, vol. 1597, pp. 167-177. cited by applicant
Munoz M., et al., “Conventional Pluripotency Markers are Unspecific for Bovine Embryonic-Derived Cell-Lines,” Theriogenology, 2008, vol. 69, pp. 1159-1
Nakamura T., et al., “Advancing Intestinal Organoid Technology toward Regenerative Medicine,” Cellular and Molecular Gastroenterology and Hepatology, 2
applicant
Nantasanti S., et al., “Concise Review: Organoids are a Powerful Tool for the Study of Liver Disease and Personalized Treatment Design in Humans and Anim
Modeling and Therapy,” Stem Cells Translational Medicine, Jan. 21, 2016, vol. 5(3), pp. 325-330. cited by applicant
Neiiendam J L., et al., “An NCAM-derived FGF-receptor Agonist, the FGL-peptide, Induces Neurite Outgrowth and Neuronal Survival in Primary Rat Neuro
2004, vol. 91, issue 4, pp. 920-935. cited by applicant
Nelson B J., et al., “Microrobots for Minimally Invasive Medicine,” Annual Review of Biomedical Engineering, 2010, vol. 15, issue 12, pp. 55-85. cited by ap
Nelson C M., “On Buckling Morphogenesis,” Journal of Biomechanical Engineering, 2016, vol. 138, pp. 021005-1-021005-6. cited by applicant
Nielsen C., et al., “Gizzard Formation and the Role of Bapx1,” Developmental Biology, 2001, vol. 231, Issue 1, pp. 164-174. cited by applicant
Noguchi T-A.K., et al., “Generation of Stomach Tissue from Mouse Embryonic Stem Cells,” Nature Cell Biology, 2015, vol. 17, Issue 8, pp. 984-993. cited by
Nomura S., et al., “Evidence for Repatteming of the Gastric Fundic Epithelium Associated With Menetrier's Disease and TGFα Overexpression,” Gastroentero
5, pp. 1292-1305. cited by applicant
Obermayr F., et al., “Development and Developmental Disorders of the Enteric Nervous System,” Nature Reviews/Gastroenterology & Hepatology, Jan. 2013
by applicant
Ogaki S., et al., “Wnt and Notch Signals Guide Embryonic Stem Cell Differentiation into the Intestinal Lineages,” Stem Cells, Jun. 2013, vol. 31, Issue 6, pp.
Okada Y., et al., “Retinoic-Acid-Concentration-Dependent Acquisition of Neural Cell Identity during in Vitro Differentiation of Mouse Embryonic Stem Cells
vol. 275, Issue 1, pp. 124-142. cited by applicant
Okita K., et al., “An Efficient Nonviral Method to Generate Integration-Free Human-Induced Pluripotent Stem Cells from Cord Blood and Peripheral Blood C
31, Issue 3, pp. 458-466. cited by applicant
Olbe L., et al., “A Mechanism by which Helicobacter Pylori Infection of the Antrum Contributes to the Development of Duodenal Ulcer,” Gastroenterology, 1
1394. cited by applicant
Ootani A., et al., “Sustained in Vitro Intestinal Epithelial Culture within a Wnt-Dependent Stem Cell Niche,” Nature Medicine, 2009, vol. 15, Issue 6, pp. 701-
Ornitz D.M., et al., “FGF Signaling Pathways in Endochondral and Intramembranous Bone Development and Human Genetic Disease,” Genes & Developme
1446-1465. cited by applicant
Ornitz D.M., et al., “The Fibroblast Growth Factor Signaling Pathway,” Wiley Interdisciplinary Reviews Developmental Biology, 2015, vol. 4, Issue 3, pp. 21
Ouchi R., et al., “Modeling Steatohepatitis in Humans with Pluripotent Stem Cell-Derived Organoids,” Cell Metabolism, Aug. 6, 2019, vol. 30, Issue 2, pp. 37
Paddison P.J., et al., “Short Hairpin Activated Gene Silencing in Mammalian Cells,” Methods in Molecular Biology, 2004, vol. 265, pp. 85-100. cited by appli
Paddison P.J., et al., “RNA interference: the new somatic cell genetics?,” Cancer Cell, 2002, vol. 2, Issue 1, pp. 17-23. cited by applicant
Pai R., et al., “Deoxycholic acid activates beta-catenin Signaling Pathway and Increases Colon Cell Cancer Growth and Invasiveness,” Molecular Biology of t
pp. 2156-2163. cited by applicant
Pan Q., “University of Science and Technology of China Press,” Physiology, Jan. 31, 2014, pp. 149-150. cited by applicant
Pardal M.L., et al., “Towards the Internet of Things: An Introduction to RFID Technology,” RFID Technology-Concepts, Applications, Challenges, Proceedin
Workshop, IWRT 2010, In conjunction with ICEIS, 2010, pp. 69-78. cited by applicant
Paris D.B.B.P., et al., “Equine Embryos and Embryonic Stem Cells: Defining Reliable Markers of Pluripotency,” Theriogenology, 2010, vol. 74, Issue 4, pp. 5
Park J.S., et al., “Differential Effects of Equiaxial and Uniaxial Strain on Mesenchymal Stem Cells,” Biotechnology and Bioengineering, 2004, vol. 88, Issue 3
Park J.S., et al., “The effect of Matrix Stiffness on the Differentiation of Mesenchymal Stem Cells in Response to TGF-β,” Biomaterials, 2011, vol. 32, Issue 1
applicant
Park K.I., et al., “Acute Injury Directs the Migration, Proliferation, and Differentiation of Solid Organ Stem Cells: Evidence for the Effect of Hypoxia-Ischem
“reporter” Neural Stem Cells,” Experimental Neurology, 2006, vol. 199, Issue 1, pp. 156-178. cited by applicant
Park Y.H., et al., “Review of Atrophic Gastritis and Intestinal Metaplasia as a Premalignant Lesion of Gastric Cancer,” Journal of Cancer Prevention, 2015, vo
applicant
Parkin D.M., “The Global Health Burden of Infection-Associated Cancers in the Year 2002,” International Journal of Cancer, 2006, vol. 118, Issue 12, pp. 303
Pastor W.A., et al., “TFAP2C Regulates Transcription in Human Naive Pluripotency by Opening Enhancers,” Nature Cell Biology, 2018, vol. 20, Issue 5, pp.
Pastula A., et al., “Three-Dimensional Gastrointestinal Organoid Culture in Combination with Nerves or Fibroblasts: A Method to Characterize the Gastrointe
Cells International, 2016, 16 pages. cited by applicant
Patankar J.V., et al., “Intestinal Deficiency of Gata4 Protects from Diet-Induced Hepatic Steatosis by Suppressing De Novo Lipogenesis and Gluconeogenesis
Posters, Abstract 1253, 2012, vol. 56, p. S496. cited by applicant
Patankar J.V., et al., “Intestinal GATA4 Deficiency Protects from Diet-Induced Hepatic Steatosis,” Journal of Hepatology, 2012, vol. 57, Issue 5, pp. 1061-106
Peek Jr., R.M., et al., “Helicobacter pylori cagA+ Strains and Dissociation of Gastric Epithelial Cell Proliferation from Apoptosis,” Journal of the National Ca
863-868. cited by applicant
Peek Jr., R.M., “Helicobacter Pylori Infection and Disease: from Humans to Animal Models,” Disease Models & Mechanisms, 2008, vol. 1, Issue 1, pp. 50-55
Pennisi C.P., et al., “Uniaxial Cyclic Strain Drives Assembly and Differentiation of Skeletal Myocytes,” Tissue Engineering: Part A, 2011, vol. 17, pp. 2543-2
Pereira C.F., et al., “Heterokaryon-Based Reprogramming of Human B Lymphocytes for Pluripotency Requires Oct4 but Not Sox2,” PLoS Genet, 2008, vol. 4
cited by applicant
Petitte J.N., et al., “Avian Pluripotent Stem Cells,” Mechanisms of Development, 2004, vol. 121, Issue 9, pp. 1159-1168. cited by applicant
Poling H.M., et al., “Mechanically Induced Development and Maturation of Human Intestinal Organoids in Vivo,” Nature Biomedical Engineering, 2018, vol.
applicant
Pompaiah M., et al., “Gastric Organoids: An Emerging Model System to Study Helicobacter pylori Pathogenesis,” Current Topics in Microbiology and Immu
168. cited by applicant

Prakash R., “Regulation of WNT Genes in Stem Cells Development and Organogenesis,” IJP, Jun. 2014, vol. 1, Issue 6, pp. 366-372. cited by applicant
Pulikkot S., “Establishment of a 3D Culture Model of Gastric Stem Cells Supporting Their Differentiation into Mucous Cells Using Microfibrous Polycaprola
United Arab Emirates University, College of Medicine and Health Sciences, May 2015, 187 pages. cited by applicant
Qi M-C., et al., “Mechanical Strain induces Osteogenic Differentiation: Cbfa1 and Ets-1 Expression in Stretched Rat Mesenchymal Stem Cells,” International
Surgery, 2008, vol. 37, pp. 453-458. cited by applicant
Que J., et al., “Morphogenesis of the Trachea and Esophagus: Current Players and New Roles for Noggin and Bmps,” Differentiation, 2006, vol. 74, pp. 422-4
Raju R., et al., “A Network Map of FGF-1/FGFR Signaling System,” Journal of Signal Transduction, Apr. 2014, Article ID 962962, 16 pages. cited by applica
Ramalingam S., et al., “Distinct Levels of Sox9 Expression Mark Colon Epithelial Stem Cells that form Colonoids in Culture,” The American Journal of Phys
Physiology, 2012, vol. 302, Issue 1, pp. G10-G20. cited by applicant
Ramirez-Weber F.A., et al., “Cytonemes: Cellular Processes that Project to the Principal Signaling Center in *Drosophila* Imaginal Discs,” Cell, May 28, 1999,
applicant
Ramsey V.G., et al., “The Maturation of Mucus-Secreting Gastric Epithelial Progenitors: into Digestive-Enzyme Secreting Zymogenic Cells Requires Mist1,”
Issue 1, pp. 211-222. cited by applicant
Rane A., et al., “Drug Metabolism in the Human Fetus and Newborn Infant,” Pediatric Clinics of North America, 1972, vol. 19, Issue 1, pp. 37-49. cited by ap
Rankin S.A., et al., “A Molecular Atlas of Xenopus Respiratory System Development,” Developmental Dynamics, 2015, vol. 244, pp. 69-85. cited by applicar
Rankin S.A., et al., Timing is everything: Reiterative Wnt, BMP and RA Signaling Regulate Developmental Competence during Endoderm Organogenesis, De
2018, vol. 434, Issue 1, pp. 121-132. cited by applicant
Rankin S.A., et al., “Suppression of Bmp4 Signaling by the Zinc-Finger Repressors Osr1 and Osr2 is required for Wnt/beta-Catenin-Mediated Lung Specifica
2012, vol. 139, Issue 16, pp. 3010-3020. cited by applicant
Rao R.R., et al., “Gene Expression Profiling of Embryonic Stem Cells Leads to Greater Understanding of Pluripotency and Early Developmental Events,” Bio
71, pp. 1772-1778. cited by applicant
Ratineau C., et al., “Endoderm- and Mesenchyme-Dependent Commitment of the Differentiated Epithelial Cell Types in the Developing Intestine of Rat,” Dif
163-169. cited by applicant
Ray K., “Engineering Human Intestinal Organoids with a Functional ENS,” Nature Reviews Gastroenterology & Hematology, Nov. 2016, 1 page. cited by app
Reilly G C., et al., “Intrinsic Extracellular Matrix Properties Regulate Stem Cell Differentiation,” Journal of Biomechanics, Jan. 2010, vol. 43, Issue 1, pp. 55-
Rennert K., et al., “A Microfluidically Perfused Three Dimensional Human Liver Model,” Biomaterials, 2015, vol. 71, pp. 119-131. cited by applicant
Ricchi M., et al., “Differential Effect of Oleic and Palmitic Acid on Lipid Accumulation and Apoptosis in Cultured Hepatocytes,” Journal of Gastroenterology
24, Issue 5, pp. 830-840. cited by applicant
Richards M. et al., “The Transcriptome Profile of Human Embryonic Stem Cells as Defined by SAGE,” Stem Cells, 2004, vol. 22, pp. 51-64. cited by applicar
Riedinger, et al., “Reversible Shutdown of Replicon Initiation by Transient Hypoxia in Ehrlich ascites Cells: Dependence of Initiation on Short-Lived Protein,
Biochemistry, Dec. 1992, vol. 210, Issue 2, pp. 389-398. cited by applicant
Roberts A., et al., “Identification of Novel Transcripts in Annotated Genomes using RNA-Seq,” Bioinformatics, 2011, vol. 27, Issue 17, pp. 2325-2329. cited b
Roberts A., et al., “Improving RNA-Seq Expression Estimates by Correcting for Fragment Bias,” Genome Biology, 2011, vol. 12, 14 pages. cited by applicant
Roberts D.J., et al., “Sonic Hedgehog is an Endodermal Signal inducing Bmp-4 and Hox genes during Induction and Regionalization of the Chick hindgut,” D
3163-3174. cited by applicant
Rodriguez, P., et al., “BMP Signaling in the Development of the Mouse Esophagus and Forestomach,” Development, 2010, vol. 137, Issue 24, pp. 4171-4176.
Rodriguez-Pineiro A.M., et al., “Studies of Mucus in Mouse Stomach, Small Intestine, and Colon. II. Gastrointestinal Mucus Proteome Reveals Muc2 and Mu
Core Proteins,” American Journal of Physiology: Gastrointestinal and Liver Physiology, 2013, vol. 305, pp. G348-G356. cited by applicant
Rohrschneider, M.R., et al., “Polarity and Cell Fate Specification in the Control of C. Elegans Gastrulation,” Developmental Dynamics, 2009, vol. 238, Issue
Roth, R.B., et al., “Gene Expression Analyses Reveal Molecular Relationships among 20 Regions of the Human CNS,” Neurogenetics, 2006, vol. 7, pp. 67-80
Rouch J.D., et al., “Scalability of an Endoluminal Spring for Distraction Enterogenesis,” Journal of Pediatric Surgery, 2016, vol. 51, pp. 1988-1992. cited by a
Roy S., et al., “Cytoneme-Mediated Contact-Dependent Transport of the *Drosophila* Decapentaplegic Signaling Protein,” Science, 2014, vol. 343, pp. 124462
applicant
Sachs N., et al., “A Living Biobank of Breast Cancer Organoids Captures Disease Heterogeneity,” Cell, 2018, vol. 172, pp. 373-386. cited by applicant
Saenz J.B., et al., “Stomach Growth in a Dish,” Nature, Jan. 2017, vol. 541, No. 7636, pp. 160-161. cited by applicant
Saffrey M J., “Cellular Changes in the Enteric Nervous System During Ageing,” Developmental Biology, 2013, vol. 382, pp. 344-355. cited by applicant
Saha S., et al., “Inhibition of Human Embryonic Stem Cell Differentiation by Medical Strain,” Journal of Cellular Physiology, 2006, vol. 206, pp. 126-137. cit
Saito M., et al., “Reconstruction of Liver Organoid using a Bioreactor,” World Journal of Gastroenterology, Mar. 2006, vol. 12, Issue 12, pp. 1881-1888. cited
Salas-Vidal E., et al., “Imaging Filopodia Dynamics in the Mouse Blastocyst,” Developmental Biology, 2004, vol. 265, pp. 75-89. cited by applicant
Sampaziotis F., et al., “Potential of Human Induced Pluripotent Stem Cells in Studies of Liver Disease,” Hepatology, 2015, vol. 62, Issue 1, pp. 303-311. cited
Sancho E., et al., “Signaling Pathways in Intestinal Development and Cancer,” Annual Review of Cell and Developmental Biology, 2004, vol. 20, pp. 695-723
Sartori-Rupp A., et al., “Correlative Cryo-Electron Microscopy Reveals the Structure of TNTs in Neuronal Cells,” Nature Communications, 2019, vol. 10, 16
Sasai Y., “Cytosystems Dynamics in Self-Organization of Tissue Architecture,” Nature, 2013, vol. 493, pp. 318-326. cited by applicant
Sasai Y., “Next-Generation Regenerative Medicine: Organogenesis from Stem Cells in 3D Culture,” Cell Stem Cell, May 2013, vol. 12, pp. 520-530. cited by
Sasselli V., et al., “The Enteric Nervous System,” Developmental Biology, Jan. 2012, vol. 366, pp. 64-73. cited by applicant
Sato T., et al., “Single Lgr5 Stem Cells Build Crypt-Villus Structures in Vitro without a Mesenchymal Niche,” Nature, 2009, vol. 459, pp. 262-265. cited by ap
Sato T., et al., “Long-term Expansion of Epithelial Organoids from Human Colon, Adenoma, Adenocarcinoma, and Barrett's Epithelium,” Gastroenterology, N
1772. cited by applicant
Sato T., et al., “Snapshot: Growing Organoids from Stem Cells,” Cell, 2015, vol. 161, pp. 1700-1700e1. cited by applicant
Savidge T.C., et al., “Human Intestinal Development in a Severe-Combined Immunodeficient Xenograft model,” Differentiation, 1995, vol. 58, pp. 361-371. c
Savin T., et al., “On the Growth and Form of the Gut,” Nature, Aug. 3, 2011, vol. 476, pp. 57-62. cited by applicant
Schlieve C.R., et al., “Created of Warm Blood and Nerves: Restoring an Enteric Nervous System in Organoids,” Cell Stem Cell, Jan. 2017, vol. 20, pp. 5-7. ci
Schmelter M., et al., “Embryonic Stem Cells Utilize Reactive Oxygen Species as Transducers of Mechanical Strain-induced Cardiovascular Differentiation,” T
vol. 20, Issue 8, pp. 1182-1184. cited by applicant
Schonhoff S E., et al., “Neurogenin 3-Expressing Progenitor Cells in the Gastrointestinal Tract Differentiate into both Endocrine and Non-Endocrine Cell Typ
2004, vol. 270, No. 2, pp. 443-454. cited by applicant
Schumacher M A., et al., “Gastric Sonic Hedgehog Acts as a Macrophage Chemoattractant during the Immune Response to Helicobacter pylori,” Gastroenter
5, pp. 1150-1159. cited by applicant
Schumacher M.A., et al., “The Use of Murine-derived Fundic Organoids in Studies of Gastric Physiology,” J. Physiol., Apr. 15, 2015, vol. 593, Issue 8, pp. 18
Shah S.B., et al., “Cellular Self-assembly and Biomaterials-based Organoid Models of Development and Diseases,” Acta Biomaterialia, Apr. 15, 2017, vol. 53
Shahbazi M N., et al., “Self-organization of the human embryo in the absence of maternal tissues,” Nature Cell Biology, May 4, 2016, vol. 18, Issue 6, pp. 700
Shan J., et al., “Identification of a Specific Inhibitor of the Dishevelled PDZ Domain,” Biochemistry, 2005, vol. 44, No. 47, pp. 15495-15503. cited by applica
Sheehan-Rooney K., et al., “Bmp and Shh Signaling Mediate the Expression of satb2 in the Pharyngeal Arches,” PLoS One, Mar. 21, 2013, vol. 8, No. 3, e59
Shekherdimian S., et al., “The Feasibility of using an Endoluminal Device for Intestinal Lengthening,” Journal of Pediatric Surgery, Aug. 2010, vol. 45, Issue
applicant

Sherwood R I., et al., "Transcriptional Dynamics of Endodermal Organ Formation," Developmental Dynamics, Jan. 2009, vol. 238, Issue 1, pp. 29-42. cited b
Sherwood R I., et al., "Wnt Signaling Specifies and Patterns Intestinal Endoderm," Mechanisms of Development, Sep. 2011, vol. 128, pp. 387-400. cited by ap
Shi X L., et al., "Evaluation of a Novel Hybrid Bioartificial Liver Based on a Multi-Layer Flat-Plate Bioreactor," World Journal of Gastroenterology, Jul. 28, 2
3760. cited by applicant
Shi X-L., et al., "Effects of Membrane Molecular Weight Cut-off on Performance of a Novel Bioartificial Liver," Artificial Organs, Mar. 2011, vol. 35, Issue, 3
Shimizu N., et al., "Cyclic Strain Induces Mouse Embryonic Stem Cell Differentiation into Vascular Smooth Muscle Cells by Activating PDGF Receptor Beta
Mar. 2008, vol. 104, pp. 766-772. cited by applicant
Shyer A.E., et al., "Bending Gradients: How the Intestinal Stem Cell Gets Its Home," Cell, Apr. 23, 2015, vol. 161, Issue 3, pp. 569-580. cited by applicant
Shyer A.E., et al., "Villification: How the Gut Gets its Villi," Science, Oct. 11, 2013, vol. 342, pp. 212-218. cited by applicant
Siegel R., et al., "Colorectal Cancer Statistics, 2014," CA Cancer Journal for Clinicians, Apr. 2014, vol. 64, Issue 2, pp. 104-117. cited by applicant
Sigalet D L., "The Role of the Enteric Neuronal System in Controlling Intestinal Function," Clinical Surgery Society Magazine, 2003, vol. 64, p. 214. cited by
Siller R., et al., "Small-Molecule-Driven Hepatocyte Differentiation of Human Pluripotent Stem Cells," Stem Cell Reports, May 2015, vol. 4, No. 5, pp. 939-9
Sim Y.J., et al., "2i Maintains a Naive Ground State in ESCs through Two Distinct Epigenetic Mechanisms," Stem Cell Reports, May 9, 2017, vol. 8, Issues. 5
applicant
Simkin J.E., et al., "Retinoic Acid Upregulates Ret and Induces Chain Migration and Population Expansion in Vagal Neural Crest Cells to Colonise the Embry
vol. 8(5), e64077, pp. 1-12. cited by applicant
Simon-Assmann P., et al., "In Vitro Models of Intestinal Epithelial Cell Differentiation," Cell Biology and Toxicology, Jul. 2007, vol. 23, No. 4, pp. 241-256. c
Sinagoga K.L., et al., "Generating Human Intestinal Tissues from Pluripotent Stem Cells to Study Development and Disease," The EMBO Journal, May 5, 20
1163. cited by applicant
Sitti M., et al., "Biomedical Applications of Untethered Mobile Milli/Microrobots," The Proceedings of the IEEE Institution of Electrical Engineers, Feb. 2015
cited by applicant
Skardal A., et al., "Organoid-on-a-Chip and Body-on-a-Chip Systems for Drug Screening and Disease Modeling," Drug Discovery Today, Sep. 2016, vol. 21,
applicant
Slaymaker I M., et al., "Rationally Engineered Cas9 Nucleases with Improved Specificity," Science, Jan. 1, 2016, vol. 351, issue 6268, pp. 84-88. cited by app
Snoeck H W., "Generation of Anterior Foregut Derivatives from Pluripotent Stem Cells," Stem Cells Handbook, S. Sell (ed.), Jul. 3, 2013, pp. 161-175. cited b
Snykers S., et al., "In Vitro Differentiation of Embryonic and Adult Stem Cells into Hepatocytes: State of the Art," Stem Cells, Mar. 2009, vol. 27, No. 3, pp. 5
Soffers J H M., et al., "The Growth Pattern of the Human Intestine and its Mesentery," BMC Developmental Biology, Aug. 22, 2015, vol. 15, Issue 31, 16 pag
Sonntag F., et al., "Design and Prototyping of a Chip-based Multi-micro-Organoid Culture System for Substance Testing, Predictive to Human (substance) Ex
Biotechnology, Jul. 1, 2010, vol. 148, Issue 1, pp. 70-75. cited by applicant
Soto-Gutierrez A., et al., "Engineering of an Hepatic Organoid to Develop Liver Assist Devices," Cell Transplant, 2010, vol. 19, No. 6, 12 pages. cited by app
Spear P.C., et al., "Interkinetic Nuclear Migration: A Mysterious Process in Search of a Function," Develop. Growth Differ, Apr. 2012, vol. 54, Issue 3, pp. 30
Speer A L., et al., "Fibroblast Growth Factor 10-Fibroblast Growth Factor Receptor 2b Mediated Signaling is not Required for Adult Glandular Stomach Hom
vol. 7, Issue 11, e49127, 12 pages. cited by applicant
Speer A L., et al., "Murine Tissue-Engineered Stomach Demonstrates Epithelial Differentiation," Journal of Surgical Research, Mar. 22, 2011, vol. 171, Issue
Spence J R., et al., "Translational Embryology: Using Embryonic Principles to Generate Pancreatic Endocrine cells from Embryonic Stem Cells," Developme
236, issue 12, pp. 3218-3227. cited by applicant
Spence J R., et al., "Vertebrate Intestinal Endoderm Development," Developmental Dynamics, Mar. 2011, vol. 240, Issue 3, pp. 501-520. cited by applicant
Stange D E., et al., "Differentiated Troy+ Chief Cells act as 'Reserve' Stem cells to Generate all Lineages of the Stomach Epithelium," Cell, Oct. 10, 2013, vo
by applicant
Stark R., et al., "Development of an Endoluminal Intestinal Lengthening Capsule," Journal of Pediatric Surgery, Jan. 2012, vol. 47, Issue 1, pp. 136-141. cited
Stuart T., et al., "Comprehensive Integration of Single-Cell Data," Cell, Jun. 13, 2019, vol. 177, pp. 1888-1902. cited by applicant
Su N., et al., "Role of FGF/FGFR Signaling in Skeletal Development and Homeostasis: Learning from Mouse Models," Bone Research, Apr. 29, 2014, vol. 2,
applicant
Sugawara T., et al., "Organoids Recapitulate Organs?," Stem Cell Investigation, Jan. 2018, vol. 5(3), 4 pages. cited by applicant
Sugimoto S., et al., "Reconstruction of the Human Colon Epithelium in Vivo," Cell Stem Cell, 2018, vol. 22, pp. 171-176, e1-e5. cited by applicant
Sui L., et al., "Signaling Pathways During Maintenance and Definitive Endoderm Differentiation of Embryonic Stem Cells," The International Journal of Dev
57(1), pp. 1-12. cited by applicant
Sun Y., et al., "Genome Engineering of Stem Cell Organoids for Disease Modeling," Protein Cell, May 2017, vol. 8(5), pp. 315-327. cited by applicant
Suzuki A., et al., "Clonal Identification and Characterization of Self-renewing Pluripotent Stem Cells in the Developing Liver," The Journal of Cell Biology, J
184. cited by applicant
Tada M., et al., "Embryonic Germ Cells Induce Epigenetic Reprogramming of Somatic Nucleus in Hybrid Cells," EMBO Journal, 1997, vol. 16(21), pp. 6510
Taipale J., et al., "The Hedgehog and Wnt signalling pathways in cancer," Nature, May 17, 2001, vol. 411, pp. 349-354. cited by applicant
Tait I.S., et al., "Colonic Mucosal Replacement by Syngeneic Small Intestinal Stem Cell Transplantation," The American Journal of Surgery, Jan. 1994, vol. 1
applicant
Tait I.S., et al., "Generation of Neomucosa in Vivo by Transplantation of Dissociated Rat Postnatal Small Intestinal Epithelium," Differentiation, Apr. 1994 vo
applicant
Takahashi K., et al., "Induction of Pluripotent Stem Cells from Mouse Embryonic and Adult Fibroblast Cultures by Defined Factors," Cell, Aug. 25, 2006, vol
applicant
Takahashi S., et al., "Epigenetic Differences between Naive and Primed Pluripotent Stem Cells," Cellular and Molecular Life Sciences, Apr. 2018, vol. 75(7),
Takaki M., et al., "In Vitro Formation of Enteric Neural Network Structure in a Gut-Like Organ Differentiated from Mouse Embryonic Stem Cells," Stem Cel
1414-1422. cited by applicant
Takashima Y., et al., "Resetting Transcription Factor Control Circuitry toward Ground-State Pluripotency in Human," Cell, Sep. 11, 2014, vol. 158(6), pp. 125
Takebe T., et al., "Massive and Reproducible Production of Liver Buds Entirely from Human Pluripotent Stem Cells," Cell Reports, Dec. 5, 2017, vol. 21(10),
applicant
Tamm C., et al., "A Comparative Study of Protocols for Mouse Embryonic Stem Cell Culturing," PLoS One, Dec. 10, 2013, vol. 8(12), e81156, 10 pages. cite
Tamminen K., et al., "Intestinal Commitment and Maturation of Human Pluripotent Stem Cells Is Independent for Exogenous FGF4 and R-spondin1," PLOS
e0134551, 19 pages. cited by applicant
Tang W., et al., "Faithful Expression of Multiple Proteins via 2A-Peptide Self-processing: a Versatile and Reliable method for Manipulating Brain Circuits," T
8, 2009, vol. 29(27), pp. 8621-8629. cited by applicant
Teo A K.K., et al., "Activin and BMP4 Synergistically Promote Formation of Definitive Endoderm in Human Embryonic Stem Cells," Stem Cells, Apr. 2012,
applicant
Terry B.S., et al., "Preliminary Mechanical Characterization of the Small Bowel for In Vivo Robotic Mobility," Journal of Biomechanical Engineering, Sep. 2
09101-7. cited by applicant
Thanasupawat T., et al., "INSL5 is a Novel Marker for Human Enteroendocrine Cells of the Large Intestine and Neuroendocrine Tumours," Oncology Reports
154. cited by applicant

The WNT Homepage, “Small molecules in Wnt signaling,” Nusse Lab, Jan. 2019, 2 pages. cited by applicant
Theunissen T.W., et al., “Systematic Identification of Culture Conditions for Induction and Maintenance of Naive Human Pluripotency,” Cell Stem Cell, Oct. 2
cited by applicant
Tiso N., et al., “BMP Signalling Regulates Anteroposterior Endoderm Patterning in Zebrafish,” Mech Dev, Oct. 2002, vol. 118, pp. 29-37. cited by applicant
Toivonen S., et al., “Activin A and Wnt-dependent Specification of Human Definitive Endoderm Cells,” Experimental Cell Research, Aug. 2013, vol. 319(17)
applicant
Tran K., et al., “Evaluation of Regional and Whole Gut Motility using the Wireless Motility Capsule: Relevance in Clinical Practice,” Therapeutic Advances i
vol. 5(4), pp. 249-260. cited by applicant
Trapnell C., et al., “Differential gene and Transcript Expression Analysis of RNA-seq Experiments with TopHat and Cufflinks,” Nature Protocols, 2013, vol. 7
applicant
Trapnell C., et al., “Transcript Assembly and Quantification by RNA-Seq reveals Unannotated Transcripts and Isoform Switching during Cell Differentiation,”
2010, vol. 28(5), pp. 511-515. cited by applicant
Trisno S.L., et al., “Esophageal Organoids from Human Pluripotent Stem Cells Delineate Sox2 Functions during Esophageal Specification,” Cell Stem Cell, C
515. cited by applicant
Tsakmaki A., et al., “3D Intestinal Organoids in Metabolic Research: Virtual Reality in a Dish,” Current Opinion in Pharmacology, 2017, vol. 37, pp. 51-58. ci
Tuschl T. et al., “Targeted mRNA degradation by Double-Stranded RNA in vitro,” Genes & Development., 1999, vol. 13, pp. 3191-3197. cited by applicant
Tyml K., et al., “Lipopolysaccharide Reduces Intercellular Coupling in Vitro and Arteriolar Conducted Response in Vivo,” American Journal of Physiology-H
2001, vol. 281, pp. H1397-H1406. cited by applicant
Uppal K., et al., “Meckel's Diverticulum: A Review,” Clinical Anatomy, 2011, vol. 24, pp. 416-422. cited by applicant
Valadi H., et al., “Exosome-Mediated Transfer of mRNAs and MicroRNAs is a Novel mechanism of Genetic Exchange between Cells,” Nature Cell Biology,
cited by applicant
Van Breemen R.B., et al., “Caco-2 Cell Permeability Assays to Measure Drug Absorption,” Expert Opinion on Drug Metabolism & Toxicology, Aug. 2005, vo
applicant
Van Dop W.A., et al., “Depletion of the Colonic Epithelial Precursor Cell Compartment upon Conditional Activation of the Hedgehog Pathway,” Gastroentero
2195-2203. cited by applicant
Van Klinken B.J.W., et al., “MUC5B is the Prominent Mucin in Human Gallbladder and is also Expressed in a Subset of Colonic Goblet Cells,” The American
vol. 274, pp. G871-G878. cited by applicant
Venick, R.S., et al., “Unique Technical and Patient Characteristics of Retransplantation: A Detailed Single-Center Analysis of Intestinal Transplantation,” Inter
Symposium 2013; Abstract 5.203, retrieved from https://www.tts.org/component/ts/?view=presentation&id=13190, accessed Jun. 12, 2017, 4 pages. cited by a
Verzi M.P., et al., “Role of the Homeodomain Transcription Factor Bapx1 in Mouse Distal Stomach Development,” Gastroenterology, 2009, vol. 136, No. 5, p
Vu J., et al., “Regulation of Appetite, Body Composition and Metabolic Hormones by Vasoactive Intestinal Polypeptide (VIP),” Journal of Molecular Neurosc
2, pp. 377-387. cited by applicant
Wakayama T., et al., “Full-term Development of Mice from Enucleated Oocytes Injected with Cumulus Cell Nuclei,” Nature, Jul. 23, 1998, vol. 394, pp. 369-
Walker E.M., et al., “GATA4 and GATA6 Regulate Intestinal Epithelial Cytodifferentiation during Development,” Developmental Biology, 2014, vol. 392, pp
Wallace A S., et al., “Development of the Enteric Nervous System, Smooth Muscle and Interstitial cells of Cajal in the Human Gastrointestinal Tract,” Cell an
vol. 319, pp. 367-382. cited by applicant
Walton K.D., et al., “Epithelial Hedgehog Signals Direct Mesenchymal Villus Patterning through BMP,” Abstracts / Developmental Biology, 2009, vol. 331, A
applicant
Walton K.D., et al., “Hedgehog-Responsive Mesenchymal Clusters Direct Patterning and Emergence of Intestinal Villi,” PNAS, Sep. 25, 2012, vol. 109, No. 3
applicant
Walton K.D., et al., “Villification in the Mouse: Bmp Signals Control Intestinal Villus Patterning,” Development, 2016, vol. 143, pp. 427-436. cited by applica
Wan W., et al., “The Role of wnt Signaling in the Development of Alzheimer's disease: A Potential Therapeutic Target?,” BioMed Research International, 201
applicant
Wang A., et al., “Generating Cells of the Gastrointestinal system: Current Approaches and Applications for the Differentiation of Human Pluripotent Stem Cel
Medicine, Jun. 20, 2012, vol. 90, pp. 763-771. cited by applicant
Wang F., et al., “Isolation and Characterization of Intestinal Stem Cells based on Surface Marker Combinations and Colony-Formation Assay,” Gastroenterolo
383-395. cited by applicant
Wang J., et al., “Mutant Neurogenin-3 in Congenital Malabsorptive Diarrhea,” New England Journal of Medicine, Jul. 20, 2006, vol. 355, pp. 270-280. cited b
Wang S., (Ed.), “The role of Homologous Genes in the Development of Appendages,” in Basis of Developmental Biology, Press of East China University of S
184-185. cited by applicant
Wang X., et al., “Cloning and Variation of Ground State Intestinal Stem Cells,” Nature, Jun. 11, 2015, vol. 522, 18 pages. cited by applicant
Wang Z., et al., “Retinoic Acid Regulates Morphogenesis and Patterning of Posterior Foregut Derivatives,” Developmental Biology, May 23, 2006, vol. 297, p
Want R., “An Introduction to RFID Technology,” IEEE Pervasive Computing, 2006, vol. 5, pp. 25-33. cited by applicant
Ward D.F Jr., et al., “Mechanical Strain Enhances Extracellular Matrix-Induced Gene Focusing and Promotes Osteogenic Differentiation of Human Mesenchy
Extracellular-Related Kinase-Dependent Pathway,” Stem Cells and Development, 2007, vol. 16, pp. 467-479. cited by applicant
Ware C.B., “Concise Review: Lessons from Naive Human Pluripotent Cells,” Stem Cells, 2017, vol. 35, pp. 35-41. cited by applicant
Warlich E., et al., “Lentiviral Vector Design and Imaging Approaches to Visualize the Early Stages of Cellular Reprogramming,” Molecular Therapy, Apr. 201
cited by applicant
Watson C.L., et al., “An In Vivo Model of Human Small Intestine Using Pluripotent Stem Cells,” Nature Medicine, Oct. 19, 2014, vol. 20, No. 11, 16 pages. c
Wehkamp J., et al., “Paneth Cell Antimicrobial Peptides: Topographical Distribution and Quantification in Human Gastrointestinal Tissues,” FEBS Letters, 20
by applicant
Weis V.G., et al., “Current Understanding of SPEM and its Standing in the Preneoplastic Process,” Gastric Cancer, 2009, vol. 12, pp. 189-197. cited by applica
Wells J.M., et al., “How to Make and Intestine,” Development, vol. 141, No. 4, Feb. 15, 2014, pp. 752-760. cited by applicant
Wen S., et al., “Helicobacter Pylori Virulence Factors in Gastric Carcinogenesis,” Cancer Letters, 2009, vol. 282, pp. 1-8. cited by applicant
Wernig M., et al., “In Vitro Reprogramming of Fibroblasts into a Pluripotent ES-cell-like State,” Nature, 2007, vol. 448, pp. 318-324. cited by applicant
Whissell G., et al., “The Transcription Factor GATA6 Enables Self-Renewal of Colon Adenoma Stem Cells by Repressing BMP Gene Expression,” Nature Ce
pp. 695-707. cited by applicant
Wieck M.M., et al., “Prolonged Absence of Mechanoluminal Stimulation in Human Intestine Alters the Transcriptome and Intestinal Stem Cell Niche,” Cell M
vol. 3, No. 3, pp. 367-388. cited by applicant
Wiley L.A., et al., “cGMP Production of Patient-Specific iPSCs and Photoreceptor Precursor Cells to Treat Retinal Degenerative Blindness,” Scientific Repor
cited by applicant
Willet S.G., et al., “Stomach Organ and Cell Lineage Differentiation: From Embryogenesis to Adult Homeostasis,” Cellular and Molecular Gastroenterology a
pp. 546-559. cited by applicant
Williamson R.C.N., et al., “Humoral Stimulation of Cell Proliferation in Small Bowel after Transection and Resection in Rats,” Gastroenterology, 1978, vol. 7
applicant

Wills A., et al., “Bmp Signaling is necessary and sufficient for Ventrolateral Endoderm Specification in Xenopus,” Developmental Dynamics, 2008, vol. 237(8
applicant
Wilmut I., et al., “Viable Offspring Derived from Fetal and Adult Mammalian Cells,” Nature, Feb. 27, 1997, vol. 385, pp. 810-813. cited by applicant
Workman M.J., “Generating 3D Human Intestinal Organoids with an Enteric Nervous System,” Thesis, Graduate School of the University of Cincinnati, 2014
Xia H.H.X., et al., “Antral-Type Mucosa in the Gastric Incisura, Body, and Fundus (Antralization): A Link Between Helicobacter pylori Infection and Intestina
Journal of Gastroenterology, 2000, vol. 95, No. 1, pp. 114-121. cited by applicant
Xinaris C., et al., “Organoid Models and Applications in Biomedical Research,” Nephron Jun. 25, 2015, Issue 130, pp. 191-199. cited by applicant
Xu R., et al. (Eds), “Retinoic Acid Receptor” in Basis and Clinic of Receptor, Shanghai Science and Technology Press, 1992, pp. 129-131. cited by applicant
Xue X., et al., “Endothelial PAS Domain Protein 1 Activates the Inflammatory Response in the Intestinal Epithelium to Promote Colitis in Mice,” Gastroenter
831-841. cited by applicant
Yahagi N., et al., “Position-Specific Expression of Hox Genes along the Gastrointestinal Tract,” Congenital Anomalies, 2004, vol. 44, pp. 18-26. cited by appl
Yamada S., et al. “Differentiation of Immature Enterocytes into Enteroendocrine Cells by Pdx1 Overexpression,” American Journal of Physiology: Gastrointes
vol. 281, No. 1, pp. G229-G236. cited by applicant
Yamaguchi Y., et al., “Purified Interleukin 5 Supports the Terminal Differentiation and Proliferation of Murine Eosinophilic Precursors,” Journal of Experimen
167, No. 1, pp. 43-56. cited by applicant
Yanagita M., “Modulator of Bone Morphogenetic Protein Activity in the Progression of Kidney Diseases,” Kidney International, 2006, vol. 70, pp. 989-993. c
Yeung E.N.W., et al., “Fibrinogen Production is Enhanced in an In-Vitro Model of Non-Alcoholic Fatty Liver Disease: An Isolated Risk Factor for Cardiovasc
and Disease, 2015, vol. 14 (86), 8 pages. cited by applicant
Yin C., et al., “Hepatic Stellate Cells in Liver Development, Regeneration, and Cancer,” The Journal of Clinical Investigation, May 2013, vol. 123, No. 5, pp.
Young H.M., et al., “Expression of Ret-, p75(NTR)-, Phox2a-, Phox2b-, and Tyrosine Hydroxylase-Immunoreactivity by Undifferentiated Neural Crest-Derive
Enteric Neurons in the Embryonic Mouse Gut,” Developmental Dynamics, 1999, vol. 216, pp. 137-152. cited by applicant
Young H.M., et al., “GDNF is a Chemoattractant for Enteric Neural Cells,” Developmental biology, Dec. 19, 2000, vol. 229, pp. 503-516. cited by applicant
Yu H., et al., “The Contributions of Human Mini-Intestines to the Study of Intestinal Physiology and Pathophysiology,” Annual Review of Physiology, Feb. 10
cited by applicant
Yuan Y., et al., “Peptic ulcer disease today,” Nature Clinical Practice Gastroenterology & Hepatology, Feb. 2006, vol. 3, No. 2, pp. 80-89. cited by applicant
Yui S., et al., “Functional Engraftment of Colon Epithelium Expanded in Vitro from a Single Adult Lgr5(+) stem cell,” Nature Medicine, Mar. 11, 2012, vol. 1
applicant
Zachos N.C., et al., “Human Enteroids/Colonoids and Intestinal Organoids Functionally Recapitulate Normal Intestinal Physiology and Pathophysiology,” The
Feb. 19, 2016, vol. 29, No. 18, pp. 3759-3766. cited by applicant
Zbuk K.M., et al., “Hamartomatous polyposis syndromes,” Nature Clinical Practice Gastroenterology & Hepatology, 2007, vol. 4, No. 9, pp. 492-502. cited by
Zhang D., et al., “Neural Crest Regionalisation for Enteric Nervous System Formation: Implications for Hirschsprung's Disease and Stem Cell Therapy,” Deve
2010, vol. 339, pp. 280-294. cited by applicant
Zhang H., et al., “The Existence of Epithelial-to-Mesenchymal Cells with the Ability to Support Hematopoiesis in Human Fetal Liver,” Cell Biology Internati
pp. 213-219. cited by applicant
Zhang Q, et al., “Small-Molecule Synergist of the Wnt/β-catenin Signaling Pathway,” PNAS, May 1, 2007, vol. 104, No. 18, pp. 7444-7448. cited by applican
Zhang R.R., et al., “Human iPSC-Derived Posterior Gut Progenitors are Expandable and Capable of Forming Gut and Liver Organoids,” Stem Cell Reports, M
793. cited by applicant
Zhang W., et al., “Elastomeric Free-Form Blood Vessels for Interconnecting Organs on Chip Systems,” Lab Chip, Apr. 26, 2016, vol. 16, No. 9, pp. 1579-158
Zhang Y., et al., “Palmitic and Linoleic Acids Induce ER Stress and Apoptosis in Hepatoma Cells,” Lipids in Health and Disease, 2012, vol. 11 (1), 8 pages. ci
Zhang Y.S., et al., “Seeking the Right Context for Evaluating Nanomedicine: from Tissue Models in Petri Dishes to Microfluidic Organs-on-a-chip,” Nanomed
5, pp. 685-688. cited by applicant
Zhang Y.S., et al., “Multisensor-Integrated Organs-on-Chips Platform for Automated and Continual in Situ Monitoring of Organoid behaviors,” Proceedings o
Sciences USA, 2017, vol. 114, pp. E2293-E2302. cited by applicant
Zhao Y., et al., “A XEN-like State Bridges Somatic Cells to Pluripotency during Chemical Reprogramming,” Cell, 2015, vol. 163, pp. 1678-1691. cited by app
Zhong J., et al., “Continuous-Wave Laser-Assisted Injection of Single Magnetic Nanobeads into Living Cells,” Sensors and Actuators B: Chemical, 2016, vol.
applicant
Zhou J., et al., “The Potential for Gut Organoid Derived Interstitial Cells of Cajal in Replacement Therapy,” International Journal of Molecular Sciences, Sep.
2059 in 17 pages. cited by applicant
Zhou Q., et al., “In Vivo Reprogramming of Adult Pancreatic Exocrine Cells to β-Cells,” Nature, 2008, vol. 455, pp. 627-632. cited by applicant
Abe T., et al., “Reporter Mouse Lines for Fluorescence Imaging,” Development, Growth & Differentiation, May 2013, vol. 55, No. 4, pp. 390-405. cited by ap
Adam M., et al., “Psychrophilic Proteases Dramatically Reduce Single-Cell RNA-Seq Artifacts: a Molecular Atlas of Kidney Development,” Development, O
3625-3632. cited by applicant
Arora R., et al., “Multiple Roles and Interactions of Tbx4 and Tbx5 in Development of the Respiratory System,” PLoS Genetics, Aug. 2, 2012, vol. 8, No. 8, e
applicant
Asahina K., et al., “Septum Transversum-Derived Mesothelium gives rise to Hepatic Stellate Cells and Perivascular Mesenchymal Cells in Developing Mouse
vol. 53, No. 3, pp. 983-995. cited by applicant
Barnes R.M., et al., “Analysis of the Hand1 Cell Lineage Reveals Novel Contributions to Cardiovascular, Neural Crest, Extra-Embryonic, and Lateral Mesode
Dynamics, vol. 239, 2010, pp. 3086-3097. cited by applicant
Baron M., et al., “A Single-Cell Transcriptomic Map of the Human and Mouse Pancreas Reveals Inter- and Intra-cell Population Structure,” Cell Systems, Oct
360. cited by applicant
Briggs J.A., et al., “The Dynamics of Gene Expression in Vertebrate Embryogenesis at Single-Cell Resolution,” Science , Jun. 1, 2018, vol. 360, No. 6392, ea
applicant
Bult C.J., et al., “Mouse Genome Database (MGD) 2019, ”Nucleic Acids Research, Jan. 8, 2019, vol. 47, No. D1, pp. D801-D806. cited by applicant
Calder, L.E., “Retinoic Acid-mediated Regulation of GLI3 Enables High Yield Motoneuron Derivation from Human Embryonic Stem Cells Independent of Ex
Signaling,” Dissertation, Jan. 2015, 24 pages. cited by applicant
Cao J., et al., “The Single-Cell Transcriptional Landscape of Mammalian Organogenesis,” Nature, Feb. 2019, vol. 566, No. 7745, pp. 496-502. cited by applic
Chambers M. S., et al., “Highly Efficient Neural Conversion of Human ES and IPS Cells by Dual Inhibition of SMAD Signaling,” Nature Biotechnol., Mar. 2
cited by applicant
Chen Y., et al., “Robust Bioengineered 3D Functional Human Intestinal Epithelium,” Scientific Reports, vol. 5 (13708), Sep. 16, 2015, XP055454950, DOI: 1
by applicant
Chua C.C., et al., “Single Luminal Epithelial Progenitors Can Generate Prostate Organoids in Culture,” Nature Cell Biology, Oct. 2014, vol. 16(10), 26 pages.
Cohen M., et al., “Lung Single-Cell Signaling Interaction Map Reveals Basophil Role in Macrophage Imprinting,” Cell, Nov. 1, 2018, vol. 175, No. 4, pp. 10
De Soysa T.Y., et al., “Single-cell Analysis of Cardiogenesis Reveals Basis for Organ-level Developmental Defects,” Nature, Aug. 2019, vol. 572, No. 7767, p
Dolle L., et al., “EpCAM and the Biology of Hepatic Stem/Progenitor Cells,” American Journal of physiology gastrointestinal liver physiology, 2015, vol. 308
applicant

Ei Sebae G.K., et al., "Single-Cell Murine Genetic Fate Mapping Reveals Bipotential Hepatoblasts and Novel Multi-organ Endoderm Progenitors," Developm
19, dev168658, 7 pages. cited by applicant
Erkan M., et al., "Organ-, Inflammation- and Cancer Specific Transcriptional Fingerprints of Pancreatic and Hepatic Stellate Cells,". Molecular Cancer, Dec. 2
by applicant
Farrell J.A., et al., "Single-Cell Reconstruction of Developmental Trajectories During Zebrafish Embryogenesis," Science, Jun. 1, 2018, vol. 360, No. 6392, e
applicant
Fattahi F., et al., "Deriving Human ENS Lineages for Cell Therapy and Drug Discovery in Hirschsprung Disease," Nature, Feb. 2016, vol. 531 (7592), pp. 105
Ferretti E., et al., "Mesoderm Specification and Diversification: From Single Cells to Emergent Tissues,". Current Opinion in Cell Biology, Dec. 2019, vol. 61
Foulke-Abel J., et al., "Human Enteroids as a Model of Upper Small Intestinal Ion Transport Physiology and Pathophysiology," Gastroenterology, Mar. 2016,
cited by applicant
Francou A., et al., "Second Heart Field Cardiac Progenitor Cells in the Early Mouse Embryo," Biochimica et Biophysica Acta, Apr. 1, 2013, vol. 1833, No. 4,
Franklin V., et al., "Regionalisation of the Endoderm Progenitors and Morphogenesis of the Gut Portals of the Mouse Embryo,". Mechanisms of Development
587-600. cited by applicant
Gissen P., et al., "Structural and Functional Hepatocyte Polarity and Liver Disease," Journal of Hepatology, 2015, vol. 63, pp. 1023-1037. cited by applicant
Graffmann N., et al., "Modeling Nonalcoholic Fatty Liver Disease With Human Pluripotent Stem Cell-Derived Immature Hepatocyte-Like Cells Reveals Acti
Regulatory Functions of Peroxisome Proliferator-Activated Receptor Alpha," Stem Cells and Development, vol. 25 (15), 2016, pp. 1119-1133. cited by applic
Grand R. J., et al., "Development of the Human Gastrointestinal Tract—A Review," Gastroenterology, May 1976, vol. 70, No. 5, pp. 790-810. cited by applica
Grapin-Botton A., "Antero-posterior Patterning of the Vertebrate Digestive Tract: 40 Years After Nicole Le Douarin's PHD Thesis," The International Journal
1, 2005, vol. 49, Nos. 2-3, pp. 335-347. cited by applicant
Griffin O.D., et al., "Human B1 Cells in Umbilical Cord and Adult Peripheral Blood Express the Novel Phenotype CD20+CD27+CD43+CD70−," Journal of
vol. 208(1), pp. 67-80. cited by applicant
Hill D R., et al., "Bacterial Colonization Stimulates a Complex Physiological Response in the Immature Human Intestinal Epithelium," Developmental Biolog
Disease, Tools and Resources, Nov. 7, 2017, XP055822977, retrieved from the Internet: https://elifesciences.org/articles/29132, 35 pages. cited by applicant
Hoffmann A.D., et al., "Sonic Hedgehog Is required in Pulmonary Endoderm for Atrial Septation," Development, 2009, vol. 136, p. 1761 1770. cited by appli
Horie M., et al., "TBX4 is involved in the Super-Enhancer-Driven Transcriptional Programs Underlying Features Specific to Lung Fibroblasts,". The America
Cellular and Molecular Physiology, Jan. 1, 2018, vol. 314, No. 1, pp. L177-L191. cited by applicant
Ibarra-Soria X. et al., "Defining Murine Organogenesis at Single-Cell Resolution Reveals a Role for the Leukotriene Pathway in Regulating Blood Progenitor
Biology, Feb. 2018, vol. 20, No. 2, pp. 127-134. cited by applicant
Khan J.A., et al., "Fetal Liver Hematopoietic Stem Cell Niches Associate With Portal Vessels," Science, Jan. 8, 2016, vol. 351(6269), pp. 176-180. cited by ap
Kharchenko V. P., et al., "Bayesian Approach to Single-cell Differential Expression Analysis," Nature Methods, Jul. 2014, vol. 11, (7), pp. 740-742. cited by a
Kim E., et al., "Isl1 Regulation of Nkx2.1 in the Early Foregut Epithelium Is Required for Trachea-Esophageal Separation and Lung Lobation," Developmenta
No. 6, pp. 675-683. cited by applicant
Kimura M., et al., "Digitalized Human Organoid for Wireless Phenotyping," iScience, cell press, XP055822469, DOI: 10.1016/j.isci.2018.05.007, retrieved fr
https://www.ncbi.nlm.nih.gov/pmc/articles/PMC6147234/, Jun. 29, 2018, vol. 4, pp. 294-301. cited by applicant
Kiselev Y. V., et al., "SCmap—A Tool for Unsupervised Projection of Single Cell RNA-seq data," Nature Methods, May 2018, vol. 15 (5), pp. 359-362. cited
Langfelder P., et al., "WGCNA: An R package for weighted correlation network analysis," BMC Bioinformatics, Dec. 2008, vol. 9 (1), pp. 1-13. cited by appl
Langmead B., et al., "Fast Gapped-read Alignment with Bowtie 2," Nature Methods, Apr. 2012, vol. 9 (4), pp. 357-359. cited by applicant
Le Douarin N., et al., "Role of the Mesoderm in the Induction of the Synthesis of Glycogen During Differentiation of the Hepatic Endoderm," CR Acad Hebd
264, pp. 1872-1874. cited by applicant
Lee G., et al., "Derivation of Neural Crest Cells From Human Pluripotent Stem Cells," Nature Protocols, Mar. 18, 2010, vol. 5(4), pp. 688-701. cited by applic
Li et al., "RSEM: Accurate Transcript Quantification from RNA-Seq data with or without a Reference Genome", BMC Bioinformatics Aug. 2011, vol. 12, No
applicant
Li L.C., et al., "Single-Cell Transcriptomic Analyses Reveal Distinct Dorsal/Ventral Pancreatic Programs,". EMBO Reports, Oct. 2018, vol. 19, No. 10, e4614
Lis R., et al., "Conversion of Adult Endothelium to Immunocompetent Haematopoietic Stem Cells," Nature, May 2017, vol. 545 (7655), pp. 439-445. cited by
Loh K. M., et al., "Mapping the Pairwise Choices Leading From Pluripotency to Human Bone, Heart, and Other Mesoderm Cell Types," Cell, Jul. 14, 2016, v
by applicant
Manno L. G., et al., "Molecular Diversity of Midbrain Development in Mouse, Human and Stem Cells," Cell, Oct. 6, 2016, vol. 167, (2), pp. 566-580. cited by
McCann C.J., et al., "Enteric Neural Stem Cell Therapies for Enteric Neuropathies," Neurogastroenterology and Motility, vol. 30, e13369, 2018, doi: 10.1111/
applicant
McGrath P.S., et al., "The Basic Helix-Loop-Helix Transcription Factor NEUROG3 Is Required for Development of the Human Endocrine Pancreas," Diabete
2505. cited by applicant
McIntyre B., et al., "Gli3-mediated hedgehog inhibition in human pluripotent stem cells initiates and augments developmental programming of adult hematop
Hematology, Feb. 28, 2013, vol. 121 (9), pp. 1543-1552. cited by applicant
McKimpson W.M., et al., "A Fluorescent Reporter Assay of Differential Gene Expression Response to Insulin in Hepatocytes," Methods in Cell Physiology, A
Cell Physiology, May 15, 2019, vol. 317, pp. C143-C151. cited by applicant
Menendez L., et al., "Directed differentiation of human pluripotent cells to neural crest stem cells", Nature Protocols, Jan. 2013, vol. 8 (1), pp. 203-212. cited
Mitaka T., "Reconstruction of Hepatic Organoid by Hepatic Stem Cells," Journal of Hepatobiliary Pancreatic Surgery, 2002, vol. 9 (6), pp. 697-703. cited by a
Moignard V., et al., "Decoding the Regulatory Network of Early Blood Development From Single-Cell Gene Expression Measurements," Nature Biotechnolo
269-276. cited by applicant
Montecino-Rodriguez E., et al., "Identification of a B-1 B Cell-Specified Progenitor," Natural Immunology, Mar. 2006, vol. 7(3), pp. 293-301. cited by applic
Morrison A. J., et al., "Single-cell transcriptome analysis of avian neural crest migration reveals signatures of invasion and molecular transitions," eLife., Dec.
applicant
Nasr T., et al., "Endosome-Mediated Epithelial Remodeling Downstream of Hedgehog-Gli Is Required for Tracheoesophageal Separation," Developmental Ce
pp. 665-674. cited by applicant
Ng S., et al., "Human iPSC-Derived Hepatocyte-Like Cells Support Plasmodium Liver-Stage Infection In Vitro," Stem cell reports, Mar. 10, 2015, vol. 4, pp.
Nowotschin S., et al., "The Emergent Landscape of the Mouse Gut Endoderm at Single-Cell Resolution," Nature, May 2019, vol. 569, No. 7756, pp. 361-367.
Pedersen J.K., et al., "Endodermal Expression of Nkx6 Genes depends differentially on Pdx1," Developmental Biology, Dec. 15, 2005, vol. 288, No. 2, pp. 48
Peng T., et al., "Coordination of Heart and Lung Co-development by a Multipotent Cardiopulmonary Progenitor," Nature, Aug. 2013, vol. 500, No. 7464, pp.
Pijuan-Sala B., et al., "A Single-Cell Molecular Map of Mouse Gastrulation and Early Organogenesis," Nature, Feb. 2019, vol. 566, No. 7745, pp. 490-495. ci
Que J., et al., "Mesothelium Contributes to Vascular Smooth Muscle and Mesenchyme During Lung Development," Proceedings of the National Academy of
vol. 105, No. 43, pp. 16626-16630. cited by applicant
Rana M.S., et al., "A Molecular and Genetic Outline of Cardiac Morphogenesis," Acta Physiologica (Oxf), Apr. 2013, vol. 207, No. 4, pp. 588-615. cited by a
Robert-Moreno A., et al., "Impaired Embryonic Haematopoiesis Yet Normal Arterial Development in the Absence of the Notch ligand Jagged1," EMBO Journ
1895. cited by applicant
Robert-Moreno A., et al., "RBPjκ-dependent Notch Function Regulates Gata2 and is Essential for the Formation of Intra-embryonic Hematopoietic Cells," De

vol. 132(5), pp. 1117-1126. cited by applicant
Rothstein L.T., et al., “Human B-1 cells take the stage,” Annals of the New York Academy of Sciences, May 2013, vol. 1285, pp. 97-114. cited by applicant
Rubin L.L., et al., “Targeting the Hedgehog Pathway in Cancer,” Nature Reviews Drug Discovery, 2006, vol. 5, pp. 1026-1033. cited by applicant
Sander M., et al, “Homeobox Gene Nkx6.1 lies Downstream of Nkx2.2 in the major Pathway of Beta-Cell formation in the Pancreas,” Development, Dec. 15,
5540. cited by applicant
Sauka-Spengler T., et al., “Snapshot: Neural Crest,” Cell, Oct. 2010, vol. 143, No. 3, 486-486.e1. cited by applicant
Scialdone A., et al., “Resolving Early Mesoderm Diversification Through Single-Cell Expression Profiling,” Nature, Jul. 2016, vol. 535, No. 7611, pp. 289-29
Semrau S., et al., “Dynamics of lineage commitment revealed by single-cell transcriptomics of differentiating embryonic stem cells”, Nature Communications
cited by applicant
Simões F.C., et al., “The Ontogeny, Activation and Function of the Epicardium During Heart Development and Regeneration,” Development, Apr. 1, 2018, vo
pages. cited by applicant
Soldatow V. Y., et al., “In Vitro Models for Liver Toxicity Testing,” Toxicology Research 2.1, 2013, vol. 2, pp. 23-39. cited by applicant
Sugimura R., et al., “Haemotopoietic Stem and Progenitor Cells from Human Pluripotent Stem Cells,” Nature, May 25, 2017, vol. 545 (7655), pp. 432-438. ci
Sweetman D., et al., “The Migration of Paraxial and Lateral Plate Mesoderm Cells Emerging From the Late Primitive Streak Is Controlled by Different Wnt S
Biology, Dec. 2008, vol. 8, No. 1, pp. 1-15. cited by applicant
Tanaka M., “Molecular and Evolutionary Basis of Limb Field Specification and Limb Initiation,” Development, Growth & Differentiation, Jan. 2013, vol. 55,
applicant
Tang X. et al. “Transcriptome Regulation and Chromatin Occupancy by E2F3 and MYC in Mice,” Scientific Data, Feb. 16, 2016, vol. 3, No. 1, pp. 1-8. cited
Testaz S., et al., “Sonic hedgehog restricts adhesion and migration of neural crest cells independently of the Patched-Smoothened-Gli signaling pathway,” PN
pp. 12521-12526. cited by applicant
Ueda T., et al., “Expansion of Human NOD/SCID-repopulating Cells by Stem Cell Factor Flk2/Flt3 ligand, thrombopoietin, IL-6, and soluble IL-6 receptor,”
2000, vol. 105(7), pp. 1013-1021. cited by applicant
Uenishi I.G., et al., “NOTCH Signaling Specifies Arterial-type Definitive Hemogenic Endothelium from Human Pluripotent Stem Cells,” Nature Communica
applicant
Wagner D.E., et al., “Lineage Tracing Meets Single-cell Omics: Opportunities and Challenges,” Nature Reviews Genetics, Jul. 2020, vol. 21, No. 7, pp. 410-4
Wang J., et al., “WebGestalt 2017: A more comprehensive, powerful, flexible and interactive gene set enrichment analysis toolkit,” Nucleic Acids Research, J
applicant
Weinreb C., et al., “Lineage tracing on transcriptional landscapes links state to fate during differentiation,” Science, Feb. 14, 2020, vol. 367, (6479), 48 pages
Weinreb C., et al., “Spring: A Kinetic Interface for Visualizing High Dimensional Single-cell Expression Data,” Bioinformatics, Apr. 2018, vol. 34 (7), pp. 12
Wilkinson C. A., et al., “Long-term Ex-vivo Haematopoietic-stem-Cell Expansion Allows Nonconditioned Transplantation,” Nature, 2019, vol. 571(7763), pp
Xie T., et al., “Single-Cell Deconvolution of Fibroblast Heterogeneity in Mouse Pulmonary Fibrosis,” Cell Reports, Mar. 27, 2018, vol. 22, No. 13, pp. 3625-3
Yao S., et al., “Long-Term Self-Renewal and Directed Differentiation of Human Embryonic Stem Cells in Chemically Defined Conditions,” PNAS, 2006, vol
cited by applicant
Yu G., et al., “ClusterProfiler: An R package for Comparing Biological Themes Among Gene Clusters,” OMICS: A Journal Integrative Biology, May 2012, vo
applicant
Zaret K.S., “From Endoderm to Liver Bud: Paradigms of Cell Type Specification and Tissue Morphogenesis,” Current Topics in Developmental Biology, Jan.
cited by applicant
Zeltner N., et al., “Feeder-free derivation of neural crest progenitor cells from human pluripotent stem cells,” Journal of Visualized Experiments, May 2014, v
Zhang C., et al., “Angiopoietin-like 5 and IGFBP2 Stimulate Ex-vivo Expansion of Human Cord Blood Hematopoietic Stem Cells as Assayed by NOD/SCID
and stem Cells, 2008, vol. 111(7), pp. 3415-3423. cited by applicant
Zhang X., et al., “A Comprehensive Structure-Function Study of Neurogenin3 Disease-Causing Alleles during Human Pancreas and Intestinal Organoid Deve
Aug. 5, 2019, vol. 50, pp. 367-380. cited by applicant
Bauwens C.L., et al., “Control of Human Embryonic Stem Cell Colony and Aggregate Size Heterogeneity Influences Differentiation Trajectories,” Stem Cells
2300-2310. cited by applicant
Brandenberg N., et al., “High-Throughput Automated Organoid Culture via Stem-Cell Aggregation in Microcavity Arrays,” Nature Biomedical Engineering, 2
applicant
Carpenedo R.L., et al., “Homogeneous and Organized Differentiation Within Embryoid Bodies Induced by Microsphere-mediated Delivery of Small Molecule
30, No. 13, pp. 2507-2515. cited by applicant
Carpenedo R.L., et al., “Rotary Suspension Culture Enhances the Efficiency, Yield, and Homogeneity of Embryoid Body Differentiation,” Stem Cells, 2007, v
applicant
Carpenedo R.L., “Microsphere-Mediated Control of Embryoid Body Microenvironments,” May 2010, 24 pages. cited by applicant
Conley B.J., et al., “Derivation, Propagation and Differentiation of Human Embryonic Stem Cells,” The International Journal of Biochemistry & Cell Biology
by applicant
Forster R., et al., “Human Intestinal Tissue with Adult Stem Cell Properties Derived from Pluripotent Stem Cells,” Stem Cell Reports, Jun. 3, 2014, vol. 2, No
applicant
Gao S., et al., Fetal Liver: An Ideal Niche for Hematopoietic Stem Cell Expansion, Science China, Life Sciences, Review, Aug. 2018, vol. 61 (8), pp. 885-892
Huss J. M., et al., “Constitutive Activities of Estrogen-Related Receptors: Transcriptional Regulation of Metabolism by the ERR Pathways in Health and Dise
Acta, 2015, vol. 1852, 2015, pp. 1912-1927. cited by applicant
Huynh N., et al., “61.06 Feasibility and Scalability of Spring Parameters in Distraction Enterogenesis in a Murine Model,” 2017, 3 pages, Retrieved from Inte
abstracts.org/abs2017/61-06-feasibility-and-scalability-of-spring-parameters-in-distraction-enterogenesis-in-a-murine-model/, Retrieved on Jun. 4, 2022. cited
Koike H., et al., “Engineering Human Hepato-Biliary-Pancreatic Organoids from Pluripotent Stem Cells,” Nature Protocols, Feb. 2021, vol. 16(2), pp. 919-93
Koike H., et al., Modeling human hepato-biliary-pancreatic organogenesis from the foregutmidgut boundary, Nature, Oct. 2019, vol. 574(7776), pp. 112-116.
Mashima H., et al., INSL5 may be a Unique Marker of Colorectal Endocrine Cells and Neuroendocrine Tumors, Biochemical and Biophysical Research Com
586-592. cited by applicant
Moschidou D., et al., “Human Mid-Trimester Amniotic Fluid Stem Cells Cultured under Embryonic Stem Cell Conditions with Valproic Acid Acquire Plurip
and Development, Feb. 1, 2013, vol. 22, No. 3, pp. 444-458. cited by applicant
Nantasanti S., et al., Disease Modeling and Gene Therapy of Copper Storage Disease in Canine Hepatic Organoids, Stem Cell Reports, 2015, vol. 5, pp. 895-
Naujok O., et al., Cytotoxicity and Activation of the WNT/Beta-Catenin Pathway in Mouse Embryonic Stem Cells Treated with Four GSK3 Inhibitors, BMC
273, pp. 1-8. cited by applicant
Ogaki S., et al., A Cost-Effective System for Differentiation of Intestinal Epithelium from Human Induced Pluripotent Stem Cells, Scientific Reports, Nov. 30
applicant
Payushina O.V., Hematopoietic Microenvironment in the Fetal Liver: Roles of Different Cell Populations, Review Article, International Scholarly Research N
pages. cited by applicant
Riehl T., et al., “CD44 and TLR4 Mediate Hyaluronic Acid Regulation of Lgr5+ Stem Cell Proliferation, Crypt Fission, and Intestinal Growth in Postnatal an
Journal of Physiology-Gastrointestinal and Liver Physiology, Dec. 1, 2015, vol. 309, No. 11, pp. G874-G887. cited by applicant

Sathananthan A.H., et al., “Human Embryonic Stem Cells and their Spontaneous Differentiation,” Italian Journal of Anatomy and Embryology, 2005, vol. 110
157. cited by applicant
Scott A., et al., “Repeated Mechanical Lengthening of Intestinal Segments in a Novel Model,” Journal of Pediatric Surgery, Jun. 2015, vol. 50, No. 6, pp. 954-
Seet C.S., et al., Generation of Mature T Cells from Human Hematopoietic Stem/Progenitor Cells in Artificial Thymic Organoids, Nature Methods, May 2017
by applicant
Sullins V. F., et al., “Intestinal Lengthening in an Innovative Rodent Surgical Model,” Journal of Pediatric Surgery, Dec. 2014, vol. 49, No. 12, pp. 1791-1794.
Wang, et al., “Spatially Monitoring Oxygen Level in 3D Microfabricated Cell Culture Systems Using Optical Oxygen Sensing Beads,” Lab on a Chip, 2013, v
applicant
Wang L., et al., “The Maintenance and Generation of Membrane Polarity in Hepatocytes,” Hepatology, 2004, vol. 39, No. 4, pp. 892-899. cited by applicant
Baptista P.M., et al., “Transplantable Liver Organoids, Too Many Cell Types to Choose: a Need for Scientific Self-Organization,” Current Transplantation Rep
18-23. cited by applicant
Duh G., et al., “EGF Regulates Early Embryonic Mouse Gut Development in Chemically Defined Organ Culture,” International Pediatric Research Foundatio
802. cited by applicant
Dye B.R., et al., “Take a Deep Breath and Digest the Material: Organoids and Biomaterials of the Respiratory and Digestive Systems,” Materials Research So
502-514. cited by applicant
Godoy P., et al., “Recent Advances in 2D and 3D in vitro Systems Using Primary Hepatocytes, Alternative Hepatocyte Sources and Non-parenchymal Liver C
Mechanisms of Hepatotoxicity Cell Signaling and ADME,” Arch Toxicol, Aug. 2013, vol. 87, 216 pages. cited by applicant
Han L., et al., Single Cell Transcriptomics Identifies a Signaling Network Coordinating Endoderm and Mesoderm Diversification during Foregut Organogene
Aug. 2020, vol. 11, No. 4158, pp. 1-16. cited by applicant
Schlieve C. R., et al., Neural Crest Cell Implantation Restores Enteric Nervous System Function and Alters the Gastrointestinal Transcriptome in Human Tissu
Stem Cell Reports, ISSCR, Sep. 12, 2017, vol. 9, pp. 883-896. cited by applicant
Weisenberg, E. M.D., “Esophagus—General Histology”; Pathology Outlines, Copyright 2003-2023, 2023,3 Pages. cited by applicant
Yuelei C., et al., “BMP Signaling Pathway and Colon Cancer,” CNKI, Oct. 15, 2009, 1 page. cited by applicant
Anderson C.M.H., et al., “Inhibition of Intestinal Dipeptide Transport by the Neuropeptide VIP is an Anti-absorptive Effect via the VPAC1 Receptor in a Hum
(Caco-2),” British Journal of Pharmacology, 2003, vol. 138, No. 4, pp. 564-573. cited by applicant
Ang L.T., et al., “A Roadmap for Human Liver Differentiation from Pluripotent Stem Cells,” Cell Reports, Feb. 20, 2018, vol. 22, pp. 2190-2205. cited by app
Baker C., et al., “Hypoganglionosis in the Gastric Antrum Causes Delayed Gastric Emptying,” Neurogastroenterology and Motility, May 2020, vol. 32(5): e13
Balbinot C., et al., “Fine-tuning and Autoregulation of the Intestinal Determinant and Tumor Suppressor Homeobox Gene CDX2 by Alternative Splicing,” Ca
2017, vol. 24, No. 12, pp. 2173-2186. cited by applicant
Barber K., et al., “Derivation of Enteric Neuron Lineages from Human Pluripotent Stem Cells,” Nature Protocols, Apr. 2019, vol. 14, No. 4, pp. 1261-1279. ci
Batterham R.L., et al., “Gut Hormone PYY3-36 Physiologically Inhibits Food Intake,” Nature, Aug. 8, 2002, vol. 418, pp. 650-654. cited by applicant
Beckett E.A.H., et al., “Inhibitory Responses Mediated by Vagal Nerve Stimulation are Diminished in Stomachs of Mice with Reduced Intramuscular Interstit
Scientific Reports, vol. 7, No. 44759, 11 pages. cited by applicant
Blanchard J.W., et al., “Reconstruction of the Human Blood-Brain Barrier in vitro reveals a Pathogenic Mechanism of APOE4 in Pericytes,” Nature Medicine
952-963. cited by applicant
Bohorquez D.V., et al., “Neuroepithelial Circuit Formed by Innervation of Sensory Enteroendocrine Cells,” The Journal of Clinical Investigation, 2015, vol. 1
applicant
Bolte C., et al., “FOXF1 Transcription Factor Promotes Lung Regeneration After Partial Pneumonectomy,” Scientific Reports, Sep. 6, 2017, vol. 7(1):10690,
Breit S., et al., “Vagus Nerve as Modulator of the Brain-Gut Axis in Psychiatric and Inflammatory Disorders,” Frontiers in Psychiatry, Mar. 13, 2018, vol. 9, A
applicant
Brooks S.J.H., et al., “Extrinsic Primary Afferent Signaling in the Gut,” Nature Reviews Gastroenterology and Hepatology, 2013, vol. 10, No. 5, pp. 286-296.
Brosch M., et al., “Epigenomic Map of Human Liver Reveals Principles of Zonated Morphogenic and Metabolic Control,” Nature Communications, 2018, vo
by applicant
Buning J.W., etaL, “Higher Hydrocortisone Dose Increases Bilirubin in Hypopituitary Patients-results from an RCT,” European Journal of Clinical Investigati
480. cited by applicant
Burleigh D.E., et al., “Stimulation of Intestinal Secretion by Vasoactive Intestinal Peptide and Cholera Toxin,” Autonomic Neuroscience: Basic and Clinical, 2
applicant
Bykov V.L., “Paneth Cells: History of Discovery, Structural and Functional Characteristics and the Role in the Maintenance of Homeostasis in the Small Intes
145, No. 1, pp. 67-80. cited by applicant
Cakir B., et al., “Development of Human Brain Organoids with Functional Vascular-like System,” Nature Methods, Nov. 2019, vol. 16, No. 11,21 pages. cited
Cao J., et al., “A Human Cell Atlas of Fetal Gene Expression,” Science, Nov. 13, 2020, vol. 370(6518), 42 pages. cited by applicant
Cardoso W.V., et al., “Regulation of Early Lung Morphogenesis: Questions, Facts and Controversies,” Development, 2006, vol. 133, pp. 1611-1624. cited by a
Chandrasekaran A., et al., “Astrocyte Differentiation of Human Pluripotent Stem Cells: New Tools for Neurological Disorder Research,” Frontiers in Cellular
10, Article. 215, 27 pages. cited by applicant
Chen M., et al., “Gene Ablation for PEPTI in Mice Abolishes the Effects of Dipeptides on Small Intestinal Fluid Absorption, Short Circuit Current and Intrace
Physiology-Gastrointestinal and Liver Physiology, Apr. 29, 2010, 33 pages. cited by applicant
Cox H.M., et aL, “Peptide YY Is Critical for Acylethanolamine Receptor Gpr119-Induced Activation of Gastrointestinal Mucosal Responses,” Cell Metabolis
542. cited by applicant
Cox H.M., “Neuroendocrine Peptide Mechanisms Controlling Intestinal Epithelial Function,” Current Opinion in Pharmacology, 2016, vol. 31, pp. 50-56. cite
Creeden J.F., et al., “Bilirubin as a Metabolic Hormone: the Physiological Relevance of Low Levels,” American Journal of Physiology-Endocrinology and Me
59 pages. cited by applicant
Daviaud N., et aL, “Vascularization and Engraftment of Transplanted Human Cerebral Organoids in Mouse Cortex,” Disorders of the Nervous System, Nov./D
cited by applicant
De Carvalho A.L.R.T et al., “The in Vitro Multi-Lineage Differentiation and Maturation of Lung and Airway Cells From Human Pluripotent Stem Cell-derive
Nature Protocols, Apr. 2021,16(4), pp. 1802-1829. cited by applicant
Egerod K.L., et al., “A Major Lineage of Enteroendocrine Cells Coexpress CCK, Secretin, GIP, GLP-1, PYY, and Neurotensin but Not Somatostatin,” Endocr
No. 12, pp. 5782-5795. cited by applicant
Egerod K.L., et al., “Profiling of G Protein-coupled Receptors in Vagal Afferents Reveals Novel Gut-to-brain Sensing Mechanisms,” Molecular Metabolism, 2
applicant
Faure S., et al., “Enteric Neural Crest Cells Regulate Vertebrate Stomach Patterning and Differentiation,” Development, 2015, vol. 142, pp. 331-342. cited by
Fomin M.E., et al., “Human Fetal Liver Cultures Support Multiple Cell Lineages That Can Engraft Immunodeficient Mice,” Open Biology, 2017, 16 pages. ci
Freddo A.M., et al., “Coordination of Signaling and Tissue Mechanics During Morphogenesis of Murine Intestinal Villi: A Role for Mitotic Cell Rounding,” In
pages. cited by applicant
Fukuda M., et al., “Small Intestinal Stem Cell Identity Is Maintained with Functional Paneth Cells in Heterotopically Grafted Epithelium Onto the Colon,” Ge
28, No. 16, pp. 1752-1757. cited by applicant

Furness J.B., et al., “The Identification of Neuronal Control Pathways Supplying Effector Tissues in the Stomach,” Cell and Tissue Research, Dec. 2020, vol.
applicant
Gage B.K., et al., “Generation of Functional Liver Sinusoidal Endothelial Cells from Human Pluripotent Stem-Cell Derived Venous Angioblasts,” Cell Stem C
254-269. cited by applicant
Galand G., “Brush Border Membrane Sucrase-Isomaltase, Maltase-Glucoamylase and Trehalase in Mammals. Comparative Development, Effects of Glucocor
and Phylogenetic Implications,” Comparative Biochemistry & Physiology, 1989, vol. 94B, No. 1, 11 pages. cited by applicant
Gerdes H-H., et al., “Intercellular Transfer Mediated by Tunneling Nanotubes,” Current Opinion in Cell Biology, 2008, vol. 20, pp. 470-475. cited by applican
Gilbert M.A., et al., “Protein-Elongating Mutations in MYH11 Are Implicated in a Dominantly Inherited Smooth Muscle Dysmotility Syndrome With Severe
Intestinal Disease,” Human Mutation, 2020, vol. 41, pp. 973-982. cited by applicant
Gillich A., et al., “Capillary Cell Type Specialization in the Alveolus,” Nature, Oct. 2020, 586(7831), pp. 785-789. cited by applicant
Gonzales L.W., et al., “Differentiation of Human Pulmonary Type II Cells in Vitro by Glucocorticoid Plus Cyclic Amp,” AJP-Lung Articles in Press, 2002, 45
Goodwin K., et al., “Smooth Muscle Differentiation Shapes Domain Branches During Mouse Lung Development,” Development, 2019, 146, 37 pages. cited b
Gribble F.M., et al., “Function and Mechanisms of Enteroendocrine Cells and Gut Hormones in Metabolism,” Reviews, Apr. 2019, vol. 15, pp. 226-237. cited
Guye P., et al., “Genetically Engineering Self-organization of Human Pluripotent Stem Cells into a Liver Bud-like Tissue Using Gata6,” Nature Communicati
by applicant
Ham O., et al., “Blood Vessel Formation in Cerebral Organoids Formed From Human Embryonic Stem Cells,” Biochemical and Biophysical Research Comm
84-90. cited by applicant
Harrison S.P., et al., “Liver Organoids: Recent Developments, Limitations and Potential,” Frontiers in Medicine, May 2021, vol. 8, 18 pages. cited by applican
Hawkins F., et al., “Prospective Isolation of NKX2-1-Expressing Human Lung Progenitors Derived From Pluripotent Stem Cells,” The Journal of Clinical Inv
2277-2294. cited by applicant
Holloway E.M., et al., “Differentiation of Human Intestinal Organoids with Endogenous Vascular Endothelial Cells,” Developmental Cell, 2020, vol. 54, pp. 5
Homan K.A., et al., “Flow-Enhanced Vascularization and Maturation of Kidney Organoids in Vitro,” Nature Methods, 2019, 16(3), pp. 255-262. cited by appl
Huang W-K., et al., “Generation of Hypothalamic Arcuate Organoids From Human Induced Pluripotent Stem Cells,” Cell Stem Cell, 2021, pp. 1657-1670. cit
Huycke T.R., et al., “Genetic and Mechanical Regulation of Intestinal Smooth Muscle Development,” 2019, Cell, vol. 179, pp. 90-105. cited by applicant
Hyland N.P., et al., “Functional Consequences of Neuropeptide Y Y2 Receptor Knockout and Y2 Antagonism in Mouse and Human Colonic Tissues,” British
vol. 139, pp. 863-871. cited by applicant
Jacob A., et al., “Derivation of Self-Renewing Lung Alveolar Epithelial Type II Cells From Human Pluripotent Stem Cells,” Nature Protocols, 2019, 14(12), p
Jacob F., et al., “Human Pluripotent Stem Cell-Derived Neural Cells and Brain Organoids Reveal SARS-CoV-2 Neurotropism Predominates in Choroid Plexu
2020, vol. 27, pp. 937-950. cited by applicant
Kaelberer M.M., et al., “A Gut-Brain Neural Circuit for Nutrient Sensory Transduction,” Science, Sep. 21, 2018, 361 (6408), 18 pages. cited by applicant
Kalucka J., et al., “Single-Cell Transcriptome Atlas of Murine Endothelial Cells,” Cell, 2020, vol. 180, pp. 764-779. cited by applicant
Khalil H.A., et al., “Intestinal Epithelial Replacement by Transplantation of Cultured Murine and Human Cells Into the Small Intestine,” Plos One, May 31, 20
cited by applicant
Kinchen J., et al., “Structural Remodeling of the Human Colonic Mesenchyme in Inflammatory Bowel Disease,” Cell, 2018, vol. 175, No. 2, pp. 372-388. cite
Kitano K., et al., “Bioengineering of Functional Human Induced Pluripotent Stem Cell-derived Intestinal Grafts,” Nature Communications, 2017, vol. 8, No. 7
Knox S.M., et al., “Parasympathetic Innervation Maintains Epithelial Progenitor Cells During Salivary Organogenesis,” Science, Sep. 24, 2010, 329(5999), pp
Koslowski M., et al., “MS4A12 Is a Colon-Selective Store-Operated Calcium Channel Promoting Malignant Cell Processes,” Cancer Research, May 1, 2008,
by applicant
Kotobank, “Encyclopedia—Basement Membrane,” Machine translated by Google, 2023, 6 pages. cited by applicant
Kuna L., et al., “Peptic Ulcer Disease: A Brief Review of Conventional Therapy and Herbal Treatment Options,” Journal of Clinical Medicine, 2019, vol. 8, N
Lanas A., et al., “Peptic Ulcer Disease,” vol. 390, Aug. 5, 2017, pp. 613-624. cited by applicant
Lasrado R., et al., “Lineage-Dependent Spatial and Functional Organization of the Mammalian Enteric Nervous System,” Science, 2017, vol. 356, pp. 722-72
Le Guen L., et al., “Mesenchymal-Epithelial Interactions During Digestive Tract Development and Epithelial Stem Cell Regeneration,” Cellular and Molecula
No. 20, pp. 3883-3896. cited by applicant
Li Z., et al., “Essential Roles of Enteric Neuronal Serotonin in Gastrointestinal Motility and the Development/Survival of Enteric Dopaminergic Neurons,” Th
15, 2011, vol. 31, No. 24, pp. 8998-9009. cited by applicant
Lian X., et al., “Robust Cardiomyocyte Differentiation From Human Pluripotent Stem Cells via Temporal Modulation of Canonical Wnt Signaling,” PNAS, M
cited by applicant
Lino S., et al., “Interstitial Cells of Cajal Are Involved in Neurotransmission in the Gastrointestinal Tract,” The Japan Society of Histochemistry and Cytochem
cited by applicant
Lippmann E.S., et al., “Human Blood-Brain Barrier Endothelial Cells Derived from Pluripotent Stem Cells,” Nature Biotechnology, Aug. 2012, 30(8), pp. 78
Little D.R., et al., “Differential Chromatin Binding of the Lung Lineage Transcription Factor NKX2-1 Resolves Opposing Murine Alveolar Cell Fates in Vivo
vol. 12, 18 pages. cited by applicant
Ma T.Y., et al., “IEC-18, A Nontransformed Small Intestinal Cell Line for Studying Epithelial Permeability,” Journal of Laboratory and Clinical Medicine, Au
341. cited by applicant
Mahe M., et al., “Establishment of Human Epithelial Enteroids and Colonoids from Whole Tissue and Biopsy,” Journal of Visualized Experiments, Mar. 6, 20
applicant
Mansour A.A., et al., “An In Vivo Model of Functional and Vascularized Human Brain Organoids,” Nature Biotechnology, Jun. 2018, 36(5), pp. 432-441. cite
McCauley H.A., “Enteroendocrine Regulation of Nutrient Absorption,” The Journal of Nutrition, 2019, pp. 10-21. cited by applicant
McCauley H.A., et al., “Enteroendocrine Cells Couple Nutrient Sensing to Nutrient Absorption by Regulating Ion Transport,” Nature Communications, 2020,
applicant
McCauley K.B., et al., “Efficient Derivation of Functional Human Airway Epithelium from Pluripotent Stem Cells via Temporal Regulation of Wnt Signaling
pp. 844-857. cited by applicant
McCauley K.B., et al., “Single-Cell Transcriptomic Profiling of Pluripotent Stem Cell-Derived SCGB3A2+ Airway Epithelium,” Stem Cell Reports, 2018, vo
applicant
Mellitzer G., et al., “Loss of Enteroendocrine Cells in Mice Alters Lipid Absorption and Glucose Homeostasis and Impairs Postnatal Survival,” The Journal o
No. 5, May 2010, pp. 1708-1721. cited by applicant
Menoret S., et al., “Generation of Immunodeficient Rats With Rag1 and 1I2rg Gene Deletions and Human Tissue Grafting Models,” Transplantation, Aug. 20
1278. cited by applicant
Mentlein R., et al., “Proteolytic Processing of Neuropeptide Y and Peptide YY by Dipeptidyl Peptidase IV,” Regulatory Peptides, 1993, vol. 49, pp. 133-144. c
Miranda J., et al., “A Novel Mutation in FOXF1 Gene Associated with Alveolar Capillary Dysplasia with Misalignment of Pulmonary Veins, Intestinal Malrot
Neonatology, 2013, vol. 103, pp. 241-245. cited by applicant
Mitaka., et al., “Characterization of Hepatic-organoid Cultures,” Drug Metabolism Reviews, 2010, vol. 42, No. 3, pp. 472-481. cited by applicant
Mizumoto H., et al., “Hybrid Artificial Liver Using Hepatocyte Organoids,” Regenerative Medicine, 2006, vol. 5 No. 3, pp. 81-86. cited by applicant
Moniot B., et al., “SOX9 Specifies the Pyloric Sphincter Epithelium Through Mesenchymal-epithelial Signals,” Development, Aug. 2004, vol. 131, No. 15, p

Moodaley R., et al., “Agonism of Free Fatty Acid Receptors 1 and 4 Generates Peptide YY-Mediated Inhibitory Responses in Mouse Colon,” British journal o
pp. 4508-4522. cited by applicant
Morrisey E.E., et al., “Preparing for the First Breath: Genetic and Cellular Mechanisms in Lung Development,” Developmental Cell, Jan. 19, 2010, vol. 18, pp
Nagy N., et al., “Enteric Nervous System Development: a Crest Cell's Journey From Neural Tube to Colon,” Seminars in Cell & Developmental Biology, 201
applicant
Nagy N., et al., “Sonic Hedgehog Controls Enteric Nervous System Development by Patterning the Extracellular Matrix,” Development, 2016, 143(2), pp. 26
Nakahara T., et al., “Human Papillomavirus Type 16 E1.SUP.∧.E4 Contributes to Multiple Facets of the Papillomavirus Life Cycle,” Journal of Virology, Oct.
13150-13165. cited by applicant
Nakamura T., et al., “Intestinal Stem Cell Transplantation,” Journal of Gastroenterology, 2017, vol. 52, pp. 151-157. cited by applicant
Nedvetsky P.I., et al., “Parasympathetic Innervation Regulates Tubulogenesis in the Developing Salivary Gland,” Developmental Cell, 2014, vol. 30, pp. 449-
Nguyen J., et al., “The Next Generation of Endothelial Differentiation: Tissue-Specific Ecs,” Cell Stem Cell, Jul. 1, 2021, vol. 28(7), pp. 1188-1204. cited by a
Norlen P., et al., “The Vagus Regulates Histamine Mobilization From Rat Stomach ECL Cells by Controlling Their Sensitivity to Gastrin,” The Journal of Phy
905. cited by applicant
Oceguera-Yanez F., et al., “Engineering the AAVS1 Locus for Consistent and Scalable Transgene Expression in Human iPSCs and their Differentiated Derivat
cited by applicant
Ohashi S., et al., “Epidermal Growth Factor Receptor and Mutant p53 Expand an Esophageal Cellular Subpopulation Capable of Epithelial-to-Mesenchymal T
Transcription Factors,” Tumor and Stem Cell Biology, Apr. 27, 2010, vol. 70, No. 10, pp. 4147-4184. cited by applicant
Paik D.T., et al., “Single-cell RNA-Seq Unveils Unique Transcriptomic Signatures of Organ-Specific Endothelial Cells,” Circulation, Nov. 10, 2020, 142(19),
Palikuqi B., et al., “Adaptable Haemodynamic Endothelial Cells for Organogenesis and Tumorigenesis,” Nature, Sep. 17, 2020, vol. 585, 33 pages. cited by ap
Panaro B.L., et al., “The Melanocortin-4 Receptor Is Expressed in Enteroendocrine L Cells and Regulates the Release of Peptide YY and Glucagon-like Peptic
Dec. 2, 2014, vol. 20, pp. 1018-1029. cited by applicant
Park B., et al., “Hematopoietic Stem Cell Expansion and Generation: the Ways to Make a Breakthrough,” Blood Research, Dec. 2015, vol. 50, No. 4, 10 pages
Penkala I.J., et al., “Age-Dependent Alveolar Epithelial Plasticity Orchestrates Lung Homeostasis and Regeneration,” Cell Stem Cell, Oct. 7, 2021, vol. 28, pp
Perriot S., et al., “Differentiation of Functional Astrocytes From Human-Induced Pluripotent Stem Cells in Chemically Defined Media,” STAR Protocols, Dec
cited by applicant
Perriot S., et al., “Human Induced Pluripotent Stem Cell-Derived Astrocytes Are Differentially Activated by Multiple Sclerosis-Associated Cytokines,” Stem C
11, pp. 1199-1210. cited by applicant
Pradhan A., et al., “The S52F FOXF1 Mutation Inhibits STAT3 Signaling and Causes Alveolar Capillary Dysplasia,” American Journal of Respiratory and Cri
2019, vol. 200, No. 8, pp. 1045-1056. cited by applicant
Qian X., et al., “Brain-Region-Specific Organoids Using Mini-bioreactors for Modeling ZIKV Exposure,” Cell, 2016, vol. 165, pp. 1238-1254. cited by applic
Qian X., et aL, “Generation of Human Brain Region-specific Organoids Using a Miniaturized Spinning Bioreactor,” Nature Protocols, Mar. 2018, 13(3), pp. 5
Qin X., “Why is Damage Limited to the Mucosa in Ulcerative Colitis but Transmural in Crohn's Disease”, World Journal of Gastrointestinal Pathophysiology,
63-64. cited by applicant
Rakhilin N., et al., “Simultaneous Optical and Electrical in Vivo Analysis of the Enteric Nervous System,” Nature Communications, Jun. 7, 2016, 7:11800, 7 p
Ran F.A., et al., “Genome Engineering using the CRISPR-Cas9 System,” Nature Protocols, Nov. 2013, 8(11), pp. 2281-2308. cited by applicant
Rice A.C., et aL, “A New Animal Model of Hemolytic Hyperbilirubinemia-Induced Bilirubin Encephalopathy (Kernicterus),” Pediatric Research, 2008, vol. 6
applicant
Roberts D.J., et al., “Epithelial-mesenchymal Signaling During the Regionalization of the Chick Gut,” Development, 1998, vol. 125, No. 15, pp. 2791-2801. c
Rydning A., et al., “Mast Cell Derived Histamine is Involved in Gastric Vasodilation During Acid Back Diffusion via Activation of Sensory Neurons,” May 15
applicant
Scavuzzo M.A., et al., “Organotypic Pancreatoids with Native Mesenchyme Develop Insulin Producing Endocrine Cells,” Scientific Reports, Sep. 7, 2017, pp
Shacham-Silverberg V., et al., “Generation of Esophageal Organoids and Organotypic Raft Cultures from Human Pluripotent Stem Cells,” Methods of Cell Bi
pp. 1-23. cited by applicant
Shaylor L.A., et al., “Convergence of Inhibitory Neural Inputs Regulate Motor Activity in the Murine and Monkey Stomach,” American Journal of Physiology
Physiology, Sep. 15, 2016, 44 pages. cited by applicant
Shi Y., et aL, “Vascularized Human Cortical Organoids (vOrganoids) Model Cortical Development in Vivo,” PloS Biology, 2020, 8(5), 29 pages. cited by appl
Shin Y., et aL, “Blood-Brain Barrier Dysfunction in a 3D In Vitro Model of Alzheimer's Disease,” Advanced Science, 2019, 6(20), 10 pages. cited by applican
Shinozawa T., et al., “High-Fidelity Drug-Induced Liver Injury Screen Using Human Pluripotent Stem Cell-Derived Organoids,” Gastroenterology. Feb. 2021
by applicant
Simian M., et aL, “Organoids: A Historical Perspective of Thinking in Three Dimensions,” Journal of Cell Biology, 2017, vol. 216, No. 1, pp. 31-40. cited by
Singh A., et al., “Evaluation of Transplantation Sites for Human Intestinal Organoids,” Plos One, Aug. 27, 2020, 15(8), 12 pages. cited by applicant
Singh A., et al., “Gastrointestinal Organoids: a Next-Generation Tool for Modeling Human Development,” American Journal of Physiology-gastrointestinal ar
319(3), pp. G375-G381. cited by applicant
Smith D.M., et al., “BMP Signalling Specifies the Pyloric Sphincter,” Nature, Dec. 16, 1999, vol. 402, pp. 748-749. cited by applicant
Song L., et al., “Assembly of Human Stem Cell Derived Cortical Spheroids and Vascular Spheroids to Model 3-D Brain-like Tissues,” 2019, Scientific Report
cited by applicant
Srinivasan B., et al., “TEER Measurement Techniques for in Vitro Barrier Model Systems,” Journal of Laboratory Automation, 2015, 20 (2), 20 pages. cited b
Stevens M.L., et al., “Genomic Integration of Wnt/-catenin and BMP/smad1 Signaling Coordinates Foregut and Hindgut Transcriptional Programs,” Develop
1295. cited by applicant
Strauss K.A., et al., “Crigler-Najjar Syndrome Type 1: Pathophysiology, Natural History, and Therapeutic Frontier,” Hepatology, 2020, 71(6), pp. 1923-1939.
Sugimoto S., et al., “An Organoid-based Organ-Repurposing Approach to Treat Short Bowel Syndrome,” Nature, Apr. 2021, vol. 99, 26 pages. cited by applic
Sun X-Y., et al., “Generation of Vascularized Brain Organoids to Study Neurovascular Interactions,” eLife, 2022, vol. 11,28 pages. cited by applicant
Sung T.S., et al., “The Cells and Conductance Mediating Cholinergic Neurotransmission in the Murine Proximal Stomach,” The Journal of Physiology, 2018,
applicant
Tan S.H., et al., “AQP5 Enriches for Stem Cells and Cancer Origins in the Distal Stomach,” Nature, 2020, 578 (7795), pp. 437-443. cited by applicant
Tanimizu N., et al., “Generation of Functional Liver Organoids on Combining Hepatocytes and Cholangiocytes with Hepatobiliary Connections Ex Vivo,” Na
12 pages. cited by applicant
Tanimizu N., et al., “Tissue Structure Formation by Liver Epithelial Cells,” 2012, vol. 84, No. 8, pp. 658-665. cited by applicant
Tcw J. et al., “An Efficient Platform for Astrocyte Differentiation from Human Induced Pluripotent Stem Cells,” Stem Cell Reports, vol. 9, 2017, pp. 600-614
Teixeira V., et al., “Neonatal Vitamin C and Cysteine Deficiencies Program Adult Hepatic Glutathione and Specific Activities of Glucokinase, Phosphofructok
Carboxylase in Guinea Pigs' Livers,” 2021, Antioxidants, 10, 953, 17 pages. cited by applicant
Theodosiou N.A., et al., “Sox9 and Nkx2. 5 Determine the Pyloric Sphincter Epithelium Under the Control of BMP Signaling,” Developmental Biology, 2005
applicant
Thompson C.A., et al., “GATA4 Is Sufficient to Establish Jejunal Versus lleal Identity in the Small Intestine,” Cellular and Molecular Gastroenterology and He
422-446. cited by applicant

Thwaites D.T., et al., "H+/Dipeptide Absorption Across the Human Intestinal Epithelium Is Controlled Indirectly via a Functional Na+/H+ Exchanger," Gastro
1322-1333. cited by applicant
Tough I.R., et al., "Endogenous Peptide YY and Neuropeptide Y Inhibit Colonic Ion Transport, Contractility and Transit Differentially via Y1 and Y2 Recepto
Pharmacology, 2011, vol. 164, pp. 471-484. cited by applicant
Traber M.G., et al., "Vitamins C and E: Beneficial Effects from a Mechanistic Perspective," Free Radical Biology and Medicine, 2011,51 (5), pp. 1000-1013.
Tsai Y-H., et al., "In Vitro Patterning of Pluripotent Stem Cell-Derived Intestine Recapitulates in Vivo Human Development," Development, 2016, 144(6), 57
Ustiyan V., et al., "FOXF1 Transcription Factor Promotes Lung Morphogenesis by Inducing Cellular Proliferation in Fetal Lung Mesenchyme," Development
63. cited by applicant
Vallicelli C., et al., "Small Bowel Emergency Surgery: Literature's Review," World Journal of Emergency Surgery, 2011, vol. 6, No. 1,8 pages. cited by applic
Van De Steeg E., et al., "Complete OATP1B1 and OATP1B3 Deficiency Causes Human Rotor Syndrome by Interrupting Conjugated Bilirubin Reuptake Into
Clinical Investigation, 2012, vol. 122, No. 2, pp. 519-528. cited by applicant
Vannucchi M.G., "The Telocytes: Ten Years after Their Introduction in the Scientific Literature. An Update on Their Morphology, Distribution, and Potential F
Journal of Molecular Sciences, 2020, vol. 21, 15 pages. cited by applicant
Vila Ellis L., et al., "Epithelial Vegfa Specifies a Distinct Endothelial Population in the Mouse Lung," Developmental Cell, 2020, 52, pp. 617-630. cited by ap
Walsh K.T., et al., "The Enteric Nervous System for Epithelial Researchers: Basic Anatomy, Techniques, and Interactions With the Epithelium," Cellular and M
Hepatology, 2019, vol. 8, No. 3, pp. 369-378. cited by applicant
Wang Y., et al., "Loss of Lrig1 Leads to Expansion of Brunner Glands Followed by Duodenal Adenomas with Gastric Metaplasia," The American Journal of P
No. 4, 12 pages. cited by applicant
Ward S.M., et al., "Involvement of Intramuscular Interstitial Cells of Cajal in Neuroeffector Transmission in the Gastrointestinal Tract," The Journal of Physio
682. cited by applicant
Westfal M.L., et al., "Pediatric Enteric Neuropathies: Diagnosis and Current Management," Current Opinion in Pediatrics, 2017, 29(3), pp. 347-353. cited by a
Wimmer R.A., et al., "Generation of Blood Vessel Organoids From Human Pluripotent Stem Cells," Nature Protocols, 2019, vol. 14, pp. 3082-3100. cited by a
Wimmer R.A., et al., "Human Blood Vessel Organoids as a Model of Diabetic Vasculopathy," Nature, 2019, 565(7740), 41 pages. cited by applicant
Wong G.L.H., et al., "High Incidence of Mortality and Recurrent Bleeding in Patients With Helicobacter Pylori-Negative Idiopathic Bleeding Ulcers," Gastroe
525-531. cited by applicant
Wright E.M., et al., "Biology of Human Sodium Glucose Transporters," Physiological Reviews, 2011, vol. 91,62 pages. cited by applicant
Wright E.M., et al., "Regulation of Na+/Glucose Cotransporters," The Journal of Experimental Biology, 1997, vol. 200, pp. 287-293. cited by applicant
Yu Q., et al., "Charting Human Development Using a Multi-Endodermal Organ Atlas and Organoid Models," Cell, 2021, vol. 184, pp. 3281-3298. cited by ap
Yun C.H.C., et al., "cAMP-mediated Inhibition of the Epithelial Brush Border Na+/H+ exchanger, NHE3, requires an Associated Regulatory Protein," PNAS,
cited by applicant
Zhang S., et al., "Vascularized Organoids on a Chip: Strategies for Engineering Organoids With Functional Vasculature," Lab Chip, 2021,21 (3), pp. 473-488.
Zhao C-M., et al., "Control of Gastric Acid Secretion in Somatostatin Receptor 2 Deficient Mice: Shift from Endocrine/Paracrine to Neurocrine Pathways," E
498-505. cited by applicant
Luzio J.P. et al., "Lysosomes: fusion and function", Nature reviews Molecular cell biology, 2007, vol. 8, No. 8, pp. 622-632. cited by applicant
Ma, R., et al., "Metabolic and non-metabolic liver zonation is established non-synchronously and requires sinusoidal Wnts," eLife, 2020, vol. 9, e46206. cited
Mabbott N.A., et al., "Microfold (M) Cells: Important Immunosurveillance Posts in the Intestinal Epithelium," Mucosal Immunology, Jul. 1, 2013, vol. 6(4), p
Mace O. J. et al., "Pharmacology and physiology of gastrointestinal enteroendocrine cells," Pharmacol Res Perspect, 2015 3(4), e00155, 26 pages. cited by ap
Maddaluno L., et al., "EndMT Contributes to the Onset and Progression of Cerebral Cavernous Malformations." Nature, vol. 498, 2013, 7 pages. cited by appl
Madisen L. et al., "A robust and high-throughput Cre reporting and characterization system for the whole mouse brain," Nat Neurosci, 2020, 13, 133-140. cite
Madsen K., et al., "Angiotensin II Promotes Development of the Renal Microcirculation through AT1 Receptors," Journal of the American Society of Nephrol
cited by applicant
Maeda Y., et al., Kras(G12D) and Nkx2-1 Haploinsufficiency Induce Mucinous Adenocarcinoma of the Lung. Journal of Clinical Investigation, Dec. 3, 2012,
cited by applicant
Mahieu-Caputo D., et al., "Twin-to-Twin Transfusion Syndrome: Role of the Fetal Renin-Angiotensin System," American Journal of Pathology, 2000, vol. 15
Mammen J. M., et al., "Mucosal Repair in the Gastrointestinal Tract," Critical Care Medicine, 2003, vol. 31, pp. S532-537. DOI: 10.1097/01.CCM.00000814
Mammoto A., et al., "Vascular Niche in Lung Alveolar Development, Homeostasis, and Regeneration," Frontiers in Bioengineering and Biotechnology, Nov. 2
10.3389/fbioe.2019.00318. cited by applicant
Mammoto, T., et al., "Mechanical control of tissue and organ development," Development, 2010, vol. 137, No. 9, pp. 1407-1420. cited by applicant
Mandegar M. A., et al., "CRISPR Interference Efficiently Induces Specific and Reversible Gene Silencing in Human iPSCs," Cell Stem Cell, 2016, 18(4), pp.
Manno., L.G., et al., "RNA Velocity of Single Cells," Nature, Aug. 2018, vol. 560 (7719), pp. 494-498. cited by applicant
Marable, S.S., et al., "Hnf4a deletion in the mouse kidney phenocopies Fanconi renotubular syndrome," JCI Insight, 2018, vol. 3, 12 Pages. cited by applicant
Mari L. et al., "A pSMAD/CDX2 complex is essential for the intestinalization of epithelial metaplasia," Cell Reports, 2014, 7(4), 1197-1210. cited by applicar
Marino G. et al., "Self-consumption: the interplay of autophagy and apoptosis", Nature reviews Molecular cell biology, 2014, vol. 15, No. 2, pp. 81-94. cited b
Mariotti, V., et al., "Animal models of biliary injury and altered bile acid metabolism," Biochimica et Biophysica Acta (BBA)—Molecular Basis of Disease, 2
cited by applicant
Martin M. "Cutadapt Removes Adapter Sequences From High-Throughput Sequencing Reads," EMBnet.journal, 2011 17, 10-12. cited by applicant
Martindale J.L., et al., "Ribonucleoprotein Immunoprecipitation (RIP) Analysis," Bio Protoc, 2020, vol. 10, No. 2, e3488. doi: 10.21769/BioProtoc.3488. cite
Maruyama E.O., et al., "Cell-Specific Cre Strains for Genetic Manipulation in Salivary Glands," Plos One, Jan. 11, 2016, vol. 11(1):e0146711,12 pages. cited
Matrka M. C., et al., "Overexpression of the Human DEK Oncogene Reprograms Cellular Metabolism and Promotes Glycolysis," PLoS One, 2017, vol. 12, e
Matsuda S., et al., "Brain-Derived Neurotrophic Factor Induces Migration of Endothelial Cells Through a TrkB-ERK-Integrin aVβ3-FAK Cascade." Journal o
pp. 2123-2129. cited by applicant
Matt N. et al., "Retinoic acid-induced developmental defects are mediated by RARI3/RXR heterodimers in the pharyngeal endoderm," Development, 2003, 1
applicant
Mayor S., et al., "Pathways of Clathrin-independent Endocytosis," Nature Reviews, Molecular Cell Biology, 2007, vol. 8(8), pp. 603-612. cited by applicant
McCarty, W. J., et al., "A Microfabricated Platform for Generating Physiologically-Relevant Hepatocyte Zonation," Scientific Reports, 2016, vol. 6, 26868, 10
McGinnis C.S., et al., "DoubletFinder: Doublet Detection in Single-Cell RNA Sequencing Data Using Artificial Nearest Neighbors," Cell Systems, Apr. 24, 2
by applicant
McMahon H.T., et al., "Molecular Mechanism and Physiological Functions of Clathrin-mediated Endocytosis," Nature Reviews Molecular Cell Biology, Aug.
cited by applicant
McNaughton, L., et al., "Distribution of nitric oxide synthase in normal and cirrhotic human liver," Proceedings of the National Academy of Sciences, 2002, v
applicant
Mendelsohn C. et al., "Developmental analsyis of the retinoic acid-inducible RARb2 promoter in transgenic animals," Development, 1991,113, 723-734. cite
Miao Y., et al., "Enhancer-Associated Long Non-Coding RNA LEENE Regulates Endothelial Nitric Oxide Synthase and Endothelial Function," Nature Comm
1, doi: 10.1038/s41467-017-02113-y. cited by applicant
Miao Y., et al., "Intrinsic Endocardial Defects in Hypoplastic Left Heart Syndrome," Cell Stem Cell, Jul. 2020. doi: 10.1016/j.stem.2020.07.015. cited by appl

Miao Z., et al., “Single Cell Regulatory Landscape of the Mouse Kidney Highlights Cellular Differentiation Programs and Disease Targets,” Nature Communi
cited by applicant
Midendorp S., et al., “Adult Stem Cells in the Small Intestine Are Intrinsically Programmed with Their Location-Specific Function,” Stem Cells, 2014, vol. 32
10.1002/stem.1655. cited by applicant
Miller J. L., et al., “Emergence of Oropharyngeal, Laryngeal and Swallowing Activity in the Developing Fetal Upper Aerodigestive Tract: An Ultrasound Eva
Development, 71, 2003, pp. 61-87. cited by applicant
Minoo P. et al., “Defects in tracheoesophageal and lung morphogenesis in Nkx2.1 (−/−) mouse embryos,” Dev. Biol., 1999, 209, 60-71. cited by applicant
Miraldi E.R., et al., Leveraging Chromatin Accessibility for Transcriptional Regulatory Network Inference in T Helper 17 Cells. Genome Research, Mar. 1, 20
by applicant
Mitani, S., et al., “Human ESC/iPSC-Derived Hepatocyte-like Cells Achieve Zone-Specific Hepatic Properties by Modulation of WNT Signaling,” Mol Ther,
cited by applicant
Miyoshi H., et al., “In Vitro Expansion and Genetic Modification of Gastrointestinal Stem Cells in Spheroid Culture,” Nature Protocols, 2013, vol. 8(12), pp.
Moffett, J.R., et al., “Acetate Revisited: A Key Biomolecule at the Nexus of Metabolism, Epigenetics and Oncogenesis-Part 1: Acetyl-CoA, Acetogenesis and
Synthetases,” Frontiers in Physiology, 2020, vol. 11. cited by applicant
Mollaoglu G., et al., “The Lineage-Defining Transcription Factors SOX2 and NKX2-1 Determine Lung Cancer Cell Fate and Shape the Tumor Immune Micro
16, 2018, vol. 49(4), pp. 764-779. cited by applicant
Moon C. et al., “Development of a primary mouse intestinal epithelial cell monolayer culture system to evaluate factors that modulate IgA transcytosis,” Muc
cited by applicant
Moorefield E.C., et al., “Generation of renewable mouse intestinal epithelial cell monolayers and organoids for functional analyses,” BMC Cell Biol, Aug. 1,
applicant
Mootha V.K. et al., “PGC-1alpha-responsive genes involved in oxidative phosphorylation are coordinately downregulated in human diabetes,” Nat Genet 34,
applicant
Morizane R., et al., “Differentiation of Murine Embryonic Stem and Induced Pluripotent Stem Cells to Renal Lineage In Vitro.” Biochemical and Biophysical
2009, pp. 1334-1339. cited by applicant
Morizane R., et al., “Generation of Nephron Progenitor Cells and Kidney Organoids from Human Pluripotent Stem Cells,” Nature Protocols, 2017, vol. 12, pp
Morizane R., et al., “Kidney Organoids: A Translational Journey,” Trends in Molecular Medicine, 2017, vol. 23, pp. 246-263. cited by applicant
Morizane R., et al., “Nephron Organoids Derived from Human Pluripotent Stem Cells Model Kidney Development and Injury.” Nature Biotechnology, 33(11)
https://doi.org/10.1038/nbt.3392. cited by applicant
Morris M. E., et al., “SLC and ABC Transporters: Expression, Localization, and Species Differences at the Blood-Brain and the Blood-Cerebrospinal Fluid Ba
19, pp. 1317-1331. cited by applicant
Mounier F., et al., “Ontogenesis of Angiotensin-I Converting Enzyme in Human Kidney,” Kidney International, 1987, vol. 32, pp. 684-690. cited by applicant
Gazzin, S., et al., “Bilirubin Accumulation and Cyp mRNA Expression in Selected Brain Regions of Jaundiced Gunn Rat Pups,” Pediatric Research, 2012, vol
applicant
German-Diaz, M., et al., “A New Case of Congenital Malabsorptive Diarrhea and Diabetes Secondary to Mutant Neurogenin,” Pediatrics, 2017, vol. 140, No.
Ghatak S. et al., “Bile acid at low pH reduces squamous differentiation and activates EGFR signaling in esophageal squamous cells in 3-D culture,” Journal o
Official Journal of the Society for Surgery of the Alimentary Tract, 2013, 17(10), 1723-31. cited by applicant
Gibbs C.S., et al., “High-performance Single-cell Gene Regulatory Network Inference at Scale: the Inferelator 3.0,” Bioinformatics, May 1, 2022, vol. 38(9),
Ginestet C., “ggplot2: Elegant Graphics for Data Analysis,” J R Stat Soc a Stat, 2011 174, 245,245. cited by applicant
Glass, L. L., et al., “Single-cell RNA-sequencing reveals a distinct population of proglucagon-expressing cells specific to the mouse upper small intestine,” M
6, pp. 1296-1303. cited by applicant
Gokey J.J., et al., “Active Epithelial Hippo Signaling in Idiopathic Pulmonary Fibrosis,” JCI insight, Mar. 3, 2018, vol. 3(6), 14 pages. cited by applicant
Gololow N., et al., “Epitheliomesenchymal Interaction in Pancreatic Morphogenesis,” Developmental Biology, 1962, vol. 4, pp. 242-255. cited by applicant
Gordillo M., et al., “Orchestrating liver development,” Development, Jun. 2015, vol. 142(12), pp. 2094-2108. cited by applicant
Gorin G., et al., “Protein Velocity and Acceleration from Single-cell Multiomics Experiments,” Genome Biology, (2020)21:39, 6 pages. cited by applicant
Gorin G., et al., “RNA Velocity Unraveled,” PLOS Computational Biology, Sep. 12, 2022, vol. 18(9):e1010492, 55 pages. cited by applicant
Goss A.M., “Wnt2/2b and beta-catenin signaling are necessary and sufficient to specify lung progenitors in the foregut,” Developmental Cell, 2009, 17(2), 29
Gotoh S. et al. “Generation of Alveolar Epithelial Spheroids via Isolated Progenitor Cells from Human Pluripotent Stem cells” Stem Cell Reports (2014) 3(3):
Granja J.M., et al., “ArchR is a Scalable Software Package for Integrative Single-Cell Chromatin Accessibility Analysis,” Nature Genetics, Mar. 2021, vol. 53
applicant
Grant R.A., et al., “Circuits Between Infected Macrophages and T cells in SARS-COV-2 Pneumonia,” Nature, Feb. 25, 2021, vol. 590(7847), pp. 635-641. cit
Green J., et al., “Diversity of Interstitial Lung Fibroblasts Is Regulated by Platelet-derived Growth Factor Receptor Alpha Kinase Activity,” American Journal
Molecular Biology, Apr. 2016, vol. 54(4), pp. 532-545. cited by applicant
Greene, A. S., et al., “Microvascular Angiogenesis and the Renin-Angiotensin System.” Current Hypertension Reports, vol. 4, No. 1, Feb. 2002, pp. 56-62. cit
Greene Y. J., et al., “Ascorbic Acid Regulation of 3-Hydroxy-3-Methylglutaryl Coenzyme A Reductase Activity and Cholesterol Synthesis in Guinea Pig Live
Acta, 834(1), 1985, pp. 134-138. cited by applicant
Greggio C., et al., “Artificial Three-Dimensional Niches Deconstruct Pancreas Development In Vitro,” Development, 2013, vol. 140, pp. 4452-4462. cited by
Gribble, F. M., et al., “Enteroendocrine Cells: Chemosensors in the Intestinal Epithelium,” Annu Rev Physiol, 2016, vol. 78, pp. 277-299. cited by applicant
Gribouval O., et al., “Mutations in Genes in the Renin-Angiotensin System Are Associated with Autosomal Recessive Renal Tubular Dysgenesis,” Nature Ge
cited by applicant
Gribouval, O., et al., “Spectrum of Mutations in the Renin-Angiotensin System Genes in Autosomal Recessive Renal Tubular Dysgenesis.” Human Mutation,
316-326. cited by applicant
Gu G., et al., “Global expression analysis of gene regulatory pathways during endocrine pancreatic development,” Development, 2004, 131, 165-179. cited by
Gu M., et al., “iPSC-Endothelial Cell Phenotypic Drug Screening and In Silico Analyses Identify Tyrphostin-AG1296 for Pulmonary Arterial Hypertension,”
2021, vol. 13, No. 592. doi: 10.1126/scitranslmed.aba6480. cited by applicant
Gu M., et al., “Microfluidic Single-Cell Analysis Shows That Porcine Induced Pluripotent Stem Cell-Derived Endothelial Cells Improve Myocardial Function
Circulation Research, 2012, vol. 111, pp. 882-893. cited by applicant
Gu M., et al., “Patient-Specific iPSC-Derived Endothelial Cells Uncover Pathways that Protect Against Pulmonary Hypertension in BMPR2 Mutation Carrier
pp. 490-504. cited by applicant
Gualdi R. et al., “Hepatic specification of the gut endoderm in vitro: cell signaling and transcriptional control,” Genes Dev, 1996, 10, 1670-1682. cited by app
Guan X. et al., “GLP-2 receptor localizes to enteric neurons and endocrine cells expressing vasoactive peptides and mediates increased blood flow,” Gastroen
cited by applicant
Guarino, M., et al., “Nicotinamide and NAFLD: Is There Nothing New Under the Sun?” Metabolites, 2019, vol. 9. cited by applicant
Gubler M. C., et al., “Renin-Angiotensin System in Kidney Development: Renal Tubular Dysgenesis,” Kidney International, 2010, vol. 77, pp. 400-406. cited
Gubler M. C., “Renal Tubular Dysgenesis,” Pediatric Nephrology, 2014, vol. 29, pp. 51-59. cited by applicant
Guo L., et al., “The Adrenal Stress Response is an Essential Host Response Against Therapy-Induced Lethal Immune Activation,” Science Signaling, 2023, vo

10.1126/scisignal.add4900. cited by applicant
Guo M., et al., “Guided Construction of Single Cell Reference for Human and Mouse Lung,” Nature Communications, Jul. 29, 2023, 14:4566, 20 pages. cited
Guo M., et al., “Single Cell RNA Analysis Identifies Cellular Heterogeneity and Adaptive Responses of the Lung at Birth,” Nature Communications, Jan. 3, 2
cited by applicant
Gupta A. et al., “The great divide: septation and malformation of the cloaca, and its implications for surgeons,” Pediatr Surg Int, 2014, 30, 1089-1095. cited by
Ha, T. Y., et al., “Ascorbate Indirectly Stimulates Fatty Acid Utilization in Primary Cultured Guinea Pig Hepatocytes by Enhancing Carnitine Synthesis,” The
124, pp. 732-737. cited by applicant
Haasdijk R. A., et al., “Cerebral Cavernous Malformations: From Molecular Pathogenesis to Genetic Counselling and Clinical Management,” European Journ
20, pp. 134-140. cited by applicant
Habib A. M. et al., “Overlap of endocrine hormone expression in the mouse intestine revealed by transcriptional profiling and flow cytometry,” Endocrinology
applicant
Haeussler M., et al., “Evaluation of Off-Target and On-Target Scoring Algorithms and Integration into the Guide RNA Selection Tool CRISPOR,” Genome Bi
pages. cited by applicant
Hagan D.M. et al., “Mutation analysis and embryonic expression of the HLXB9 Currarino syndrome gene,” Am. J. Hum. Genet., 2000, 66, 1504-1515. cited b
Haigh J. J., et al., “Cortical and Retinal Defects Caused by Dosage-Dependent Reductions in VEGF-A Paracrine Signaling,” Developmental Biology, 2003, vo
applicant
Hajal C., et al., “Biology and Models of the Blood-Brain Barrier,” Annual Review of Biomedical Engineering, 2021, vol. 23, pp. 359-384. cited by applicant
Hale, C., et al., “Molecular Targeting of the GK-GKRP Pathway in Diabetes.” Expert Opinion on Therapeutic Targets, vol. 19, No. 1, 2015, pp. 129-139. cited
Hamilton T.G., et al., “Evolutionary Divergence of Platelet-derived Growth Factor Alpha Receptor Signaling Mechanisms,” Molecular and Cellular Biology, J
4013-4025. cited by applicant
Han L., et al., “Osr1 Functions Downstream of Hedgehog Pathway to Regulate Foregut Development,” Developmental Biology, 2017, 427, pp. 72-83. cited by
Hansmann G., et al., “Pulmonary Hypertension in Bronchopulmonary Dysplasia,” Pediatric Research, Jun. 2020, No. 10.1038/s41390-020-0993-4. cited by ap
Hao Y., et al., “Integrated Analysis of Multimodal Single-cell Data,” Cell, Jun. 24, 2021, ;vol. 184(13), pp. 3573-3587. cited by applicant
Harris-Johnson K.S. et al., “13-Catenin promotes respiratory progenitor identity in mouse foregut,” Proc. Natl. Acad. Sci. U. S. A., 2009, 106, 16287-16292. c
Harrison S.A., et al., “Selonsertib for Patients with Bridging Fibrosis or Compensated Cirrhosis Due to NASH: Results from Randomized Phase III Stellar Tri
2020, vol. 73, pp. 26-39. cited by applicant
Haussinger, D., “Nitrogen metabolism in liver: structural and functional organization and physiological relevance,” Biochem J, 1990, vol. 267, pp. 281-290. c
Pollin T. I., et al., “Triglyceride Response to an Intensive Lifestyle Intervention Is Enhanced in Carriers of the GCKR Pro446Leu Polymorphism,” Journal of
Metabolism, 2011, vol. 96, pp. E1142-1147. cited by applicant
Powell D. W., et al., “Myofibroblasts. II. Intestinal Subepithelial Myofibroblasts,” American Journal of Physiology, 1999, vol. 277, pp. C183-201. DOI: 10.11
by applicant
Prakash Y., et al., “Neurotrophins in Lung Health and Disease,” Expert Review of Respiratory Medicine, 2010, vol. 4, pp. 395-411. cited by applicant
Prakash Y. S., et al., “Brain-Derived Neurotrophic Factor in the Airways,” Pharmacology & Therapeutics, 2014, vol. 143, pp. 74-86. cited by applicant
Pupilli C., et al., “Angiotensin II Stimulates the Synthesis and Secretion of Vascular Permeability Factor/Vascular Endothelial Growth Factor in Human Mesa
American Society of Nephrology, 1999, vol. 10, pp. 245-255. cited by applicant
Pyke C., et al., “GLP-1 Receptor Localization in Monkey and Human Tissue: Novel Distribution Revealed With Extensively Validated Monoclonal Antibody,
1280-1290. cited by applicant
Qian T., et al., “Directed Differentiation of Human Pluripotent Stem Cells to Blood-Brain Barrier Endothelial Cells,” Science Advances, 2017, vol. 3, e170167
Qiu X., et al., “Single-Cell mRNA Quantification and Differential Analysis with Census,” Nature Methods, 2017, vol. 14, pp. 309-315. cited by applicant
Qiu X., et al., “Reversed Graph Embedding Resolves Complex Single-Cell Trajectories,” Nature Methods, 2017, vol. 14, pp. 979-982. cited by applicant
Que J. et al., “Multiple dose-dependent roles for Sox2 in the patterning and differentiation of anterior foregut endoderm,” Development (Cambridge, England)
applicant
Que J. et al., “Multiple roles for Sox2 in the developing and adult mouse trachea,” Development (Cambridge, England), 2009,136(11), 1899-1907. cited by ap
Que J., “The initial establishment and epithelial morphogenesis of the esophagus: a new model of tracheal-esophageal separation and transition of simple colu
epithelium in the developing esophagus,” Wiley Interdiscip. Rev. Dev. Biol., 2015, 4(4), 419-430. cited by applicant
Raisner, R., et al., “Enhancer Activity Requires CBP/P300 Bromodomain-Dependent Histone H3K27 Acetylation,” Cell Reports, 2018, vol. 24, pp. 1722-172
Ramilowski J. A., et al., “A Draft Network of Ligand-Receptor-Mediated Multicellular Signalling in Human,” Nature Communications, 2015, vol. 6, Article 7
Ranganathan S., et al., “Evaluating Shigella Flexneri Pathogenesis in the Human Enteroid Model,” Infection and Immunity, Apr. 2019, vol. 87(4), 14 pages. ci
Raredon M.S.B., et al., “Computation and Visualization of Cell-Cell Signaling Topologies in Single-Cell Systems Data Using Connectome,” Scientific Report
10.1038/s41598-022-07959-x. cited by applicant
Ren, X., et al., “Postnatal Alveologenesis Depends on FOXF1 Signaling in c-KIT+ Endothelial Progenitor Cells.” American Journal of Respiratory and Critica
2019, pp. 1164-1176. cited by applicant
Revencu N. et al. “Cerebral cavernous malformation: new molecular and clinical insights,” J Med Genet, 2006, 43, 716-721. cited by applicant
Reyes-Palomares A., et al., “Remodeling of Active Endothelial Enhancers is Associated with Aberrant Gene Regulatory Networks in Pulmonary Arterial Hype
Communications, Apr. 2020, vol. 11, No. 1, 1673. doi: 10.1038/s41467-020-15463-x. cited by applicant
Reza H.A., et al., “Organoid Transplant Approaches for the Liver,” Transplant International, Nov. 2021, vol. 34, No. 11, pp. 2031-2045. cited by applicant
Reza, H.A., et al., “Synthetic augmentation of bilirubin metabolism in human pluripotent stem cell-derived liver organoids,” Stem Cell Reports, 2023. cited by
Rhoads, K., et al., “A Role for Hox A5 in Regulating Angiogenesis and Vascular Patterning.” Lymphatic Research and Biology, vol. 3, No. 4, 2005, pp. 240-2
Rich, N.E., et al., “Racial and Ethnic Disparities in Nonalcoholic Fatty Liver Disease Prevalence, Severity, and Outcomes in the United States: A Systematic R
Clinical Gastroenterology and Hepatology, 2018, vol. 16, pp. 198-210 e192. cited by applicant
Riemondy K. A., et al., “Single Cell RNA Sequencing Identifies TGF-β as a Key Regenerative Cue Following LPS-induced Lung Injury,” JCI Insight, Apr. 4,
by applicant
Rindler T.N., et al., “Efficient Transduction of Alveolar Type 2 Cells with Adeno-associated Virus for the Study of Lung Regeneration,” American Journal of
Biology, Jul. 2021, vol. 65(1), pp. 118-121. cited by applicant
Robbins D. J. et al., “The Hedgehog Signal Transduction Network,” Science Signaling 2012, 5(246), re6-re6, 28 pages. cited by applicant
Robinson B.D., et al., “Measurement of Microvascular Endothelial Barrier Dysfunction and Hyperpermeability In Vitro,” Methods in Molecular Biology, Feb
cited by applicant
Rochman M., et al., “Profound Loss of Esophageal Tissue Differentiation in Patients with Eosinophilic Esophagitis,” Journal of Allergy and Clinical Immuno
749.e3. cited by applicant
Rodriguez-Castillo J. A., et al., “Understanding Alveolarization to Induce Lung Regeneration,” Respiratory Research, Dec. 2018, vol. 19:1-1, 11 pages. cited
Roitbak T., et al., “Neural Stem/Progenitor Cells Promote Endothelial Cell Morphogenesis and Protect Endothelial Cells against Ischemia via HIF-1a-Regulat
Cerebral Blood Flow Metabolism, 2008, vol. 28, pp. 1530-1542. cited by applicant
Rosekrans S. L. et al., “Esophageal development and epithelial homeostasis,” American Journal of Physiology—Gastrointestinal and Liver Physiology, 2015,
applicant
Ross M.G. et al., “Development of ingestive behavior.,” Am J Physiol, 1998, 274, R879-893. cited by applicant

Rossi J.M. et al., “Distinct mesodermal signals, including BMPs from the septum transversum mesenchyme, arc required in combination for hepatogenesis fro
2001, 15, 1998-2009. cited by applicant
Rouch J.D., et al., “Development of Functional Microfold (M) Cells from Intestinal Stem Cells in Primary Human Enteroids,” PLOS One . Jan. 28, 2016, vol.
applicant
Rubio-Cabezas O., et al., “Permanent Neonatal Diabetes and Enteric Anendocrinosis Associated with Biallelic Mutations in NEUROG3,” Diabetes, 2011, vol.
applicant
Ruppert C. et al., “Role of HGF in the healthy and injured lung,” European Respiratory Journal, 2015, vol. 46, 2 pages. cited by applicant
Sahdeo S., et al., “High-Throughput Screening of FDA-Approved Drugs Using Oxygen Biosensor Plates Reveals Secondary Mitofunctional Effects,” Mitocho
125. cited by applicant
Saili K. S., et al., “Blood-Brain Barrier Development: Systems Modeling and Predictive Toxicology,” Birth Defects Research, 2017, vol. 109, pp. 1680-1710.
Sajiki T., et al., “Transmission Electron Microscopic Study of Hepatocytes in Bioartificial Liver,” Tissue Engineering, 2000, vol. 6, No. 6, pp. 627-640. cited b
Salahudeen A. A., et al., “Progenitor Identification and SARS-CoV-2 Infection in Human Distal Lung Organoids,” Nature, Dec. 24, 2020, vol. 588(7839), pp.
Samson A. et al., “Effect of somatostatin on electrogenic ion transport in the duodenum and colon of the mouse, Mus domesticus,” Comparative Biochemistry
Molecular Integrative Physiology, 2000, 125, 459-468. cited by applicant
Samuel, V.T., et al., “Nonalcoholic Fatty Liver Disease, Insulin Resistance, and Ceramides,” New England Journal of Medicine, 2019, vol. 381, pp. 1866-186
Sankoda N., et al., “Epithelial Expression of Gata4 and Sox2 Regulates Specification of the Squamous-columnar Junction via MAPK/ERK Signaling in Mice,
vol. 12, pp. 1-15. cited by applicant
Santoro N., et al., “Variant in the Glucokinase Regulatory Protein (GCKR) Gene Is Associated with Fatty Liver in Obese Children and Adolescents,” Hepatolo
cited by applicant
Sarkar A., et al., “Sox2 Suppresses Gastric Tumorigenesis in Mice,” Cell Reports, 2016, 16(7), pp. 1929-1941. cited by applicant
Sato T., et al., “Growing Self-Organizing Mini-Guts from a Single Intestinal Stem Cell: Mechanism and Applications,” Science, 2013, vol. 340, pp. 1190-119
cited by applicant
Saunders N. R., et al., “Barrier Mechanisms in the Developing Brain,” Frontiers in Pharmacology, 2012, vol. 3, Article 46, 18 pages. cited by applicant
Sayar E., et al., “Chromogranin-A Staining Reveals Enteric Anendocrinosis in Unexplained Congenital Diarrhea,” Journal of Pediatric Gastroenterology and N
e21. cited by applicant
Sayar E., et al., “Extremely Rare Cause of Congenital Diarrhea: Enteric Anendocrinosis,” Pediatrics International, 2013, vol. 55, pp. 661-663. cited by applica
Scheidecker, B., et al., “Induction of in vitro Metabolic Zonation in Primary Hepatocytes Requires Both Near-Physiological Oxygen Concentration and Flux,”
Biotechnology, 2020, vol. 8. cited by applicant
Wahlicht, T., et al., “Controlled Functional Zonation of Hepatocytes In Vitro by Engineering of Wnt Signaling.” ACS Synthetic Biology, vol. 9, 2020, pp. 163
Walker E.M. et al., “Characterization of the developing small intestine in the absence of either GATA4 or GATA6,” BMC Res Notes, 2014, 7, 902, 12 pages. c
Wang D. H., et al., “Regulation of Angiotensin Type 1 Receptor and Its Gene Expression: Role in Renal Growth,” Journal of the American Society of Nephrol
cited by applicant
Wang D.H. et al., “Aberrant Epithelial-Mesenchymal Hedgehog Signaling Characterizes Barrett's Metaplasia,” Gastroenterology 2010, 138(5), 1810-1822.e2.
Wang H., et al., “Recent Progress in microRNA Delivery for Cancer Therapy by Non-Viral Synthetic Vectors,” Advanced Drug Delivery Reviews, 2015, vol.
applicant
Wang K., et al., “ANNOVAR: Functional Annotation of Genetic Variants from High-Throughput Sequencing Data,” Nucleic Acids Research, 2010, vol. 38, e1
Wang Q. et al., “Regulatable in vivo biotinylation expression system in mouse embryonic stem cells,” PloS One, 2013, 8, 5, e63532, 7 pages. cited by applica
Wang S., “Fundamentals of developmental biology”, edited by , East China University of 25 Technology Press, Feb. 2014, 1st edition, pp. 184-185 “Role of h
of appendages”, published on Feb. 28, 2014). cited by applicant
Wang T. et al “Polypeptide Growth Factor and Spinal Cord Injury”. Xinjiang Science and Technology Press, Yunnan Science and Technology Press, “Biologic
published on Apr. 30, 2003). cited by applicant
Wang X., et al., “A Tropomyosin Receptor Kinase Family Protein, NTRK2, is a Potential Predictive Biomarker for Lung Adenocarcinoma,” PeerJ, Jun. 2019,
cited by applicant
Wang, Y., et al., “Metformin Improves Mitochondrial Respiratory Activity through Activation of AMPK,” Cell Reports, 2019, vol. 29, pp. 1511-1523 e1515. c
Wang, Y., et al., “Transcriptional regulation of hepatic lipogenesis,” Nat Rev Mol Cell Biol, 2015, vol. 16, pp. 678-689. cited by applicant
Watanabe H., et al., “SOX2 and p63 colocalize at genetic loci in squamous cell carcinomas,” Journal of Clinical Investigation, 2014, 124(4), 1636-1645. cited
Watanabe M., et al., “Feasibility Study of NMR-Based Serum Metabolomic Profiling to Animal Health Monitoring: A Case Study on Iron Storage Disease in
(*Dicerorhinus sumatrensis*),” PLoS One, 2016, vol. 11, e0156318. cited by applicant
Weber R.J., et al., “Efficient Targeting of Fatty-acid modified Oligonucleotides to live Cell Membranes through Stepwise Assembly,” Biomacromolecules, 20
cited by applicant
Wei Y., et al., “Liver Homeostasis is Maintained by Midlobular Zone 2 Hepatocytes,” Science, Feb. 2021, vol. 371, No. eabb1625. cited by applicant
Weigmann B., et al., Isolation and Subsequent Analysis of Murine Lamina Propria Mononuclear Cells from Colonic Tissue, Nature Protocols, Oct. 2007, vol.
applicant
Weirauch M. T. et al., “Determination and inference of eukaryotic transcription factor sequence specificity,” Cell, 2014, 158, 1431-1443. cited by applicant
Wells J.M. et al., “Wnt/beta-catenin signaling is required for development of the exocrine pancreas,” BMC Dev Biol, 2007, 7, 4, 18 pages. cited by applicant
Weng A., et al., “Lung Injury Induces Alveolar Type 2 Cell Hypertrophy and Polyploidy with Implications for Repair and Regeneration,” American Journal o
Biology, May 2022, vol. 66(5), pp. 564-576. cited by applicant
Wesley, B. T., et al., “Single-Cell Atlas of Human Liver Development Reveals Pathways Directing Hepatic Cell Fates,” Nature Cell Biology, 2022, vol. 24, No
applicant
Wessel J., et al., “Do Genes Determine Our Health? Implications for Designing Lifestyle Interventions and Drug Trials,” Circulation: Cardiovascular Genetics
applicant
Wesson, D., et al., “The effect of intrauterine esophageal ligation on growth of fetal rabbits,” J Pediatr Surg, 1984, vol. 19, pp. 398-399. cited by applicant
Whitehead K. J., et al., “The Cerebral Cavernous Malformation Signaling Pathway Promotes Vascular Integrity via Rho GTPases,” Nature Medicine, 2009, vo
applicant
Wiel C., et al., “BACH1 Stabilization by Antioxidants Stimulates Lung Cancer Metastasis,” Cell, Jul. 11, 2019, vol. 178 (2), pp. 330-345. cited by applicant
Wieland H.A., et al., “Subtype selectivity of the novel nonpeptide neuropeptide YY1 receptor antagonist BIBO3304 and its effect on feeding in rodents,” Br J
125(3), pp. 549-555. cited by applicant
Williamson K. A., et al., “Mutations in SOX2 Cause Anophthalmia-Esophageal-Genital (AEG) Syndrome,” Human Molecular Genetics, 2006, 15(9), pp. 141
Woo J. et al., “Band-mediated inhibition of Wnt signaling in the mouse thoracic foregut controls tracheo-esophageal septation and epithelial differentiation,”
pages. cited by applicant
Wu H., et al., “Advantages of Single-Nucleus over Single-Cell RNA Sequencing of Adult Kidney: Rare Cell Types and Novel Cell States Revealed in Fibrosis
Society of Nephrology, Jan. 2019, vol. 30, No. 1, pp. 23-32. cited by applicant
Wu H., et al., “Comparative Analysis and Refinement of Human PSC-Derived Kidney Organoid Differentiation with Single-Cell Transcriptomics,” Cell Stem
881.e868. cited by applicant
Wu X., et al., “Modeling Drug-induced Liver Injury and Screening for Anti-hepatofibrotic Compounds Using Human PSC-derived Organoids,”, Cell Regener

No. 1, Mar. 3, 2023, pp. 1-13. cited by applicant
Wunderlich M., et al., “AML Xenograft Efficiency Is Significantly Improved in NOD/SCID-IL2RG Mice Constitutively Expressing Human SCF, GM-CSF ar
vol. 24(10) pp. 1785-1788. cited by applicant
Wunderlich M., et al., “Improved Multilineage Human Hematopoietic Reconstitution and Function in NSGS Mice,” PLOS One, Dec. 12, 2018, vol. 13(12), 2
Xia, M.F., et al., “NAFLD and Diabetes: Two Sides of the Same Coin? Rationale for Gene-Based Personalized NAFLD Treatment,” Frontiers in Pharmacolog
applicant
Xia Y., et al., “Angiotensin Receptors, Autoimmunity, and Preeclampsia,” Journal of Immunology, 2007, vol. 179, pp. 3391-3395. cited by applicant
Xiang Y., et al., “Fusion of Regionally Specified hPSC-Derived Organoids Models Human Brain Development and Interneuron Migration,”. Cell Stem Cell, 2
by applicant
Xiao C. et al., “Gut peptides are novel regulators of intestinal lipoprotein secretion: experimental and pharmacological manipulation of lipoprotein metabolism
2318. cited by applicant
Xiao S., et al., “Gastric Stem Cells: Physiological and Pathological Perspectives,” Frontiers in Cell and Developmental Biology, 2020, vol. 8, pp. 1-13. cited b
Xu, C.-R., et al., “Chromatin ‘Prepattern’ and Histone Modifiers in a Fate Choice for Liver and Pancreas,” Science, 2011, vol. 332, pp. 963-966. cited by appl
Xu R., “Basis and Clinical Applications of Receptors”, edited by et al. Shanghai Science and Technology Press, 1st edition, Feb. 1992, Section of “Retinoic A
published on Feb. 29, 1992. cited by applicant
Xu, W., et al., “Hypoxia activates Wnt/ß-catenin signaling by regulating the expression of BCL9 in human hepatocellular carcinoma,” Scientific Reports, 201
by applicant
Xu, Y., et al., “Ascorbate protects liver from metabolic disorder through inhibition of lipogenesis and suppressor of cytokine signaling 3 (SOCS3),” Nutrition
cited by applicant
Yamaguchi T., et al., “NKX2-1/TTF-1: An Enigmatic Oncogene That Functions as a Doubleedged Sword for Cancer Cell Survival and Progression,” Cancer C
718-723. cited by applicant
Yanan Y., et al., “Research Progress on Hedgehog Signaling Pathway and Liver Fibrosis,” Chinese Journal of Anatomy, 06, Dec. 25, 2019, pp. 589-592. cited
Yang M., et al., “Angiogenesis-Related Genes May Be a More Important Factor than Matrix Metalloproteinases in Bronchopulmonary Dysplasia Developmen
18670-18679. cited by applicant
Yang Y., et al., “Transcription Factor C/EBP Homologous Protein in Health and Diseases,” Frontiers in Immunology, Nov. 27, 2017, vol. 8:1612, 18 pages. cit
Ye D.Z. et al., “Foxa1 and Foxa2 control the differentiation of goblet and enteroendocrine L- and D-cells in mice,” Gastroenterology, 2009, 137, 2052-2062. c
Ye F., et al., “Fibroblast Growth Factors 7 and 10 Are Expressed in the Human Embryonic Pancreatic Mesenchyme and Promote the Proliferation of Embryon
Diabetologia, 2005, vol. 48, pp. 277-281. cited by applicant
Yin H., et al., “Non-viral Vectors for Gene-based Therapy,” Nature Reviews Genetics, 2014, vol. 15(8), pp. 541-555. cited by applicant
Yokobori, T., et al., “Intestinal epithelial culture under an air-liquid interface: a tool for studying human and mouse esophagi,” Dis. Esophagus, 2016, vol. 29,
Aakerlund, L., et al., “Y1 receptors for neuropeptide Y are coupled to mobilization of intracellular calcium and inhibition of adenylate cyclase,” FEBS Letters
by applicant
Abbott N.J., “Astrocyte-Endothelial Interactions and Blood-Brain Barrier Permeability,” Journal of Anatomy, 2002, vol. 200, pp. 629-638. cited by applicant
Abo., K.M., et al., Human iPSC-Derived Alveolar and Airway Epithelial Cells Can Be Cultured at Air-liquid Interface and Express SARS-CoV-2 Host Factors
cited by applicant
Adachi S., et al., “Three Distinctive Steps in Peyer's Patch Formation of Murine Embryo,” International Immunology, Apr. 1997, vol. 9(4), pp. 507-514. cited
Adams S. H. et al., “Effects of peptide YY [3-36] on short-term food intake in mice are not affected by prevailing plasma ghrelin levels,” Endocrinology, 200
applicant
Aday S., et al., “Stem Cell-Based Human Blood-Brain Barrier Models for Drug Discovery and Delivery,” Trends in Biotechnology, 2016, vol. 34, pp. 382-39
Afgan E., et al., “The Galaxy Platform for Accessible, Reproducible and Collaborative Biomedical Analyses: 2016 Update,” Nucleic Acids Research, 2016, vo
applicant
Ager E. I., et al., “The Renin-Angiotensin System and Malignancy,” Carcinogenesis, 2008, vol. 29, pp. 1675-1684. cited by applicant
Aird W.C. et al., “Endothelial cell heterogeneity,” Cold Spring Harb Perspect Med, 2012, 2, a006429, 14 pages. cited by applicant
Aizarani N., et al., “A Human Liver Cell Atlas Reveals Heterogeneity and Epithelial Progenitors,” Nature, Aug. 2019, vol. 572, No. 7770, pp. 199-204. cited b
Akbari S., et al., “Next-Generation Liver Medicine Using Organoid Models”, Frontiers in Cell and Developmental Biology, vol. 7, Dec. 20, 2019. cited by app
Akers A., et al., “Synopsis of Guidelines for the Clinical Management of Cerebral Cavernous Malformations: Consensus Recommendations Based on Systema
Angioma Alliance Scientific Advisory Board Clinical Experts Panel,” Neurosurgery, 2017, vol. 80, pp. 665-680. cited by applicant
Al Alam D., et al., “Contrasting expression of canonical Wnt signaling reporters TOPGAL, BATGAL and Axin2(LacZ) during murine lung development and r
e23139, 11 pages. cited by applicant
Alber A.B., et al., “Directed Differentiation of Mouse Pluripotent Stem Cells into Functional Lung-Specific Mesenchyme,” bioRxiv, Aug. 2022, doi: 10.1101/
applicant
Alber A.B., et al., “Directed Differentiation of Mouse Pluripotent Stem Cells into Functional Lung-specific Mesenchyme,” Nature Communications, Jun. 13, 2
cited by applicant
Allanson J.E., et al., “Possible New Autosomal Recessive Syndrome With Unusual Renal Histopathological Changes,” American Journal of Medical Genetics
by applicant
Almeida, L. F., et al., “Role of the Renin-Angiotensin System in Kidney Development and Programming of Adult Blood Pressure.” Clinical Science, vol. 134,
656. cited by applicant
Alvira C. M., “Aberrant Pulmonary Vascular Growth and Remodeling in Bronchopulmonary Dysplasia,” Frontiers in Medicine (Lausanne), 2016, vol. 3, p. 21
Alvira C. M., “Nuclear Factor-Kappa-B Signaling in Lung Development and Disease: One Pathway, Numerous Functions,” Birth Defects Research Part A: Cl
2014, vol. 100, pp. 202-216. cited by applicant
Amir et al., “Comparing the Cellular Phenotype of Naïve and Primed Human Embryonic Stem Cells,” Fertility and Sterility, Sep. 2018 100(4):e36 Abstract. c
Amir M., et al., “Hepatic Autonomic Nervous System and Neurotrophic Factors Regulate the Pathogenesis and Progression of Non-Alcoholic Fatty Liver Dis
(Lausanne), 2020, vol. 7, Article 62. doi: 10.3389/fmed.2020.00062. cited by applicant
Amireddy N., et al., “The Unintended Mitochondrial Uncoupling Effects of the FDA-Approved Anti-Helminth Drug Nitazoxanide Mitigates Experimental Pa
Biological Chemistry, 2017, vol. 292, pp. 15731-15743. cited by applicant
Andl C.D. et al., “Epidermal growth factor receptor mediates increased cell proliferation, migration, and aggregation in esophageal keratinocytes in vitro and i
Chemistry, 2003, 278(3), 1824-1830. cited by applicant
Andrews T.S., et al., “Single-Cell, Single-Nucleus, and Spatial RNA Sequencing of the Human Liver Identifies Cholangiocyte and Mesenchymal Heterogenei
Nov. 2022, vol. 6, No. 11, pp. 821-840. cited by applicant
Anstee, Q.M., et al., “Genome-wide association study of non-alcoholic fatty liver and steatohepatitis in a histologically characterised cohort,” Journal of Hepa
515. cited by applicant
Appuhn S.V., et al., “Capillary Changes Precede Disordered Alveolarization in a Mouse Model of Bronchopulmonary Dysplasia,” American Journal of Respir
Biology, Mar. 2021, vol. 65, No. 1, pp. 81-91. doi: 10.1165/rcmb.2021-0004OC. cited by applicant
Arnold K. et al., “Sox2+ Adult Stem and Progenitor Cells Are Important for Tissue Regeneration and Survival of Mice,” Cell Stem Cell, 2011, 9(4), 317-329.
Artegiani B., et al., “Fast and Efficient Generation of Knock-in Human Organoids Using Homology-independent CRISPR-Cas9 Precision Genome Editing”, N

22, No. 3, pp. 321-331. cited by applicant
Auerbach A. D., “Fanconi anemia and its diagnosis,” Mutation Research—Fundamental and Molecular Mechanisms of Mutagenesis, 2009, 668(1-2), 4-10. cit
Aven L., et al., “An NT4/TrkB Dependent Increase in Innervation Links Early-Life Allergen Exposure to Persistent Airway Hyperreactivity,” FASEB Journal,
by applicant
Bagnat M. et al., “Genetic control of single lumen formation in the zebrafish gut,” Nat Cell Biol, 2007, 9, 954-960. cited by applicant
Bakker S. T. et al., “Learning from a paradox: recent insights into Fanconi anaemia through studying mouse models,” Disease Models Mechanisms, 2013, 6(1
Baldelli, S., et al., “Glutathione and Nitric Oxide: Key Team Players in Use and Disuse of Skeletal Muscle,” Nutrients, 2019, vol. 11. cited by applicant
Ballermann B. J., “Dependence of Renal Microvessel Density on Angiotensin II: Only in the Fetus?” Journal of the American Society of Nephrology, 2010, vo
applicant
Bamberger C. et al., “Retinoic acid inhibits downregulation of DeltaNp63alpha expression during terminal differentiation of human primary keratinocytes,” T
Dermatology, 2002, 118(1), 133-8. cited by applicant
Bandara, N., et al., “Molecular Control of Nitric Oxide Synthesis through eNOS and Caveolin-1 Interaction Regulates Osteogenic Differentiation of Adipose-
Modulation of Wnt/ß-Catenin Signaling,” Stem Cell Research & Therapy, 2016, vol. 7, No. 182, pp. 1-15. cited by applicant
Barbera M. et al., “The human squamous oesophagus has widespread capacity for clonal expansion from cells at diverse stages of differentiation,” Gut, 2015,
Bar-Ephraim Y.E., et al., “Organoids in Immunological Research,”. Nature Reviews Immunology, May 2020, vol. 20(5), pp. 279-293. cited by applicant
Barkauskas C. E. et al. “Lung Organoids: Current Uses and Future Promise,” Development, Mar. 15; 2017, vol. 144(6), pp. 986-997. cited by applicant
Barkauskas C.E., et al., Type 2 alveolar Cells Are Stem Cells in Adult Lung. The Journal of Clinical Investigation, Jul. 1, 2013, vol. 123(7), pp. 3025-3036. cit
Barnes P.J., et al., “Chronic Lung Diseases: Prospects for Regeneration and Repair,” European Respiratory Review, Mar. 31, 2021, vol. 30(159), 14 pages. cite
Bartl, M., et al., “Optimality in the Zonation of Ammonia Detoxification in Rodent Liver.” Archives of Toxicology, vol. 89, 2015, pp. 2069-2078. cited by app
Basil, M. C., et al., “The Cellular and Physiological Basis for Lung Repair and Regeneration: Past, Present, and Future,” Apr. 2, 2020, vol. 26(4, pp. 482-502.
Basu-Roy U. et al., “Sox2 maintains self renewal of tumor-initiating cells in osteosarcomas,” Oncogene, 2012, 31(18), 2270-2282. cited by applicant
Batra S., et al., “Cavernous Malformations: Natural History, Diagnosis and Treatment.” Nature Reviews Neurology, 2009, vol. 5, pp. 659-670. cited by applica
Beer N. L., et al., “The P446L Variant in GCKR Associated with Fasting Plasma Glucose and Triglyceride Levels Exerts Its Effect through Increased Glucoki
Molecular Genetics, 2009, vol. 18, pp. 4081-4088. cited by applicant
Beers M.F., et al., “Alveolar Type 2 Epithelial Cell Quality Control Responses to Pulmonary Fibrosis Related SFTPC Mutations Are Dysfunctional And Subst
Journal, Apr. 2020, vol. 34(S1), 1 page (Abstract Only). cited by applicant
Belalcazar L. M., et al., “Lifestyle Intervention for Weight Loss and Cardiometabolic Changes in the Setting of Glucokinase Regulatory Protein Inhibition: Gl
Leu446Pro Variant in Look AHEAD,” Circulation: Cardiovascular Genetics, 2016, vol. 9, pp. 71-78. cited by applicant
Bellentani, S., et al., “Epidemiology of Non-Alcoholic Fatty Liver Disease.” Digestive Diseases, vol. 28, 2010, pp. 155-161. cited by applicant
Bergen V., et al., “Generalizing RNA Velocity to Transient Cell States Through Dynamical Modeling,” Nature Biotechnology, Oct. 28, 2019, vol. 38(12), 26 p
Bergers G. et al., “The role of pericytes in blood-vessel formation and maintenance,” Neuro Oncol, 2005, 7, 452-464. cited by applicant
Besserer-Offroy, E., et al., “The signaling signature of the neurotensin type 1 receptor with endogenous ligands,” Eur J Pharmacol, 2017, vol. 805, pp. 1-13. ci
Beucher A., et al., “The homeodomain-containing transcription factors Arx and Pax4 control enteroendocrine subtype specification in mice,” PLoS One, 2012
applicant
Bharat A., et al., “Lung Transplantation for Patients with Severe COVID-19,” Science Translational Medicine, Dec. 16, 2020, vol. 12(574):eabe4282, 13 page
Bhatt A. J., et al., “Disrupted Pulmonary Vasculature and Decreased Vascular Endothelial Growth Factor, Flt-1, and TIE-2 in Human Infants Dying with Bron
American Journal of Respiratory and Critical Care Medicine, 2001, vol. 164, pp. 1971-1980. cited by applicant
Biancalani T., et al., “Deep Learning and Alignment of Spatially Resolved Single-Cell Transcriptomes with Tangram,” Nature Methods, Nov. 2021, vol. 18, N
applicant
Bilchik A. J., et al., “Peptide YY Augments Postprandial Small Intestinal Absorption in the Conscious Dog,” The American Journal of Surgery, 1994, vol. 167
Biology Stack Exchange., “Are there situations where in vivo results work better than in vitro results would have shown?”, Forum post , reply on Sep. 28, 201
https://biology.stackexchange.com/questions/77736/are-there-situations-where-in-vivo-results-work-better-th (Year: 2018). cited by applicant
Blair T.A., et al., “Mass cytometry reveals distinct platelet subtypes in healthy subjects and novel alterations in surface glycoproteins in Glanzmann thrombast
2018; 8(1):10300 in 13 pages. cited by applicant
Blakenberg D. et al., “Manipulation of FASTQ data with Galaxy,” Bioinformatics, 2010, 26(14), 1783-1785. cited by applicant
Blanchard C., et al., “Coordinate Interaction between IL-13 and Epithelial Differentiation Cluster Genes in Eosinophilic Esophagitis,” The Journal of Immuno
cited by applicant
Bochkis I.M. et al., “Genome-wide location analysis reveals distinct transcriptional circuitry by paralogous regulators Foxa1 and Foxa2,” PLoS genetics, 2012
by applicant
Boj S.F., et al., “Forskolin-induced Swelling in Intestinal Organoids: An In Vitro Assay for Assessing Drug Response in Cystic Fibrosis Patients,” J Vis Exp, F
by applicant
Bolger A. M. et al., “Trimmomatic: a flexible trimmer for Illumina sequence data,” Bioinformatics, 2014, 30, 2114-2120. cited by applicant
Bolte C., et al., “Nanoparticle Delivery of Proangiogenic Transcription Factors into the Neonatal Neurotrophic Factor-Mediated Alveolar Capillary Injury and
Alveolar Simplification Caused by Hyperoxia,” American Journal of Respiratory and Critical Care Medicine, Jul. 2020, vol. 202, No. 1, pp. 100-111. doi: 10.1
by applicant
Boon et al., “Amino Acid Levels Determine Metabolism and CYP450 Function in Hepatocytes and Hepatoma Cell Lines,” Nature Communications, 2020, vo
Bordi C. et al., “Classification of gastric endocrine cells at the light and electron microscopical levels,” Microsc. Res. Tech., 2000, 48, 258-271. cited by applic
Braegger C.P., et al., “Ontogenetic Aspects of the Intestinal Immune System in Man,” International Journal of Clinical and Laboratory Research, 1992, vol. 22
Bray N. L., et al., “Near-Optimal Probabilistic RNA-Seq Quantification,” Nature Biotechnology, 2016, vol. 34, pp. 525-527. cited by applicant
Buettner et al., “Computational Analysis of Cell-to-cell Heterogeneity in Single-cell RNA-sequencing Data reveals Hidden Subpopulations of cells”, Nature B
160. cited by applicant
Buske P., et al., “On The Biomechanics of Stem Cell Niche Formation in the Gut-Modelling Growing Organoids,” The FEBS Journal, 2012, vol. 279, pp. 347
Butler, A., et al. Integrating Single-Cell Transcriptomic Data Across Different Conditions, Technologies, and Species,. Nature Biotechnology, 2018, 36(4), pp.
Cai, W., et al., “Genetic polymorphisms associated with nonalcoholic fatty liver disease in Uyghur population: a case-control study and meta-analysis,” Lipids
vol. 18, 14. cited by applicant
Cain M.P., et al., “Quantitative Single-Cell Interactomes in Normal and Virus-Infected Mouse Lungs,” Disease Models & Mechanisms, May 2020, vol. 1
10.1242/dmm.044404. cited by applicant
Cakir, et al., “Engineering of Human Brain Organoids with a Functional Vascular-Like System,” Nature Methods, 2019, vol. 16, No. 11, 1169-1175. cited by a
Caldwell J. M., et al., “Novel Immunologic Mechanisms in Eosinophilic Esophagitis,” Current Opinion in Immunology, 2017, vol. 48, pp. 114-121. cited by a
Candi E. et al., “Differential roles of p63 isoforms in epidermal development: selective genetic complementation in p63 null mice,” Cell Death Differ, 2006, 1
Capeling M. M. et al., “Suspension culture promotes serosal mesothelial development in human intestingal organoids,” Cell Reports, 2002, 38, 110379, 33 pag
Cardenas-Diaz F. L., et al., Temporal and Spatial Staging of Lung Alveolar Regeneration Is Determined by the Grainyhead Transcription Factor Tfcp211Cell F
21 pages. cited by applicant
Carmona R., et al., “Conditional Deletion of WT1 in the Septum Transversum Mesenchyme Causes Congenital Diaphragmatic Hernia in Mice,”. eLife, Sep. 1
1-17. cited by applicant

Chambers J. C., et al., "Genome-Wide Association Study Identifies Loci Influencing Concentrations of Liver Enzymes in Plasma," Nature Genetics, 2011, vol
applicant
Chance W.T., et al., "Preservation of Intestine Protein by Peptide YY During Total Parenteral Nutrition," Life Sciences 1996, vol. 58, No. 21, pp. 1785-1794. c
Chandran S., et al., "Necrotising Enterocolitis in a Newborn Infant Treated with Octreotide for Chylous Effusion: Is Octreotide Safe?," BMJ Case Reports, 20
10.1136/bcr-2019-232062. cited by applicant
Char V.C. et al., "Digestion and absorption of carbohydrates by the fetal lamb in utero," Pediatr Res, 1979, 13, 1018-1023. cited by applicant
Charlton, V. E., et al., "Effects of Gastric Nutritional Supplementation on Fetal Umbilical Uptake of Nutrients," Am J Physiol, 1981, vol. 241, pp. E178-185. c
Chassaing B., et al., "Mammalian Gut Immunity," Biomedical Journal, Sep. 2014, vol. 37(5) p. 246 in 22 pages. cited by applicant
Chatterjee S., et al., "Tissue-Specific Gene Expression during Productive Human Papillomavirus 16 Infection of Cervical, Foreskin, and Tonsil Epithelium." J
e00915-19. cited by applicant
Chen et al., "A Versatile Polypharmacology Platform Promotes Cryoprotection and Viability of Human Pluripotent and Differentiated Cells," Nature Methods
by applicant
Chen, F., et al., "Inhibition of Tgf beta signaling by endogenous retinoic acid is essential for primary lung bud induction," Development, 2007, vol. 134, pp. 2
Chen H., et al., "Single-Cell Trajectories Reconstruction, Exploration and Mapping of Omics Data with STREAM," Nature Communications, 2019, vol. 10, A
019-09670-4, 14 pages. cited by applicant
Chen H. et al., "Transcript profiling identifies dynamic gene expression patterns and an important role for Nrf2/Keap1 pathway in the developing mouse esop
e36504, 10 pages. cited by applicant
Chen J., et al., "Improved Human Disease Candidate Gene Prioritization Using Mouse Phenotype," BMC Bioinformatics, 2007, vol. 8, p. 392. cited by applic
Chen J., et al., "ToppGene Suite for Gene List Enrichment Analysis and Candidate Gene Prioritization," Nucleic Acids Research, 2009, vol. 37, pp. W305-W3
Chen S., et al., "fastp: An Ultra-Fast All-in-One FASTQ Preprocessor," Bioinformatics, 2018, vol. 34, pp. i884-i890. cited by applicant
Chen X., "Aberrant expression of Wnt and Notch signal pathways in Barrett's esophagus," Clinics and Research in Hepatology and Gastroenterology, 2012, 3
Chen Y., et al., "A Three-Dimensional Model of Human Lung Development and Disease from Pluripotent Stem Cells," Nature Cell Biology, May 2017, vol. 1
applicant
Chen Y. et al., "BMP Signaling pathway and colon cancer," Journal of Cell Biology 2009, 5, 6 pages (Chinese with machine translation). cited by applicant
Chen, Y., et al., "SOX2 expression inhibits terminal epidermal differentiation," Exp. Dermatol., 2015, vol. 24, pp. 966-982. cited by applicant
Chen Y., et al., "Regulation of Angiogenesis Through a MicroRNA (miR-130a) That Down-Regulates Antiangiogenic Homeobox Genes GAX and HOXA5,"
1226. cited by applicant
Chen Y. et al., "The Molecular Mechanism Governing the Oncogenic Potential of SOX2 in Breast Cancer," Journal of Biological Chemistry 2008, 283(26), 17
Younossi, Z.M., et al., "Economic and Clinical Burden of Nonalcoholic Steatohepatitis in Patients With Type 2 Diabetes in the U.S.," Diabetes Care, 2020, vo
applicant
Yu J., et al., "Induced Pluripotent Stem Cell Lines Derived from Human Somatic Cells," Science, 2007, vol. 318, pp. 1917-1920. cited by applicant
Yu X. et al., "Lentiviral vectors with two independent internal promoters transfer highlevel expression of multiple transgenes to human hematopoietic stem-pr
7, 827-838. cited by applicant
Yu, Y., Chinese Studies on Disease Signaling Pathway and Targeted Therapy, Anhui Science and Technology Press, May 31, 2013, p. 363 [Reference unavaila
Report, 3 pgs.]. cited by applicant
Yu, Y., et al., "A comparative analysis of liver transcriptome suggests divergent liver function among human, mouse and rat," Genomics, 2010, vol. 96, pp. 28
Yuan T., et al., "Fgf10 Signaling in Lung Development, Homeostasis, Disease, and Repair After Injury," Frontiers in Genetics, Sep. 25, 2018, vol. 9(418), 8 pa
Yusta B. et al., "Enteroendocrine localization of GLP-2 receptor expression in humans and rodents," Gastroenterology, 2000, 119, 744-755. cited by applicant
Zacharias W.J., et al., "Regeneration of the Lung Alveolus by an Evolutionarily Conserved Epithelial Progenitor," Nature, Mar. 8, 2018, vol. 555(7695), pp. 2
Zanini F., et al., "Developmental Diversity and Unique Sensitivity to Injury of Lung Endothelial Subtypes During Postnatal Growth," iScience, Mar. 2023, vo
10.1016/j.isci.2023.106097. cited by applicant
Zanini F., et al., "Phenotypic Diversity and Sensitivity to Injury of the Pulmonary Endothelium During a Period of Rapid Postnatal Growth," bioRxiv, Apr. 20
10.1101/2021.04.27.441649. cited by applicant
Zeng, Q., et al., "O-Linked GlcNAcylation Elevated by HPV E6 Mediates Viral Oncogenesis." Proceedings of the National Academy of Sciences, vol. 113, N
9338. cited by applicant
Zepp J.A., et al., "Distinct Mesenchymal Lineages and Niches Promote Epithelial Self-Renewal and Myofibrogenesis in the Lung," Cell, Sep. 7, 2017, vol. 17
applicant
Zhang, D., et al., "Highly efficient differentiation of human ES cells and iPS cells into mature pancreatic insulin-producing cells," Cell Res., 2009, vol. 19, pp
Zhang H., et al., "Generation of Quiescent Cardiac Fibroblasts From Human Induced Pluripotent Stem Cells for In Vitro Modeling of Cardiac Fibrosis," Circu
125, No. 5, pp. 552-566. cited by applicant
Zhang S. L., et al., "Angiotensin II Stimulates Pax-2 in Rat Kidney Proximal Tubular Cells: Impact on Proliferation and Apoptosis," Kidney International, 200
by applicant
Zhao Z., et al., "Establishment and Dysfunction of the Blood-Brain Barrier," Cell, 2015, vol. 163(5), pp. 1064-1078. cited by applicant
Zheng G.X.Y., et al., "Massively Parallel Digital Transcriptional Profiling of Single Cells," Nature Communications, 2017, vol. 8(1), pp. 1-12. cited by applica
Zheng, Y., et al., pH-and Temperature-Senstive PCL-Grafted Poly (R-amino ester)-Poly (ethylene glycol)-Poly (R-amino ester) Copolymer Hydrogels, Macro
18, No. 11, pp. 1096-1102. cited by applicant
Zhou B., et al., "Comprehensive Epigenomic Profiling of Human Alveolar Epithelial Differentiation Identifies Key Epigenetic States and Transcription Facto
Maintenance of Distal Lung Identity," BMC Genomics, Dec. 2021, vol. 22(906), 25 pages. cited by applicant
Zhou C., et al., "Comprehensive Profiling Reveals Mechanisms of SOX2-Mediated Cell Fate Specification in Human ESCs and NPCs," Cell Research, 2016,
10.1038/cr.2016.15. cited by applicant
Zhou H. J., et al., "Endothelial Exocytosis of Angiopoietin-2 Resulting from CCM3 Deficiency Contributes to Cerebral Cavernous Malformation," Nature Me
1042. cited by applicant
Zhou Y., et al., "A Subtype of Oral, Laryngeal, Esophageal, and Lung Squamous Cell Carcinoma with High Levels of TrkB-T1 Neurotrophin Receptor mRNA
19, No. 1, doi: 10.1186/s12885-019-5789-8. cited by applicant
Zhou Z., et al., "Cerebral Cavernous Malformations Arise from Endothelial Gain of MEKK3-KLF2/4 Signaling," Nature, 2016, vol. 532, pp. 122-126. cited b
Zhu, S., et al., "Liver Endothelial Heg Regulates Vascular/Biliary Network Patterning and Metabolic Zonation via Wnt Signaling," Cell Molecular Gastroente
13, pp. 1757-1783. cited by applicant
Zhu Z. et al., "Human pluripotent stem cells: an emerging model in developmental biology," Development 140, 705-717 (2013). cited by applicant
Zhuo J. L., et al., "Proximal Nephron," Comprehensive Physiology, 2013, vol. 3, No. 3, pp. 1079-1123. cited by applicant
Ziegler B. L., et al., "KDR Receptor: A Key Marker Defining Hematopoietic Stem Cells," Science, vol. 285, No. 5433, Sep. 3, 1999, pp. 1553-1558. cited by
Zwerschke, W., et al., "Modulation of Type M2 Pyruvate Kinase Activity by the Human Papillomavirus Type 16 E7 Oncoprotein." Proceedings of the Nationa
96, Feb. 1999, pp. 1291-1296. cited by applicant
Dong R., et al., "SpatialDWLS: Accurate Deconvolution of Spatial Transcriptomic Data." Genome Biology, 22, 145, 2021, 10 pages. https://doi.org/10.1186/s
applicant
Dorison A., et al., "What Can We Learn from Kidney Organoids?" Kidney International, 102, 2022, pp. 1013-1029. https://doi.org/10.1016/j.kint.2022.06.032

Dougherty, E., "Tackling the common denominator in liver disease," Novartis, Jun. 16, 2016, https://www.novartis.com/stories/tackling-common-denominato
Doupe D. P. et al., "A Single Progenitor Population Switches Behavior to Maintain and Repair Esophageal Epithelium," Science, 2012, 337(6098), 1091-1093
Draheim K. M., et al., "Cerebral Cavernous Malformation Proteins at a Glance," Journal of Cell Science, 2014, vol. 127(4), pp. 701-707. cited by applicant
Drukcer, D. J., "Evolving Concepts and Translational Relevance of Enteroendocrine Cell Biology," J Clin Endocrinol Metab, 2016, vol. 101, No. 3, pp. 778-7
Du A. et al., "Arx is required for normal enteroendocrine cell development in mice and humans," Developmental biology, 2012, 365, 175-188. cited by applica
Du Y., et al., "Lung Gene Expression Analysis (LGEA): An Integrative Web Portal for Comprehensive Gene Expression Data Analysis in Lung Development,
cited by applicant
Du Y. et al., "'LungGENS': a web-based tool for mapping single-cell gene expression in the developing lung," Thorax, 2015, 70, 1092-1094. cited by applica
Dubrovskyi O., et al., "Measurement of Local Permeability at Subcellular Level in Cell Models of Agonist- and Ventilator-Induced Lung Injury," Laboratory
254-263. cited by applicant
Duluc I., et al., "Changing Intestinal Connective Tissue Interactions Alters Homeobox Gene Expression in Epithelial Cells," Journal of Cell Science, 1997, vo
applicant
Dunn A., et al., "POLY seq: A poly(J3-amino ester)-based Vector for Multifunctional Cellular Barcoding," Stem Cell Reports. Sep. 2021, vol. 14(2), pp. 2149-
Dunn A., et al., "Polymeric Vectors for Strategic Delivery of Nucleic Acids," Nano Life, 2017, vol. 7, No. 02, 1730003, 12 pages. cited by applicant
Dunn et al., "Highly Efficient In Vivo Targeting of the Pulmonary Endothelium Using Novel Modifications of Polyethylenimine: An Importance of Charge",
7(23): 1800876, 25 pages. cited by applicant
Duren Z., et al., "Modeling Gene Regulation from Paired Expression and Chromatin Accessibility Data,". Proceedings of the National Academy of Sciences o
Jun. 2, 2017, vol. 114(25), pp. E4914-E4923. cited by applicant
Duval, K., et al., "Revisiting the role of Notch in nephron segmentation confirms a role for proximal fate selection during mouse and human nephrogenesis,"
cited by applicant
Dye B. R., et al., "A Bioengineered Niche Promotes In Vivo Engraftment and Maturation of Pluripotent Stem Cell Derived Human Lung Organoids," eLife 5,
applicant
Dye B.R. et al., "In vitro generation of human pluripotent stem cell derived lung organoids," Elife 4:e05098 (2015), 25 pages. cited by applicant
Eberl G., et al., "An Essential Function for the Nuclear Receptor RORgamma(t) in the Generation of Fetal Lymphoid Tissue Inducer Cells," Nature Immunol
64-73. cited by applicant
Efremova I., et al., "CellPhoneDB: Inferring Cell-Cell Communication from Combined Expression of Multi-Subunit Receptor-Ligand Complexes," Nature Pr
1506. cited by applicant
Eicher A.K., et al., "Functional Human Gastrointestinal Organoids Can Be Engineered from Three Primary Germ Layers Derived Separately from Pluripotent
2022, vol. 29, pp. 36-51.e6. doi: 10.1016/j.stem.2021.10.010. cited by applicant
Engelstoft, M. S. et al., "Enteroendocrine Cell Types Revisited". Current Opinion in Pharmacology, 2013, vol. 13, pp. 912-921. cited by applicant
Everhart J. E., et al., "Fatty Liver: Think Globally," Hepatology, 2010, vol. 51, pp. 1491-1493. cited by applicant
Fan Y., et al., "Bioengineering Thymus Organoids to Restore Thymic Function and Induce Donor-Specific Immune Tolerance to Allografts", The American So
23, No. 7, Jul. 2015, pp. 1262-1277. cited by applicant
Fang M., et al., "Ulinastatin Ameliorates Pulmonary Capillary Endothelial Permeability Induced by Sepsis Through Protection of Tight Junctions via Inhibitio
Pathways," Frontiers in Pharmacology, Sep. 2018, vol. 9, doi: 10.3389/fphar.2018.00823. cited by applicant
Fantes J. et al., "Mutations in SOX2 cause anophthalmia," Nature Genetics, 2003, 33(4), 461-463. cited by applicant
Faure S., et al., "Endogenous Patterns of BMP Signaling During Early Chick Development," Developmental Biology, Apr. 1, 2002, vol. 244(1), pp. 44-65. cite
Faure S., et al., "Expression Pattern of the Homeotic Gene Bapx1 During Early Chick Gastrointestinal Tract Development," Gene Expression Patterns, Dec. 2
applicant
Fausett S. R. et al., "Compartmentalization of the foregut tube: developmental origins of the trachea and esophagus," Wiley Interdisciplinary eviews. Develop
202. cited by applicant
Fausett S.R., et al., "BMP antagonism by Noggin is required in presumptive notochord cells for mammalian foregut morphogenesis," Developmental Biology,
applicant
Feliers D., et al., "Mechanism of VEGF Expression by High Glucose in Proximal Tubule Epithelial Cells," Molecular and Cellular Endocrinology, 2010, vol.
applicant
Ferguson, D., et al., "Emerging Therapeutic Approaches for the Treatment of NAFLD and Type 2 Diabetes Mellitus," Nature Reviews Endocrinology, 2021, v
applicant
Fermini, B., et al., "Clinical Trials in a Dish: A Perspective on the Coming Revolution in Drug Development," SLAS Discovery, 2018, vol. 23, pp. 765-776. c
Finn J., et al., "Dlk1-Mediated Temporal Regulation of Notch Signaling Is Required for Differentiation of Alveolar Type II to Type I Cells during Repair," Cel
26(11), pp. 2942-2954. cited by applicant
Fischer B., et al., "Oxygen Tension in the Oviduct and Uterus of Rhesus Monkeys, Hamsters and Rabbits," Journal of Reproduction and Fertility, 1993, vol. 9
Flodby P., et al., "Cell-Specific Expression of Aquaporin-5 (Aqp5) in Alveolar Epithelium Is Directed by GATA6/Spl via Histone Acetylation," Scientific Rep
7(1):3473, 12 pages. cited by applicant
Frank D. B., et al., "Emergence of a Wave of Wnt Signaling that Regulates Lung Alveologenesis by Controlling Epithelial Self-Renewal and Differentiation,"
17(9), pp. 2312-2325. cited by applicant
Freedman B.S., "Physiology Assays in Human Kidney Organoids," American Journal of Physiology—Renal Physiology, 2022, vol. 322, pp. F625-F638. cited
Freund J.B. et al., "Fluid flows and forces in development: functions, features and biophysical principles," Development, 2012, 139(7), 1229-1245. cited by ap
Fujita Y. et al., "Pax6 and Pdx1 are required for production of glucose-dependent insulinotropic polypeptide in proglucagon-expressing L cells," Am J Physiol
E648-657. cited by applicant
Funakoshi, K., et al., "Highly Sensitive and Specific Alu-Based Quantification of Human Cells Among Rodent Cells," Scientific Reports, 2017, vol. 7, Article
017-13402-3. cited by applicant
Furuta G. T. et al., "Eosinophilic Esophagitis," New England Journal of Medicine, 2015, 373(17), 1640-1648. cited by applicant
Gang, X., et al., "P300 Acetyltransferase Regulates Fatty Acid Synthase Expression, Lipid Metabolism and Prostate Cancer Growth," Oncotarget, 2016, vol. 7
by applicant
Gao C., et al., "RBFox1-Mediated RNA Splicing Regulates Cardiac Hypertrophy and Heart Failure," Journal of Clinical Investigation, 2016, vol. 126, pp. 195
Gao et al., "Highly Branched Poly (β-amino esters) for Non-Viral Gene Delivery: High Transfection Efficiency and Low Toxicity Achieved by Increasing Mo
Biomacromolecules, 2016, vol. 17(11), pp. 3640-3647. cited by applicant
Gao, H., et al., "Association of GCKR Gene Polymorphisms with the Risk of Nonalcoholic Fatty Liver Disease and Coronary Artery Disease in a Chinese No
of Clinical and Translational Hepatology, 2019, vol. 7, pp. 297-303. cited by applicant
García-Suárez, O., et al., "TrkB is Necessary for the Normal Development of the Lung." Respiratory Physiology & Neurobiology, vol. 167, No. 3, Jul. 31
applicant
Garcia-Martinez, S., et al., "Mimicking physiological 02 tension in the female reproductive tract improves assisted reproduction outcomes in pig," Molecular
24, pp. 260-270. cited by applicant
Garlanda C. et al., "Heterogeneity of endothelial cells. Specific markers" Arterioscler Thromb Vasc Biol, 1997, 17, 1193-1202. cited by applicant
Gaskill C.F., et al., "Disruption of Lineage Specification in Adult Pulmonary Mesenchymal Progenitor Cells Promotes Microvascular Dysfunction," Journal o

2017, vol. 127(6), pp. 2262-2276. cited by applicant
Cheng Y., et al., “Current Development Status of MEK Inhibitors,” Molecules, 2017, vol. 22, pp. 1-20. cited by applicant
Cheung K.C.P., et al., “Preservation of Microvascular Barrier Function Requires CD31 Receptor-Induced Metabolic Reprogramming,” Nature Communicatio
10.1038/s41467-020-17329-8. cited by applicant
Chey, W. Y., et al., “Secretin: historical perspective and current status,” Pancreas, 2014, vol. 43, pp. 162-182. cited by applicant
Chin, A. M., et al., “Morphogenesis and maturation of the embryonic and postnatal intestine,” Seminars in Cell & Developmental Biology, 2017, vol. 66,
Cho C., et al., “Reck and Gpr124 Are Essential Receptor Cofactors for Wnt7a/Wnt7b-Specific Signaling in Mammalian CNS Angiogenesis and Blood-Brain B
2017, vol. 95, pp. 1221-1225. cited by applicant
Cho C. F., et al., “Blood-Brain-Barrier Spheroids as an In Vitro Screening Platform for Brain-Penetrating Agents,” Nature Communications, 2017, vol. 8, p. 15
Choi J., et al., “Inflammatory Signals Induce AT2 Cell-Derived Damage-Associated Transient Progenitors that Mediate Alveolar Regeneration,” Cell stem cell
366-382. cited by applicant
Choi K., et al., “iGEAK: An Interactive Gene Expression Analysis Kit for Seamless Workflow Using the R/Shiny Platform,” BMC Genomics, 2019, vol. 20, p
Choudhary S.,et al., “Comparison and Evaluation of Statistical Error Models for scRNA-seq,” Genome Biology, 2022, vol. 23, 27. doi: 10.1186/s13059-021-0
Chung C., et al., “Hippo-Foxa2 Signaling Pathway Plays a Role in Peripheral Lung Maturation and Surfactant Homeostasis.” Proceedings of the National Aca
States of America, May 7, 2013, vol. 110(19), pp. 7732-7737. cited by applicant
Chung, M. I., et al., “Niche-mediated BMP/SMAD Signaling Regulates Lung Alveolar Stem Cell Proliferation and Differentiation,” Development, May 1, 20
pages. cited by applicant
Claesson-Welsh L., et al., “Permeability of the Endothelial Barrier: Identifying and Reconciling Controversies,” Trends in Molecular Medicine, Apr. 2021, vol
applicant
Claeys, W., et al., “A mouse model of hepatic encephalopathy: bile duct ligation induces brain ammonia overload, glial cell activation and neuroinflammation,
12, 17558. cited by applicant
Clemmensen, C. et al., “Emerging Hormonal-Based Combination Pharmacotherapies for the Treatment of Metabolic Diseases”. Nat Rev Endocrinol, 2018, 14
applicant
Collier et al., “Identifying Human Naïve Pluripotent Stem Cells—Evaluating State-Specific Reporter Lines and Cell-Surface Markers”, BioEssays. May 2018,
by applicant
Concepcion J. P., et al., “Neonatal Diabetes, Gallbladder Agenesis, Duodenal Atresia, and Intestinal Malrotation Caused by a Novel Homozygous Mutation in
vol. 15, pp. 67-72. cited by applicant
Coon S. D. et al., “Glucose-dependent insulinotropic polypeptide-mediated signaling pathways enhance apical PepT1 expression in intestinal epithelial cells,”
Physiol, 2015, 308, G56-62. cited by applicant
Cortina, G., et al., “Enteroendocrine Cell Dysgenesis and Malabsorption, a Histopathologic and Immunohistochemical Characterization,” Human Pathology, 2
by applicant
Coskun T. et al., “Activation of Prostaglandin E Receptor 4 Triggers Secretion of Gut Hormone Peptides GLP-1, GLP-2, and PYY,” Endocrinology, 2013, 154
Cox H. M., “Endogenous PYY and NPY mediate tonic Y(1)- and Y(2)-mediated absorption in human and mouse colon,” Nutrition, 2008, 24, 900-906. cited b
Creane M., et al., “Biodistribution and Retention of Locally Administered Human Mesenchymal Stromal Cells: Quantitative Polymerase Chain Reaction-Base
Murine Organs,” Cytotherapy, 2017, vol. 19, pp. 384-394. DOI: 10.1016/j.jcyt.2016.12.003. cited by applicant
Crisera C. A., et al., “Expression and Role of Laminin-1 in Mouse Pancreatic Organogenesis,” Diabetes, 2000, vol. 49, pp. 936-944. cited by applicant
Cruz, N. M., et al., “Differentiation of Human Kidney Organoids from Pluripotent Stem Cells.” In Methods in Cell Biology, vol. 153, Chapter 7, 2019, pp. 13
Cucullo L., et al., “The role of shear stress in Blood-Brain Barrier endothelial physiology,” BMC Neurosci, 2011, 40, 15 pages. cited by applicant
Cuevas I., et al., “Sustained Endothelial Expression of HoxA5 In Vivo Impairs Pathological Angiogenesis and Tumor Progression,” PLoS One, 2015, vol. 10,
Cui, J., et al., “Progressive Pseudogenization: Vitamin C Synthesis and Its Loss in Bats,” Molecular Biology and Evolution, 2011, vol. 28, No. 4, pp. 1025-103
Cunningham, R.P., et al., “Liver Zonation—Revisiting Old Questions With New Technologies,” Frontiers in Physiology, 2021, vol. 12, 732929. cited by appli
Dahlman et al., “Barcoded Nanoparticles for High Throughput in Vivo Discovery of Targeted Therapeutics”, PNAS, U.S.A., 2017, vol. 114(8), pp. 2060-2065
Daneman R., et al., “The Blood-Brain Barrier,” Cold Spring Harbor Perspectives in Biology, 2015, vol. 7, a020412. cited by applicant
Daniely Y. et al., “Critical role of p63 in the development of a normal esophageal and tracheobronchial epithelium,” American Journal of Physiology, Cell Phy
C181. cited by applicant
Dathan N. et al., “Distribution of the titf2/foxe1 gene product is consistent with an important role in the development of foregut endoderm, palate, and hair,” D
cited by applicant
Davidson L. M., et al., “Bronchopulmonary Dysplasia: Chronic Lung Disease of Infancy and Long-Term Pulmonary Outcomes,” Journal of Clinical Medicine
applicant
Davis B. P., et al., “Eosinophilic Esophagitis-Linked Calpain 14 is an IL-13-Induced Protease that Mediates Esophageal Epithelial Barrier Impairment,” JCI In
by applicant
Dawkins H.J.S., et al., “Progress in rare diseases research 2010-2016: An IRDIRC Perspective,” Clinical and Translational Science. Jan. 2018;11(1):11-20. cit
De Carvalho A.L.R.T., et al., “Glycogen Synthase Kinase 3 Induces Multilineage Maturation of Human Pluripotent Stem Cell-derived Lung Progenitors in 3D
2019, vol. 146(2):dev171652, 34 pages. cited by applicant
De Felice, M. et al., “A mouse model for hereditary thyroid dysgenesis and cleft palate,” Nat. Genet., 1998, 19, 395-398. cited by applicant
De Jong E. M. et al., Etiology of esophageal atresia and tracheoesophageal fistula: ‘Mind the gap’, Current Gastroenterology Reports, 2010, 12(3), 215-222. c
De Lau W., et al., “Peyer's Patch M Cells Derived from Lgr5+ Stem Cells Require SpiB and are Induced by RankL in Cultured Miniguts,”. Molecular and Cel
32(18), pp. 3639-3647. cited by applicant
De Paepe M. E., et al., “Growth of Pulmonary Microvasculature in Ventilated Preterm Infants,” American Journal of Respiratory and Critical Care Medicine, 2
by applicant
De Santa Barbara P., et al., “Molecular Etiology of Gut Malformations and Diseases,” American Journal of Medical Genetics, Dec. 30, 2002; vol. 115(4), pp. 2
De Santa Barbara, P., et al., “Tail gut endoderm and gut/genitourinary/tail development: a new tissue-specific role for Hoxa13,” Development, 2002, vol. 129,
De Souza H.S.P., et al., “Immunopathogenesis of IBD: Current State of the Art,” Nature Reviews Gastroenterology Hepatology, Jan. 2016, vol. 13(1), pp. 13-2
Dekkers J.F., et al., “High-resolution 3D Imaging of Fixed and Cleared Organoids,” Nature protocol, Jun. 2019, vol. 14(6), pp. 1756-1771. cited by applicant
Demirgan E.B., et al., “AGTR1-related Renal Tubular Dysgeneses May Not Be Fatal,” Kidney International Reports, 2021, vol. 6, pp. 846-852. cited by appli
Distefano P.V. et al., “KRIT1 protein depletion modifies endothelial cell behavior via increased vascular endothelial growth factor (VEGF)signaling,” J Biol C
cited by applicant
D'Mello R. J., et al., “LRRC31 is Induced by IL-13 and Regulates Kallikrein Expression and Barrier Function in the Esophageal Epithelium,” Mucosal Immu
cited by applicant
Dolinay T., et al., “Integrated Stress Response Mediates Epithelial Injury in Mechanical Ventilation,” American Journal of Respiratory Cell and Molecular Bio
cited by applicant
Dollard S. C., et al., “Production of Human Papillomavirus and Modulation of the Infectious Program in Epithelial Raft Cultures.” Genes & Developmen
by applicant
Domyan E.T. et al., “Signaling through BMP receptors promotes respiratory identity in the foregut via repression of Sox2,” Development (Cambridge, Englan
by applicant

Donati, B., et al., "The rs2294918 E434K Variant Modulates Patatin-Like Phospholipase Domain-Containing 3 Expression and Liver Damage." Hepatology, v
798. cited by applicant
Mowat A., et al., "Regional Specialization Within the Intestinal Immune System," Nature Reviews Immunology, Oct. 2014, vol. 14(10), pp. 667-685. cited by
Murphy C. L., et al., "HIF-Mediated Articular Chondrocyte Function: Prospects for Cartilage Repair," Arthritis Research Therapy, 2009, vol. 11, p. 213. DOI:
applicant
Murphy P.A., et al., "Alternative RNA Splicing in the Endothelium Mediated in Part by Rbfox2 Regulates the Arterial Response to Low Flow," eLife, Jan. 20
10.7554/eLife.29494. cited by applicant
Nabhan A., et al., "A Single Cell Wnt Signaling Niche Maintains Stemness of Alveolar Type 2 Cells," Science, Mar. 9, 2018; vol. 359(6380), pp. 1118-1123. c
Navin N., et al., "Tumor Evolution Inferred by Single-cell Sequencing," Nature, Apr. 7, 2011, vol. 472(7341), pp. 90-94. cited by applicant
Neal E.H., et al., "A Simplified, Fully Defined Differentiation Scheme for Producing Blood-Brain Barrier Endothelial Cells from Human iPSCs," Stem Cell R
1388. cited by applicant
Nebert D. W., et al., "Letter to the Editor for 'Update of the Human and Mouse Fanconi Anemia Genes,'" Human Genomics, 2016, vol. 10, No. 1, 25 pages. c
Negretti N. M., et al., "A Single-cell Atlas of Mouse Lung Development," Development Dec. 15, 2021, vol. 148(24), 30 pages. cited by applicant
Nejak-Bowen, K., et al., "Beta-catenin regulates vitamin C biosynthesis and cell survival in murine liver," J Biol Chem, 2009, vol. 284, pp. 28115-28127. cite
Nelson L.J., et al., "Low-Shear Modelled Microgravity Environment Maintains Morphology and Differentiated Functionality of Primary Porcine Hepatocyte C
2010, vol. 192, pp. 125-140. cited by applicant
Niederreither K. "Embryonic retinoic acid synthesis is essential for early mouse post-implantation development," Nature Genetics, 1999, 21(4), 444-448. cite
Niethamer T.K., et al., "Defining the Role of Pulmonary Endothelial Cell Heterogeneity in the Response to Acute Lung Injury," eLife, Feb. 2020, vol. 9, No. e
cited by applicant
Nishinakamura R., "Human Kidney Organoids: Progress and Remaining Challenges," Nature Reviews Nephrology, 2019, vol. 15, pp. 613-624. cited by appli
Nochi T., et al., "Cryptopatches are essential for the development of human GALT," Cell Reports, Jun. 27, 2013, vol. 3(6), vol. 1874-1884. cited by applicant
Noel G., et al., "A Primary Human Macrophage-enteroid Co-culture Model to Investigate Mucosal Gut Physiology and Host-pathogen Interactions," Scientifi
7(45270), 13 pages. cited by applicant
Nonn, O., et al., "Maternal Angiotensin Increases Placental Leptin in Early Gestation via an Alternative Renin-Angiotensin System Pathway: Suggesting a Lin
Hypertension, 2021, vol. 77, pp. 1723-1736. cited by applicant
Nozaki, Y., et al., "Metabolic Control Analysis of Hepatic Glycogen Synthesis In Vivo," Proceedings of the National Academy of Sciences of the United State
8166-8176. cited by applicant
Nyeng P. et al., FGF10 signaling controls stomach morphogenesis. Developmental Biology, 2007, 303, 295-310. cited by applicant
Oberg, K. C., et al., "Renal Tubular Dysgenesis in Twin-Twin Transfusion Syndrome." Pediatric Developmental Pathology, vol. 2, No. 1, 1999, pp. 25-32. cite
Offield M.F. et al., "PDX-1 is required for pancreatic outgrowth and differentiation of the rostral duodenum," Development, 1996, 122(3), 983-995. cited by a
Ohashi T., "Enzyme replacement therapy for lysosomal storage diseases," Pediatr Endocrinol Rev. Oct. 1, 2012;10(supp 1):26 34. cited by applicant
Ohmori T et al. "Efficient expression of a transgene in platelets using simian immunodeficiency virus based vector harboring glycoprotein Iba promoter: in viv
gene therapy," FASEB J. (2006);20(9):1522 4. cited by applicant
Ohsie S. et al., "A paucity of colonic enteroendocrine and/or enterochromaftin cells characterizes a subset of patients with chronic unexplained diarrhea/malab
40(7), 1006-1014. cited by applicant
Ohta et al., "Hemogenic endothelium differentiation from human pluripotent stem cells in a feeder and xeno free defined condition," Journal of Visualized Exp
2019;148:e59823 in 6 pages. cited by applicant
Oliverio M. I., et al., "Reduced Growth, Abnormal Kidney Structure, and Type 2 (AT2) Angiotensin Receptor-Mediated Blood Pressure Regulation in Mice La
Receptors for Angiotensin II," Proceedings of the National Academy of Sciences USA, 1998, vol. 95, pp. 15496-15501. cited by applicant
Omer, D., et al., "Human Kidney Spheroids and Monolayers Provide Insights into SARS-CoV-2 Renal Interactions," Journal of the American Society of Neph
2254. cited by applicant
Onaga T., et al., "Multiple Regulation of Peptide YY Secretion in the Digestive Tract," Peptides, 2002, vol. 23, pp. 279-290. cited by applicant
Onlilsoy Aksu A., et al., "Mutant Neurogenin-3 in a Turkish Boy with Congenital Malabsorptive Diarrhea," Pediatrics International, 2016, vol. 58, pp. 379-38
Orho-Melander M., et al., "Common Missense Variant in the Glucokinase Regulatory Protein Gene Is Associated with Increased Plasma Triglyceride and C-R
Glucose Concentrations," Diabetes, 2008, vol. 57, pp. 3112-3121. cited by applicant
Orskov C. et al., "GLP-2 stimulates colonic growth via KGF, released by subepithelial myofibroblasts with GLP-2 receptors," Regulatory peptides, 2005, 124
Ortiz-Meoz, R. F., et al., "A Small Molecule that Inhibits OGT Activity in Cells." ACS Chemical Biology, vol. 10, No. 6, Jun. 19, 2015, pp. 1392-1397. cited
Ostrin E. J., et al., "β-Catenin Maintains Lung Epithelial Progenitors After Lung Specification," Development, Mar. 1, 2018, vol. 145(5), 32 pages. cited by ap
Pan S., "Physiology," Science and Technology of China Press, Chapter 6, "Digestion within Large Intestine," 149-150, Jan. 2014. cited by applicant
Pankevich D.E. et al., "Improving and accelerating drug development for nervous system disorders," Neuron, 2014, 84, 546-553. cited by applicant
Paris A.J., et al., "STAT3BDNF-TrkB Signaling Promotes Alveolar Epithelial Regeneration After Lung Injury," Nature Cell Biology, Oct. 2020, vol. 22(10), p
Paris, J., et al., "Liver zonation, revisited," Hepatology, 2022, vol. 76. cited by applicant
Park E.J. et al., "System for tamoxifen-inducible expression of Cre-recombinase from the Foxa2 locus in mice," Developmental Dynamics, 2008, 237(2), 447-
Patel Y. C., "Somatostatin and Its Receptor Family," Frontiers in Neuroendocrinology, 1999, 20, 157-198. cited by applicant
Patro R., et al., "Salmon Provides Fast and Bias-Aware Quantification of Transcript Expression using Dual-Phase Inference," Nature Methods, 2017, vol. 14, p
Pedersen J. et al., "The glucagon-like peptide 2 receptor is expressed in enteric neurons and not in the epithelium of the intestine," Peptides, 2015, 67, 20-28. c
Peng K., et al., "Regulation of O-Linked N-Acetyl Glucosamine Transferase (OGT) Through E6 Stimulation of the Ubiquitin Ligase Activity of E6AP." Journ
2021, 10286. https://doi.org/10.3390/ijms221910286. cited by applicant
Perdomo J., et al., "Megakaryocyte differentiation and platelet formation from human cord blood derived CD34+ cells," Journal of Visualized Experiments. D
pages. cited by applicant
Petta S., et al., "Glucokinase Regulatory Protein Gene Polymorphism Affects Liver Fibrosis in Non-Alcoholic Fatty Liver Disease," PLoS One, 2014, vol. 9, e
Pham D., et al., "stLearn: Integrating Spatial Location, Tissue Morphology and Gene Expression to Find Cell Types, Cell-Cell Interactions and Spatial Traject
Tissues," bioRxiv, May 2020, doi: 10.1101/2020.05.31.125658. cited by applicant
Picelli S., et al., "Smart-seq2 for Sensitive Full-length Transcriptome Profiling in Single Cells," Nature Methods, Nov. 2013, vol. 10(11), pp. 1096-1098. cited
Pierre C., et al., "Can We Live Without a Functional Renin-Angiotensin System?" Clinical and Experimental Pharmacology and Physiology, 2008, vol. 35, pp
Pinney S. E. et al., "Neonatal diabetes and congenital malabsorptive diarrhea attributable to a novel mutation in the human neurogenin-3 gene coding sequenc
endocrinology and metabolism, 2011, 96, 1960-1965. cited by applicant
Pirola, C.J., et al., "A Rare Nonsense Mutation in the Glucokinase Regulator Gene Is Associated with a Rapidly Progressive Clinical Form of Nonalcoholic St
Communications, 2018, vol. 2, pp. 1030-1036. cited by applicant
Pless G., "Artificial and Bioartificial Liver Support," Organogenesis, Jan. 2007, vol. 3(1), pp. 20-24. cited by applicant
Podolsky D. K., "Healing the Epithelium: Solving the Problem from Two Sides," Journal of Gastroenterology, 1997, vol. 32, pp. 122-126. DOI: 10.1007/BF01
Jiang H., et al., "Tyrosine Kinase Receptor B Protects Against Coronary Artery Disease and Promotes Adult Vasculature Integrity by Regulating Ets1-Mediate
Arteriosclerosis, Thrombosis, and Vascular Biology, 2015, vol. 35, pp. 580-588. cited by applicant
Jiang M. et al., "BMP-driven NRF2 activation in esophageal basal cell differentiation and eosinophilic esophagitis," The Journal of Clinical Investigation,201
applicant

Jiang M., et al., “Transitional Basal Cells at the Squamous-Columnar Junction Generate Barrett's Oesophagus,” Nature, 2017, 550(7677), pp. 529-533. cited b
Jin S., et al., “Inference and Analysis of Cell-cell Communication Using Cellchat,” Nature Communications, Feb. 17, 2021, vol. 12(1):1088, 20 pages. cited b
Jin W., et al., “Regulation of BDNF-TrkB Signaling and Potential Therapeutic Strategies for Parkinson's Disease,” Journal of Clinical Medicine, Jan. 2020, vo
10.3390/jcm9010257. cited by applicant
Jonatan D., et al., “Sox17 regulates insulin secretion in the normal and pathologic mouse beta cell,” PloS one, 2014, 9, e104675, 16 pages. cited by applicant
Kaczmarek J. C., et al., “Polymer-Lipid Nanoparticles for Systemic Delivery of mRNA to the Lungs,” Angewandte Chemie International Edition, 2016, vol. 5
applicant
Kaczmarek J.C., et al., “Optimization of a Degradable Polymer-Lipid Nanoparticle for Potent Systemic Delivery of mRNA to the Lung Endothelium and Imm
vol. 18, No. 10, 6449-6454. doi: 10.1021/acs.nanolett.8b02917. cited by applicant
Kaiser J., Virus used in gene therapies may pose cancer risk, dog study hints, Science—Jan. 6, 2020 doi:10.1126/science.aba7696 in 3 pages. cited by applican
Kajiwara, K., et al., “Molecular Mechanisms Underlying Twin-to-Twin Transfusion Syndrome,” Cells, 2022, vol. 11, 18 pages. cited by applicant
Kalabis J. et al., “A subpopulation of mouse esophageal basal cells has properties of stem cells with the capacity for self-renewal and lineage specification,” Jo
2008, (118), 3860-3869. cited by applicant
Kalabis J. et al., “Isolation and characterization of mouse and human esophageal epithelial cells in 3D organotypic culture,” Nature Protocols, 2012, 7(2), 235
Kang, J., et al., “Simultaneous deletion of the methylcytosine oxidases Tet1 and Tet3 increases transcriptome variability in early embryogenesis,” Proceedings
Sciences, 2015, vol. 112, pp. E4236-E4245. cited by applicant
Kang, S. D., et al., “Effect of Productive Human Papillomavirus 16 Infection on Global Gene Expression in Cervical Epithelium.” Journal of Virology, vol. 92
18. cited by applicant
Kapadia, B., et al., “PIMT regulates Hepatic Gluconeogenesis in Mice,” iScience, 2023, 106120. cited by applicant
Kathiriya, J.J., et al., “Human Alveolar Type 2 Epithelium Transdifferentiates into Metaplastic KRT5+ Basal Cells,” Nature Cell Biology, Jan. 2022, vol. 24(1
Kawaguchi T., et al., “Genetic Polymorphisms of the Human PNPLA3 Gene Are Strongly Associated with Severity of Non-Alcoholic Fatty Liver Disease in J
7, e38322. cited by applicant
Kazumori H. et al., “Bile acids directly augment caudal related homeobox gene Cdx2 expression in oesophageal keratinocytes in Barrett's epithelium,” Gut, 2
applicant
Kazumori H. et al., “Roles of caudal-related homeobox gene Cdx1 in oesophageal epithelial cells in Barrett's epithelium development,” Gut, 2009, 58(5), 620
Kc K. et al., “In vitro model for studying esophageal epithelial differentiation and allergic inflammatory responses identifies keratin involvement in eosinophi
10(6), e0127755. cited by applicant
Kearns N. A. et al., “Generation of organized anterior foregut epithelia from pluripotent stem cells using small molecules,” Stem Cell Res., 2013, 11, 1003-10
Kebschull et al., “High-throughput mapping of single-neuron projections by sequencing of barcoded RNA”, Neuron, 2016, vol. 91(5), pp. 975-987. cited by a
Keebler M. E., et al., “Fine-Mapping in African Americans of 8 Recently Discovered Genetic Loci for Plasma Lipids: The Jackson Heart Study,” Circulation:
vol. 3, pp. 358-364. cited by applicant
Keeley, T.P., et al., “Defining Physiological Normoxia for Improved Translation of Cell Physiology to Animal Models and Humans,” Physiological Reviews, 2
by applicant
Kennedy D., et al., “Optimal Absorptive Transport of the Dipeptide Glycylsarcosine Is Dependent on Functional Na?/H? Exchange Activity,” Pflugers Archiv,
cited by applicant
Kermani P., et al., “Neurotrophins Promote Revascularization by Local Recruitment of TrkB+ Endothelial Cells and Systemic Mobilization of Hematopoietic
Investigation, 2005, vol. 115, pp. 653-663. cited by applicant
Kietzmann, T., et al., “Metabolic zonation of the liver: The oxygen gradient revisited,” Redox Biol, 2017, vol. 11, pp. 622-630. cited by applicant
Kim et al., “Recent progress in development of siRNA delivery vehicles for cancer therapy”, Advanced Drug Delivery Reviews, 2016, vol. 104, pp. 61-77. cit
Kim M., et al., “O-Linked N-Acetylglucosamine Transferase Promotes Cervical Cancer Tumorigenesis through Human Papillomavirus E6 and E7 Oncogenes
44596-44607. cited by applicant
Kim S., et al., “Engraftment Potential of Spheroid-Forming Hepatic Endoderm Derived from Human Embryonic Stem Cells,” Stem Cells Development. Jun. 1
1829. cited by applicant
Kim, S. G., et al., “Bilirubin Activates Transcription of HIF-1a in Human Proximal Tubular Cells Cultured in the Physiologic Oxygen Content,” J Korean Me
S154. cited by applicant
Kim Y. K., et al., “Gene-Edited Human Kidney Organoids Reveal Mechanisms of Disease in Podocyte Development,” Stem Cells, 2017, vol. 35, pp. 2366-23
Kimura M., et al., “En Masse Organoid Phenotyping Informs Metabolic-Associated Genetic Susceptibility to NASH,” Cell, Jun. 2022, vol. 185, No. 12, pp. 4
applicant
Kitamoto A., et al., “Association of Polymorphisms in GCKR and TRIB1 with Nonalcoholic Fatty Liver Disease and Metabolic Syndrome Traits,” Endocrine
689. cited by applicant
Kleshchevnikov V., et al., “Comprehensive Mapping of Tissue Cell Architecture via Integrated Single Cell and Spatial Transcriptomics,” bioRxiv, Nov. 2020,
cited by applicant
Kligerman S. J., et al., “From the Radiologic Pathology Archives: Organization and Fibrosis as a Response to Lung Injury in Diffuse Alveolar Damage, Organ
Fibrinous and Organizing Pneumonia,” Radiographics, 2013, 33, doi:10.1148/rg.337130057. PMID-24224590. cited by applicant
Kobayashi Y., et al., “Persistence of a Regeneration-associated, Transitional Alveolar Epithelial Cell State in Pulmonary Fibrosis,” Nature Cell Biology, Aug. 2
cited by applicant
Koboziev I., et al., “Use of Humanized Mice to Study the Pathogenesis of Autoimmune and Inflammatory Diseases,” Inflammatory Bowel Diseases, Jul. 1, 20
by applicant
Kolbe E., et al., “Mutual Zonated Interactions of Wnt and Hh Signaling Are Orchestrating the Metabolism of the Adult Liver in Mice and Human,” Cell Repor
pp. 4553-4567.e4557. cited by applicant
Kong J. et al., “Ectopic Cdx2 expression in murine esophagus models an intermediate stage in the emergence of Barrett's esophagus,” PLoS One, 2011, 6(4),
Kong J. et al., “Induction of intestinalization in human esophageal keratinocytes is a multistep process,” Carcinogenesis, 2009, 30(1), 122-130. cited by applic
Kormish J.D. et al., “Interactions between SOX factors and Wnt/beta-catenin signaling in development and disease,” Developmental Dynamics: An Official P
Association of Anatomists, 2010, 239, 56-68. cited by applicant
Koui Y., et al., “An In Vitro Human Liver Model by iPSC-Derived Parenchymal and Non-Parenchymal Cells,” Stem Cell Reports, 2017, 9, pp. 490-498. cited
Kouznetsova I. et al., Self-renewal of the human gastric epithelium: new insights from expression profiling using laser microdissection. Mol Biosyst, 2011, 7,
Kowalski P.S., et al., “Delivering the messenger: Advances in Technologies for Therapeutic mRNA delivery,” Molecular Therapy. Apr. 10, 2019;27(4):710-72
Kozyra M., et al., “Human Hepatic D Spheroids As a Model for Steatosis and Insulin Resistance”, Scientific Reports, vol. 8, No. 1, Sep. 24, 2018, Retrieved f
https://www.nature.com/articles/s41598-018-32722-6. cited by applicant
Krishnan, U., et al., “Evaluation and Management of Pulmonary Hypertension in Children with Bronchopulmonary Dysplasia.” The Journal of Pediatrics, vol
cited by applicant
Kuhnert F. et al., “Essential regulation of CNS angiogenesis by the orphan G protein-coupled receptor GPR124,” Science, 2010, 330, 985-989. 10.1126/scienc
Kumagai et al., “A bilirubin-inducible fluorescent protein from eel muscle,” Cell (2013) 153(7):1602-11. cited by applicant
Kumar A., et al., “Specification and Diversification of Pericytes and Smooth Muscle Cells from Mesenchymoangioblasts,” Cell Reports, 2017, vol. 19, pp. 19
Schreiber R., et al., “Inherited Renal Tubular Dysgenesis May Not Be Universally Fatal,” Pediatric Nephrology, 2010, vol. 25, pp. 2531-2534. cited by applica

Schreiber R., et al., "Renal Tubular Dysgenesis Secondary to Mutations in Genes Encoding the Renin-Angiotensin System," Harefuah, 2021, vol. 160, pp. 822
Schuldiner et al. "Induced Neuronal Differentiation of Human Embryonic Stem Cells," Brain Research 2001, Sep. 21, 2001, vol. 913(2):201-205. cited by app
Schupp J.C., et al., "Integrated Single-Cell Atlas of Endothelial Cells of the Human Lung," Circulation, May 2021, vol. 144, No. 4, 286-302. doi: 10.1161/CIR
cited by applicant
Sebrell T.A., et al., "Live Imaging Analysis of Human Gastric Epithelial Spheroids Reveals Spontaneous Rupture, Rotation and Fusion Events,". Cell and Tiss
293-307. cited by applicant
Sekar R. and Chow B. K. C., "Secrelin Receptor-Knockout Mice Are Resistant to High-Fat Diet-Induced Obesity and Exhibit Impaired Intestinal Lipid Absor
2014, vol. 28, pp. 3494-3505. cited by applicant
Self M. et al., "Six2 activity is required for the formation of the mammalian pyloric sphincter," Dev Biol, 2009, 334, 409-417. cited by applicant
Serra M., et al., "Pluripotent Stem Cell Differentiation Reveals Distinct Developmental Pathways Regulating Lung-Versus Thyroid-lineage Specification," De
144(21), pp. 3879-3893. cited by applicant
Shaham O. et al., "Pax6 is essential for lens fiber cell differentiation," Development (Cambridge, England), 2009, 136(15), 2567- 2578. cited by applicant
Shankar A.S., et al., "Human Kidney Organoids Produce Functional Renin," Kidney International, 2021, vol. 99, pp. 134-147. cited by applicant
Shao Z. et al., "MAnorm: a robust model for quantitative comparison of ChIP-Seq data sets," Genome Biol, 2012, 13, R16, 17 pages. cited by applicant
Shapiro E., et al., "Single-cell Sequencing-based Technologies will Revolutionize Whole Organism Science," Nature Reviews Genetics, 2013, vol. 14(9), pp. 6
Shen H., et al., "Glucokinase Regulatory Protein Gene Polymorphism Affects Postprandial Lipemic Response in a Dietary Intervention Study," Human Geneti
cited by applicant
Shoyaib A.A., et al., "Intraperitoneal Route of Drug Administration: Should it Be Used in Experimental Animal Studies?," Pharm Res, Dec. 23, 2019, vol. 37(
Simon, M., et al., "Expression of Vascular Endothelial Growth Factor and Its Receptors in Human Renal Ontogenesis and in Adult Kidney." American Journal
Feb. 1995, pp. F240-F250. cited by applicant
Simon T.G., et al., "Mortality in Biopsy-Confirmed Nonalcoholic Fatty Liver Disease: Results from a Nationwide Cohort," Gut, 2021, vol. 70, pp. 1375-1382.
322786. cited by applicant
Sinagoga K. L., et al., "Distinct Roles for the mTOR Pathway in Postnatal Morphogenesis, Maturation and Function of Pancreatic Islets," Development, 2017
by applicant
Sinagoga K.L., et al., "Deriving Functional Human Enteroendocrine Cells from Pluripotent Stem Cells," Development, 2018, vol. 145. cited by applicant
Singh A., et al., "Transplanted Human Intestinal Organoids: A Resource for Modeling Human Intestinal Development," Development, 2023, vol. 150, dev201
cited by applicant
Singh S. K., et al., "Glucose-Dependent Insulinotropic Polypeptide (GIP) Stimulates Transepithelial Glucose Transport," Obesity, 2008, vol. 16, pp. 2412-241
Sinner D. et al. "Sox17 and Sox4 differentially regulate beta-catenin/T-cell factor activity and proliferation of colon carcinoma cells," Molecular and Cellular
7815. cited by applicant
Sloan S. A., et al., "Human Astrocyte Maturation Captured in 3D Cerebral Cortical Spheroids Derived from Pluripotent Stem Cells," Neuron, 2017, vol. 95, pp
applicant
Sloth B. et al., "Effect of subcutaneous injections of PYY1-36 and PYY3-36 on appetite, ad libitum energy intake, and plasma free fatty acid concentration in
of Physiology Endocrinology and Metabolism, 2007, 293, E604-E609. cited by applicant
Sluch V.M., et al., "Highly Efficient Scarless Knock-in of Reporter Genes into Human and Mouse Pluripotent Stem Cells via Transient Antibiotic Selection",
29, 2018. Retrieved from the Internet: URL :https://www.ncbi.nlm.nih.gov/pmc/articles/PMC6264506/pdf/ pone.0201683.pdf. cited by applicant
Smith, S.M., et al., "Obeticholic Acid: A Farnesoid X Receptor Agonist for Primary Biliary Cholangitis," Journal of Pharmacy Technology, 2017, vol. 33 (2),
Snellings D.A., et al., "Cerebral Cavernous Malformation: From Mechanism to Therapy," Circulation Research, 2021, vol. 129, pp. 195-215. doi: 10.1161/CIR
by applicant
Snowball J. et al., "Endodermal Wnt signaling is required for tracheal cartilage formation," Dev Biol 405, 56-70 (2015). cited by applicant
Soneson C., et al., "Differential Analyses for RNA-Seq: Transcript-Level Estimates Improve Gene-Level Inferences," F1000Research, 2015, vol. 4, p. 1521. c
Song H.W., et al., "Transcriptomic Comparison of Human and Mouse Brain Microvessels," Scientific Reports, 2020, vol. 10, 12358. doi: 10.1038/s41598-020
Spangle, J. M., et al., "The Human Papillomavirus Type 16 E6 Oncoprotein Activates mTORC1 Signaling and Increases Protein Synthesis." Journal of Virolo
9398-9407. cited by applicant
Sparrow D. B., et al., "A Mechanism for Gene-Environment Interaction in the Etiology of Congenital Scoliosis," Cell, 2012, vol. 149, pp. 295-306. cited by ap
Speliotes E. K., et al., "Genome-Wide Association Analysis Identifies Variants Associated with Nonalcoholic Fatty Liver Disease That Have Distinct Effects o
Genetics, 2011, vol. 7, e1001324. cited by applicant
Spencer J., et al., "T Cell Subclasses in Fetal Human lleum," Clinical and Experimental Immunology, 1986, pp. 553-558. cited by applicant
Spencer J., et al., "The Development of Gut Associated Lymphoid Tissue in the Terminal lleum of Fetal Human Intestine," Clinical and Experimental Immuno
applicant
Spencer-Dene B. et al.,"Stomach development is dependent on fibroblast growth factor 10/fibroblast growth factor receptor 2b-mediated signaling," Gastroen
cited by applicant
Sreter K.B., et al., "Plasma Brain-Derived Neurotrophic Factor (BDNF) Concentration and BDNF/TrkB Gene Polymorphisms in Croatian Adults with Asthma
Medicine, Oct. 2020, vol. 10, No. 4, doi: 10.3390/jpm10040189. cited by applicant
Srinivas S., et al., "Cre Reporter Strains Produced by Targeted Insertion of EYFP and ECFP into the ROSA26 Locus," BMC Developmental Biology, Dec. 20
applicant
Staab J.F., et al., "Co-Culture System of Human Enteroids/Colonoids with Innate Immune Cells,". Current Protocols in Immunology, Dec. 2020, vol. 131(1),
Stoeckius M., et al., "Cell Hashing with Barcoded antibodies enables Multiplexing and Doublet Detection for Single Cell Genomics," Genome Biology, 2018,
applicant
Stoffers D. A., et al., "Pancreatic Agenesis Attributable to a Single Nucleotide Deletion in the Human IPF1 Gene Coding Sequence," Nature Genetics, 1997, v
applicant
Stoll B. J., et al., "Neonatal Outcomes of Extremely Preterm Infants from the NICHD Neonatal Research Network," Pediatrics, 2010, vol. 126, pp. 443-456. c
Stras., et al., "Maturation of the Human Intestinal Immune System Occurs Early in Fetal Development," Developmental Cell, Nov. 4, 2019, vol. 51(3), pp. 35
Street K., et al., "Slingshot: Cell Lineage and Pseudotime Inference for Single-cell Transcriptomics," BMC Genomics, Dec. 2018, vol. 19, pp. 1-16. cited by a
Strikoudis A., et al., "Modeling of Fibrotic Lung Disease Using 3D Organoids Derived from Human Pluripotent Stem Cells," Cell Reports, Jun. 18, 2019, vol.
applicant
Strunz M., et al., "Alveolar Regeneration Through a Krt8+ Transitional Stem Cell State That Persists in Human Lung Fibrosis," Nature Communications, Jul.
pages. cited by applicant
Subramanian, A. et al., "Gene set enrichment analysis: A knowledge-based approach for interpreting genome-wide expression profiles," Proceedings of the Na
2005, 102(43), 15545-15550. cited by applicant
Sucre J. M.S., et al., "Hyperoxia Injury in the Developing Lung is Mediated by Mesenchymal Expression of Wnt5A," American Journal of Respiratory and Cr
2020, vol. 201(10), pp. 1249-1262. cited by applicant
Sun X., et al., "A Census of the Lung: CellCards from LungMAP," Developmental Cell, Jan. 10, 2022, vol. 57(1), pp. 112-145. cited by applicant
Sunshine J.C., "Effects of Base Polymer Hydrophobicity and End-Group Modification on Polymeric Gene Delivery," Biomacromolecules 2011, 12, pp. 3592-
Suprynowicz, F. A., et al., "HPV-16 E5 Oncoprotein Upregulates Lipid Raft Components Caveolin-1 and Ganglioside GM1 at the Plasma Membrane of Cervi

2008, pp. 1071-1078. cited by applicant
He B., et al., "Understanding Transcriptional Regulatory Networks Using Computational Models," Current Opinion in Genetics Development, Apr. 1, 2016, v
applicant
He, L., et al., "Proliferation tracing reveals regional hepatocyte generation in liver homeostasis and repair," Science, 2021, vol. 371, eabc4346. cited by applic
Heinz S. et al., "Simple combinations of lineage-determining transcription factors prime cis-regulatory elements required for macrophage and B cell identities
cited by applicant
Hernaez R., et al., "Association Between Variants in or near PNPLA3, GCKR, and PPP1R3B with Ultrasound-Defined Steatosis Based on Data from the Thir
Examination Survey," Clinical Gastroenterology and Hepatology, 2013, vol. 11, pp. 1183-1190.e1182. cited by applicant
Herriges M.J., et al., "Long Noncoding RNAs are Spatially Correlated with Transcription Factors and Regulate Lung Development," Genes Development, Jur
1379. cited by applicant
Hilgers K. F., et al., "Aberrant Renal Vascular Morphology and Renin Expression in Mutant Mice Lacking Angiotensin-Converting Enzyme," Hypertension, 1
by applicant
Hill, D. R., et al., "Gastrointestinal Organoids: Understanding the Molecular Basis of the Host-Microbe Interface," Cell Mol Gastroenterol Hepatol, 2017, vol.
applicant
Hirschhorn, J. N., et al., "Genome-Wide Association Studies for Common Diseases and Complex Traits." Nature Reviews Genetics, vol. 6, 2005, pp. 95-108.
Hirsh A. J., et al., "Effect of Cholecystokinin and Related Peptides on Jejunal Transepithelial Hexose Transport in the Sprague-Dawley Rat," American Journa
and Liver Physiology, 1996, vol. 271, No. G755-G761. cited by applicant
Hoffmeister K.M., "Desialylated Platelets: A Missing Link in Hepatic Thrombopoietin Regulation," The Hematologist 2015; 12(3), 7 pages. cited by applican
Hoffmeister K.M., "The role of lectins and glycans in platelet clearance," Journal of Thrombosis and Haemostasis. Jul. 2011;9 (supp1), pp. 35-43. cited by app
Hohwieler H., et al., "Human Pluripotent Stem Cell-Derived Acinar/Ductal Organoids Generate Human Pancreas upon Orthotopic Transplantation and Allow
vol. 66, pp. 473-486. cited by applicant
Holt L.M., et al., "Astrocyte Morphogenesis is Dependent on BDNF Signaling via Astrocytic TrkB.T1," eLife, Aug. 2019, vol. 8, No. e44667. doi: 10.7554/eL
Homayun B., et al., "Challenges and Recent Progress in Oral Drug Delivery Systems for Biopharmaceuticals," Pharmaceutics, Mar. 19, 2019, vol. 11(3):129.
Hoskins E. E., et al., Fanconi anemia deficiency stimulates HPV-associated hyperplastic growth in organotypic epithelial raft culture. Oncogene, 2009, 28(5),
Hotta K., et al., "Association of the rs738409 Polymorphism in PNPLA3 with Liver Damage and the Development of Nonalcoholic Fatty Liver Disease," BM
11, p. 172. cited by applicant
Hu H., et al., "AnimalTFDB 3.0: A Comprehensive Resource for Annotation and Prediction of Animal Transcription Factors," Nucleic Acids Research, 2019,
applicant
Hu S., et al., "Wnt/-Catenin Signaling and Liver Regeneration: Circuit, Biology, and Opportunities," Gene expression, 2021, vol. 20(3), pp. 189-199. cited by
Hu Y. et al., "Targeted disruption of peptide transporter Pept1 gene in mice significantly reduces dipeptide absorption in intestine," Molecular pharmaceutics,
applicant
Hu Y., et al., "Wnt/-Catenin Signaling Is Critical for Regenerative Potential of Distal Lung Epithelial Progenitor Cells in Homeostasis and Emphysema," Stem
pp. 1467-1478. cited by applicant
Hu Z. et al., "Generation of Naivetropic Induced Pluripotent Stem Cells from Parkinson's Desease Patients for High-Efficiency Genetic Manipulation oand Di
Development, 2015, vol. 24, No. 21, 2591-2604. cited by applicant
Huang H., et al., "p300-Mediated Lysine 2-Hydroxyisobutyrylation Regulates Glycolysis," Molecular Cell, 2018, vol. 70, pp. 663-678.e666. doi: 10.1016/j.m
applicant
Huang J., et al., "Activation of Wnt/ß-Catenin Signalling via GSK3 Inhibitors Direct Differentiation of Human Adipose Stem Cells into Functional Hepatocyt
2017, 7, Article No. 40716, 12 pages. cited by applicant
Huang, S. X. L., et al., "Efficient generation of lung and airway epithelial cells from human pluripotent stem cells," Nat. Biotechnol., 2014, vol. 32, No. 1, pp.
Hudert, C. A., et al., "Genetic Determinants of Steatosis and Fibrosis Progression in Paediatric Non-Alcoholic Fatty Liver Disease." Liver International, vol. 3
applicant
Huo X. et al., "Acid and Bile Salt-Induced CDX2 Expression Differs in Esophageal Squamous Cells From Patients With and Without Barrett's Esophagus," G
194-203.e1. cited by applicant
Hurr, C., et al., "Liver Sympathetic Denervation Reverses Obesity-Induced Hepatic Steatosis." The Journal of Physiology, vol. 597, No. 17, Sep. 2019, pp. 45
Hurskainen M., et al., "Single Cell Transcriptomic Analysis of Murine Lung Development on Hyperoxia-Induced Damage," Nature Communications, Mar. 20
10.1038/s41467-021-21865-2. cited by applicant
Husson, A., et al., "Argininosuccinate synthetase from the urea cycle to the citrulline-NO cycle," European Journal of Biochemistry, 2003, vol. 270, pp. 1887-
Huynh N., et al., "Feasibility and Scalability of Spring Parameters in Distraction Enterogenesis in a Murine Model," Journal of Surgical Research, 2017, vol. 2
applicant
Iansante, V., et al., "Human hepatocyte transplantation for liver disease: current status and future perspectives," Pediatric Research, 2018, vol. 83, pp. 232-240
Ikeda, Y., et al., "Bilirubin exerts pro-angiogenic property through Akt-eNOS787 dependent pathway," Hypertension Research, 2015, vol. 38, pp. 733-740. cit
Ikegami., M. et al. "Surfactant Protein D Influences Surfactant Ultrastructure and Uptake by Alveolar Type Ii Cells," American Journal of Physiology-Lung C
Mar. 2005, vol. 288(3), pp. L552-L561. cited by applicant
Illig R. et al., "Spatio-temporal expression of HOX genes in human hindgut development," Developmental dynamics: an official publication of the American
242, 53-66. cited by applicant
Isosaari L., et al., "Simultaneous Induction of Vasculature and Neuronal Network Formation on a Chip Reveals a Dynamic Interrelationship Between Cell Typ
Signaling, 2023, vol. 21, 132. doi: 10.1186/s12964-023-01159-4. cited by applicant
Iwafuchi-Doi, M. et al., "Pioneer transcription factors in cell reprogramming," Genes Dev 2014, 28, 2679-2692. cited by applicant
Iwasawa K., et al., "Organogenesis In Vitro," Current Opinion in Cell Biology, 2021, 73, pp. 84-91. cited by applicant
Jackerott M. et al., "Immunocytochemical localization of the NPY/PYY Y1 receptor in enteric neurons, endothelial cells, and endocrine-like cells of the rat in
Cytochem, 1997, 45(12), 1643-1650. cited by applicant
Jacob A., et al., "Differentiation of Human Pluripotent Stem Cells into Functional Lung Alveolar Epithelial Cells," Cell Stem Cell, Oct. 5, 2017, vol. 21(4), pp
Jain R., et al., "Plasticity of Hopx (+) Type I Alveolar Cells to Regenerate Type Ii Cells in the Lung," Nature Communications, 2015, vol. 13;6(1):6727, 20 pa
Jang S. W., et al., "A Selective TrkB Agonist with Potent Neurotrophic Activities by 7,8-Dihydroxyflavone," Proceedings of the National Academy of Science
2010, vol. 107, pp. 2687-2692. cited by applicant
Jaramillo M., et al., "Endothelial Cells Mediate Islet-Specific Maturation of Human Embryonic Stem Cell-Derived Pancreatic Progenitor Cells," Tissue Engin
14-25. cited by applicant
Jarmas, A.E., et al., "Progenitor translatome changes coordinated by Tsc1 increase perception of Wnt signals to end nephrogenesis," Nature Communications,
"Generation of Human PSC-Derived Kidney Organoids with Patterned Nephron Segments and a De Novo Vascular Network," Cell Stem Cell, 2019, vol. 25, p
Jennings R. E., et al., "Development of the Human Pancreas from Foregut to Endocrine Commitment," Diabetes, 2013, vol. 62, pp. 3514-3522. cited by applic
Jennings R. E., et al., "Human Pancreas Development," Development, 2015, vol. 142, pp. 3126-3137. cited by applicant
Jensen E. A., et al., "Epidemiology of Bronchopulmonary Dysplasia," Birth Defects Research Part A: Clinical and Molecular Teratology, 2014, vol. 100, pp. 1
Jensen K. J., et al., "Hepatic Nervous System and Neurobiology of the Liver," Comprehensive Physiology, 2013, vol. 3, pp. 655-665. cited by applicant
Jeong, Y., et al., "Identification and genetic manipulation of human and mouse oesophageal stem cells," Gut, 2015, pp. 1-10. cited by applicant

Jho E. et al, "Wnt/beta-catenin/Tcf signaling induces the transcription of Axin2, a negative regulator of the signaling pathway," Molecular and Cellular Biolog
applicant
Jiang C., et al., "Comparative Transcriptomics Analyses in Livers of Mice, Humans, and Humanized Mice Define Human-Specific Gene Networks," Cells, No
by applicant
Suzuki, et al., "Directed differentiation of human induced pluripotent stem cells into mature stratified bladder urothelium," Scientific Reports, 2019, vol. 9, 10
Takasato M., et al., "Kidney Organoids from Human iPS Cells Contain Multiple Lineages and Model Human Nephrogenesis," Nature, 2016, vol. 536, p. 238.
Tam P. P., and Loebel D. A., "Gene Function in Mouse Embryogenesis: Get Set for Gastrulation," Nature Reviews Genetics, 2007, 8, pp. 368-381. cited by ap
Tanaka K., et al., "Structure and Functional Expression of the Cloned Rat Neurotensin Receptor," Neuron, 1990, vol. 4, pp. 847-854. cited by applicant
Tanii, H., et al., "Induction of Cytochrome P450 2A6 by Bilirubin in Human Hepatocytes," Pharmacology & Pharmacy, 2013, vol. 04, pp. 182-190. cited
Tannenbaum, S.E. et al., "Derivation of Xeno-Free and GMP-Grade Human Embryonic Stem Cells—Platforms for Future Clinical Applications," PLoS One,
cited by applicant
Tanwar S. et al., "Validation of terminal peptide of procollagen III for the detection and assessment of nonalcoholic steatohepatitis in patients with nonalcohol
Hepatology, 2013, 57, 103-111. cited by applicant
Tarlungeanu D. C., et al., "Impaired Amino Acid Transport at the Blood Brain Barrier Is a Cause of Autism Spectrum Disorder," Cell, 2016, vol. 167, pp. 1481
Tata P.R., et al., "Developmental History Provides a Roadmap for the Emergence of Tumor Plasticity," Developmental Cell, Mar. 26, 2018, vol. 44(6), pp. 679
Terry, N. A. et al., "Lipid Malabsorption from Altered Hormonal Signaling Changes Early Gut Microbial Responses". Am J Physiol-Gastrointest Liver Physio
cited by applicant
Terry N.A. et al., "Dysgenesis of enteroendocrine cells in Aristaless-Related Homeobox polyalanine expansion mutationsm," J Pediatr Gastroenterol Nutr, 201
applicant
Tessarollo L., et al., "TrkB Truncated Isoform Receptors as Transducers and Determinants of BDNF Functions," Frontiers in Neuroscience, Mar. 2022, vol. 16
10.3389/fnins.2022.847572. cited by applicant
Thakur, A., et al., "Hepatocyte Nuclear Factor 4-Alpha Is Essential for the Active Epigenetic State at Enhancers in Mouse Liver," Hepatology, 2019, vol. 70, p
The Lancet Gastroenterology, "Headway and hurdles in non-alcoholic fatty liver disease," Lancet Gastroenterology Hepatology, 2020, vol. 5, 93. cited by app
Thebaud B., et al., "Vascular Endothelial Growth Factor Gene Therapy Increases Survival, Promotes Lung Angiogenesis, and Prevents Alveolar Damage in H
Evidence that Angiogenesis Participates in Alveolarization," Circulation, 2005, vol. 112, pp. 2477-2486. cited by applicant
Thommensen L. et al., "Molecular mechanisms involved in gastrin-mediated regulation of cAMP-responsive promoter elements," Am J Physiol Endocrinol M
cited by applicant
Thompson F.M., et al., "Epithelial Growth of the Small Intestine Occurs by Both Crypt Fission and Crypt Hyperplasia During Infancy and Childhood in Huma
and Hepatology, 2001, 2 Pages. cited by applicant
Tomassoni-Ardori F., et al., "Rbfox1 Up-Regulation Impairs BDNF-Dependent Hippocampal LTP by Dysregulating TrkB Isoform Expression Levels," eLife,
10.7554/eLife.49673. cited by applicant
Toth A., et al., "Alveolar Epithelial Progenitor Cells Drive Lung Regeneration via Dynamic Changes in Chromatin Topology Modulated by Lineage-specific N
31 pages. cited by applicant
Toth A., et al., "Alveolar Epithelial Stem Cells in Homeostasis and Repair," Chapter 10 in Lung Stem Cells in Development, Health and Disease, European Re
133. cited by applicant
Totoson P., et al., "Activation of endothelial TrkB receptors induces relaxation of resistance arteries." Vascular Pharmacology, 106, 2018, pp. 46-53. cited by a
Touboul T. et al., "Generation of functional hepatocytes from human embryonic stem cells under chemically defined conditions that recapitulate liver develop
1754-1765. cited by applicant
Traag V.A., et al., "From Louvain to Leiden: Guaranteeing Well-Connected Communities," Scientific Reports, 2019, vol. 9, 5233. doi: 10.1038/s41598-019-4
Tran M., et al., "Spatial Analysis of Ligand-Receptor Interaction in Skin Cancer at Genome-Wide and Single-Cell Resolution," bioRxiv, Sep. 2021, doi: 10.11
applicant
Trapnell C. et al., "The dynamics and regulators of cell fate decisions are revealed by pseudotemporal ordering of single cells," Nat Biotechnol, 2014, 32, 381
Travaglini K.J., et al., "A Molecular Cell Atlas of the Human Lung from Single-cell RNA Sequencing," Nature, Nov. 26, 2020, vol. 587(7835), pp. 619-625. c
Tsakmaki A., et al., "Diabetes Through a 3D Lens: Organoid Models," Diabetologia, Springer Berlin Heidelberg, Berlin/heidelberg, vol. 63, No. 6, Mar. 27, 20
applicant
Tsankov A. M. et al., "Transcription factor binding dynamics during human ES cell differentiation," Nature, 2015, 518 (7539), 344-9. cited by applicant
Tufro-McReddie, A., et al., "Angiotensin II Regulates Nephrogenesis and Renal Vascular Development." American Journal of Physiology, vol. 269, No. 1, 19
applicant
Uchida H., et al., "A Xenogeneic-free System Generating Functional Human Gut Organoids from Pluripotent Stem Cells," JCI Insight, Jan. 12, 2017, vol. 2, N
Uchimura, K., et al., "Human Pluripotent Stem Cell-Derived Kidney Organoids with Improved Collecting Duct Maturation and Injury Modeling," Cell Repor
applicant
Vales, S., et al., "In Vivo Human PSC-Derived Intestinal Organoids to Study Stem Cell Maintenance." In Methods in Molecular Biology, vol. 2171, Chapter 1
applicant
Van Den Berg C.W., et al., "Renal Subcapsular Transplantation of PSC-Derived Kidney Organoids Induces Neo-vasculogenesis and Significant Glomerular ar
Stem Cell Reports, 2018, vol. 10, pp. 751-765. doi: 10.1016/j.stemcr.2018.01.041. cited by applicant
Van Dop W. A., et al., "Hedgehog Signalling Stimulates Precursor Cell Accumulation and Impairs Epithelial Maturation in the Murine Oesophagus," Gut, 201
applicant
Van Hoecke et al., "How mRNA therapeutics are entering the monoclonal antibody field," Journal of Translational Medicine . Feb. 22, 2019;17(1):54 in 14 pa
Van Lieshout., L.P., et al., "A Novel Triple-Mutant AAV6 Capsid Induces Rapid and Potent Transgene Expression in the Muscle and Respiratory Tract of Mic
and Clinical Development, Open Access Jun. 15, 2018, vol. 9, pp. 323-329. cited by applicant
Van Raay T. J. et al., "Frizzled 5 signaling governs the neural potential of progenitors in the developing Xenopus retina," Neuron, 2005, 46(1), 23-36. cited by
Van Straten, G., et al., "Aberrant Expression and Distribution of Enzymes of the Urea Cycle and Other Ammonia Metabolizing Pathways in Dogs with Conge
One, 2014, vol. 9, e100077, 11 pages. cited by applicant
Vatine G. D., et al., "Modeling Psychomotor Retardation Using iPSCs from MCT8-Deficient Patients Indicates a Prominent Role for the Blood-Brain Barrier,
pp. 831-843.e835. cited by applicant
Vatine G.D., et al., "Human iPSC-Derived Blood-Brain Barrier Chips Enable Disease Modeling and Personalized Medicine Applications," Cell Stem Cell, 201
by applicant
Vega M. E. et al., "Inhibition of notch signaling enhances transdifferentiation of the esophageal squamous epithelium towards a Barrett's-like metaplasia via K
3857-3866. cited by applicant
Veldman, T., et al., "Human Papillomavirus E6 and Myc Proteins Associate In Vivo and Bind to and Cooperatively Activate the Telomerase Reverse Transcrip
the National Academy of Sciences, vol. 100, No. 14, Jul. 8, 2003, pp. 8211-8216. cited by applicant
Verdera H,C., et al., AAV vector immunogenicity in humans: A long journey to successful gene transfer, Mol Thera. Mar. 4, 2020;28(3):723-746. cited by appl
Verheyden J.M., et al., "A Transitional Stem Cell State in the Lung," Nature Cell Biology, Sep. 2020, vol. 22(9), pp. 1025-1026. cited by applicant
Verma S.K., et al., "RBFOX2 is Required for Establishing RNA Regulatory Networks Essential for Heart Development," Nucleic Acids Research, 2022, vol. 5
10.1093/nar/gkac055. cited by applicant

Verscheijden L.F.M., et al., “Differences in P-Glycoprotein Activity in Human and Rodent Blood-Brain Barrier Assessed by Mechanistic Modelling,” Archive
pp. 3015-3029. cited by applicant
Vincent K.M., et al., “Expanding the Clinical Spectrum of Autosomal-Recessive Renal Tubular Dysgenesis: Two Siblings with Neonatal Survival and Review
Genetics and Genomic Medicine, 2022, vol. 10, e1920. cited by applicant
Vohwinkel C.U., et al., “Bronchopulmonary Dysplasia: Endothelial Cells in the Driver's Seat,” American Journal of Respiratory Cell and Molecular Biology,
doi: 10.1165/rcmb.2021-0145ED. cited by applicant
Wagner N., et al., “Coronary Vessel Development Requires Activation of the TrkB Neurotrophin Receptor by the Wilms' Tumor Transcription Factor Wt1,” G
vol. 19, pp. 2631-2642. cited by applicant
Wahlestedt C. et al., “Neuropeptide Y Receptor Subtypes, Y1 and Y2,” Annals of the New York Academy of Sciences, 1990, 611, 7-26. cited by applicant
Kumari, D., “States of Pluripotency: Naïve and Primed Pluripotent Stem Cells,” InTech Open, vol. 1, Chapter 3, 2016, pp. 31-45. cited by applicant
Kurz H., “Cell Lineages and Early Patterns of Embryonic CNS Vascularization,” Cell Adhesion & Migration, 2009, vol. 3, pp. 205-210. cited by applica
Kusakabe T., et al., “Thyroid-Specific Enhancer-binding Protein/NKX2.1 is Required for the Maintenance of Ordered Architecture and Function of the Differe
Endocrinology, Aug. 2006, vol. 20(8), pp. 1796-1809. cited by applicant
Kuzmichev A. N.et al., “Sox2 acts through Sox21 to regulate transcription in pluripotent and differentiated cells,” Current Biology, 22(18), 2012, 1705-1710.
Kwapiszewska G., et al., “BDNF/TrkB Signaling Augments Smooth Muscle Cell Proliferation in Pulmonary Hypertension,” American Journal of Pathology, 2
cited by applicant
L. Landsman, et al., “Pancreatic Mesenchyme Regulates Epithelial Organogenesis Throughout Development,” PLoS Biology, 2011, vol. 9, Article e1001143,
Lacanna R., et al., “Yap/Taz Regulate Alveolar Regeneration and Resolution of Lung Inflammation,” Journal of Clinical Investigation, May 1, 2019, vol. 129(
applicant
Lammert E., “Induction of Pancreatic Differentiation by Signals from Blood Vessels,” Science, 2001, vol. 294, pp. 564-567. cited by applicant
Lancaster M. A., et al., “Cerebral Organoids Model Human Brain Development and Microcephaly,” Nature, 2013, vol. 501(7467), pp. 373-379. cited by appli
Landin, B. H., et al., “Labeled Lectin Studies of Renal Tubular Dysgenesis and Renal Tubular Atrophy of Postnatal Renal Ischemia and End-Stage Kidney Dis
14, No. 1, 1994, pp. 87-99. cited by applicant
Lange M., et al., “CellRank for Directed Single-cell Fate Mapping,” Nature Methods, Feb. 2022, vol. 19(2), pp. 159-170. cited by applicant
Langen U.H., et al., “Development and Cell Biology of the Blood-Brain Barrier,” Annual Review of Cell and Developmental Biology, 2019, vol. 35, pp. 591-
Langer R., “Tissue Engineering,” Science, 1990, vol. 249, pp. 1527-1533. cited by applicant
Lau J. Y., et al., “Systematic Review of the Epidemiology of Complicated Peptic Ulcer Disease: Incidence, Recurrence, Risk Factors and Mortality,” Digestio
DOI: 10.1159/000323958. cited by applicant
Laughney A.M., et al., “Regenerative Lineages and Immune-mediated Pruning in Lung Cancer Metastasis,” Nature Medicine, Feb. 2020, vol. 26(2), pp. 259-
Leblanc G. G., et al., “Biology of Vascular Malformations of the Brain,” Stroke, 2009, vol. 40, pp. e694-702. cited by applicant
Lee J., et al., “IL-25 and CD4(+) TH2 Cells Enhance Type 2 Innate Lymphoid Cell-derived IL-13 Production, Which Promotes IgE-mediated Experimental Fo
Allergy and Clinical Immunology, Apr. 1, 2016, vol. 137(4), pp. 1216-1225. cited by applicant
Lee J.H. et al., “Anatomically and Functionally Distinct Lung Mesenchymal Populations Marked by Lgr5 and Lgr6,” Cell, Sep. 7, 2017, vol. 170(6), pp. 1149
Leedham S. J. et al., “Individual crypt genetic heterogeneity and the origin of metaplastic glandular epithelium in human Barrett's oesophagus,” Gut, 2008, 57
Lehner, R. et al., “A Comparison of Plasmid DNA Delivery Efficiency and Cytotoxicity of Two Cationic Diblock Polyoxazoline Copolymers”, Nanotechnolog
applicant
Li B., et al., “Benchmarking Spatial and Single-Cell Transcriptomics Integration Methods for Transcript Distribution Prediction and Cell Type Deconvolution
vol. 19, No. 6, pp. 662-670. cited by applicant
Li H., et al., “Directed Differentiation of Human Embryonic Stem Cells into Keratinocyte Progenitors In Vitro: An Attempt with Promise of Clinical Use,” In
Biology—Animal, 2016, 52(8), pp. 885-893. cited by applicant
Li H. et al., “Fast and accurate short read alignment with Burrows-Wheeler transform,” Bioinformatics, 2009, 25(14), 1754-1760. cited by applicant
Li H.J. et al., Basic helix-loop-helix transcription factors and enteroendocrine cell differentiation. Diabetes Obes Metab, 2011, 13(01), Suppl 1, 5-12, 16 pages
Li J., et al., “An Obligatory Role for Neurotensin in High Fat Diet-Induced Obesity,” Nature, 2016, vol. 533, No. 7603, pp. 411-415. cited by applicant
Li N., et al., Early-Life Compartmentalization of Immune Cells in Human Fetal Tissues Revealed by High-Dimensional Mass Cytometry, Frontiers in Immun
10(1932), 13 pages. cited by applicant
Li N., et al., Mass cytometry reveals innate lymphoid cell differentiation pathways in the human fetal intestine. Journal of Experimental Medicine, May 7, 201
cited by applicant
Li N., et al., “Memory CD4+ T Cells Are Generated in the Human Fetal Intestine,” Nature Immunology, Mar. 2019, vol. 20(3), pp. 301-312. cited by applican
Li, Y., et al., “The Renin-Angiotensin-Aldosterone System (RAAS) Is One of the Effectors by Which Vascular Endothelial Growth Factor (VEGF)/Anti-VEG
Barrier,” American Journal of Pathology, 2020, vol. 190, pp. 1971-1981. cited by applicant
Li Y., et al., “Synthesis and Characterization of an Amphiphilic Graft Polymer and its Potential as a pH-Sensitive Drug Carrier”, Polymer, vol. 58, No. 15, Jul.
applicant
Liang, W., et al., MEF2C Alleviates Acute Lung Injury in Cecal Ligation and Puncture (CLP)-induced Sepsis Rats by Up-regulating AQP1Allergologia et Imm
vol. 49(5), pp. 117-124. cited by applicant
Lignitto L., et al., Nrf2 Activation Promotes Lung Cancer Metastasis by Inhibiting the Degradation of Bach 1Cell, Jul. 11, 2019, vol. 178(2), pp. 316-329. cite
Lin Y. C., et al., “Genetic Variants in GCKR and PNPLA3 Confer Susceptibility to Nonalcoholic Fatty Liver Disease in Obese Individuals,” American Journal
99, pp. 869-874. cited by applicant
Lindström N. O., et al., “Integrated B-catenin, BMP, PTEN, and Notch signalling patterns the nephron.” eLife, 4, e04000. 2015, 29 pages. https://doi.org/10.7
applicant
Lindström N. O., et al., “Spatial Transcriptional Mapping of the Human Nephrogenic Program.” Developmental Cell, 56(16), 2021, pp. 2381-2398.e6.
https://doi.org/10.1016/j.devcel.2021.07.017. cited by applicant
Lindstrom N.O., et al., “Integrated Beta-Catenin, BMP, PTEN, and Notch Signalling Patterns the Nephron,” eLife, 2015, vol. 3, e04000. cited by applicant
Liu D., Chinese Encyclopedia of Medicine—Pathophysiology, “China Signal Pathway and Targeted Therapeutics”, edited by Yu Yuanxun, Anhui Science and
edition. cited by applicant
Liu K., et al., “Sox2 Cooperates with Inflammation-Mediated Stat3 Activation in the Malignant Transformation of Foregut Basal Progenitor Cells,” Cell Stem
by applicant
Liu T., et al., “Regulation of Cdx2 expression by promoter methylation, and effects of Cdx2 transfection on morphology and gene expression of human esoph
Carcinogenesis, 2007, 28(2), 488-496. cited by applicant
Lloyd D. J., et al., “Antidiabetic Effects of Glucokinase Regulatory Protein Small-Molecule Disruptors.” Nature, 504, 2013, 16 pages. cited by applicant
Loh K.M., et al., “Efficient Endoderm Induction from Human Pluripotent Stem Cells by Logically Directing Signals Controlling Lineage Bifurcations,” Cell S
14(2), pp. 237-252. cited by applicant
Lois C. et al., “Germline transmission and tissue-specific expression of transgenes delivered by lentiviral vectors,” Science, 2002, 295, 868-872. cited by appl
Loomba, R., et al., “Combination Therapies Including Cilofexor and Firsocostat for Bridging Fibrosis and Cirrhosis Attributable to NASH.” Hepatology, vol.
643. cited by applicant
Loomba R., et al., “Heritability of Hepatic Fibrosis and Steatosis Based on a Prospective Twin Study,” Gastroenterology, 2015, vol. 149, pp. 1784-1793. cited

Loquet Ph., et al., “Influence of Raising Maternal Blood Pressure With Angiotensin II on Utero-Placental and Feto-Placental Blood Velocity Indices in the Hu
vol. 78, pp. 95-100. cited by applicant
Low, J.H., et al., “Generation of Human PSC-Derived Kidney Organoids with Patterned Nephron Segments and a De Novo Vascular Network,” Cell Stem Cel
e379. cited by applicant
Lu T.M., et al., “Pluripotent Stem Cell-Derived Epithelium Misidentified as Brain Microvascular Endothelium Requires ETS Factors to Acquire Vascular Fate
Academy of Sciences of the United States of America, 2021, vol. 118. cited by applicant
Lubinsky M. Sonic Hedgehog, VACTERL, and Fanconi anemia: Pathogenetic connections and therapeutic implications. American Journal of Medical Genetic
2598. cited by applicant
Lucía Selfa, I., et al., “Directed Differentiation of Human Pluripotent Stem Cells for the Generation of High-Order Kidney Organoids.” Methods in Molecular
171-189. cited by applicant
Lustig B. et al., “Negative feedback loop of Wnt signaling through upregulation of conductin/axin2 in colorectal and liver tumors,” Molecular and Cellular Bi
by applicant
Ahn Y., et al., “Human Blood Vessel Organoids Penetrate Human Cerebral Organoids and Form a Vessel-Like System,” Cells, Aug. 9, 2021, vol. 10, No. 2036
Internet URL: https://doi.org/10.3390/cells10082036. cited by applicant
Andang M. et al., “Optimized Mouse ES Cell Culture System By Suspension Growth in a Fully Defined Medium”, Nature Protocols, May 1, 2008, vol. 3, No
10.1038/nprot.2008.65. cited by applicant
Asano H., et al., “Astrocyte Differentiation of Neural Precursor Cells is Enhanced by Retinoic Acid Through a Change in Epigenetic Modification,” Stem Cel
Retrieved from the Internet URL: https://academic.oup.com/stmcls/article/27/11/2744/6401847. cited by applicant
Bakhti. M., et al., “Modelling the Endocrine Pancreas in Health and Disease,” Nat Rev Endocrinol, 2019, vol. 15, pp. 155-171. cited by applicant
Balboa. D., et al., “Functional, Metabolic and Transcriptional Maturation of Human Pancreatic Islets Derived from Stem Cells,” Nat Biotechnol, 2022, vol. 40
applicant
Bar-Nur O., et al., “Small Molecules Facilitate Rapid and Synchronous iPSC Generation” Nature Methods, Nov. 2024, vol. 11, No. 11, pp. 1170-1179. cited b
Binek. A., et al., “Flow Cytometry Has a Significant Impact on the Cellular Metabolome,” Journal of Proteome Research, 2019, vol. 18, pp. 169-181. cited by
Bipat. R., et al., “Drinking Water with Consumption of a Jelly Filled Doughnut Has a Time Dependent Effect on the Postprandial Blood Glucose Level in Hea
Nutrition ESPEN, 2018, vol. 27, pp. 20-23. cited by applicant
Box. A., et al., “Evaluating the Effects of Cell Sorting on Gene Expression,” Journal of Biomolecular Techniques, 2020, vol. 31, pp. 100-111. cited by applica
Brafman D.A., et al., “Analysis of SOX2-Expressing Cell Populations Derived from Human Pluripotent Stem Cells,” Stem Cell Reports, Nov. 19, 2013, vol. 1
Campinho. P., et al., “Blood Flow Forces in Shaping the Vascular System: A Focus on Endothelial Cell Behavior,” Frontiers in Physiology, 2020, vol. 11, pp. 5
Cervantes D. C., et al., “Peering Into Tunneling Nanotubes—The Path Forward,” The EMBO Journal, 2021, vol. 40, 22 pages. cited by applicant
Coll. M., et al., “Generation of Hepatic Stellate Cells from Human Pluripotent Stem Cells Enables In Vitro Modeling of Liver Fibrosis,” Cell Stem Cell, 2018,
applicant
Cormier J.T., et al., “Expansion of Undifferentiated Murine Embryonic Stem Cells as Aggregates in Suspension Culture Bioreactors,” Tissue Engineering, No
3233-3245, Retrieved from the Internet URL: https://www.liebertpub.com/doi/epdf/10.1089/ten.2006.12.3233. cited by applicant
Crouch. E.E., et al., “Ensembles of Endothelial and Mural Cells Promote Angiogenesis in Prenatal Human Brain,” Cell, 2022, vol. 185, pp. 3753-3769. cited b
Dao. L., et al., “Modeling Blood-Brain Barrier Formation and Cerebral Cavernous Malformations in Human PSC-Derived Organoids,” Cell Stem Cell, Jun. 6,
by applicant
Darrigrand. J. F., et al., “Acinar-Ductal Cell Rearrangement Drives Branching Morphogenesis of the Murine Pancreas in an IGF/PI3K-Dependent Manner,” D
pp. 326-338. cited by applicant
Duester G., “Retinoic Acid Synthesis and Signaling During Early Organogenesis,” Cell, Sep. 19, 2008, vol. 134, No. 6, pp. 921-931. cited by applicant
Duggal G., et al., “Alternative Routes to Induce Naive Pluripotency in Human Embryonic Stem Cells,” Stem Cells, 2015, vol. 33, pp. 2686-2698. cited by app
Edwards. N.A., et al., “Developmental Basis of Trachea-Esophageal Birth Defects,” Developmental Biology, 2021, vol. 477, pp. 85-97. cited by applicant
Eicher. A.K., et al., “Engineering Functional Human Gastrointestinal Organoid Tissues Using the Three Primary Germ Layers Separately Derived from Plurip
41 pages. cited by applicant
Elmentaite. R., et al., “Single-Cell Sequencing of Developing Human Gut Reveals Transcriptional Links to Childhood Crohn's Disease,” Developmental Cell,
by applicant
Eubelen. M., et al., “A Molecular Mechanism for Wnt Ligand-Specific Signaling,” Science, Jul. 19, 2018, vol. 361, 14 pages. cited by applicant
Eze. U.C., et al., “Single-cell atlas of early human brain development highlights heterogeneity of human neuroepithelial cells and early radial glia,” Nature Ne
584-594. cited by applicant
Fan. X., et al., “Single-Cell Transcriptome Analysis Reveals Cell Lineage Specification in Temporal-Spatial Patterns in Human Cortical Development,” Scien
pages. cited by applicant
Feng. Y., et al., “Identification of Rare Heterozygous Missense Mutations in FANCA in Esophageal Atresia Patients Using Next Generation Sequencing,” Gen
cited by applicant
Gehart. H., et al., “Identification of Enteroendocrine Regulators by Real-Time Single-Cell Differentiation Mapping,” Cell, 2019, vol. 176, pp. 1158-1173. cite
Geudens. I., et al., “Artery-Vein Specification in the Zebrafish Trunk is Pre-Patterned by Heterogeneous Notch Activity and Balanced by Flow-Mediated Fine-
vol. 146, No. 16, 26 pages. cited by applicant
Gieseck R. L., et al., “Maturation of Induced Pluripotent Stem Cell Derived Hepatocytes by 3D-Culture,” PLoS One, vol. 9, No. 1, Jan. 22, 2014, vol. 9, No. 1
10.1371/journal.pone.0086372. cited by applicant
Goodwin. K., et al., “Branching Morphogenesis,” Development, 2020, vol. 147, 6 Pages. cited by applicant
Grenier K., et al., “Three-Dimensional Modeling of Human Neurodegeneration: Brain Organoids Coming of Age,” Molecular Psychiatry, 2020, vol. 25, pp. 25
019-0500-7. cited by applicant
Guo, Z., et al., “Human Vascularized Brain Organoids with Blood-Brain-Barrier Formation for Study of Brain Vascular Disorders,” Center for Stem Cell Orga
Division of Developmental Biology Cincinnati Children's Hospital Medical Center, Frontiers in Stem Cell Organoid Medicine Symposium, Mar. 24, 2022. cite
Gupta A. K., et al., “An Efficient Method to Generate Kidney Organoids at the Air-Liquid Interface,” Journal of Biological Methods, 2021, vol. 8, No. 2, 11 pa
Hagey. D. W., et al., “SOX2 Regulates Common and Specific Stem Cell Features in the CNS and Endoderm-Derived Organs,” PLOS Genetics, 2018, vol. 14,
applicant
Han. X., et al., “Construction of a Human Cell Landscape at Single-Cell Level,” Nature, 2020, vol. 581, pp. 303-309. cited by applicant
Hancili S., et al., “A Novel NEUROG3 Mutation in Neonatal Diabetes Associated with a Neuro-Intestinal Syndrome,” Pediatric Diabetes, 2018, vol. 19, pp. 3
Hao Y., et al., “Dictionary Learning for Integrative, Multimodal and Scalable Single-Cell Analysis,” Nature Biotechnology, 2024, vol. 42, pp. 293-304. cited b
Harley J. B., et al., “Transcription Factors Operate Across Disease Loci, with EBNA2 Implicated in Autoimmunity,” Nature Genetics, 2018, vol. 50, No. 5, pp
He S. et al., “Single-Cell Transcriptome Profiling of an Adult Human Cell Atlas of 15 Major Organs,” Genome Biology, 2020, vol. 21, No. 294, 34 pages. cite
Herring C. A., et al., “Human Prefrontal Cortex Gene Regulatory Dynamics from Gestation to Adulthood at Single-Cell Resolution,” Cell, 2020, vol. 185, pp.
Honda A., et al., “Discrimination of Stem Cell Status After Subjecting Cynomolgus Monkey Pluripotent Stem Cells to Naive Conversion,” Scientific Reports,
DOI: 10.1038/srep45285. cited by applicant
Huang L., et al., “Commitment and Oncogene-Induced Plasticity of Human Stem Cell-Derived Pancreatic Acinar and Ductal Organoids,” Cell Stem Cell, 202
by applicant

Huang L., et al., "Revealing the Structure and Organization of Intercellular Tunneling Nanotubes (TNTs) by STORM Imaging", Nanoscale Advances, 2022, v
applicant
Jaeseo L., et al., "A 3D Alcoholic Liver Disease Model on a Chip", Integrative Biology, Jan. 1, 2016, vol. 8, No. 3, pp. 302-308. cited by applicant
Kamada T., et al., "Evidence-Based Clinical Practice Guidelines for Peptic Ulcer Disease 2020," Journal of Gastroenterology, 2021, vol. 56, pp. 303-322. cite
Kamal K., et al., "Bioengineering an Artificial Human Blood-Brain Barrier in Rodents," Bioengineering, Apr. 30, 2019, vol. 6, 14 pages, doi: 10.3390/bioengi
applicant
Kasagi Y., et al., "The Esophageal Organoid System Reveals Functional Interplay Between Notch and Cytokines in Reactive Epithelial Changes," Cellular an
Hepatology, 2018, vol. 5(3), pp. 333-352. cited by applicant
Katsura H., et al., "Human Lung Stem Cell-Based Alveolospheres Provide Insights into SARS-CoV-2 Mediated Interferon Responses and Pneumocyte Dysfu
doi: 10.1016/j.stem.2020.10.005. cited by applicant
Kechele D.O., et al., "Recent Advances in Deriving Human Endodermal Tissues from Pluripotent Stem Cells," Current Opinion in Cell Biology, 2019, vol. 6
Kehoe D. E et al., "Scalable Stirred-Suspension Bioreactor Culture of Human Pluripotent Stem Cells," Tissue Engineering, Jan. 1, 2010, vol. 16, No. 2, pp. 4
10.1089/ten.tea.2009.0454. cited by applicant
Kendig K.I., et al., "Sentieon DNASeq Variant Calling Workflow Demonstrates Strong Computational Performance and Accuracy," Frontiers in Genetics, 201
applicant
Kishimoto K., et al., "Bidirectional Wnt Signaling Between Endoderm and Mesoderm Confers Tracheal Identity in Mouse and Human Cells," Nature Commu
applicant
Kishimoto K., et al., "Directed Differentiation of Human Pluripotent Stem Cells into Diverse Organ-Specific Mesenchyme of the Digestive and Respiratory S
vol. 17, pp. 2699-2719. cited by applicant
Kishimoto. K., et al., "Mammalian tracheal development and reconstruction: insights from in vivo and in vitro studies," Development, 2021, vol. 148, No. 13,
Kitazawa T., et al., "Regulation of Gastrointestinal Motility by Motilin and Ghrelin in Vertebrates," Frontiers in Endocrinology (Lausanne), 2019, vol. 10, 17
Krefft O., et al., "Generation of Standardized and Reproducible Forebrain-type Cerebral Organoids from Human Induced Pluripotent Stem Cells," Journal of
2018, vol. 131, 8 pages, doi: 10.3791/56768, Retrieved from the Internet URL: https://www.ncbi.nlm.nih.gov/pmc/articles/PMC5908685/. cited by applicant
Leeman K.T., et al., "Mesenchymal Stem Cells Increase Alveolar Differentiation in Lung Progenitor Organoid Cultures," Scientific reports, Apr. 23, 2019, vo
applicant
Li R., et al., "Myofibroblast Contraction is Essential for Generating and Regenerating the Gas-Exchange Surface," Journal of Clinical Investigation, 2020, vo
applicant
Llurfrio E.M., et al., "Sorting Cells Alters Their Redox State and Cellular Metabolome," Redox Biology, 2018, vol. 16, pp. 381-387. cited by applicant
Loomans N., et al., "Expansion of Adult Human Pancreatic Tissue Yields Organoids Harboring Progenitor Cells with Endocrine Differentiation Potential," Ste
pp. 712-724. cited by applicant
Lopez-Ramirez. M.A., et al., "Astrocytes Propel Neurovascular Dysfunction During Cerebral Cavernous Malformation Lesion Formation," Journal of Clinica
pages. cited by applicant
Lyu H., et al., "The Role of Bone-Derived Exosomes in Regulating Skeletal Metabolism and Extraosseous Diseases," Frontiers in Cell and Developmental Bi
by applicant
Matthis A.L., et al., "Deficient Active Transport Activity in Healing Mucosa After Mild Gastric Epithelial Damage," Digestive Diseases and Sciences, 2020, v
by applicant
Maxwell K.G., et al., "Gene-edited Human Stem Cell-Derived ß Cells from a Patient with Monogenic Diabetes Reverse Preexisting Diabetes in Mice," Scien
vol. 12, No. 540, 23 pages. cited by applicant
Mayran A., et al., "Pioneer Transcription Factors Shape the Epigenetic Landscape," Journal of Biological Chemistry, 2018, vol. 293, pp. 13795-13804. cited b
Mead B.E., et al., "All Models are Wrong, but Some Organoids May be Useful," Genome Biology, 2019, vol. 20, 3 pages. cited by applicant
Meng G., et al., "Optimizing Human Induced Pluripotent Stem Cell Expansion in Stirred-Suspension Culture," Stem Cells and Development, Dec. 15, 2017, v
DOI: 10.1089/scd.2017.0090, Retrieved from the Internet URL:
https://www.liebertpub.com/doi/pdf/10.1089/scd.2017.0090casatoken=4jmtPMYfDEcAAAAA:3nd2OwOrb6Kyltkq641ZOmaNdxD4fHqdAI8it6DjaU7EuxX
cited by applicant
Mitchell M.J., et al., "Engineering Precision Nanoparticles for Drug Delivery," Nature Reviews Drug Discovery, 2021, vol. 20, pp. 101-124. cited by applican
Mulvihill E.E., "Regulation of Intestinal Lipid and Lipoprotein Metabolism by the Proglucagon-Derived Peptides Glucagon Like Peptide 1 and Glucagon like
Lipidology, 2018, vol. 29, No. 2, pp. 95-103. cited by applicant
Munera. J.O., "Development of functional resident macrophages in human pluripotent stem cell-derived colonic organoids and human fetal colon," Cell Stem
1451.e9. cited by applicant
Neuschulz A., et al., "A Single-Cell RNA Labeling Strategy for Measuring Stress Response Upon Tissue Dissociation," Molecular Systems Biology, 2023, vo
Ng W.H., et al., "Recapitulate Human Cardio-pulmonary Co-development Using Simultaneous Multilineage Differentiation of Pluripotent Stem Cells," bioR
by applicant
Oberg H-H., et al., "Differential Expression of CD126 and CD130 Mediates Different STAT-3 Phosphorylation in CD4 + CD25- and CD25high Regulatory T
Immunology, 2006, vol. 18, No. 4, pp. 555-563. cited by applicant
Ouelette J., et al., "Vascular Contributions to 16p11.2 Deletion Autism Syndrome Modeled in Mice," Nature Neuroscience, 2020, vol. 23, pp. 1090-1101. cite
Oz-Levi D., et al., "Noncoding Deletions Reveal a Gene that is Critical for Intestinal Function," Nature, 2019, vol. 571, pp. 107-111. cited by applicant
Pera M. F., et al., The Exploration of Pluripotency Space: Charting Cell State Transitions in Peri-Implantation Development, Cell Stem Cell, Nov. 4, 2021, vo
by applicant
Pfister G., et al., "An Evaluation of Sorter Induced Cell Stress (SICS) on Peripheral Blood Mononuclear Cells (PBMCs) After Different Sort Conditions—Ar
Journal of Immunological Methods, 2020, vol. 487, 7 pages. cited by applicant
Pode-Shakked N., et al., "RAAS-Deficient Organoids Indicate Delayed Angiogenesis as a Possible Cause for Autosomal Recessive Renal Tubular Dysgenesis
vol. 14. cited by applicant
Pradhan. S., et al., "Tissue Responses to Shiga Toxin in Human Intestinal Organoids," Cellular and Molecular Gastroenterology Hepatology, 2020, vol. 10, pp
Protze. S.I., et al., "Human Pluripotent Stem Cell-Derived Cardiovascular Cells: From Developmental Biology to Therapeutic Applications," Cell Stem Cell,
by applicant
Ramaiahgari S. C., et al., "A 3D in Vitro Model of Differentiated HepG2 Cell Spheroids with Improved Liver-Like Properties for Repeated Dose High-Throu
of Toxicology, Mar. 6, 2014, pp. 1083-1095, DOI: 10.1007/s00204-014-1215-9. cited by applicant
Raouf Z., et al., "Colitis-Induced Small Intestinal Hypomotility Is Dependent on Enteroendocrine Cell Loss in Mice," Cellular and Molecular Gastroenterolo
No. 1, pp. 53-70, DOI: 10.1016/j.jcmgh.2024.02.017. cited by applicant
Ren. H., et al., "Identifying multicellular spatiotemporal organization of cells with SpaceFlow," Nature Communications, 2022, vol. 13, Article 4076, 14 page
Romayor I., et al., "A Comparative Study of Cell Culture Conditions During Conversion from Primed to Naive Human Pluripotent Stem Cells," Biomedicine
cited by applicant
Rutherford. D., et al., "Therapeutic Potential of Human Intestinal Organoids in Tissue Repair Approaches in Inflammatory Bowel Diseases," Inflammatory B
1488-1498. cited by applicant
Ryu. S., et al., "Enhancing the Fitness of Embryoid Bodies and Organoids by Chemical Cytoprotection," bioRxiv preprint, Mar. 23, 2022, 40 pages. cited by a

Selvaggi. G., et al., “Overview of intestinal and multivisceral transplantation,” UpToDate, Jul. 3, 2024, https://www. uptodate.com/contents/overview-of-intes
transplantation/print, accessed Sep. 11, 2024, 3 pages. cited by applicant
Shafa M., et al., “Expansion and Long-Term Maintenance of Induced Pluripotent Stem Cells in Stirred Suspension Bioreactors”, Journal of Tissue Engineering
Jun. 1, 2012, vol. 6, No. 6, pp. 462-472, DOI: 10.1002/term.450. cited by applicant
Shi M., et al., “Directed differentiation of ureteric bud and collecting duct organoids from human pluripotent stem cells,” Nature Protocols, 2023, vol. 18, pp.
Shi. M., et al., “Human ureteric bud organoids recapitulate branching morphogenesis and differentiate into functional collecting duct cell types,” Nature Biote
261. cited by applicant
Simunovic M., et al., “Embryoids, Organoids and Gastruloids: New Approaches to Understanding Embryogenesis,” Development, 2017, vol. 144, pp. 976-98
by applicant
Song L., “Modelling 3-D Brain-like Tissues Using Human Stem Cell-Derived Vascular Spheroids, Cortical Spheroids and Microglia-like Cells,” Florida State
pages, Retrieved from Internet URL: https://www.proquest.com/docview/2124413137?pq-origsite=gscholar&fromopenview=true. cited by applicant
Stirparo. G.G., et al., “Integrated analysis of single-cell embryo data yields a unified transcriptome signature for the human pre-implantation epiblast,” Develo
pages. cited by applicant
Stresser. D., et al., “Validation of Pooled Cryopreserved Human Hepatocytes as a Model for Metabolism Studies,” ResearchGate, Jan. 1, 2004, Retrieved from
https://www.researchgate.net/publication/268359224, accessed Jun. 28, 2024. cited by applicant
Sweeney. M., et al., “It Takes Two: Endothelial-Perivascular Cell Cross-Talk in Vascular Development and Disease,” Cell Reports, 2018, vol. 24, pp. 2705-27
Sweeney, M.D. et al., “Blood-brain barrier breakdown in Alzheimer disease and other neurodegenerative disorders,” Nature Reviews Neurology, 2018, 14, pp
Takahashi J., et al., Suspension Culture in a Rotating Bioreactor for Efficient Generation of Human Intestinal Organoids, Cell Reports Methods, Nov. 1, 2022,
101-110 10.1016/j.crmeth.2022.100337, Retrieved from the Internet URL: https://www.ncbi.nlm.nih.gov/pmc/articles/PMC9701612/pdf/main.pdf. cited by ap
Takahashi, Y. et al., “Organoid-derived intestinal epithelial cells are a suitable model for preclinical toxicology and pharmacokinetic studies,” iScience, 2022,
Takebe, T. et al., “Organoids by design,” Science, 2019, vol. 364, pp. 956-959. cited by applicant
The Tabula Muris Consortium, “Single-cell transcriptomics of 20 mouse organs creates a Tabula Muris,” Nature, 2018, 562, pp. 367-372. cited by applicant
Tian A., et al., “Studying Human Neurodevelopment and Diseases Using 3D Brain Organoids,” The Journal of Neuroscience, Feb. 5, 2020, vol. 40, No. 6, pp.
Internet URL: https://www.ncbi.nlm.nih.gov/pmc/articles/PMC7002141/. cited by applicant
Trujillo, C.A. et al., “Complex Oscillatory Waves Emerging from Cortical Organoids Model Early Human Brain Network Development,” Cell Stem Cell, 201
applicant
Uzquiano, A. et al., “Proper acquisition of cell class identity in organoids allows definition of fate specification programs of the human cerebral cortex,” Cell,
cited by applicant
Valdoz, J.C. et al., “Soluble extracellular matrix promotes organotypic formation in lung alveolar model,” Biomaterials, 2022, 283, pp. 121464. cited by applic
Van Der Meijden, P.E.J et al., “Platelet biology and functions: new concepts and clinical perspectives,” Nature Reviews Cardiology, 2018, 14, pp. 14 pages. ci
Vazquez-Armendariz, A.I. et al., “From Clones to Buds and Branches: The Use of Lung Organoids to Model Branching Morphogenesis Ex Vivo,” Frontiers in
Biology, 2021, 9, pp. 631579. cited by applicant
Velasco, S. et al., “Individual brain organoids reproducibly form cell diversity of the human cerebral cortex,” Nature, 2019, 1317, 570, pp. 523-527. cited by a
Venema, W.T.C.U. et al., “Gut mucosa dissociation protocols influence cell type proportions and single-cell gene expression levels,” Scientific Reports, 2022,
Vincent C. C., et al., “Scalable GMP Compliant Suspension Culture System for Human ES cells,” Stem Cell Research, May 1, 2012, vol. 8, No. 3, DOI: 10.10
402. cited by applicant
Wang, C. et al., “A critical role of RUNX1 in governing megakaryocyte-primed hematopoietic stem cell differentiation,” Blood Advances, 2023, 7, pp. 2590-2
Wang, X. et al., “Genome-wide analysis of PDX1 target genes in human pancreatic progenitors,” Molecular Metabolism, 2018, 9, pp. 57-68. cited by applican
Watanabe, S. et al., “Transplantation of intestinal organoids into a mouse model of colitis,” Nature Protocols, 2022, 17, pp. 649-671. cited by applicant
Weijts, B. et al., “Blood flow-induced Notch activation and endothelial migration enable vascular remodeling in zebrafish embryos,” Nature Communications,
by applicant
Wiedenmann, S. et al., “Single-cell-resolved differentiation of human induced pluripotent stem cells into pancreatic duct-like organoids on a microwell chip,”
2021, 5, pp. 897-913. cited by applicant
Worsdorfer, P. et al., “Generation of complex human organoid models including vascular networks by incorporation of mesodermal progenitor cells,” Scientifi
cited by applicant
Wu, H. et al., “Progressive Pulmonary Fibrosis Is Caused by Elevated Mechanical Tension on Alveolar Stem Cells,” Cell, 2020, 180, pp. 107-121.e117. cited b
Wu, L. et al., “Filaggrin and tight junction proteins are crucial for IL-13-mediated esophageal barrier dysfunction,” American Journal of Physiology-Gastroint
2018, ajpgi.00404.2017, pp. 39 pages. cited by applicant
Wynne, K. et al., “Diabetes of the exocrine pancreas,” Journal of Gastroenterology and Hepatology, 2019, 34, pp. 346-354. cited by applicant
Xu T.Y., et al., “HiFi-Slide Spatial RNA Sequencing,” Protocol.io, 2023, pp. 12 pages. cited by applicant
Yu I., et al., “A Novel 96-Well Plate Cell Culture Assay for Lineage-Specific Hematopoietic Cell Toxicity Screening,” Stemcell Technologies, Retreived from
https://www.stemcell.com/media/files/poster/SP00174-A_Novel_96-well_Plate_Cell_Culture_Assay_for_Lineage-Specific_Hematopoietic_Cell_Toxicity_Sc
Yu, Y. et al., “Research progress of Hedgehog signaling pathway in liver fibrosis,” Chinese Journal of Anatomy, 2019, 42, No. 6, pp. Machine Translated Abst
Zepp, J.A. et al., “Genomic, epigenomic, and biophysical cues controlling the emergence of the lung alveolus,” Science, 2021, 371, pp. doi:10.1126/science.a
Zhang T., et al., “The Role of Glycosphingolipids in Immune Cell Functions,” Frontiers in Immunology, Jan. 29, 2019, vol. 10, 22 pages. cited by applicant
Basak O., et al., “Induced Quiescence of Lgr5+ Stem Cells in Intestinal Organoids Enables Differentiation of Hormone-Producing Enteroendocrine Cells”, Ce
20, No. 2, pp. 177-190. cited by applicant
Breslin S., et al., “Receptor Tyrosine Kinase Targeting in Multicellular Spheroids,” Methods and Protocols, Methods in Molecular Biology, Jan. 2015, vol. 12
applicant
Dana-Farber Cancer Institute, “Clinical Trials for Relapsed Cancer,” 2025, [retrieved on Mar. 28, 2025], 4 pages, Retrieved from the Internet URL:
https://www.danafarberbostonchildrens.org/our-services/innovative-approaches/clinical-trials-relapsed-cancer. cited by applicant
Granato A., et al., “Bilirubin Inhibits Bile Acid Induced Apoptosis in Rat Hepatocytes,” Gut, Dec. 2003, vol. 52, No. 12, pp. 1774-1778, DOI: 10.1136/gut.52.
Harrison S.P., et al., “Scalable Production of Tissue-like Vascularized Liver Organoids from Human PSCs,” bioRxiv, Dec. 2, 2020, 31 pages, DOI: 10.1101/20
by applicant
International Search Report and Written Opinion for Application No. PCT/US2023/067716, mailed Nov. 29, 2023, 12pages. cited by applicant
Macea M.M.I., et al., “Quantitative Study of Brunner's Galnds in the Human Duodenal Submucosa,” Int. J. Morphol, Jan. 1, 2006, vol. 24, No. 1, pp. 7-12, Re
https://www.scielo.cl/pdf/ijmorphol/v24n1/art02.pdf. cited by applicant
Menard D., et al., “Explant Culture of Human Fetal Small Intestine,” Gastroenterology, Mar. 1985, vol. 88, No. 3, pp. 691-700. cited by applicant
Ni X., et al., “A Region-Resolved Mucosa Proteome of the Human Stomach,” Nature Communications, 2019, vol. 10, pp. 1-11. cited by applicant
Nostro M. C., et al., Stage-Specific Signaling Through TGF Family Members and WNT Regulates patterning and Pancreatic Specification of Human Pluripot
Mar. 2011, vol. 138, No. 5, pp. 861-871. cited by applicant
Schmidt S., et al., “A Blood Vessel Organoid Model Recapitulating Aspects of Vasculogenesis, Angiogenesis and Vessel Wall Maturation” Organoids, Apr. 28,
DOI: 10.3390/organoids1010005. cited by applicant
Ungrin M.D., et al., “Rational Bioprocess Design for Human Pluripotent Stem Cell Expansion and Endoderm Differentiation Based on Cellular Dynamics,” B
Bioengineering, Apr. 2012, vol. 109, No. 4, pp. 853-866. cited by applicant

Willenbring H., et al., “Transplantable Liver Organoids Made From Only Three Ingredients,” Cell Stem Cell, Aug. 1, 2013, vol. 13, No. 2, 4 pages, DOI: 10.1
applicant
Wong H. R., et al., “Biomarkers for Estimating Risk of Hospital Mortality and Long-Term Quality of Life Morbidity after Surviving Pediatric Septic Shock: A
LAPSE Investigation” Pediatric Critical Care Medicine, Jan. 1, 2021, vol. 22, No. 1, pp. 8-15. cited by applicant
Xinaris C., et al., “In Vivo Maturation of Functional Renal Organoids Formed from Embryonic Cell Suspensions,” Journal of the American Society of Nephro
11, pp. 1857-1868. cited by applicant

Primary Examiner: Valenrod; Yevgeny

Attorney, Agent or Firm: Norton Rose Fulbright US LLP

Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS (1) This application claims priority to and benefit of U.S. Provisional Patent Application No. 62/471,371, filed Nov. 4, 2016, and 62/517,414, filed Jun. 9, 2016, the contents of each are incorporated by reference in their entirety for all purposes.

BACKGROUND
(1) PNALD is the most devastating complication of long-term PN that occurs in the majority of children receiving PN (Kumar and Teckman, 2015; Orso et al., 2016; Zambrano et al., 2004). Because its progression is typically insidious and its long-term consequences are generally underappreciated, PNALD is often recognized too late, when liver injury is irreversible. It is histologically characterized by intrahepatic cholestasis but can progress to fibrosis and cirrhotic liver failure with continued exposure to PN. There are no established ameliorative strategies for PNALD and PNALD is the leading indication for liver transplantation in infants (El Kasmi et al., 2013).
BRIEF SUMMARY
(2) Disclosed are methods of treating or reducing the occurrence of a steatohepatitis disorder. The disorder may include, for example, NASH, parenteral nutrition associated liver disease (PNALD), or genetic forms of liver disease. The method may comprise the step of administering a composition comprising obeticholic acid to an individual in need thereof.

Description

BRIEF DESCRIPTION OF THE DRAWING
(1) Those of skill in the art will understand that the drawings, described below, are for illustrative purposes only. The drawings are not intended to limit the scope of the present teachings in any way.
(2) FIG. **1**. Obeticholic acid treatment rescue PNALD like pathology in vitro A. Bright-field image of and survival rate (%) of HLO treated 0 (black line), 400 μM Intralipid (Int; light blue line), 400 μM Int with OCA (gray line), and 400 μM Int with 40 ng/ml FGF19 (pink line). Arrowheads indicate the dead organoids with clumped morphology in the images. Survival rate (%) of 0 (black bar), 800 μM OA (blue bar), 800 μM OA+FGF19 (pink bar) at the same time point. The survival of 800 μM OA HLO was significantly improved by FGF19 addition. B. Live-cell imaging of ROS (green) and nuclear (blue). OCA (gray bar) and FGF19 (pink bar) inhibit the ROS production caused by 400 μM Int (light blue bar).C. Intralipid. The stiffness of 0 (gray bar), 800 μM OA (blue bar), 800 μM OA+FGF19 (pink bar) treated HLO was assessed by AFM. The stiffness of 800 μM OA HLO was significantly softened by FGF19 addition.
(3) FIG. **2**. Lipids accumulation in the organoid in the presence and absence of 400 μM Intralipid.
(4) FIG. **3**. Bright-field image of liver organoids in the culture of non-treatment, 800 μM of OA alone, and 800 μM of OA with 40 ng/m**1** FGF19.
DETAILED DESCRIPTION OF THE INVENTION
(5) Unless otherwise defined, all technical and scientific terms used herein have the same meaning as commonly understood by one of ordinary skill in the art. In case of conflict, the present document, including definitions, will control. Preferred methods and materials are described below, although methods and materials similar or equivalent to those described herein can be used in practice or testing of the present invention. All publications, patent applications, patents and other references mentioned herein are incorporated by reference in their entirety. The materials, methods, and examples disclosed herein are illustrative only and not intended to be limiting.
(6) The terms and expressions used herein have the ordinary meaning as is accorded to such terms and expressions with respect to their corresponding respective areas of inquiry and study except where specific meanings have otherwise been set forth herein.
(7) As used herein and in the appended claims, the singular forms “a,” “and,” and “the” include plural referents unless the context clearly dictates otherwise. Thus, for example, reference to “a method” includes a plurality of such methods and reference to “a dose” includes reference to one or more doses and equivalents thereof known to those skilled in the art, and so forth.
(8) The term “about” or “approximately” means within an acceptable error range for the particular value as determined by one of ordinary skill in the art, which will depend in part on how the value is measured or determined, e.g., the limitations of the measurement system. For example, “about” can mean within 1 or more than 1 standard deviation, per the practice in the art. Alternatively, “about” can mean a range of up to 20%, or up to 10%, or up to 5%, or up to 1% of a given value. Alternatively, particularly with respect to biological systems or processes, the term can mean within an order of magnitude, preferably within 5-fold, and more preferably within 2-fold, of a value. Where particular values are described in the application and claims, unless otherwise stated the term “about” meaning within an acceptable error range for the particular value should be assumed.
(9) The terms “individual,” “host,” “subject,” and “patient” are used interchangeably to refer to an animal that is the object of treatment, observation and/or experiment. Generally, the term refers to a human patient, but the methods and compositions may be equally applicable to non-human subjects such as other mammals. In some embodiments, the terms refer to humans. In further embodiments, the terms refer to children.
(10) “Therapeutically effective amount” relates to the amount or dose of an active compound or composition described herein that will lead to one or more therapeutic effect, in particular, a desired beneficial effect. A therapeutically effective amount of a substance can vary according to factors such as the disease state, age, sex, and weight of the subject, and the ability of the substance to elicit a desired response in the subject. Dosage regime may be adjusted to provide the optimum therapeutic response. For example, several divided doses may be administered daily or the dose may be proportionally reduced as indicated by the exigencies of the therapeutic situation.
(11) The term “parenteral nutrition,” or “intravenous feeding,” is a method of providing nutrition into an individual via intravenous administration. Depending on which vein is used, this procedure is often referred to as either total parenteral nutrition (TPN) or peripheral parenteral nutrition (PPN).
(12) The phrase “pharmaceutically acceptable,” as used in connection with compositions of the disclosure, refers to molecular entities and other ingredients of such compositions that are physiologically tolerable and do not typically produce untoward reactions when administered to a subject (e.g., human) In certain embodiments, as used herein, the term “pharmaceutically acceptable” means approved by a regulatory agency of a Federal or a state government or listed in the U.S. Pharmacopeia or other generally recognized pharmacopeia for use in mammals (e.g., humans).
(13) The terms “pharmaceutically acceptable salts” or “a pharmaceutically acceptable salt thereof” refer to salts prepared from pharmaceutically

acceptable, non-toxic acids or bases. Suitable pharmaceutically acceptable salts include metallic salts, e.g., salts of aluminum, zinc, alkali metal salts such as lithium, sodium, and potassium salts, alkaline earth metal salts such as calcium and magnesium salts; organic salts; salts of free acids and bases; inorganic salts, e.g., sulfate, hydrochloride, and hydrobromide; and other salts which are currently in widespread pharmaceutical use and are listed in sources well known to those of skill in the art, such as The Merck Index. Any suitable constituent can be selected to make a salt of an active drug discussed herein, provided that it is non-toxic and does not substantially interfere with the desired activity. In addition to salts, pharmaceutically acceptable precursors and derivatives of the compounds can be employed. Pharmaceutically acceptable amides, lower alkyl esters, and protected derivatives of the disclosed actives can also be suitable for use in the compositions and methods disclosed herein. A salt of a compound of this disclosure may be formed between an acid and a basic group of the compound, such as an amino functional group, or a base and an acidic group of the compound, such as a carboxyl functional group. According to another embodiment, the compound is a pharmaceutically acceptable acid addition salt. Acids commonly employed to form pharmaceutically acceptable salts include inorganic acids such as hydrogen bisulfide, hydrochloric acid, hydrobromic acid, hydroiodic acid, sulfuric acid and phosphoric acid, as well as organic acids such as para-toluenesulfonic acid, salicylic acid, tartaric acid, bitartaric acid, ascorbic acid, maleic acid, besylic acid, fumaric acid, gluconic acid, glucuronic acid, formic acid, glutamic acid, methanesulfonic acid, ethanesulfonic acid, benzenes ulfonic acid, lactic acid, oxalic acid, para-bromophenylsulfonic acid, carbonic acid, succinic acid, citric acid, benzoic acid and acetic acid, as well as related inorganic and organic acids. Such pharmaceutically acceptable salts thus include sulfate, pyrosulfate, bisulfate, sulfite, bisulfite, phosphate, monohydrogenphosphate, dihydrogenphosphate, metaphosphate, pyrophosphate, chloride, bromide, iodide, acetate, propionate, decanoate, caprylate, acrylate, formate, isobutyrate, caprate, heptanoate, propiolate, oxalate, malonate, succinate, suberate, sebacate, fumarate, maleate, butyne-1,4-dioate, hexyne-1,6-dioate, benzoate, chlorobenzoate, methylbenzoate, dinitrobenzoate, hydroxybenzoate, methoxybenzoate, phthalate, terephthalate, sulfonate, xylene sulfonate, phenylacetate, phenylpropionate, phenylbutyrate, citrate, lactate, β-hydroxybutyrate, glycolate, maleate, tartrate, methanesulfonate, propanesulfonate, naphthalene-1-sulfonate, naphthalene-2-sulfonate, mandelate and other salts. In one embodiment, pharmaceutically acceptable acid addition salts include those formed with mineral acids such as hydrochloric acid and hydrobromic acid, and especially those formed with organic acids such as maleic acid.
(14) The terms “treat,” “treating” or “treatment,” as used herein, refers to methods of alleviating, abating or ameliorating a disease or condition symptoms, preventing additional symptoms, ameliorating or preventing the underlying metabolic causes of symptoms, inhibiting the disease or condition, arresting the development of the disease or condition, relieving the disease or condition, causing regression of the disease or condition, relieving a condition caused by the disease or condition, or stopping the symptoms of the disease or condition either prophylactically and/or therapeutically.
(15) The term “carrier” applied to pharmaceutical compositions of the disclosure refers to a diluent, excipient, or vehicle with which an active compound is administered. Such pharmaceutical carriers can be sterile liquids, such as water, saline solutions, aqueous dextrose solutions, aqueous glycerol solutions, and oils, including those of petroleum, animal, vegetable, or synthetic origin, such as peanut oil, soybean oil, mineral oil, sesame oil and the like. Suitable pharmaceutical carriers are described in “Remington's Pharmaceutical Sciences” (any edition).
(16) The term “compound,” as used herein, is also intended to include any salts, solvates, or hydrates thereof.
(17) Applicant has discovered novel methods for the treatment of liver disease in an individual in need thereof, which may include administration of a disclosed compound to an individual in need thereof.
(18) In one aspect, a method of treating or reducing the occurrence of a steatohepatitis disorder is disclosed. The disorder may include, for example, NASH, parenteral nutrition associated liver disease (PNALD), or genetic forms of liver disease. The method may comprise the step of administering a composition comprising obeticholic acid to an individual in need thereof.
(19) In one aspect, the composition may be administered in an amount sufficient to reduce or prevent the occurrence of liver cell fibrosis.
(20) In one aspect, the composition may be administered prior to, following, or during administration of parenteral nutrition to said individual in need thereof.
(21) In one aspect, a composition is disclosed, wherein the composition may comprise total parenteral nutrition (TPN) composition or peripheral parenteral nutrition (PPN) composition; and obeticholic acid.
(22) In a further aspect, an article of manufacture is disclosed. The article of manufacture may comprise a container, a composition as described herein, the composition being in a dosage form, and instructions for administering the dosage form to an individual diagnosed or suspected of having or developing a liver disease as disclosed herein.
(23) Dosage
(24) In one aspect, an agent disclosed herein may be present in an amount of from about 0.5% to about 95%, or from about 1% to about 90%, or from about 2% to about 85%, or from about 3% to about 80%, or from about 4%, about 75%, or from about 5% to about 70%, or from about 6%, about 65%, or from about 7% to about 60%, or from about 8% to about 55%, or from about 9% to about 50%, or from about 10% to about 40%, by weight of the composition.
(25) The compositions may be administered in oral dosage forms such as tablets, capsules (each of which includes sustained release or timed release formulations), pills, powders, granules, elixirs, tinctures, suspensions, syrups, and emulsions. They may also be administered in intravenous (bolus or infusion), intraperitoneal, subcutaneous, intralesional, or intramuscular forms all utilizing dosage forms well known to those of ordinary skill in the pharmaceutical arts. The compositions may be administered by intranasal route via topical use of suitable intranasal vehicles, or via a transdermal route, for example using conventional transdermal skin patches. A dosage protocol for administration using a transdermal delivery system may be continuous rather than intermittent throughout the dosage regimen.
(26) In one aspect, the compounds may be administered at the rate of 100 μg to 1000 mg per day per kg of body weight. Orally, the compounds may be suitably administered at the rate of about 100, 150, 200, 250, 300, 350, 400, 450, or 500 μg to about 1, 5, 10, 25, 50, 75, 100, 200, 300, 400, 500, 600, 700, 800, 900, 1000 mg per day per kg of body weight. The required dose can be administered in one or more portions. For oral administration, suitable forms are, for example, tablets, gel, aerosols, pills, dragees, syrups, suspensions, emulsions, solutions, powders and granules; one method of administration includes using a suitable form containing from 1 mg to about 500 mg of active substance. In one aspect, administration may comprise using a suitable form containing from about 1, 2, 5, 10, 25, or 50 mg to about 100, 200, 300, 400, 500 mg of active substance.
(27) A dosage regimen will vary depending upon known factors such as the pharmacodynamic characteristics of the agents and their mode and route of administration; the species, age, sex, health, medical condition, and weight of the patient, the nature and extent of the symptoms, the kind of concurrent treatment, the frequency of treatment, the route of administration, the renal and hepatic function of the patient, and the desired effect. The effective amount of a drug required to prevent, counter, or arrest progression of a symptom or effect of a muscle contracture can be readily determined by an ordinarily skilled physician.
(28) The pharmaceutical compositions may include suitable dosage forms for oral, parenteral (including subcutaneous, intramuscular, intradermal and intravenous), transdermal, sublingual, bronchial or nasal administration. Thus, if a solid carrier is used, the preparation may be tableted, placed in a hard gelatin capsule in powder or pellet form, or in the form of a troche or lozenge. The solid carrier may contain conventional excipients such as binding agents, fillers, tableting lubricants, disintegrants, wetting agents and the like. The tablet may, if desired, be film coated by conventional techniques. Oral preparations include push-fit capsules made of gelatin, as well as soft, scaled capsules made of gelatin and a coating, such as glycerol or sorbitol. Push-fit capsules can contain active ingredients mixed with a filler or binders, such as lactose or starches, lubricants, such as talc or magnesium stearate, and, optionally, stabilizers. In soft capsules, the active compounds may be dissolved or suspended in suitable liquids, such as fatty oils, liquid, or liquid polyethylene glycol with or without stabilizers. If a liquid carrier is employed, the preparation may be in the form of a syrup, emulsion, soft gelatin capsule, sterile vehicle for injection, an aqueous or non-aqueous liquid suspension, or may be a dry product for

reconstitution with water or other suitable vehicle before use. Liquid preparations may contain conventional additives such as suspending agents, emulsifying agents, wetting agents, non-aqueous vehicle (including edible oils), preservatives, as well as flavoring and/or coloring agents. For parenteral administration, a vehicle normally will comprise sterile water, at least in large part, although saline solutions, glucose solutions and like may be utilized. Injectable suspensions also may be used, in which case conventional suspending agents may be employed. Conventional preservatives, buffering agents and the like also may be added to the parenteral dosage forms. For topical or nasal administration, penetrants or permeation agents that are appropriate to the particular barrier to be permeated are used in the formulation. Such penetrants are generally known in the art. The pharmaceutical compositions are prepared by conventional techniques appropriate to the desired preparation containing appropriate amounts of the active ingredient, that is, one or more of the disclosed active agents or a pharmaceutically acceptable salt thereof according to the invention.
(29) The dosage of an agent disclosed herein used to achieve a therapeutic effect will depend not only on such factors as the age, weight and sex of the patient and mode of administration, but also on the degree of inhibition desired and the potency of an agent disclosed herein for the particular disorder or disease concerned. It is also contemplated that the treatment and dosage of an agent disclosed herein may be administered in unit dosage form and that the unit dosage form would be adjusted accordingly by one skilled in the art to reflect the relative level of activity. The decision as to the particular dosage to be employed (and the number of times to be administered per day) is within the discretion of the physician, and may be varied by titration of the dosage to the particular circumstances of this invention to produce the desired therapeutic effect.
(30) Routes of Administration
(31) Any suitable route of administration can be employed for providing the patient with an effective dosage of the disclosed compositions. For example, oral, rectal, transdermal, parenteral (subcutaneous, intramuscular, intravenous), intrathecal, topical, inhalable, and like forms of administration can be employed. Suitable dosage forms include tablets, troches, dispersions, suspensions, solutions, capsules, patches, and the like. Administration of medicaments prepared from the compounds described herein can be by any suitable method capable of introducing the compounds into the bloodstream. In some embodiments, the formulations can contain a mixture of active compounds with pharmaceutically acceptable carriers or diluents known to those of skill in the art.
(32) The compositions can be prepared in any desired form, for example, tables, powders, capsules, injectables, suspensions, sachets, cachets, patches, solutions, elixirs, and aerosols. Carriers such as starches, sugars, microcrystalline cellulose, diluents, granulating agents, lubricants, binders, disintegrating agents, and the like can be used in oral solid preparations. In certain embodiments, the compositions are prepared as oral solid preparations (such as powders, capsules, and tablets). In certain embodiments, the compositions are prepared as oral liquid preparations. In some embodiments, the oral solid preparations are tablets. If desired, tablets can be coated by standard aqueous or non-aqueous techniques.
(33) In addition to the dosage forms set out above, one or more disclosed compounds may also be administered by sustained release, delayed release, or controlled release compositions and/or delivery devices.
(34) Pharmaceutical compositions suitable for oral administration can be provided as discrete units such as capsules, cachets, sachets, patches, injectables, tablets, and aerosol sprays, each containing predetermined amounts of the active ingredients, as powder or granules, or as a solution or a suspension in an aqueous liquid, a non-aqueous liquid, an oil-in-water emulsion, or a water-in-oil liquid emulsion. Such compositions can be prepared by any of the conventional methods of pharmacy, and may include the step of bringing into association the active ingredients with a carrier which constitutes one or more ingredients. In general, the compositions may be prepared by uniformly and intimately admixing the active ingredients with liquid carriers, finely divided solid carriers, or both, and then, optionally, shaping the product into the desired presentation.
(35) For example, a tablet can be prepared by compression or molding, optionally, with one or more additional ingredients. Compressed tablets can be prepared by compressing in a suitable machine the active ingredient in a free-flowing form such as powder or granules, optionally mixed with a binder, lubricant, inert diluent, surface active or dispersing agent. Molded tablets can be made by molding, in a suitable machine, a mixture of the powdered compound moistened with an inert liquid diluent.
(36) A composition or formulation may be administered to a subject continuously or periodically.
(37) The compositions or fractions thereof typically comprise suitable pharmaceutical diluents, excipients, vehicles, or carriers selected based on the intended form of administration, and consistent with conventional pharmaceutical practices. The carriers, vehicles etc. may be adapted to provide an additive, synergistically effective or therapeutically effective amount of the active compounds. Suitable pharmaceutical diluents, excipients, vehicles, and carriers are described in the standard text, Remington: The Science and Practice of Pharmacy, and in The United States Pharmacopeia: The National Formulary (USP 24 NF 19) published in 1999. By way of example, for oral administration in the form of a capsule or tablet, the active components can be combined with an oral, non-toxic pharmaceutically acceptable inert carrier such as lactose, starch, sucrose, methyl cellulose, magnesium stearate, glucose, calcium, sulfate, dicalcium phosphate, mannitol, sorbital, and the like. For oral administration in a liquid form, the agents may be combined with any oral, non-toxic, pharmaceutically acceptable inert carrier such as ethanol, glycerol, water, and the like. Suitable binders (e.g. gelatin, starch, corn sweeteners, natural sugars including glucose; natural and synthetic gums, and waxes), lubricants (e.g. sodium oleate, sodium stearate, magnesium stearate, sodium benzoate, sodium acetate, and sodium chloride), disintegrating agents (e.g. starch, methyl cellulose, agar, bentonite, and xanthan gum), flavoring agents, and coloring agents may also be combined in the compositions or components thereof.
(38) In one aspect, a pharmaceutical composition may have pH from about 7 to 10.
(39) Formulations for parenteral administration of a composition may include aqueous solutions, syrups, aqueous or oil suspensions and emulsions with edible oil such as cottonseed oil, coconut oil or peanut oil. Dispersing or suspending agents that can be used for aqueous suspensions include synthetic or natural gums, such as tragacanth, alginate, acacia, dextran, sodium carboxymethylcellulose, gelatin, methylcellulose, and polyvinylpyrrolidone.
(40) Compositions for parenteral administration may include sterile aqueous or non-aqueous solvents, such as water, isotonic saline, isotonic glucose solution, buffer solution, or other solvents conveniently used for parenteral administration of therapeutically active agents. A composition intended for parenteral administration may also include conventional additives such as stabilizers, buffers, or preservatives.
(41) In an embodiment, a solid form pharmaceutical composition is provided (e.g. tablets, capsules, powdered, or pulverized form) comprising one or more disclosed compounds or salt thereof.
(42) After pharmaceutical compositions have been prepared, they can be placed in an appropriate container and labeled for treatment of an indicated condition. For administration of a composition, such labeling would include amount, frequency, and method of administration.
(43) Kits
(44) Kits are also provided. In one aspect, a kit may comprise or consist essentially of agents or compositions described herein. The kit may be a package that houses a container which may contain a composition comprising an oxime or pharmaceutically acceptable salt thereof as disclosed herein, and also houses instructions for administering the agent or composition to a subject. In one aspect, a pharmaceutical pack or kit is provided comprising one or more containers filled with one or more composition as disclosed herein. Associated with such container(s) can be various written materials such as instructions for use, or a notice in the form prescribed by a governmental agency regulating the manufacture, use or sale of pharmaceuticals or biological products, which notice reflects approval by the agency of manufacture, use, or sale for human administration.
(45) As there may be advantages to mixing a component of a composition described herein and a pharmaceutically acceptable carrier, excipient or vehicle near the time of use, kits in which components of the compositions are packaged separately are disclosed. For example, the kit can contain an active ingredient in a powdered or other dry form in, for example, a sterile vial or ampule and, in a separate container within the kit, a carrier, excipient, or vehicle, or a component of a carrier, excipient, or vehicle (in liquid or dry form). In one aspect, the kit can contain a component in a dry form, typically as a powder, often in a lyophilized form in, for example, a sterile vial or ampule and, in a separate container within the kit, a carrier,

excipient, or vehicle, or a component of a carrier, excipient, or vehicle. Alternatively, the kit may contain a component in the form of a concentrated solution that is diluted prior to administration. Any of the components described herein, any of the carriers, excipients or vehicles described herein, and any combination of components and carriers, excipients or vehicles can be included in a kit.
(46) Optionally, a kit may also contain instructions for preparation or use (e.g., written instructions printed on the outer container or on a leaflet placed therein) and one or more devices to aid the preparation of the solution and/or its administration to a patient (e.g., one or a plurality of syringes, needles, filters, tape, tubing, alcohol swabs, or the like). Compositions which are more concentrated than those administered to a subject can be prepared. Accordingly, such compositions can be included in the kits with, optionally, suitable materials (e.g., water, saline, or other physiologically acceptable solutions) for dilution. Instructions included with the kit can include, where appropriate, instructions for dilution.
(47) In other embodiments, the kits can include pre-mixed compositions and instructions for solubilizing any precipitate that may have formed during shipping or storage. Kits containing solutions of one or more of the aforementioned active agents, or pharmaceutically acceptable salts thereof, and one or more carriers, excipients or vehicles may also contain any of the materials mentioned above (e.g., any device to aid in preparing the composition for administration or in the administration per se). The instructions in these kits may describe suitable indications (e.g., a description of patients amenable to treatment) and instructions for administering the solution to a patient.
EXAMPLES
(48) Repurposing OCA for PNALD Therapy
(49) In this study, Applicant found that OCA and FGF19 inhibited ROS production caused by Intralipid exposure in HLO, leading to an improvement in survival rate of HLO. Moreover, the stiffness of OA treated HLO, measured by AFM, was decreased by FGF19 exposure. These results provide direct experimental evidence that OCA and FGF19 play a preventive role in the pathogenesis of PNALD organoid model and that the suppression of ROS production and fibrosis by FXR agonism is the likely mechanism of protection against PNALD. Our study thus highlighted the potential repurposing of OCA for the treatment of patients with PNALD.
(50) Obeticholic Acid Treatment Improved PNALD-Like Pathology of HLO
(51) PNALD is an iatrogenic fatty liver disease which occurs frequently in infants provided with parental nutrition (PN) solutions employing a lipid emulsion (Intralipid) that provide life-sustaining calories in the setting of inadequate absorption of enteral nutrients (Kumar and Teckman, 2015; Orso et al., 2016). PNALD has a histological evidence of steatosis, cell death (apoptosis), and fibrosis (Nandivada et al., 2013). To recapitulate PNALD pathology in HLO, we reconstituted components of Intralipid and placed it directly into HLO culture. Similar to OA treatment, significant lipid accumulation was confirmed in 400 μM Intralipid treated HLO reflecting a steatosis-like condition (FIG. **2**). In parallel, we established an organoid survival assessment algorithm by morphological analysis backed by viability assessment (FIGS. **3** and **2**, panel A.) and demonstrated that HLO survival of 400 μM Intralipid treated group was severely compromised by 37% after 12 days of culture (FIG. **2**, panel A).
(52) Because there are no established ameliorative strategies, and PNALD is the leading indication for liver transplantation in infants (El Kasmi et al., 2013), we further investigated preventive and therapeutic effects of FXR agonist obeticholic acid (OCA) and FGF19 hormone that are activated by OCA, which have shown great potential in nonalcoholic steatohepatitis treatment (Dash et al., 2017; Kumar and Teckman, 2015; Neuschwander-Tetri et al., 2015). To implicate the therapeutic significance of OCA and FGF19 for PNALD like pathology in HLO, OCA and FGF19 were added to 400 μM Intralipid exposed HLO. By addition of OCA and FGF19 into PNALD-HLO, the number of surviving organoids remarkably improved around 83 and 78% in total, respectively. Mechanistically, the accumulation of reactive oxygen species (ROS) is strongly associated with the fatty liver related hepatocyte death (Nandivada et al., 2013; Tsedensodnom and Sadler, 2013). We thus wondered whether OCA and FGF19 would inhibit the production of ROS so we sought to confirm that theory with live-cell imaging with CellROX dye. Consistent with HLO survival improvements, ROS production was 60% in 400 μM Intralipid treated HLO but reduced to 20 and 30% by OCA and FGF19 treatments, respectively (FIG. **7**, B), suggesting that OCA and FGF19 might have therapeutic potential against PNALD through suppressing ROS production.
(53) To further determine whether OCA and FGF19 have an impact on fibrosis, HLO stiffness was assessed. Organoid damage induced by Intralipid is extremely significant and treated HLOs for 5 days are not able to be stabilized on a dish, which is a minimal requirement for assessing the rigidity by AFM. To circumvent this limitation, the stiffness was assessed at Dya3. These observations indicate that OCA and FGF19 reduce the progression of fibrosis, potentially alleviating PNALD phenotype in an organoid.
(54) Methods
(55) hPSCs Maintenance. The TkDA3 human iPSC clone used in this study was kindly provided by K. Eto and H. Nakauchi. Human iPSC lines were maintained as described previously (Takebe et al., 2015; Takebe et al., 2014). Undifferentiated hiPSCs were maintained on feeder-free conditions in mTeSR1 medium (StemCell technologies, Vancouver, Canada) on plates coated with Matrigel (Corning Inc., NY, USA) at 1/30 dilution at 37° C. in 5% CO.sub.2 with 95% air.
(56) Definitive endoderm induction. Human iPSCs into definitive endoderm was differentiated using previously described methods with slight modifications (Spence et al., 2011). In brief, colonies of human iPSCs were isolated in Accutase (Thermo Fisher Scientific Inc., MA, USA) and 150,000 cells/mL, were plated on Matrigel coated tissue culture plate (VWR Scientific Products, West Chester, PA). Medium was changed to RPMI 1640 medium (Life Technologies) containing 100 ng/mL Activin A (R&D Systems, MN, USA) and 50 ng/mL bone morphogenetic protein 4 (BMP4; R&D Systems) at day 1, 100 ng/mL Activin A and 0.2% fetal calf serum (FCS; Thermo Fisher Scientific Inc.) at day 2 and 100 ng/mL Activin A and 2% FCS at day 3. Day 4-6 cells were cultured in Advanced DMEM/F12 (Thermo Fisher Scientific Inc.) with B27 (Life Technologies) and N2 (Gibco, CA, USA) containing 500 ng/ml fibroblast growth factor (FGF4; R&D Systems) and 3 uM CHIR99021 (Stemgent, MA, USA). Cells were maintained at 37° C. in 5% CO.sub.2 with 95% air and the medium was replaced every day. Spheroids appeared on the plate at day 7 of differentiation.
(57) HLO induction. At day 7, spheroids and attached cells are gently pipetted to be delaminated from dishes. They were centrifuged at 800 rpm for 3 minutes, embedded in a 100% Matrigel drop on the dishes in Advanced DMEM/F12 with B27, N2 and 2 uM retinoic acid (RA; Sigma, MO, USA) after removing supernatant, and cultured for 4 days. After RA treatment, spheroids embedded in the Matrigel drop were cultured in Hepatocytes Culture Medium (HCM; Lonza, MD, USA) with 10 ng/mL hepatocyte growth factor (HGF; PeproTech, NJ, USA), 0.1 uM Dexamethasone (Dex; Sigma) and 20 ng/mL Oncostatin M (OSM; R&D Systems). Cultures for HLO induction were maintained at 37° C. in 5% CO.sub.2 with 95% air and the medium was replaced every 2-3 days. To analyze HLO (day 20-30), organoids were isolated from Matrigel by scratching and pipetting.
(58) Albumin, IL-6, and P3NP ELISA. To measure albumin secretion level of HLO, 200 μL of culture supernatant was collected from HLO embedded in Matrigel. For IL-6 and P3NP, 20-30 organoids were seeded and cultured on an ultra-low attachment multiwell plates **96** well plate (Corning). To define the exact number of organoids in each well and lastly normalize the secreted level for IL-6 and P3NP by the number, the organoids were captured on The KEYENCE BZ-X710 Fluorescence Microscope. The culture supernatants were collected at 24 hrs (for albumin), 96 hrs (for IL-6) and 120 hrs (P3NP) time points after the culture and stored at −80° C. until use. The supernatant was centrifuged at 1,500 rpm for 3 min and to pellet debris, and the resulting supernatant was assayed with Human Albumin ELISA Quantitation Set (Bethyl Laboratories, Inc., TX, USA), Human IL-6 ELISA Kit (Biolegend, CA, USA), and Human N-terminal procollagen III propeptide, PIIINP ELISA Kit (My BioSource, CA, USA) according to the manufacturer's instructions. Significance testing was conducted by Student's t-test.
(59) Bile transport activity. Fluorescein diacetate was used for evaluating bile transport activity in organoids. 10 mg/mL fluorescein diacetate (Sigma) was added into HCM media cultured with HLO and allowed to sit for 5 minutes and captured using fluorescent microscopy BZ-X710 (Keyence, Osaka, Japan).
(60) Phagocyte, lipids, ROS live-cell imaging. After being cultured in an ultra-low attachment 6 multi-well plate, 5-10 HLO were picked up and seeded in a Microslide 8 Well Glass Bottom plate (Ibidi, WI, USA) and subjected to live-cell staining. The following antibodies were used: pHrodo®

Red S. aureus Bioparticles® Conjugate for Phagocyte activity (Thermo Fisher Scientific Inc.), BODIPY® 493/503 for lipids (Thermo Fisher Scientific Inc.), Di-8-ANEPPS for membrane (Thermo Fisher Scientific Inc.), and CellROX green reagent for ROS detection (Thermo Fisher Scientific Inc.). Nuclear staining was marked by NucBlue Live ReadyProbes Reagent (Thermo Fisher Scientific Inc.). HLO was visualized and scanned on a Nikon A1 Inverted Confocal Microscope (Japan) using 60× water immersion objectives. The final lipid droplet volume was calculated by IMARIS8 and normalized by each organoid size. Significance testing for lipid droplet volume and ROS production (%) was conducted by Student's t-test.
(61) HE staining and immunohistochemistry. HLO were isolated from Matrigel and fixed in 4% paraformaldehyde and embedded in paraffin. Sections were subjected to HE and immunohistochemical staining. The following primary antibodies were used: anti-alpha smooth muscle actin antibody (1:200 dilution; abcam, Cambridge, UK), Desmin antibody (Pre-diluted; Roche, Basel, Switzerland), and CD68 antibody (1:200 dilution; abcam).
(62) Flow cytometry. HLO were isolated from 10 Matrigel droplets and washed by 1×PBS. HLO were dissociated to single cells by the treatment of Trypsin-EDTA (0.05%), phenol red (Gibco) for 10 min After PBS wash, the single cells were subjected to flow cytometry with BV421-conjugated Epcam antibody (BioLegend), PE-conjugated CD166 antibody (eBioscience, CA, USA), and PE/Cy7-CD68 (eBioscience). DNA was measured by propidium iodide staining.
(63) LPS and FFA exposure and OCA and FGF19 treatment. 20-30 hLOHLO, which had been isolated from Matrigel and washed by 1× PBS, were divided into each condition and cultured on an ultra-low attachment 6 multi-well plates (Corning). HLO were cultured with LPS (Sigma), OA (Sigma), LA (Sigma), SA (Sigma), or PA (Sigma) and collected at day 1 and 3 (for LPS HLO) and at day 3 and 5 (for OA) after the culture. To test the inhibitory effect of OCA (INT-747, MedChem Express, NJ, USA) and human FGF19 recombinant (Sigma) on HLO, 20-30 HLO were cultured in HCM media in the presence or absence of oleic acid (800 μM), and 1 μM OCA and 40 ng/ml FGF19 were added into 800 μM OA condition. HLO were collected at day 3 for lipids live-cell imaging and at day 5 for stiffness measurement.
(64) Whole mount immunofluorescence. HLO were fixed for 30 min in 4% paraformaldehyde and permeabilized for 15 min with 0.5% Nonidet P-40. HLO were washed by 1×PBS three times and incubated with blocking buffer for 1 h at room temperature. HLO were then incubated with primary antibody; anti-alpha smooth muscle actin antibody (1:50 dilution; abcam) overnight at 4° C. HLO were washed by 1×PBS and incubated in secondary antibody in blocking buffer for 30 min at room temperature. HLO were washed and mounted using Fluoroshield mounting medium with DAPI (abcam). The stained HLO were visualized and scanned on a Nikon A1 Inverted Confocal Microscope (Japan) using 60× water immersion objectives.
(65) RNA isolation, RT-qPCR. RNA was isolated using the RNeasy mini kit (Qiagen, Hilden, Germany) Reverse transcription was carried out using the SuperScriptlll First-Strand Synthesis System for RT-PCR (Invitrogen, CA, USA) according to manufacturer's protocol. qPCR was carried out using TaqMan gene expression master mix (Applied Biosystems) on a QuantStudio 3 Real-Time PCR System (Thermo Fisher Scientific Inc.). All primers and probe information for each target gene was obtained from the Universal ProbeLibrary Assay Design Center (https://qpcr.probefinder.com/organism.jsp). Significance testing was conducted by Student's t-test.
(66) HLO stiffness measurement by AFM. HLOs treated with 0, 50200, 1400, 2800 ng/mLμM LPSOA were used for stiffness measurement with a AFM (NanoWizard IV, JPK Instruments, Germany). The AFM head with a silicon nitride cantilever (CSC37, k=0.3 N/m, MikroMasch, Bulgaria) was mounted on a fluorescence stereo microscope (M205 FA, Leica, Germany) coupled with a Z-axis piezo stage (JPK CellHesion module, JPK Instruments, Germany), which allows the indentation measurement up to the depth of ˜100 μm. As a substrate for organoids, a fibronectin-coated dish was used. The tissue culture dish (φ=34 mm, TPP Techno Plastic Products, Switzerland) was first incubated with a 1 μg/mL fibronectin solution (Sigma)) at 4° C. for overnight. Then, the tissue culture dish was washed twice by distilled water and dried for 1 hour. Thereafter, HLOs incubated with OALPS for 51 days were deposited to the fibronectin-coated dish and incubated for 1 hour at 37° C. The sample dish was then placed onto the AFM stage, and force-distance curves in a 14×14 matrix in a 25×25 μm square were measured from each HLO. Finally, Young's moduli (E, Pa) of HLOs were determined by fitting the obtained force-distance curves with the modified Hertz model (Sneddon, 1965). Dunn-Holland-Wolfe test was performed for significance testings.
(67) THP-1 cell migration assay. THP-1 cell, which was gifted from T. Suzuki, was maintained in Advanced DMEM/F12 (Thermo Fisher Scientific Inc.) containing 10% FBS. THP-1 floating cells were collected, and 200,000 cells were added with serum-free Advanced DMEM/F12 to the membrane chamber of the CytoSelect™96-Well Cell Migration Assay (5 μm, Fluorometric Format; Cell Biolabs, CA, USA). 10-20 HLO were cultured in HCM media including 0, 400, 800 uM OA with an ultra-low attachment **96** multi-well plate (Corning) for three days. To define the exact number of organoids in each well and lastly normalize the final migrated cells by the number, the organoids were captured on The KEYENCE BZ-X710 Fluorescence Microscope. 150 μL of culture supernatant of HLO was collected and added to the feeder tray of the kit. The kit was incubated at 37° C. for 24 h in a 5% CO.sub.2 cell culture incubator. Cells that had migrated were counted using Countess II FL Automated Cell Counter (Thermo Fisher Scientific Inc.). Significance testing was conducted by Student's t-test.
(68) Triglyceride assay. For quantitative determination of triglycerides, HLO were isolated from one Matrigel drop and divided into HCM media in the presence or absence of oleic acid (800 μM). They were cultured on an ultra-Low attachment 6 multi-well plate for three days. Quantitative estimation of hepatic triglyceride accumulation was performed by an enzymatic assay of triglyceride mass using the EnzyChrom Triglyceride assay kit (Bioassay Systems, CA, USA).
(69) HLO survival assay. HLO were collected from Matrigel and washed by 1×PBS. 30-40 organoids were cultured on an ultra-low attachment 6 multi-well plate (Corning). HLO were captured on The KEYENCE BZ-X710 Fluorescence Microscope every day. The surviving and dead organoids were counted manually from the photo. HLO with a rounded configuration was counted as the surviving while the organoids with out of shape is counted as dead. To assess the survival rate of OA treated HLO at the same time point, 3D cell titer glo assay was used (Promega, Wi, USA).
(70) Statistics and reproducibility. Statistical analysis was performed using unpaired two-tailed Student's t-test, Dunn-Holland-Wolfe test, or Welch's t-test. Results were shown mean±s.e.m.; P values<0.05 were considered statistically significant. N-value refers to biologically independent replicates, unless noted otherwise.
(71) All percentages and ratios are calculated by weight unless otherwise indicated.
(72) All percentages and ratios are calculated based on the total composition unless otherwise indicated.
(73) It should be understood that every maximum numerical limitation given throughout this specification includes every lower numerical limitation, as if such lower numerical limitations were expressly written herein. Every minimum numerical limitation given throughout this specification will include every higher numerical limitation, as if such higher numerical limitations were expressly written herein. Every numerical range given throughout this specification will include every narrower numerical range that falls within such broader numerical range, as if such narrower numerical ranges were all expressly written herein.
(74) The dimensions and values disclosed herein are not to be understood as being strictly limited to the exact numerical values recited. Instead, unless otherwise specified, each such dimension is intended to mean both the recited value and a functionally equivalent range surrounding that value. For example, a dimension disclosed as “20 mm” is intended to mean “about 20 mm.”
(75) Every document cited herein, including any cross referenced or related patent or application, is hereby incorporated herein by reference in its entirety unless expressly excluded or otherwise limited. The citation of any document is not an admission that it is prior art with respect to any invention disclosed or claimed herein or that it alone, or in any combination with any other reference or references, teaches, suggests or discloses any such invention. Further, to the extent that any meaning or definition of a term in this document conflicts with any meaning or definition of the same term in a document incorporated by reference, the meaning or definition assigned to that term in this document shall govern.

(76) While particular embodiments of the present invention have been illustrated and described, it would be obvious to those skilled in the art that various other changes and modifications can be made without departing from the spirit and scope of the invention. It is therefore intended to cover in the appended claims all such changes and modifications that are within the scope of this invention.

Claims

1. A method of treating or reducing the occurrence of a steatohepatitis disorder, including NASH, parenteral nutrition associated liver disease (PNALD), and a genetic form of liver disease, comprising the step of administering a composition comprising obeticholic acid and total parenteral nutrition or peripheral parenteral nutrition to an individual in need thereof, wherein the total parenteral nutrition or peripheral parenteral nutrition comprises a lipid emulsion, and wherein said composition is administered parenterally.
2. The method of claim 1, wherein said composition is administered in an amount sufficient to reduce or prevent the occurrence of liver cell fibrosis.
3. The method of claim 1 wherein said composition **1***s* administered prior to, following, or during administration of parenteral nutrition to said individual in need thereof.
4. The method of claim 1 wherein said composition further comprises a buffer.
5. The method of claim 1 wherein said composition further comprises a carrier.
6. The method of claim 1 wherein said composition **1***s* administered intravenously.
7. A parenteral composition comprising a. total parenteral nutrition (TPN) or peripheral parenteral nutrition (PPN), wherein the total parenteral nutrition or peripheral parenteral nutrition comprises a lipid emulsion; and b. obeticholic acid.
8. A kit comprising a container, the composition of claim 7 in a dosage form disposed in the container, and instructions for administering said dosage form to an individual diagnosed or suspected of having or developing PNALD.